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Introduction

1

IN THIS CHAPTER

Give a model an untreated cell and the name of a drug, and ask it what that drug will do to that cell. That is the task this thesis evaluates. It is worth answering because the answer is otherwise obtainable only by running the experiment.

The thesis takes one published cell-sentence checkpoint, fine-tunes it on a hundred-million-cell drug atlas, and asks whether the predictions it produces identify the drug that produced them. They do not. Most of what follows is the apparatus that makes that a finding and not an anecdote: a metric calibrated before it was trusted, controls a drug-blind predictor cannot pass, and a bar that needs no model at all.

- 1.1 The prediction task
- 1.2 The atlas and the unit of inference
- 1.3 Cell-sentence models and drug-blindness
- 1.4 Research questions
- 1.5 Evaluation as identification
- 1.6 The baseline: a per-drug lookup

■ EXPR / ▲ RESID
each section declares the frame in force
beside its heading

1.1 The prediction task

A drug that enters a cell changes what that cell transcribes. Which genes move, by how much, and in which direction is the readout the field treats as its most direct account of what a drug does. That is a working premise, not an established fact. A substantial share of the treated conditions in the atlas used here are statistically inert or not empirically discriminable at the tested pseudobulk and cell budget (chapter 3), and cannot support this evaluation.

Reading one response is routine; reading every combination that might matter is not, because a drug behaves differently across doses and cellular backgrounds, and even the largest screens sample a small corner of the space. A computed response would make the untested

combinations inspectable, which is what prediction is for. The task is simple to state: given the transcriptome of an untreated cell and the identity of a drug, predict the transcriptome that cell would have shown had the drug been applied.

It is harder to grade than to state, because the ground truth is counterfactual. Sequencing destroys the cell it reads, so a prediction can only be checked against a *different* cell that received the drug. Paired training data must therefore be assembled by matching cells, evaluation is framed at population level, and distinguishing a drug effect from ordinary cell-to-cell variation requires explicit controls. The units, the frames and the controls that follow all descend from that one fact.

1.2 The atlas and the unit of inference

Tahoe-100M [1] makes that evaluation possible at unusual scale, with roughly one hundred million single-cell transcriptomes spanning about 1,100 small-molecule perturbations, 379 compounds at several doses each, across 50 cancer cell lines. Each condition, one drug at one dose in one cell line on one plate, is a population and not a single summary.

A population per condition admits two questions. The first, the conditional one, is what the task asks: what a drug does in a particular cellular context. The second is the prior one, whether the apparent response is there to be predicted at all, and how much of it reproduces across disjoint halves of the same treated well. Standard evaluations blur the two; much of this thesis keeps them apart.

Scale does not remove design limitations. Conditions are spread unevenly across drugs, doses, cell lines and treated wells, and plate membership can change which alternatives a rank-based score sees. Cells from the same well are not independent biological replicates, and neither are drug rows sharing a cell line.

SEE chapter 3: the atlas layout, and which unit each interval is clustered on

1.3 Cell-sentence models and drug-blindness

Cell2Sentence [2], our main object of study, writes a cell as its gene names, in English, in descending order of expression. This turns a sequencing readout into a form that any ordinary language model can consume. It discards expression magnitudes and keeps ranks; in exchange it needs no specialised architecture and any off-the-shelf language model can be used. C2S-Scale [3] extends it to models pretrained on more than 50 million cells from 825 single-cell datasets, fine-tuned across classification, generation, captioning and perturbation tasks. Its published perturbation work covers cytokine stimulation, scored as distributions rather than condition by condition, and, for small molecules, bulk L1000 profiles in a transfer test across cell lines.

cell sentence A cell written as its expressed genes, named in descending order of expression.

The checkpoint studied here is C2S-Scale-Pythia-1b-pt, the pretrained Pythia-1b model [4] from that family, with no drug-treated cells in its declared pretraining corpus. Fine-tuning uses several hundred thousand Tahoe cells, not the entire atlas. Every experiment arm we perform that requires different training is cold-started from that checkpoint with identical hyperparameters, so within an experiment the training target or objective is the only variable. The prompt supplies the cell line, the drug, the dose, a mechanism annotation and the untreated control cell's own sentence, and what the model returns for that prompt is the prediction that gets scored.

The training target separates the two principal arms that we analyze throughout this work, and with it the frame each is scored in. The first arm predicts the treated cell's own sentence and is scored in what we call the expression frame (■ **EXPR**), where a prediction is a cell profile ranked against other drugs on the same plate. The second arm predicts the drug-specific residual, meaning what survives once the drug-

agnostic mean response, the *generic*, is subtracted from the treated-minus-control shift. It is scored in the residual frame ($\blacktriangle$ RESID), where a prediction is a drug-specific signature ranked against other drugs in the same cell line. Both frames put chance in the same place, and neither is interchangeable with the other. Which is in force is declared at the head of every section.

The appeal of the encoding is also the scientific risk. A ranked profile leads with its most abundant genes, and those change little with the drug: between two drugs tested in the same context, only 34.6 of the 200 highest-ranked genes differ on average (Table 4.6). Treated cells retain most of the state of their untreated lineage, and many perturbations share a generic response programme. A model can therefore produce a plausible treated cell while remaining almost indifferent to which drug was named.

A held-out drug is not unseen by the whole pipeline. In natural-language pretraining the language backbone may have encountered drug names and their pharmacology, and every prompt supplies the drug name plus a Mechanism field, an annotation where one exists and the literal string unclear otherwise. “Held out” therefore means absent from Tahoe fine-tuning, not absent from the model’s prior world knowledge or from the inference prompt.

SCOPE LIMIT carried into every held-out result in chapter 4 and chapter 5

Holding data out comes at three levels of difficulty, not one. Trained conditions test whether the model can reproduce information available during fine-tuning. Unseen combinations hold out a whole treated well but often recombine a drug and a cell line each encountered separately. Unseen drugs remove all retained fine-tuning conditions for those drugs. These are not rungs on one clean learning curve. They differ in support and in empirical reliability, and only the last asks for a response without a Tahoe-trained per-drug memory. The thesis therefore avoids calling all held-out examples zero-shot and never pools the three strata into one transfer number.

1.4 Research questions

This thesis extends the cell-sentence line of work to chemical perturbation on Tahoe-100M and asks whether a model fine-tuned on this atlas learns what a particular drug does to a particular cell, that is, does the model learn drug-specific responses or does it simply learn a generic perturbation programme. A negative answer earns its place only if it can be made specific, so the investigation asks four questions in order, each one raised by the answer to the one before it.

- 1 **What is the best way to evaluate the model's predictions?** The training objective cannot answer this question: next-token cross-entropy against the true cell sentence rewards a plausible cell and says nothing about whether it is *this* drug's, so the whole burden falls on the evaluation. To find the right metric we keep this question in mind: Can a predictor carrying no drug information score well on the field's standard metrics, and which metrics survive that control?
- 2 **Does the model read the drug?** Measured on whichever metric survives the first question, with the input cell fixed and only the prompt's drug changed, does the prediction move at all?
- 3 **Why does it behave that way?** Is the drug absent from the data, absent from the model's internal representation, or present in the representation but not consulted when the model generates?
- 4 **Does redefining the effect repair it?** If the target is re-encoded as the drug-specific residual instead of the treated cell's own sentence, does the model become sensitive to the drug in its prompt?

The order is not presentational. A metric a zero-information predictor can saturate cannot grade anything asked after it, and four of the five metrics we tried fail the corresponding control: the field's standard

score is topped outright by a predictor carrying no drug information at all. The survivor is a retrieval test, defined next, and it is the repaired metric every later answer is read from. The null the second question returns on it is worth little until the third locates it: drug identity is written into the activations and barely consulted when the model generates. That diagnosis is what motivates the fourth, which changes neither the model nor the data but the definition of what the model is asked to produce.

1.5 Evaluation as identification

Distributional similarity and token likelihood reward a prediction for looking like a treated cell, even when the condition identity is wrong. The test left standing is harder, and it is a retrieval test: among the other drugs measured in the same context, does a prediction identify its own treatment?

DEFINITION Normalised inverse rank (NIR)

A prediction is compared with its own condition's truth and with the truths of the other conditions in a declared comparison set. NIR is the fraction of that set whose truth the prediction resembles *less* than it resembles its own. The comparison set is what fixes chance at 0.50, in both measurement frames and under an exchangeability condition that § 4.1 audits with empirical control arms. The score is tie-aware, and any component shared by every candidate cancels.

The metric is not infallible: its value depends on the gallery a prediction is ranked against, it can compress near one, and its uncertainty must respect the cell lines, drugs and wells that repeat in the design. What it will not do is pass a prediction for resembling a generic treated profile, because every treated profile does that.

1.6 The baseline: a per-drug lookup

Two questions are routinely conflated in perturbation prediction. One is whether a drug has a characteristic response that transfers across cellular contexts. The other is whether a model uses the untreated cell to predict how *this* context departs from that average.

The first has a model-free answer. A per-drug lookup predicts a drug’s residual as its mean residual over training conditions in other cell lines. It needs no checkpoint and no prompt, and it measures the shared component directly. The residual target has substantial observed squared norm. Normalising by the treated-minus-control shift’s squared norm gives an observed residual share of 0.621, an estimated generic share of 0.390 and a cross term of -0.011 , which sum to one. The 62.1% residual and 39.0% generic shares describe this squared-norm scale, not identified fractions of a biological response: the residual can also carry sampling noise, baseline-estimation error and context effects. The small cross term supports approximate orthogonality on this aggregate measure, rather than an exact biological decomposition.

FRAME ▲ RESID scores here are on the residual after generic subtraction, § 4.7

The second is what a lookup cannot answer, because a per-drug constant makes the same prediction in every context. How much of a drug’s residual is shared across contexts is measured in § 4.8 by a disattenuated transfer coefficient, $\mathcal{T} = 0.553 [0.514, 0.593]$. The complementary 44.7% [40.7%, 48.6%] of that similarity scale is not shared across the compared contexts. It is not a drug-by-cell-line interaction and not a variance share; the estimator mixes cell line, dose, well and target-estimation differences at once. It bounds what a per-drug constant leaves unresolved without establishing that a model could recover any of it.

The win condition, fixed in advance, is therefore demanding. A conditional model must beat the training-only drug lookup on common support; only then would its access to the control cell have supplied information the dictionary lacked. Beating chance would not do it, and neither would beating a drug-agnostic control.

It was not met. On the 1,192 conditions where model and lookup are both defined, the lookup scores 0.913 against the model's 0.549. Its advantage is not merely that it averages many cell lines; a lookup restricted to one other cell line still scores 0.822. Re-encoding the target does make the model sensitive to the drug named in its prompt, which is the project's first measurable trained-drug effect. Responding to the drug is not the same as predicting what it does: on held-out treated wells the prompt sensitivity survives, and biological transfer is not established. Nothing built here reaches the dictionary.

SEE § 4.11: the full baseline ladder, scored through one identical pipeline

SCOPE LIMIT carried into § 4.1, § 4.11
and chapter 5

The within-well reference is a measuring instrument, not an oracle ceiling. It asks how well one half of a treated population retrieves the other among competing drugs. A model or a lookup may exceed it because it averages more cells, borrows information across lines, or is compared with a noisy half-sample; NIR also compresses close to one. The reference establishes empirical discriminability at a stated cell budget and comparison set. It does not state the maximum predictable fraction of biology, and it is recomputed within each stratum and never transported across them.

Every conclusion that follows is re-tested under four changes to the measurement: comparison sets are plate-matched, uncertainty is re-clustered on drug, token budgets are equalised, and generic components are fitted without the holdout they later grade. Not every conclusion survives them. The protein-target channel, for one, clears the pre-set relevance margin as a route to an unseen drug while uncertainty is clustered on cell line and well, and stops clearing it once it is re-clustered on the drug. A larger backbone was not tested, and a replication with one is informative only if it receives the same split, targets, sentinels and token budgets; is scored against the same within-plate galleries and training-only lookups; and carries the same validity and completion checks. Otherwise architecture, tokenisation, optimisation and evaluation change together. This thesis does not supply that replication, but it leaves an auditable ladder into which one can be inserted without weakening the controls that produced the negative result.

THE CHAPTERS. Chapter 2 reviews single-cell foundation models and whether deep perturbation models beat simple baselines. Chapter 3 describes the atlas, model and measurement rules; chapter 4 presents the investigation chronologically; chapter 5 distinguishes measured conclusions from untested questions; and chapter 6 concludes. Symbols, terms and conventions are collected in Reference apparatus at the back, for reference as needed.

Background

2

IN THIS CHAPTER

This chapter is not a survey, but the shortest route to what this field measures, what those measurements assume, and where they give way^{2.1}

2.1 Transformers on single-cell data

A recipe built for text is now used to write cells. What made it portable,^{2.6} and what does adopting it commit you to? It asks almost nothing of^{2.7} language, and everything it does ask falls on how the data are written down.^{2.8}

The transformer recipe requires only sequences of discrete tokens, some predictable from others. Nothing there mentions words, which invited the transfer into biology. scGPT [5] and Geneformer [6] pretrain on tens of millions of cells written as gene-token sequences, toward some version of the *virtual cell* [7], whose simulated responses could substitute for an experiment.

That sets the bar for everything measured later, but it is not where the models differ. A cell is a vector over some twenty thousand genes; a transformer consumes discrete tokens. What must be discarded in between sets a bound on what the model can learn.

Transformers on single-cell data

2.2 Cell2Sentence and C2S-Scale

2.3 Latent-space perturbation models

2.4 Baselines and metric calibration

2.5 DrEval: pseudoreplication, bias and splits

Cross-context transfer of drug responses

Probing versus intervention

Optimal transport as a target constructor

■ EXPR / ▲ RESID

each section declares the frame in force beside its heading

2.2 Cell2Sentence and C2S-Scale

If a cell has to be written as tokens, something must go. Which choice does the encoding make, and what argument says the loss is safe? Rank is kept, magnitude is thrown away, and the defence is a reconstruction argument.

Cell2Sentence [2] takes the most literal route available. Rank a cell's genes by expression, write the symbols out in descending order, and the cell is a string of ordinary text, a *cell sentence*. Rank is the whole of the representation, which symbols appear and in what order. Magnitude is not written down at all.

```
GAPDH RPS6 KTN1 EXT1 PTK2 MACF1 HSPD1 ALDOA GNAS CD44  
. . .
```

The same literalness extends to everything the model is given. Because a cell sentence is ordinary text, the prompt that surrounds it is ordinary text too (Figure 2.1): one English instruction naming the cell line, the drug, the dose and the mechanism annotation, then the untreated cell written as a sentence, then an open-ended cue that the model completes with a sentence of its own. Nothing about the cell is handed to the model as a vector, and no part of the architecture is reserved for it. That is what makes an off-the-shelf language model usable here without modification, and it is the property the encoding buys: the model can bring to bear whatever it learned about drug names, mechanisms and gene symbols in natural-language pretraining, and it reasons about the cell's sequencing in the same medium it reasons about anything else, as a sequence of words. Which string it is trained to complete the prompt with is the one thing that varies across this thesis, and Figure 2.1 shows the two it is asked for.

PROMPT

```
Predict the response of PANC-1 to Docetaxel at 0.05 uM . Mechanism:
Microtubule inhibitor .
Control cell: GAPDH RPS6 KTN1 EXT1 PTK2 MACF1 HSPD1 ALDOA GNAS CD44 . . . [END_CELL]
Response cell:
```

one instruction line in English, the untreated cell as a sentence, and an open-ended cue the model completes

STANDARD TARGET (■ EXPR)

```
GAPDH RPS6 HSPD1 CDKN1A ALDOA GNAS . . .
[END_CELL]
```

the treated cell's own sentence: ranked genes, most abundant first

↓ *the model continues the text; the target decides with what*

RESIDUAL TARGET (▲ RESID)

```
CDKN1A DDB2 MDM2 . . . [DOWN] HMOX1 TXNIP . . .
[END_CELL]
```

the drug-specific residual of § 3.5: up-regulated genes, a separator, down-regulated genes

FIGURE 2.1 The model sees plain text and returns plain text. The prompt is the literal template every arm uses: an instruction line whose four shaded spans carry the cell line, the drug, the dose and the mechanism annotation, the control cell as a sentence closed by [END_CELL], and the cue Response cell:. The four values are real atlas entries; the gene lists are illustrative and carry no measurement. Below, the two strings the model is trained to continue with: the treated cell's own sentence in the expression frame, and the signed drug-specific residual of § 3.5 in the residual frame, whose two blocks are split by [DOWN]. Both frames share the prompt to the byte; only the target differs.

WHAT THE DEFENCE COVERS. C2S defends the discarded magnitude by reconstruction. Expression can be recovered from rank through a fitted log-linear relationship $e_i = a_d \log(r_i) + b_d$, achieving $R^2 = 0.815$ against the original values on peripheral blood data. Rank, they argue, therefore encodes expression reversibly enough for the loss not to matter downstream.

The criterion is absolute, not discriminative. Reconstruction fidelity asks how close the recovered profile is to the true one, which is not the same as preserving what distinguishes one condition from another. That distinction runs through the whole thesis. The map is not only C2S's defence of the discarded magnitude but also part of this thesis's own measurement path: in the expression frame a prediction and every truth it is scored against are decoded through the same fitted relationship before any distance between them is taken (§ 4.1).

THREE PROPERTIES THAT MATTER LATER. C2S-Scale builds on decoder-only transformers of the GPT family, the checkpoint used here on Pythia [4]. Token embeddings form a *residual stream*, one vector per position that every layer reads from and writes back into additively, so a representation can be read at depth and modified. Training weights every token position equally, so a target largely shared between

SEE § 4.5 performs the intervention this design permits

conditions gives any one condition only a small share of the gradient. And conditioning runs entirely through attention across a causal mask.

THE FAMILY AND THE CHECKPOINT. C2S-Scale [3] spans backbones from Pythia at 410 million and 1 billion parameters up to Gemma-2 at 27 billion, pretrained on over 50 million human and mouse cells drawn from CELLxGENE and the Human Cell Atlas. None of that pretraining is a perturbation task: the objectives are cell sentence generation, cell type and tissue prediction, and text tasks over the accompanying metadata. Perturbation prediction enters only afterwards, as a separate fine-tuning stage, and it is evaluated on two benchmarks that ask different questions.

The first is single-cell. Immune cells are exposed to cytokines; the model receives an untreated cell's sentence together with a cell type, a cytokine and an exposure duration, and must generate the treated cell's sentence while recovering those three labels. Held-out conditions are graded into difficulty tiers by how far they sit from what training contained.

The second is bulk, and it is a transfer test across cell lines rather than across compounds. L1000 reads a fixed panel of 978 landmark genes from a treated well, one profile per compound and cell line pair. The profiles are quantile-normalised, and for each compound half of the cell lines are held out. The compound itself is therefore present in training, applied to other lines, and the question is whether the model can carry its effect onto a line it has never seen treated with it.

Two features of that design bear on the comparison later. Pairing is by group and not by cell: in the bulk arm treated and untreated profiles are matched on cell line name alone, which is unavoidable once sequencing destroys the cell it reads. And the single-cell results are scored as distributions, generated population against observed and

assessed collectively rather than condition by condition, which asks whether the model produces a plausible population and never whether it tells one condition apart from another.

The checkpoint fine-tuned below, C2S-Scale-Pythia-1b-pt, is the pretrained model rather than either of those fine-tunes, and no perturbation fine-tune of the family has been released. The starting point here is the authors' own, with none of their perturbation training already in it.

2.3 Latent-space perturbation models

The obvious way to predict a treated cell is not this one. What do the standard methods do differently? In every one except the model this thesis fine-tunes, the drug arrives through a channel the architecture was built to receive it on.

The direct approach predicts a treated cell as a displacement in a learned latent space. scGen [8] estimates the vector between control and perturbed populations in a variational autoencoder's latent space and adds it to a context where the perturbation was never applied. CPA [9] makes that arithmetic learnable, composing a basal cell state with additive embeddings for drug, dose and covariates; chemCPA swaps its one-hot drug embedding for a molecular-structure encoder, letting an unseen drug be embedded at all [10]. GEARS [11] reaches unseen genes through a gene–gene knowledge graph. CellOT instead fits a neural optimal-transport map from control to treated per perturbation [12, 13].

TWO PROPERTIES THEY SHARE. The prediction is a vector in a continuous space, emitted in one pass by a decoder trained on a loss written directly on that vector. And the drug enters on a channel built

to receive it, an embedding in CPA, chemCPA and GEARS, a latent offset in scGen, a fitted map in CellOT.

A cell-sentence model does neither. It samples a token sequence left to right, carries the drug as a few words of prompt, and weights every token position equally in its loss. In the standard cell-sentence target, 83% of the top-200 gene positions are identical between any two drugs in the same context (§ 4.7); a gene spans several sub-tokens, so that count is not a count over loss positions. The string to be emitted is nearly the same string whichever drug was named, and a latent-transition model has no target with that property.

Whether that property is the binding constraint is not settled here. § 4.9 reports what re-encoding the target does, and stops short of establishing that tokenisation is what binds.

SCORED DIFFERENTLY. The latent-transition literature is evaluated condition by condition against the observed population: distributional distances, R^2 on pseudobulk, DE-gene overlap. Figure 2.2 draws where that diverges from a discriminative question.

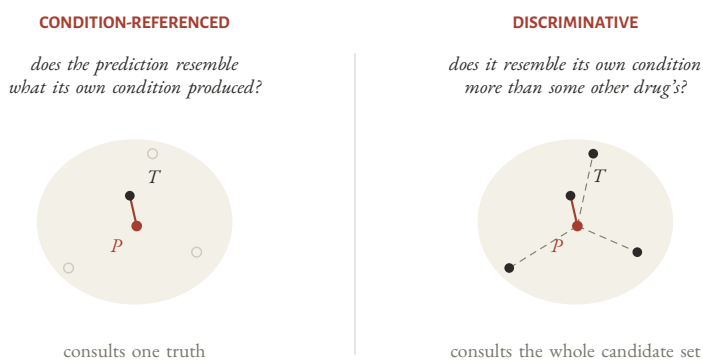


FIGURE 2.2 The same prediction, the same truths, two different questions. P is a prediction for one condition, T is that condition's own truth, and the other three points are rival candidates, all inside the pale region when the part of the shift shared across drugs dominates. A condition-referenced metric consults T alone, so the rivals are hollow on the left; a discriminative metric grades where T ranks among them. Point positions are identical in both panels. No axis carries a value.

WHERE THE FIELD HAS CONSOLIDATED. scPerturb [14] harmonises the datasets and standardises distances between populations, and PerturBench [15] supplies the exception this thesis relies on, a rank-based score grading a prediction against a set of candidate conditions

and not one truth. None of these is run as a baseline here; the results chapter concedes the missing head-to-head as a real gap.

What is reported below is not evidence about CPA, chemCPA or CellOT, none of which was run here. The architectures named in this section supply the *motivation* for the baseline built in § 4.11 — they tell us what a drug-specific channel typically carries — and they are not competitors that have been scored and beaten.

WHAT IS OFFERED INSTEAD. Something narrower, measured and not argued. In scGen and in CPA the drug-specific content carried into a new context is a single per-drug vector, dose-scaled in CPA's case. § 4.11 measures what such a vector is worth by looking the drug up: the mean residual of the same drug in other cell lines, fitted from training conditions only.

On the conditions where both are defined it scores 0.913(NIR). On the held-out wells, where the lookup cannot borrow the target's own well, it scores 0.890(NIR). Chance is 0.50 under the exchangeability condition audited in § 4.1, so both comparisons sit far from chance, and what the section turns on is the ordering, not the size of either support.

That is not a score for CPA or chemCPA, whose decoders make the emitted shift context-dependent even when the drug channel is one vector. But it does say which question is informative. Not which architecture wins. Whether anything improves on a per-drug constant.

SCOPE LIMIT carried into § 4.11 and chapter 5

FRAME ▲ RESID the scores below are residual-frame; see § 4.7

SEE § 4.11 builds the ladder these scores sit on

2.4 Baselines and metric calibration

Does any of these deep learning models work? The field has answered both ways, and what decides which answer to believe is not another model but an audit of the metrics. Both camps agree the standard metrics are defective; they disagree about what a repaired measurement would reveal.

ONE SIDE, TRIVIAL PREDICTORS ARE NOT BEATEN. Ahlmann-Eltze, Huber and Anders [16] find that deep-learning prediction of genetic perturbation effects does not outperform linear baselines. Csendes et al. [17] reach the same conclusion for foundation cell models, Bendidi et al. [18] report that a single PCA still rules them all, and Kernfeld et al. [19] concur across expression-forecasting methods. Nor is this confined to transcriptomic readouts. Hauptmann and Kramer [20] hold the protocol fixed and watch the differences between architectures dissolve; DrEval [21] finds deep models barely beating a mean drug and cell-line predictor.

THE OTHER SIDE, THE METRICS WERE AT FAULT. Miller et al. [22] attribute much of the apparent underperformance to weaknesses in the evaluation metrics rather than to the models' predictive capabilities. They formalise positive and negative controls in perturbation benchmarking and introduce the *interpolated duplicate*, blending a technical duplicate with the mean baseline by per-gene differential-expression significance.

interpolated duplicate A positive control of known and tunable quality: a technical duplicate blended with the mean baseline.

DEFINITION Dynamic range fraction (DRF)

A quantitative measure of whether a metric can see quality at all. Given reference points whose quality is known and tunable — the interpolated duplicate supplies them — DRF grades the metric on how much of that known difference in quality it registers. It is a property of the metric, not of any model scored with it.

Miller et al. apply it across 14 Perturb-seq datasets and 13 metrics. MSE and control-referenced Pearson correlation come out poorly calibrated; weighted MSE, weighted $R^2\Delta$ [23] and Normalised Inverse Rank [15] are consistently calibrated. On the metrics that survive, they report, deep models *do* beat mean, control and linear baselines.

A THIRD RESULT, ON THE SAME SIDE OF THE FIELD. Viñas Torné et al. [24] agree with the first camp about what is found and with the second about how to look, which is what makes their result more than another vote. Across ten genetic-perturbation datasets — three technologies, five cell lines, from 81 to 1,813 perturbations — two trivial baselines match or beat CPA, GEARS and scGPT. The *perturbed mean* predicts the average of every perturbed cell in the training set, the same profile for all perturbations, and it wins on Pearson- Δ across every dataset they test.

The diagnosis is what matters here. Those baselines win because perturbed and control cells differ in a direction that is shared across perturbations, so a prediction reproducing only the shared direction already scores well against a control-referenced target.

DEFINITION Systematic variation

The consistent transcriptional difference between perturbed and control cells that is common to perturbations rather than specific to any one of them. Viñas Torné et al. measure it as the cosine between a perturbation's own shift from the control centroid and the average perturbation effect, averaged over perturbations. It is a property of a dataset, not of a model.

In their data it is neither small nor incidental. It ranges from about 0.2 to 0.7 across the ten datasets and traces to concrete causes: panel selection — Norman et al. [25] chose genes in cell-cycle and erythroid

pathways — and the perturbation technology itself, with CRISPR-induced chromosomal instability placing 46% of perturbed RPE1 cells in G1 against 25% of controls. Across datasets the measured systematic variation tracks reported performance at $r = 0.91$ for GEARS and 0.95 for fine-tuned scGPT. To that extent a benchmark ranking is reporting a property of the data.

Their repair is one change, and it is a change to the measurement rather than to any model: score against the *perturbed centroid*, the average of the per-perturbation centroids, instead of against the control centroid. On that reference the numbers collapse — scGPT on Norman from Pearson Δ 0.69 to 0.13, GEARS from 0.65 to 0.13 — and their correlation with systematic variation goes with them. Repairing the reference did not vindicate the models. It is again, though, a genetic-perturbation answer.

WHAT IS ACTUALLY IN DISPUTE. One reading is that the models were always capturing real biology and the broken metrics hid it; the other is that they were never capturing much, and better measurement will show that failure more clearly.

Either way the question is empirical, and settled only in the setting where it is tested. Miller et al. and Ahlmann-Eltze et al. both study **genetic** perturbation, a designed near-binary knockout with a named target gene. DrEval studies **drug sensitivity** as a scalar potency such as IC_{50} . None takes as its object a full transcriptomic response to a drug, which acts at a dose and engages several proteins at once, at single-cell resolution, from an autoregressive generative model. The models of § 2.3 do work there. What we are not aware of anyone having done is to run the calibration audit first, letting it decide which metrics are admissible before any model is scored.

THE POSITION TAKEN HERE. § 4.1 ports the DRF framework onto drug perturbation, so the verdict cannot be dismissed by the objection Miller et al. themselves raise. The framework and its interpolated-duplicate positive control are inherited; not every choice inside them is. The surviving metric uses a similarity function we picked, Euclidean distance in expression space and not over rank vectors, and the two variants disagree. The family graded is our selection of five, including our own implementation of their weighted R^2 variant.

TWO SEPARATIONS FROM MILLER AND COLLEAGUES. Only the first is a metric-level disagreement, and it begins with weighted $R^2\Delta$, which they endorse as calibrated but which is negative in both arms under our implementation. Its sign tracks cell count, so § 4.1 draws no directional conclusion from that negative, but this member of their calibrated set does not transfer intact. Their endorsed NIR does survive. `panel- $\tau$` and `spearman_expr`, neither in that set, both invert in this field-wide audit; that failure does not generalise to NIR or to rank-based metrics as a family.

The second separation concerns models. Miller et al. report that deep models beat uninformative baselines once the metric is fixed; on the metric their analysis endorses, the checkpoint evaluated here does not (§ 4.2). Both can stand. A repaired metric is a precondition for seeing skill, not a guarantee that there is skill to see. Which difference accounts for the split, genetic against chemical perturbation, regression against generation, or the per-drug baseline, is taken up in § 4.1 and § 4.11.

THE TARGET THIS THESIS INHERITS FROM SYSTEMA. § 4.7 is Viñas Torné et al.’s reference change, carried into the training target as well as the score. Carrying it there invites an objection better answered

before the construction is used than after. A checkpoint trained to emit residuals is then scored in the residual frame, which is to say graded in the representation it was optimised for. This looks like a problem and is not one, because the claim resting on this frame is not a level but a gap: one checkpoint against its own scrambled prompt, on the same conditions, in the same frame, with the same generic subtracted and the same scoring rule applied, differing only in which drug the prompt names. Whatever a shared frame confers is present in both terms and subtracts out. The metric supplies that protection a second time, since NIR ranks a prediction against the other drugs of its own group and any component common to that comparison set cancels from the ranking — the generic among them, and the error in estimating it, because the generic is fitted on exactly the group the comparison set is drawn from. A model that had learned what a residual looks like and nothing about which drug it was named would sit at 0.500 and not above it. What the shared frame does buy is precision and not position: the drug-specific part is a larger share of what is being compared, so the same ranking is read with less noise. § 4.1 fixes the metric before any arm is scored, and § 4.9 states the scope the reliability filter imposes on the population it is read over.

Their perturbation-specific effect is the perturbation’s own centroid measured against the average of the per-perturbation centroids, fitted over the training perturbations,

$$\Delta_X = O(X) - O_{\text{pert}}, \quad O_{\text{pert}} = \frac{1}{|\mathcal{P}_{\text{train}}|} \sum_{X' \in \mathcal{P}_{\text{train}}} O(X'). \quad (2.1)$$

The residual target we use is written as a control-referenced shift with a generic subtracted, $r(d, c) = (\bar{y}_{d,c} - \bar{y}_{\text{ctrl},c}) - \text{generic}(c)$, but the control cancels and what is left is the same object,

$$r(d, c) = \bar{y}_{d,c} - \frac{1}{D-1} \sum_{d' \neq d} \bar{y}_{d',c}. \quad (2.2)$$

Two further choices are theirs as well. The reference averages over *perturbations* and not over cells, so a drug run at five doses does not carry five times the weight of a drug run at one — the same distinction they draw between O_{pert} and the cell-weighted perturbed mean they use as a baseline. And it is fitted on the training split only, which is what § 4.7 enforces as split-before-fit.

THREE SEPARATIONS FROM VIÑAS TORNÉ AND COLLEAGUES.

Each is forced by the setting rather than preferred, and the setting differs in three ways that matter: many cell lines instead of one, scoring on trained conditions as well as held-out ones, and a generative model that emits text rather than a vector.

The first is scope. Their reference is one centroid for a dataset, which is well specified when the dataset is one cell line and one screen; Norman is K562, Replogle is RPE1 or K562. This atlas is 50 cell lines across many plates, and the shared response is context-dependent: a cytotoxic dose does not leave the same common signature in every line. A single global reference would subtract the wrong quantity nearly everywhere and leave cell-line and plate structure *inside* the residual — which is disqualifying here in particular, because NIR ranks a drug against other drugs *within* a context, so retained context structure would survive into the comparison the metric makes. The generic is therefore estimated within a (cell line, plate) group.

The second is that the drug is always excluded from its own reference. Their O_{pert} runs over training perturbations while their evaluation is on held-out ones, so the perturbation being scored is excluded already. This thesis scores trained conditions too (§ 4.9), and without an explicit leave-one-drug-out a trained condition would be centred on a mean containing itself while a held-out condition is not, so the two splits would be scored against differently defined targets. Since the compari-

son between them is the claim (§ 4.10), the two must be built the same way. The bias removed is of order $1/D$, which is larger here than in their datasets because a plate group holds fewer drugs than their panels hold perturbations.

The third is where the correction is applied, and it is the substantive one. Systema is an evaluation framework: the reference moves at scoring time and the models are left alone. That is sufficient when the question is whether reported scores are inflated. It is not sufficient here, for two reasons. § 4.7 finds that the shared component does not only inflate the score but dominates the *target*: 83% of the top-200 gene positions of a standard target string are identical between any two drugs in the same context, against 40% under the residual. A model trained on a target that barely encodes the drug cannot be repaired by rescoreing its output. And a cell sentence is a rank order of gene names, not a vector, so it does not support subtraction at all; recentring it at scoring time would require decoding back to expression through a channel that keeps rank and discards magnitude (§ 2.2). The arithmetic has to happen where arithmetic is still defined, which is on the target before it is tokenised.

A LIMITATION OF THEIRS, AND THE FILTER IT MOTIVATES.

Viñas Torné et al. note that a reference-sensitive metric is poorly suited to weak perturbations, because the shift has small norm and a few cells can swing its orientation, and they leave this open. Moving the reference into the target inherits the problem and sharpens it: a residual too small to estimate stably is not merely scored noisily, it is *trained on*. § 4.7 answers it with a reliability criterion — a condition enters only if its residual, estimated on two disjoint halves of its own cells, reproduces above a threshold fixed at a matched null. That filter is a measurement-

quality gate inherited from their limitation, not a performance one, and its retention rates are reported per split so the selection stays visible.

HOW MUCH SYSTEMATIC VARIATION THIS ATLAS CARRIES.

Their statistic is defined on a dataset, so it could in principle be computed here and read on the same scale as their ten. It was not computed for this thesis, and no value is quoted for it. What this thesis measures instead is the share of the treated-minus-control shift's squared norm that the generic programme carries, 0.390 across the cache (§ 4.8), read within the (cell line, plate) group that the first separation above argues the reference must be scoped to. That number matters for the argument rather than decorating it: the case for re-encoding the target rests on the shared component being large in *this* data, and that is a measurement, not an inheritance from their datasets.

2.5 DrEval: pseudoreplication, bias and splits

A benchmark can report a number that is arithmetically correct and still mislead about everything a reader cares about. Which failure modes are catalogued? Four of DrEval's six shape the apparatus directly.

DrEval [21] identifies six obstacles to clinical translation: non-reproducible models, data leakage producing weak generalisation, pseudoreplication, biased evaluation metrics, missing ablation studies, and inconsistent response data.

PSEUDOREPLICATION. In Hurlbert's original sense [26], this is treating repeated measurements of one unit as if they were independent replicates. GDSC2's 395,025 measurements descend from only 969 cell lines and 287 drugs, so an interval computed over measurements is badly overconfident.

Our data has the same shape, hundreds of thousands of cells resolving to a few thousand conditions across 50 cell lines, and no interval here is computed over cells. Three estimators stand in their place. An interval on a mean is clustered on the two units that repeat and are crossed, the cell line and the treatment well. A statistic over *pairs* of conditions uses a dyadic estimator, since neither grouping partitions a pair. And the two ratio statistics the two-way form is not written for use a one-way cluster bootstrap over cell lines, flagged where they appear.

BIASED EVALUATION. A version of Simpson’s paradox. Because drugs act at vastly different concentrations, most of the variance lies *between* drugs, so a model that learns nothing but each drug’s mean explains most of it. Their illustration is stark, a global Pearson of 0.90 alongside an average within-drug Pearson of 0.01. § 4.1 explains why a discrimination metric scored *within* a comparison set is immune by construction.

SPLITS. DrEval names four by application: Leave-Pairs-Out for imputation, Leave-Cell-Lines-Out for personalised medicine, Leave-Drugs-Out for drug design, Leave-Tissue-Out for repurposing.

Our holdout lands on one of those and deliberately misses another. `unseen_drug` is Leave-Drugs-Out. `unseen_combo` is *not* Leave-Pairs-Out, and that name is reserved here for the estimand this thesis declines: holding out a (drug, cell line) pair leaves the pair’s own treated well in training through its other ≈ 48 cell lines, measuring within-atlas matrix completion and not generalisation. Our split withholds the whole treatment well, strictly the stronger of the two, and answers a narrower question, an unseen well of a seen drug.

SEE § 4.10 names the estimand this thesis declines, and why

Citing Partin et al. [27], DrEval reports that good Leave-Pairs-Out and Leave-Cell-Lines-Out performance is reachable by predicting each drug’s average response across the training cell lines. A strong result on those splits therefore demands the per-drug average be measured explicitly, which is what § 4.11 discharges.

MISSING ABLATIONS. Their remedy is randomisation and robustness testing. The scramble ablation of § 4.2 is one, specialised to the drug. Hold every byte of the prompt fixed except the drug and see whether anything downstream notices.

2.6 Cross-context transfer of drug responses

An entire drug-discovery methodology rests on the premise that a drug’s characteristic transcriptional response travels between cellular contexts. The premise is normally assumed. Here it is measured, with the attenuation from sampling noise corrected.

The Connectivity Map programme [28, 29] built that methodology: a drug’s transcriptional response is taken to be stable enough across contexts that a reference database can be queried by similarity to it. Its L1000 platform [29] ran the idea at scale, and its 978-gene landmark set defines the gene panel used here, intersected with Tahoe’s measured genes to give 946.

How much of that holds determines whether a conditional model has anything left to predict once a per-drug constant is accounted for.

§ 4.8 measures it on condition-level pseudobulk residuals. On a disattenuated cross-context similarity scale the shared part is $\mathcal{T} = 0.553$ [0.514, 0.593], a little over half, far above the structure-matched cross-line negative control (−0.008) and far below reproduction. The earlier

transfer coefficient How much of a drug’s residual in one context is recovered in another, on a disattenuated similarity scale.

same-plate, different-drug control (-0.017) is not a second matched control. It differs structurally from the estimand and was voided.

CORRECTION the same-plate control, voided in § 4.8

WHAT THE REMAINDER IS NOT. The remaining 44.7% [40.7%, 48.6%] is not thereby shown to be biology. The compared pairs differ in cell line, in dose and in treatment well at once, so the figure is not a variance share and not a drug $\times$ cell-line interaction, and the section says so where it reports it. What it delivers is a calibration of the transfer premise, not a verdict on it, and we are aware of no prior estimate of that quantity for chemical single-cell perturbation data.

2.7 Probing versus intervention

A probe can read a drug's identity out of a model's internals. Does that mean the model uses it? To learn whether a property is used, change it and watch what the model emits.

Linear probing establishes that information is *present* in a representation, not that it is *used*. Elazar et al. [30] make the case for keeping the two apart. Their amnesic-probing design is adopted here with a matched control added, since an intervention that fails to move the prediction proves nothing unless a comparable intervention does.

A SHARPER FORM OF THE SAME DISTINCTION. Gurnee et al. [31] report that language models maintain a small privileged set of representations, the *J-space*, found by a Jacobian Lens, and argue it functions as a global workspace. Roughly ten to twenty-five concepts are active at any moment, typically under 10% of activation variance. The defining property is causal. Swapping one active workspace vector for another changes the answer to match, while content outside may be richly represented and still inert.

WHAT THIS THESIS TAKES. The design is the swap. § 4.5 builds a domain analogue, substituting a drug’s learned representation for another drug’s and measuring whether the prediction moves toward the substituted drug. A property can be recoverable from the residual stream and still sit outside whatever the readout consults, in which case making it more explicit need not change what the model emits, an expectation § 4.6 tests and § 4.9 bounds. The thesis also borrows the lens, in § 4.14.

What the thesis does not do is test the workspace claim itself. § 4.14 measures how much of the change a drug swap makes lies along the lens’s token directions, which is geometric co-location and not use, and no workspace vector is ever swapped. The substitution result of § 4.5 is consistent with the workspace reading and equally consistent with a readout that reads the drug at low gain for some other reason; the co-location measurement is consistent with the first and does not exclude the second. The framework is cited for the instruments it justifies and the distinction it sharpens, not as a hypothesis this thesis adjudicates.

SCOPE LIMIT carried into § 4.5 and
§ 4.14

WHY ALLOCATE REPRESENTATIONS THIS WAY. Elhage et al. [32] show that networks reserve dedicated directions for features needed on every forward pass and push the rest into superposition, still linearly recoverable but without an axis. A property required to emit any plausible prediction should sit in the causally active set; one needed only for occasional composition need not. That asymmetry motivates building the instrument. It is not something the instrument returns.

2.8 Optimal transport as a target constructor

Sequencing destroys the cell it reads, so no cell is seen both before and after treatment and there is no paired training data. Where, then, does a training target come from? The instrument is optimal transport, used not as the predictor but as the thing that builds the target.

There is no ground-truth correspondence between the control and treated populations. Here the two clouds come from treated wells and their matched DMSO controls within the same cell line and plate, so the correspondence must be *inferred*.

COUPLINGS. A deterministic Monge map, one destination per source point, cannot split mass and may not exist for discrete distributions with incompatible atom masses. Kantorovich’s relaxation replaces it with a *coupling*, a joint distribution $\pi \in \Pi(\mu, \nu)$ with marginals μ and ν ,

$$\min_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y), \quad (2.3)$$

for a ground cost c , typically squared Euclidean distance. A coupling may split a source point across destinations, which makes the problem solvable for noisy, overlapping empirical distributions. When c is the p -th power of a metric the optimal value is $\mathcal{W}_p(\mu, \nu)^p$; taking its p -th root gives the Wasserstein distance. Here $p = 2$.

REGULARISATION. Solved exactly, the discrete problem costs roughly $O(n^3 \log n)$, prohibitive at single-cell scale. Adding a negative entropy term,

$$\min_{\pi \in \Pi(\mu, \nu)} \langle C, \pi \rangle - \varepsilon H(\pi), \quad (2.4)$$

renders it strictly convex, so the optimum is unique and takes a scaling form recovered by the Sinkhorn–Knopp algorithm through alternating matrix–vector products [33].

The parameter ε is not merely a computational convenience. It controls how sharply mass is assigned. As $\varepsilon \rightarrow 0$ the plan concentrates toward a sparse coupling; without further conditions, that limit need not send each control cell to one observed treated cell. As ε grows the plan blurs, until every control cell couples equally to every treated cell.

The *barycentric projection* maps each control cell x_i to the π -weighted average of its partners,

$$\hat{y}(x_i) = \frac{\sum_j \pi_{ij} y_j}{\sum_j \pi_{ij}}, \quad (2.5)$$

a convex combination inside the observed treated convex hull, not necessarily a point on the data manifold. Small ε therefore yields a target dominated by a sparse set of observed treated cells, large ε the group mean. Varying ε interpolates between these limits; § 4.6 uses one OT setting and compares it with single-cell and rank-proxy consensus targets, which are analogous constructions rather than the exact endpoints.

SEE § 4.6 compares three target constructions

A THIRD USE. Applied to biology this machinery mostly does trajectory inference, as in Waddington-OT [34], or prediction, taking the fitted map *as* the predictor. The use made here is neither. Transport supplies the correspondence from which a training target is built, and what then has to generalise is the language model trained on that target, not the coupling. Which is why what matters below is ε , the parameter deciding which target the coupling defines.

TWO PRACTICAL POINTS. The ground cost needs a space where Euclidean distance is meaningful, and raw counts are not one, being

sparse, overdispersed and dominated by library size. Couplings are therefore computed in the released Tahoe scVI latent [35]. And because a full coupling costs quadratically, transport is solved per condition, coupling its treated cells to the controls shared within its cell-line/plate group. Both choices, the shipped scVI latent as the transport space and one coupling per condition against that condition's own controls, follow the optimal-transport pipeline of Kaan [36], a companion thesis on the same atlas from which this construction was taken. A control cell from one cell line is not a plausible counterfactual partner for a treated cell from another.

Materials and methods

3

IN THIS CHAPTER

What has to be settled before a score on this atlas means anything? Four things. What is measured, on what population, against what reference, with what uncertainty. Tahoe answers none by convention, so each is settled below by a measurement, and each narrows a claim later.

3.1 Gene panel and normalisation

Tahoe-100M [1] holds drug-treated single-cell transcriptomes across cancer cell lines, with plate-matched DMSO controls. Metadata supplies cell line (Cellosaurus ID), drug, dose (drugname_drugconc), plate, and a fine-grained mechanism-of-action annotation (moa-fine), five fields; all but the plate reach the prompt, and with the control cell's own sentence they are the whole of what any prompt below can carry.

All analysis is restricted to the $G = 946$ genes in the intersection of the L1000 landmark set [29] with Tahoe's measured genes, library-normalised to counts per 10,000 and transformed as $\log_{10}(1 + x)$. The restriction is a budget choice, not a biological one: the context is 8192 tokens and a prompt carries an instruction line and two cell sentences, so the transcriptome does not fit. Normalisation multiplies every gene in a cell by one constant and the logarithm is monotone, so within a cell the ordering, and therefore the sentence, is invariant to both.

- 3.1 Gene panel and normalisation
- 3.2 Conditions, treated wells and batch units
- 3.3 Evaluation populations and identifiability
- 3.4 Backbone checkpoint and cell-sentence format
- 3.5 Training targets and measurement frames
- 3.6 The within-well split-half reference
- 3.7 Uncertainty: two-way cluster-robust intervals
- 3.8 Generation budget and validity checks

■ EXPR / ▲ RESID

each section declares the frame in force beside its heading

WHAT IT DOES CHANGE. Everything downstream of a single cell. Scaling by library size improves comparability across cells of different depth, but the profiles remain compositional, not absolute, so pseudobulk means, the drug-specific residual, and every cosine and Euclidean quantity built on them depend on the normalisation even though the sentences do not. Depth also shapes which genes are detected, though never their order.

3.2 Conditions, treated wells and batch units

DEFINITION Condition, treated well, batch unit

A *condition* is a (drug, cell line, plate, dose) tuple, and it is the unit everything below is scored at. The physical assignment is the *treated well*, so a condition sits inside an assignment and is not one. A plate carries wells from many cell lines at once, and a control pseudobulk is matched within both, so the batch unit is the pair (cell line, plate), never the plate label alone. A *pseudobulk* is a group's mean expression profile; a *control pseudobulk* is the same over plate-matched DMSO cells.

Tahoe does not treat cell lines one at a time. Roughly 49 of them are pooled into a single *well* and one drug at one dose is applied to the pool, so one well yields around 49 (drug, dose, cell line) conditions at once and whatever happened to that well happened to all of them.

The conditions this thesis evaluates — one drug, at one dose, on one cell line, on one plate, which is the unit every number below is attached to — all carry the plate labels `plate4`, `plate5` and `plate6`. Labels run within cell line rather than globally, so these are three labels and not three plates. The cache holds further labels, none of them evaluated. Across it are 289 treated wells, recovering a mean of 48.6 cell lines each (median 49, range 43–50). Almost none of them repeat: 286 of the 287 distinct (drug, dose) treatments occur in exactly one well.

Those numbers describe this cache, not the atlas: Tahoe includes further plates, among them a replication plate, so “one well” below

means one well *here*. The single exception, one treatment spread over three wells, is also the whole of the cross-plate structure at condition level, and it accounts for the 49 (drug, dose, cell line) groups the unit audit finds on more than one plate. Every other condition on more than one plate differs in dose, and each drug is assayed on its own plate or plates with plate-matched controls.

THREE CONSEQUENCES. Conditions from one well share a physical assignment, so they are not independent (§ 3.7); drug and plate are partially confounded, so discrimination is measured within plate (§ 4.2); and a (drug, cell line) pair cannot be held out cleanly (§ 4.10).

Figure 3.1 draws the three in that order.

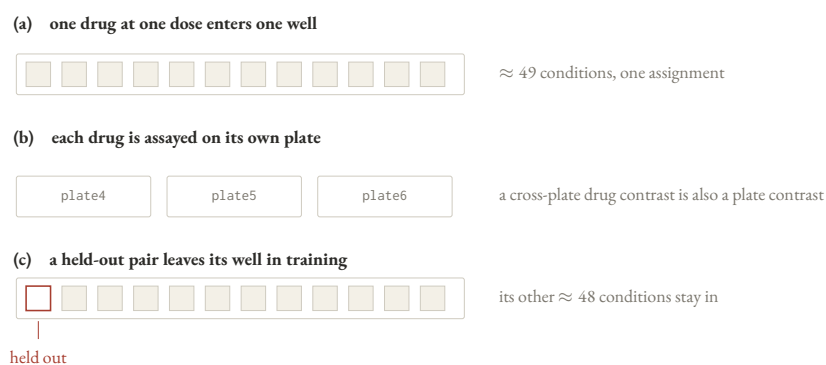


FIGURE 3.1 One assignment, many conditions. The roughly 49 conditions from one well share an assignment and a batch (a). Each drug is assayed on its own plate or plates, so a drug contrast across plates is also a plate contrast (b). Removing a (drug, cell line) pair leaves its well in training through the other ≈ 48 lines (c). Squares are conditions, twelve per well for legibility.

Dose is a molar concentration resolved from `drugname_drugconc`; an inspection of all 289 treated wells found 289 usable molar doses with zero unit collisions, so the defect was a demonstrated possibility that did not materialise.

3.3 Evaluation populations and identifiability

Table 3.1 states the quantities every reported sample size, n , in the investigation should be located against. One symbol in it runs ahead of its derivation. NIR, the normalised inverse rank, is the fraction of a declared comparison set whose truth a prediction resembles less than its own, so chance is 0.50 under the exchangeability condition of § 4.1.

SEE § 4.1 derives NIR and audits its chance level with empirical control arms.

TABLE 3.1 The populations behind every n quoted in the investigation. A is a streamed sample of the atlas, B the cache built once and reused. C and D are two different evaluation schemes, keyed at different units; neither is a subset of the other.

A. TASK CHARACTERISATION streamed sample of the atlas		B. RESIDUAL-FRAME CACHE rebuilt targets, § 4.7		
conditions characterised	6,628	cells, treated / control	620,564	(600,000 / 20,564)
statistically active against DMSO ^a	3,647 / 5,014 (72.7%)	drugs / cell lines	106	/ 50
identifiable at the tested pseudobulk budget ^b	3,064 (46.2%)	(cell line, plate) batch groups behind them	119	
with a plate-mate closer than their own split-half	78.7%	conditions with ≥ 40 treated and ≥ 20 control cells	6,617	
median cells per condition	44	retained by the data-resolved reliability line	2,523	of 4,999 train (50.5%)
median #DEGs per condition (BH $q < 0.05$)	0	training / well-held-out validation examples	147,710	/ 3,600
median SNR (effect $\div$ within-well split-half noise)	0.75			
drugs with a real dose series	218 / 351 (62.1%)			
drugs seen in ≥ 3 cell lines	227			
C. EVALUATION, GENERATION AND EXPRESSION FRAMES keyed at the (drug, cell line) pair				
		TIER 1	TIER 2	TIER 3
withheld from training ^c		nothing	the drug	the pairing
conditions available		4,325	2,188	99
scored, generation evaluations				400 per tier
scored, expression-frame NIR, within plate			606	
D. EVALUATION, RESIDUAL FRAME keyed at the treated well, § 4.10				
		train	unseen_combo	unseen_drug
withheld from training		nothing	whole wells	all its wells
conditions after inventory filters		4,999	900	711
conditions scored, 1,394 in all		300	895	199
treated wells behind them ^d		136	36	28
drugs, 90 distinct in all		76	34	10
cell lines, 42 distinct in all ^e		39	42	40

^aOver the 5,014 conditions whose (cell line, plate) group carries plate-matched DMSO cells, not all 6,628.

^bOwn within-plate split-half NIR ≥ 0.8 ; per-condition, and distinct from the aggregate ceilings of Table 3.2.

^cTier 2 is DrEval's Leave-Drugs-Out. Tier 3 is Leave-Pairs-Out, which § 4.10 declines as an estimand: withholding a pairing leaves that pair's own well in training through its other ≈ 48 cell lines. Block D exists because of it.

^dNo well contributes conditions to two splits.

^eNearly every cell line appears in all three splits, by design: D withholds wells, not lines, so it asks about an unseen treatment well of a seen drug and never about transfer to an unseen line.

Of the 895 unseen_combo conditions, 893 carry a training-only drug lookup; with 299 of the 300 train conditions they form the common support of 1,192 in Table 4.12.

SCOPE LIMIT carried into § 4.2 and § 4.9

Two populations are used and they are not interchangeable. The broad characterisation (§ 4.2) streams a wide sample of the atlas, over more plates and drugs than cached, so the drug and condition counts in its block of Table 3.1 count that sample, not the cache. The residual arm (§ 4.9 onward) works from a smaller cached subset. A count from one block does not bound a count in the other.

THE UPPER BLOCK IS A RESULT. It describes Tahoe-100M, not this project. Across the 5,014 control-reachable conditions the median has no differentially expressed gene at BH $q < 0.05$ and a signal-to-noise ratio of 0.75 against its own within-well split-half noise, while 72.7% are active on the whole profile, leaving 27.3% statistically inert. A per-gene test is far harsher, and 27.3% alone would make the atlas sound better resolved than it is. The typical condition carries a real response measured with too few cells, and not no response (§ 4.2).

The identifiability bar is per-condition: a condition's own split-half NIR against its plate-mates has to clear 0.8. Of the 6,628 conditions, 46.2% clear it, and 78.7% have a plate-mate lying closer to them than their own second half does. Identifiability is a property of the pairing, not of the drug: over the 227 drugs assayed in at least three cell lines, the median drug's split-half reference spans 0.83 between its most and least identifiable context. A benchmark omitting those figures quotes accuracies over a population barely half of which is empirically identifiable at the tested pseudobulk budget, and a per-drug leaderboard on it ranks pairings.

The three tiers of the last block are defined by what training contained. *Tier 1* holds conditions whose (drug, cell line) pairing was seen in training, so it measures fit and not generalisation. *Tier 2* holds conditions whose drug never appeared, Leave-Drugs-Out in the vocabulary of § 2.5. *Tier 3* holds pairings whose drug and cell line were each seen but never together, Leave-Pairs-Out; it is small (99 conditions) and is

not reported: § 4.10 declines it as an estimand, because withholding a pairing leaves that pair’s own well in training (Table 3.1, note c). The residual arm builds its own, larger holdout instead: Table 3.1 (block D) names its three splits, and § 4.10 settles the unit they are keyed at.

3.4 Backbone checkpoint and cell-sentence format

A cell is written as a *cell sentence*, its expressed panel genes in descending order, terminated by [END_CELL], which is registered as an atomic special token. The prompt supplies cell line, drug, dose, mechanism and the control cell’s sentence; what the model must produce is the treated cell’s.

The terminator is not inherited. A cell expresses only part of the panel, so sentences differ in length, and a model with no way to mark where its expressed list ends leaves the boundary between what it ranked and what it omitted to be inferred from the output. An earlier build here removed the problem rather than solving it, writing all 946 panel genes with the unexpressed appended in a fixed tail: nothing was ever absent, so every convention for handling absence collapsed into the same one, and most of the generation budget went on a tail that carried no information. This build lists only what is expressed and closes it with [END_CELL], an atomic special token registered in the trainer, so a sentence is variable in length and its end is declared rather than deduced. Stripping the tail without a marker would have left the same inference to be made silently. Absence is now a real property of a prediction, which is also why the rank-based metrics of § 4.1 need a stated convention for it.

The backbone is C2S-Scale-Pythia-1b-pt [3], a C2S-Scale checkpoint on the Pythia-1b decoder [4], cold-started for every arm from the

same base checkpoint with identical hyperparameters (learning rate 10^{-5} , gradient accumulation 16, maximum length 8192, bf16) and a one-epoch schedule, of which the consensus arm completed $\approx 60\%$ before stopping on a flat loss. Cold-starting makes the arms comparable; the training target or objective is the only variable.

SEE Table A.3 tabulates the training arms.

WHAT THE CHECKPOINT HAS SEEN. It is pretrained on over 50 million human and mouse cells from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas, an atlas corpus with no chemical or genetic perturbation data. Its model card lists cell and tissue prediction, conditioned generation and gene-set conversion, but not perturbation prediction. The base model arrives at fine-tuning here having never seen a drug-treated cell.

TWO CHANNELS STAY OPEN. The decoder underneath is Pythia-1b, pretrained on text in which drug names, targets and mechanisms occur, and every prompt supplies the drug's moa-fine annotation. A drug the fine-tuning never saw is not a drug the system knows nothing about.

“Held out” therefore means held out from Tahoe fine-tuning, with mechanism supplied, and every unseen-drug result below is read under that narrower scope.

SCOPE LIMIT carried into § 4.12 and chapter 5

WHICH LEAVES THE TARGET. Two facts sit together here. The checkpoint has never seen a drug-treated cell, so whatever it comes to know about perturbation it learns during fine-tuning and learns from the target it is given. And the formulation it inherits was written for a different job, producing a plausible cell rather than producing a drug specific treated cell. Cold-starting fixes the model, so the target is the only thing left that can be moved, and whether it should be is the question § 3.5 takes up.

3.5 Training targets and measurement frames

The model and the prompt are now fixed, so the string the model is trained to produce is the only thing left that varies. Two constructions of it run through the investigation. They are choices about how the target is built and not about the model, and each carries with it the object a score is taken on, the competitors it is taken against, and the function that compares them. That triple is what is called a *measurement frame* here.

THE INHERITED TARGET. The first is the formulation C2S-Scale defines for perturbation prediction, kept unchanged. The target is the treated cell’s own sentence, its expressed panel genes in descending order, and nothing is subtracted from it. It is a general-purpose construction, written for a model asked to produce a plausible cell rather than to tell two treatments apart, and it is what makes the comparison of § 2.2 possible at all.

THE PERTURBATION-SPECIFIC TARGET. The second is built for this question, because the first one hides the drug: two different drugs on the same cells give almost the same target string, so the failure to give drug-specific predictions is a fault of the data before it is a fault of the model. § 4.7 is the measurement that motivates replacing it. Two subtractions turn a treated profile into a drug-specific one.

DEFINITION The shift, the generic and the drug-specific residual

For a condition c , a (drug, cell line, plate, dose) tuple with at least 40 treated and 20 plate-matched control cells, write d for its drug and

$$r(d, c) = \underbrace{(\bar{y}_{\text{treated}} - \bar{y}_{\text{control}})}_{s(d, c)} - \underbrace{\frac{1}{|D_c \setminus \{d\}|} \sum_{d' \in D_c, d' \neq d} s(d', c)}_{\text{generic}(c)}. \quad (3.1)$$

Both bars are pseudobulks: $\bar{y}_{\text{treated}}$ averages the condition's own treated cells and $\bar{y}_{\text{control}}$ its plate-matched controls, so $r(d, c)$ is one vector for the whole condition and not a quantity carried by any single cell. The control subtraction removes the state those cells share, leaving the *shift*. The mean over the other drugs removes the *generic* programme every drug triggers, conventionally read as a shared stress and cell-cycle response, a reading used here for intuition and nowhere tested. What survives both subtractions is the *drug-specific residual*, and taking it as the string the model is trained to emit makes it the *residual target*. A condition therefore supplies many training examples but only one target: the prompt varies from one plate-matched control cell to the next, and every one of them is paired with the same residual string.

D_c is the training drugs sharing c 's (cell line, plate) group, each drug entering once whatever its doses. That grouping is the generic's *scope*, and this build takes it at plate scope; § 4.8 reports the cell-line-scoped alternative alongside it. Two properties of D_c are fundamental. It is *split-before-fit*: the holdout is fixed from metadata before any statistic is estimated, the generic included. And it is *leave-one-drug-out*: a condition's own drug never enters its own generic, so the subtraction removes what the other drugs share and cannot remove the drug being predicted.

The scope is also the limit. What remains is not what makes this drug different from all drugs. It is what makes it different *from the other training drugs that happen to share its plate and cell line*.

WRITING THE RESIDUAL AS A STRING. $r(d, c)$ is a signed vector over the panel, and a decoder emits tokens rather than vectors, so the target has to be written down before it can be trained on. The obvious form is the wrong one. A single list in descending order buries the most strongly repressed genes at the tail, among the genes the drug did not move at all, and repression is exactly what the control subtraction was performed to expose. The sign is the biology, so the encoding keeps it: the target is two ordered blocks, the most up-regulated genes first, a [DOWN] separator, then the most down-regulated genes, closed by [END_CELL]. Up to 100 genes enter each block, which is the 200 target positions Table 4.6 counts over, and a gene enters a block only if its residual actually carries that sign, so a condition with few movers emits a short one. [DOWN] is registered as an atomic special token alongside [END_CELL]; unregistered it splits into sub-words and the up/down boundary stops being a clean symbol. What the model is asked to produce is therefore a ranked signed signature, not a cell.

READING A GENERATION BACK, IN EACH FRAME. What the model emits is a string, and a similarity needs vectors, so a generation — and the truth it is scored against — has to be read back into one. Three read-backs are used below, and which one applies is a property of the metric and not of the frame: the frame names the object scored, a cell profile or a residual, and not the units it is scored in.

The first leaves the sentence where it is. A gene's value is its position in the sentence, counting from one, and a panel gene the sentence omits takes the rank the absent-gene convention assigns (§ A.2). $DE-\Delta r$ and $panel-\tau$ are computed here and never leave rank space.

The second decodes to a reconstructed expression profile. Write p_g for the position at which gene g appears. Then

$$\psi(s)_g = \begin{cases} \max(0, a \log_{10} p_g + b) & \text{if } g \text{ appears in } s, \\ 0 & \text{if } g \text{ does not,} \end{cases} \quad (3.2)$$

with a and b a single slope and intercept fitted once for the whole run (§ 2.2). This is what `nir_expr` reads, prediction and truth both decoded, and it is the only read-back that invents a magnitude. Note that the absent-gene convention does not reach it: ψ fills an omitted gene with zero whatever the convention says.

The third keeps only sign and order, because a residual has no expression units to decode into. Write s for a signed signature of either kind, and let $U(s)$ and $D(s)$ be its up and its down block: for a generation, the two lists the model emitted, in the order it emitted them; for a continuous residual, its most positive and its most negative entries

under the same cut of 100. Let r_g be the position of gene g within whichever block holds it, counting from one. Then

$$\phi(s)_g = \begin{cases} +1/\log_2(r_g + 1) & \text{if } g \in U(s), \\ -1/\log_2(r_g + 1) & \text{if } g \in D(s), \\ 0 & \text{if } g \text{ is in neither block.} \end{cases} \quad (3.3)$$

The leading gene of each block carries weight one and the weight falls away down the block, so a gene's rank is all that reaches the vector and the value that earned it that rank is discarded. The discount is what makes the ordering inside a block count for something — agreeing on the most up-regulated gene is worth more than agreeing on the hundredth, where one rank is barely distinguishable from its neighbours, and a flat encoding would score those two agreements alike.

All three read position and nothing else. A gene's identity decides *where* a value lands in the vector, never *what* that value is, and all three fill with zero or with a convention whatever the string leaves out. They differ in what they then claim: ψ writes position into expression units through a map fitted to data, so an expression-frame distance taken on it is a distance between two reconstructions; ϕ writes position into fixed weights that no fitting produced, so a residual-frame comparison is a rank-weighted overlap of gene sets and carries no units at all. Table 4.2 records which read-back each quoted figure rests on.

WHY THE TARGET CARRIES A FRAME WITH IT. A model trained on the first target emits a cell profile; a model trained on the second emits a signed up-and-down signature. The two are not the same kind of object, and neither scoring path accepts the other's output: the rank-to-expression decode is fitted for expression ranks and its truths are full pseudobulks, while a residual signature has no

expression units to decode into. So the choice of target settles the space a prediction lives in, and a score has to be taken in that space or not at all. Comparing a number from one frame with a number from the other is the most expensive misreading this thesis allows.

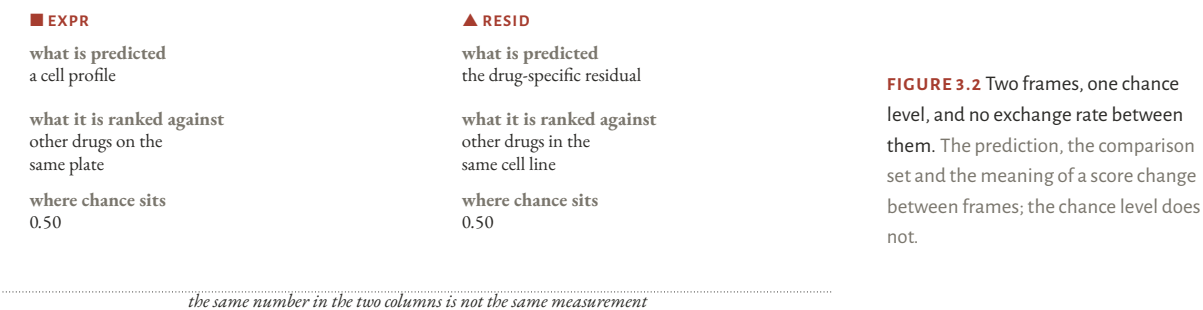
Both frames score by discrimination and not by proximity. A prediction is never asked how close it came to its own truth. It is set beside a lineup of other conditions' truths and asked whether it resembles its own more than it resembles any of theirs. That fraction is the *normalised inverse rank* NIR, and the lineup is the *comparison set* J . Choosing J is a substantive decision rather than a formality: anything every candidate in J has in common cannot help a prediction win, because it is equally true of all of them. Narrowing J is how a nuisance cue is switched off.

SEE § 4.1 derives NIR and audits its chance level with empirical control arms.

The two frames narrow it differently. The ■ **EXPR** *expression frame* scores the inherited target's output, a predicted cell profile, against the other drugs *on its own plate*, so every candidate carries the same batch and no prediction can score by recognising the plate it came from — a leak § 4.2 measures rather than assumes. The ▲ **RESID** *residual frame* scores the residual target's output against the other drugs *in its own cell line*, more widely scoped than the generic, so every candidate carries the same cellular background and the drug is the only thing left to separate them. Chance sits at 0.50 in both, under an exchangeability condition § 4.1 tests with control arms instead of assuming, and J is declared at every score below. A score without its comparison set is not a score.

The *similarity* σ , the function by which a prediction is judged closer to one truth than to another, is the third thing the target settles, and it follows neither from the scored object nor from the comparison set on its own: a negated Euclidean distance on decoded expression in the expression frame, a cosine on the signed rank-discounted residual in the

residual frame, the two instances Table 4.2 fixes with their comparison sets. A score without its similarity is not a score either, and both are to be declared before a number is read.



3.6 The within-well split-half reference

Almost every negative result below is stated as a distance from a *ceiling*.

DEFINITION Within-well split-half precision reference

The score obtained when a disjoint cell subset of the same treated well is substituted for the prediction, recomputed for every stratum scored, so its value moves with the stratum. It is not a replicate ceiling. What varies across the rows of Table 3.2 is the frame, the comparison set and the cell budget, never the estimator.

Two properties limit how the ceilings read. First, they are *split-half* estimates and not biological replicates, because this cache places 286 of 287 treatments in exactly one well. Second, a split-half ceiling estimates within-well sampling precision, which is optimistic relative to true biological reproducibility, and every coverage figure in § 4.11 inherits that optimism.

THE ONE EXCEPTION. One ceiling is not that estimator and not a row in Table 3.2. The positive control inside the expression-frame DRF audit of § 4.1 is a denoised interpolated duplicate, not a raw split-half, and that section turns on the difference.

TABLE 3.2 One estimator, seven settings. Every “ceiling” here is a within-well split-half precision reference, the score of a disjoint subset of the same treated well’s cells. They differ in frame, comparison set, cell budget and split contents.

CEILING	FRAME AND METRIC	COMPARISON SET AND BUDGET	WHERE USED
0.76	expression, DE- Δr	two real cells	metric audit (§ 4.1)
0.625 / 0.575	expression, NIR (negated Euclidean on decoded expression)	leak audit; cross-plate / within plate	plate leak (§ 4.2)
0.67–0.83	expression, forced choice	within plate, by cell budget	§ 4.2
0.576	expression, NIR (negated Euclidean on decoded expression)	within plate, tier 2	the blindness verdict
0.61 / 0.76 / 0.85	expression, NIR (negated Euclidean on raw pseudobulk, undecoded)	<i>cross-plate</i> ; 10/40/120 cells	identifiability curve
0.956/0.824/0.836	residual, NIR (cosine on the signed rank-discounted residual)	within cell line; train/combo/drug	§ 4.11
0.742 / 0.7467	retrieval, negated Euclidean on undecoded log-expression	within condition, split-half; full profile / residual (plate scope)	§ 4.7

The DRF positive control of § 4.1 is the investigation’s one exception and is not a row here.

None of these is a biological replicate (286 of 287 treatments sit in a single well).

In the residual frame the raw reference is not constant across splits (a spread of 0.132), but that spread is an artefact of how the splits were filtered and not a difference in recoverable signal; matched at the training reliability line the three agree to 0.005 (chapter 5).

A distance to the reference is comparable within a split and not between them, because the splits are scored on populations selected by different rules. And because the reference is a sampling-precision estimate and not a biological replicate, every coverage figure computed against it is a distance to an optimistic bar by an unknown amount. Where one is quoted, the ordering carries the claim.

SCOPE LIMIT carried into § 4.11 and chapter 5

3.7 Uncertainty: two-way cluster-robust intervals

Two rules apply to every result below. Conditions are not independent, and not independent in two crossed ways at once (Figure 3.3): two from one treated well share a physical assignment and a batch, two from one cell line share the biology. Treating them as independent gives intervals about 1.5 times too narrow on the scramble strata and 1.7 to 2.5 times too narrow on the contrasts against the lookup baselines. Clustering on the assignment unit alone uses the larger cluster count (289 wells against ≈ 50 lines) and would tighten every interval while sounding more rigorous. That is the failure the estimator below avoids.

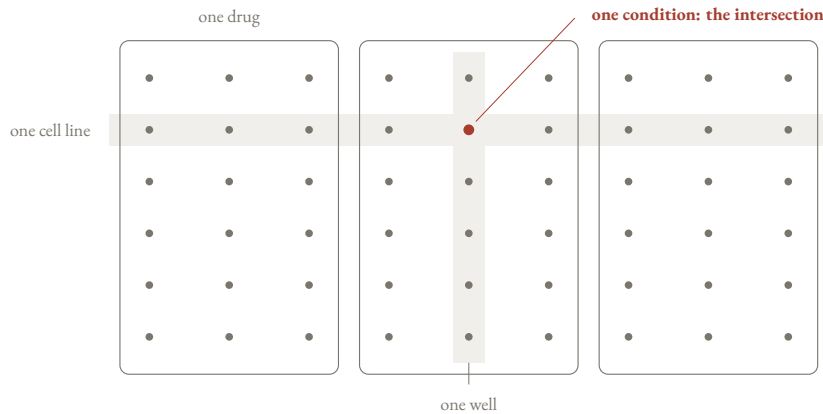


FIGURE 3.3 A condition belongs to a well and to a cell line at once, and the two groupings cross. Each dot is a condition; the horizontal band is one cell line, the vertical band one well. The dot where they meet is one condition, which is why the intersection term is the ordinary independent-sampling variance. The rounded envelopes are drugs: a well carries one drug at one dose, so wells nest inside drugs, while the cell-line band crosses every drug and neither of the two bands is nested in the other. The lattice is schematic — nine columns of six, for legibility, and not a count of anything in the cache.

Every interval on a *mean* is therefore two-way cluster-robust (Cameron–Gelbach–Miller [37]),

$$V = V_{\text{line}} + V_{\text{well}} - V_{\text{intersection}}, \quad (3.4)$$

where the intersection of the two groupings is the single condition, so $V_{\text{intersection}}$ is the ordinary independent-sampling variance. When it goes non-positive in a finite sample, the wider one-way variance is used and the branch recorded. The formula above is valid only where the two groupings genuinely cross, which cell line and well do: a treated well

spans ≈ 49 cell lines and a cell line recurs across wells. Drug and well do *not* cross, because one drug is assigned per well and wells therefore nest strictly inside drugs. Where an interval is reported on the drug axis the estimator tests the two groupings for nesting first and, finding it, clusters one-way on the drug rather than applying the expression above to a nested pair; the branch taken is recorded beside the interval. Reading “two-way” as “the equation above was applied to drug and well” would therefore be wrong. It is asymptotic in clusters, so the reference distribution is Student’s t with one fewer degree of freedom than the smaller cluster count. The `unseen_drug` arm shows what that costs: Table 3.1 puts 199 scored conditions behind it, drawn from 28 treated wells and 40 cell lines, so the smaller cluster count is the 28 wells, the reference is t_{27} and not the normal, and the interval widens by 4.7%.

For statistics over pairs of conditions, such as the transfer coefficient, neither grouping partitions the pairs, so those intervals use the Fafchamps–Gubert dyadic estimator [38], under which two pairs are dependent when they share either member.

Throughout, an estimate is quoted as value [lower, upper], a two-way cluster-robust 95% interval unless stated otherwise. DrEval [21] identifies the same structure more broadly, citing Hurlbert [26]: hundreds of thousands of cells resolving to a few thousand conditions over 50 cell lines.

3.8 Generation budget and validity checks

The second rule concerns generation validity. The budget is a claim-level variable, not a runtime setting.

A 600-token cap truncated 26% of predictions in an early residual-frame run, silently, because nothing measured them; that run's validity log was not retained, and the figure is quoted from the note made at the time. A prediction cut at the cap never reaches [DOWN] and is scored with its whole down-block missing, and no completion rate was quoted beside the score.

WHAT IT COST. The optimal-transport arm of § 4.6, scored at that cap, gave a gap of +0.0144 [−0.0032, +0.0323] against its geometry-defined or th partner, whose observed score is near chance. That interval spanned zero and the arm was printed as inert. Re-run at the 1400 tokens the residual arm had, it gives +0.0292 [+0.0144, +0.0441], and the negative had to be withdrawn (§ 4.9). The same cap had already halved the residual arm's own effect. Wherever a paired comparison is scored on variable-length predictions, then, the token budget is part of the claim, and a null quoted without its budget cannot be read.

CORRECTION The printed negative is retracted in § 4.9, in full and where it was made.

THE BUDGETS USED. Four generation budgets appear in this thesis: 600 tokens in the early residual-frame runs, 1200 in the swap-distance sweep of § A.3, 1400 for every canonical residual-frame result, and 3800 for the output-invariance instrument of § 4.2, where truncation would cut the comparison itself. Residual-frame arms draw four sampled generations per condition and expression-frame arms eight, at temperature 0.8, and the sampled generations of one condition are averaged as vectors before they are scored (§ 4.7).

THE RULE THAT FOLLOWS. Every generation-based evaluation computes the `[END_CELL]/[DOWN]` completion rate, the fraction of emitted symbols that are valid panel genes, and the duplicate rate before any score is read from it, and no score below comes from a run whose validity block was not inspected.

That is the apparatus. It settles what is measured, on what population, against what reference and with what uncertainty, and reaches no verdict about the model.

Experiments and results

4

IN THIS CHAPTER

Does a model that has learned something about a drug answer differently when it is told a different drug? The question is cheap to ask and expensive to ask honestly, because the instrument had to be measured before the model could be. Each beat below builds the tool that the measurement before it made necessary.

The order is chronological, not the order that would argue best. That

is fundamental: a design choice appears as the failure it avoids, and a

dead end appears where it was walked into. The apparatus is chapter 3;

the terms, symbols and conventions are fixed in Reference apparatus.

THE FINDINGS, BEFORE THE BEATS THAT PRODUCED THEM.

Six, in the order they arrived, with no numbers attached. Each one

is established, qualified and scoped in the beat that owns it, and is

quotable only from there. What follows is a skeleton, not a record.

- The metric was wrong. The field’s standard perturbation metric is saturated by a predictor that carries no drug information at all, so the instrument was repaired before the model was measured.
- On the repaired instrument the model is drug-blind in the expression frame. It returns the same answer whether told the true drug or a different one, and a baseline that copies the untreated cell matches it.

- 4.1 Metric calibration: the DRF audit
- 4.2 Drug-blindness in the expression frame
- 4.3 Decoding drug identity from activations
- 4.4 Ablating the drug subspace
- 4.5 Activation substitution: the identity test
- 4.6 Denoised targets: consensus and OT
- 4.7 The residual target
- 4.8 Shift decomposition and the transfer coefficient
- 4.9 Prompt sensitivity on trained conditions
- 4.10 Held-out wells and held-out drugs
- 4.11 The baseline ladder and the per-drug lookup
- 4.12 Channels for unseen drugs
- 4.13 Drug name versus mechanism field
- 4.14 Coda: a Jacobian-lens view of the drug swap

■ EXPR / ▲ RESID

each section declares the frame in force beside its heading

- The blindness is not a failure to encode. Drug identity is written into the activations at every layer measured, and generation barely consults it. This is the chapter’s least well-evidenced measurement.
- Re-encoding the training target as the drug-specific residual makes the model demonstrably sensitive to the drug in its prompt. That is the project’s first non-null drug effect: first in sequence, and not the only target construction that produces one.
- Responding to the drug is not predicting it. Prompt sensitivity survives a holdout of whole treated wells; biological transfer to those wells is not established on the axis a drug-specific claim has to generalise across, and a lookup table of training averages outscores the model. That dissociation is this chapter’s argument.
- For a drug the model has never seen, same-target retrieval is the strongest observed lead on its own restricted support; mechanism and chemical structure are inconclusive, not dead.

Figure 4.1 is the map. Nine beats produce those six findings; the ninth produces none, and that is a decision, not work left undone. A coda after the ninth returns to the workspace account the background drew on (§ 4.14). It is not a beat and adds no seventh finding: it reads the third and fourth through that account’s lens.

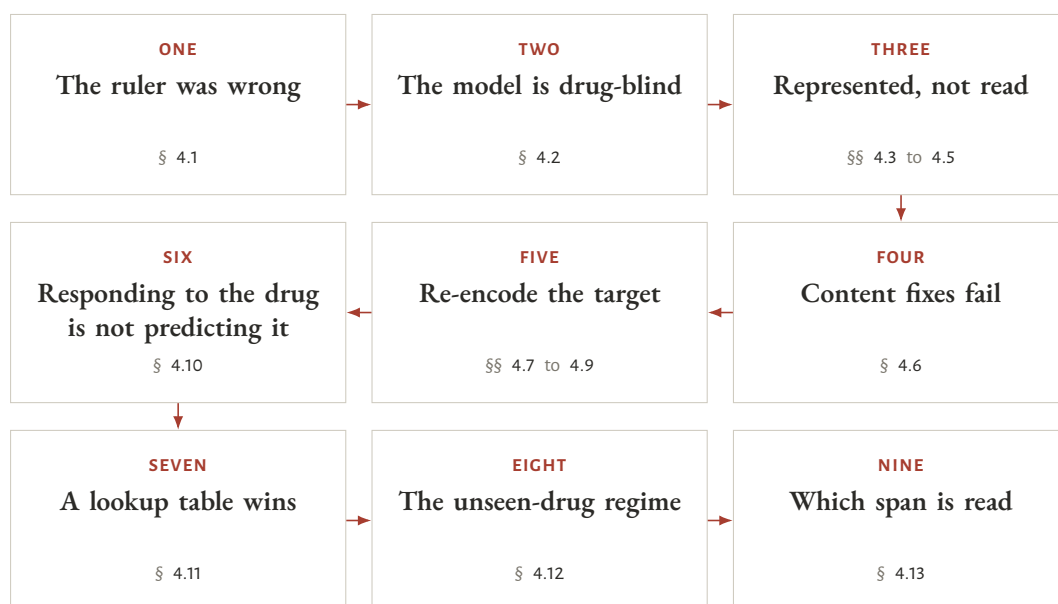


FIGURE 4.1 The investigation in nine beats, each told with the instrument it required. The arrows are the order of discovery. The ordinal on each beat names the cell the strip fills in that beat's opening box. The apparatus (chapter 3) is not a beat. The ninth beat is the one the chapter does not close.

4.1 Metric calibration: the DRF audit ■ EXPR

Every claim in this chapter is a number handed down by a scoring rule, so the scoring rule is what has to be audited first. The instrument grades metrics, not models. It asks each candidate to separate a positive control from a predictor that knows no drug, and four of the five candidates fail.

VERDICT The field's standard score is saturated by a predictor that carries no information, and of the five metrics audited only NIR grades in the direction its name implies.

A reported score is two claims at once, one about the model and one about the rule that graded it. Nobody checks the second. Two questions decide it. Does the score reward the thing it is named after, or can it be earned another way? And does it register a true positive when there is one?

THE SCORE THE FIELD REPORTS, AND TWO COMPANIONS.

That score is a correlation between predicted and true change from control, restricted to the genes that actually move. The benchmarks of § 2.4 report it, Miller et al. [22] find it poorly calibrated, DrEval [21] counts it among six standing obstacles. Write x for the control profile, y for the treated truth and $\hat{y}$ for the prediction, and let

SEE chapter 3 fixes G .

$$S_K = \arg \text{top-}K_g |y_g - x_g| \quad (4.1)$$

be the K genes with the largest true change. Then

$$\text{DE-}\Delta r = \text{corr}((\hat{y} - x)_{|S_K}, (y - x)_{|S_K}). \quad (4.2)$$

All three are the panel profiles converted to ranks, absent genes tied at the worst rank G (§ A.2), so S_K is selected on rank change too; the *expression frame* label used elsewhere names the object scored, a cell profile rather than a residual, and not the units. We report $K = 50$,

Pearson; $K \in \{20, 100, 200\}$ and Spearman move the numbers by under 0.02, except at $K = 200$, where the top- K set starts to include genes that do not move.

Two companions to it are reported below, and neither is that score. Writing $s = (y - x)|_{S_K}$ and $\hat{s} = (\hat{y} - x)|_{S_K}$ for the true and the predicted shift, and r for Pearson correlation on S_K , *partial-DE- Δr* regresses the control out of both arguments first:

$$\text{partial-DE-}\Delta r = \text{corr}(\hat{s}, s \mid x|_{S_K}) = \frac{r_{\hat{s}s} - r_{\hat{s}x} r_{sx}}{\sqrt{(1 - r_{\hat{s}x}^2)(1 - r_{sx}^2)}}. \quad (4.3)$$

Both arguments of DE- Δr carry the same $-x$, so a predictor that merely reverts toward the control inherits agreement it did not earn; conditioning on x leaves only what a prediction has beyond control-reversion, and the quotient is undefined rather than small when either shift is collinear with the control on S_K , its denominator vanishing. The second companion moves the other lever. *panel- τ* is Kendall's τ between the predicted and the true rank vector over all G panel genes,

$$\text{panel-}\tau = \tau(\hat{y}, y), \quad (4.4)$$

with no top- K restriction and no control subtracted: where DE- Δr picks its genes from the truth and scores a change, panel- τ scores the ordering itself on a gene set fixed before any prediction is read. It depends on the absent-gene convention rather than escaping it, every omitted gene being tied at G , so it registers omissions that DE- Δr sees only when an omitted gene falls in S_K , and it is undefined when a prediction ties every gene.

WHAT THE METRIC ACTUALLY REWARDS. Score a predictor that contains nothing. Place every gene at the middle rank. It scores DE- $\Delta r(K=50) = 0.999913$ [0.999908, 0.999919], above the two-real-cell positive control of 0.76 and above the model's 0.73. Less informa-

revert_center Predicts every panel gene at the panel's middle rank, defined from the metric alone and needing no model, training set or atlas.

tion yields a higher score, the hallmark of an artefact; Figure 4.3, panel a, sets the five arms against that reference on one axis.

Its partial-DE- Δr and panel- τ are undefined, not merely poor, for two separate reasons. The prediction is the constant midpoint rank $\hat{y} = G/2$, so the shift from control is $G/2 - x$, an exact affine function of x from which nothing survives partialling. And every predicted gene being tied, no rank ordering exists whose Kendall τ can be computed. The zero-shift construction, easily confused with this one, is control_copy, $\hat{y} = x$; revert_center is not it.

PREDICTOR	INFORMATION	DE- Δr (K50)	PARTIAL-DE- Δr
revert_center (genes $\rightarrow G/2$)	none	0.9999	undefined
revert_mean (predict mean control)	none, no fit	0.961	0.25
ridge linear (control $\rightarrow$ shift)	control	0.947	0.28
positive control (two real cells)	–	0.76	–
model (1B LLM)	control + drug	0.73	not measured

TABLE 4.1 DE- Δr is saturated by a control artefact. The two arms needing no drug information sit above the positive control and above the model; regressing the control out leaves ≈ 0.26 of drug-agnostic skill in the two rows where the partial is defined.

The DE- Δr column is tier 2; the partial column, tier 1.

TWO INNOCENT PROPERTIES COMPOSE INTO THE FAILURE.

S_K is selected using the truth, and treated cells in this atlas regress toward a shared response, so a prediction reverting toward the population centre correlates with $y - x$ by construction. Partial-DE- Δr regresses the control out of both arguments first, closing that route and collapsing the raw 0.95 to ≈ 0.26 in the two baselines with a defined partial value, one of which performs no fitting at all.

The model’s own partial-DE- Δr was never computed,¹ so the comparison anchors to panel- τ , Kendall’s τ over the full 946-gene panel, measured for every arm. The calibration finds panel- τ inverted, so neither an absolute claim nor an ordering read off it counts as evidence of skill. What it can still show is an *absence* of separation. revert_mean

¹ It requires a GPU pass that was not run, so the cell in Table 4.1 is marked, not filled with an expected value.

scores 0.268, ridge 0.265 and the model 0.257 (tier 2, worst convention), three arms inside a spread of 0.011.

The shift therefore decomposes into three parts: control reversion, an artefact; a generic drug programme worth ≈ 0.26 , trivially matched; and a drug-specific component no arm is shown capturing. Later residual-frame instruments show the checkpoint reads trained-drug information while still failing condition-specific prediction.

THE SPIKE-IN, IN BRIEF. One instrument recurs from here on and is worth naming before it is used. The *spike-in* is a forced choice between two drug populations measured within a plate, and no model enters it: a reference pseudobulk is drawn from drug A’s cells, and a metric must score a disjoint sample of A’s own cells closer to that reference than a sample of drug B’s, chance 0.50. Difficulty is then dialled by titration — B’s population is contaminated with a known fraction of A’s cells — so the task runs from easy at zero contamination to impossible once B has been replaced by A and the two candidates are one population. Because the right answer is fixed by construction, the curve grades the metric and never a prediction, which is what makes it the place to ask whether an arbitrary scoring convention changes a score. Six metrics are run on it below.

ONE CONVENTION, DISCLOSED AND THEN DROPPED. A cell sentence lists only expressed genes, so a rank-based metric needs a convention for the unranked panel genes: worst gives them the last rank G , middle a fixed middle rank $G/2$. In that one-off diagnostic, an earlier spike-in run and not the same-plate titration below, whose zero-contamination $DE-\Delta r$ is 0.995, the score reads 0.952 under worst against 0.954 under middle, and nothing moves by more than 0.01, so

all figures are quoted under worst. That diagnostic went to no artifact, so it is a check that was run, not a reproducible result.

THE METRIC THIS INVESTIGATION RELIES ON INSTEAD. NIR is Wu et al.’s “transposed-rank” metric from PerturBench [15], which Miller et al. [22] identify as one of the few consistently calibrated metrics across 14 Perturb-seq datasets. It asks not how closely a prediction matches its own truth, but whether it matches its own truth *more closely than it matches anyone else’s* — much harder to answer by accident.

DEFINITION Normalised Inverse Rank (NIR)

For a prediction $\hat{y}$ of condition i , with truths $\{y_j\}$ over a declared comparison set J ,

$$\text{NIR}(\hat{y}, i) = \frac{1}{|J|} \sum_{j \in J} \left[\mathbf{1}\{\sigma(\hat{y}, y_i) > \sigma(\hat{y}, y_j)\} + \frac{1}{2} \mathbf{1}\{\sigma(\hat{y}, y_i) = \sigma(\hat{y}, y_j)\} \right], \quad (4.5)$$

the fraction of the other conditions in J whose truth the prediction resembles less than its own, ties counted at one half. σ is a *similarity*, so higher means closer; one built from a distance is that distance negated. The score counts comparisons, so it reads σ only through the ordering that σ induces on J , and no strictly increasing transform of σ changes it. Only σ and J change across frames, both declared per frame in Table 4.2, with the single exception the table’s note records: the token-retrieval row is computed with a strict inequality and no tie term.

Figure 4.2 draws one comparison set under this definition, with the three reading rules that follow from it, the half-tie rule among them.

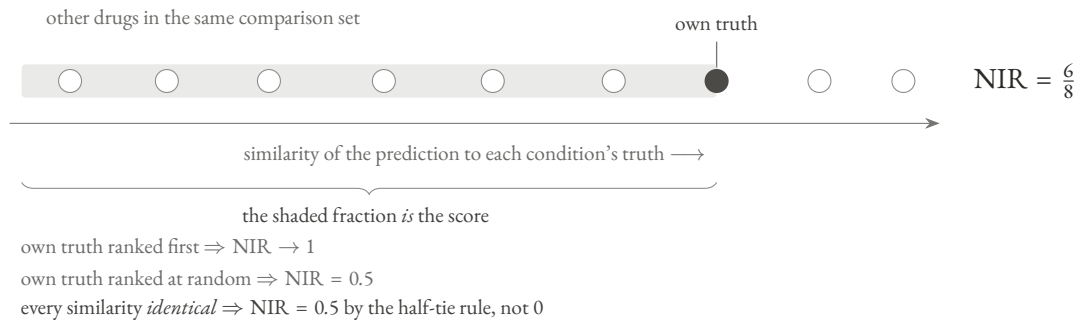


FIGURE 4.2 NIR asks whether a prediction resembles its own condition’s truth more than it resembles the truths of the other conditions in the comparison set J . Own truth ranked first gives NIR $\rightarrow$ 1; ranked at random gives 0.5; and a predictor whose similarities are all identical — the zero vector, which is what “predict the average drug response” becomes in residual space — scores 0.5 by the half-tie rule, not 0.

WHY CHANCE IS 0.50, AND WHAT THAT CLAIM RESTS ON.

Chance sits at 0.50 under a symmetry assumption, not as a consequence of using the same truth set, which is necessary but insufficient. Three things must hold for an uninformative prediction: y_i exchangeable with the comparison truths $\{y_j\}$; its label conferring no privileged relation to the prediction; and symmetric tie handling. The reference is assumed, not proved, so it is checked empirically in each frame. In the expression frame the drug-agnostic linear map and control-copy score 0.500 and 0.504; in the residual frame: random, control-copy and orth score 0.501, 0.506 and 0.501. In the residual frame one asymmetry survives *between* arms: the split-half reference is scored against half-sample truths, every baseline against full-sample truths, which makes the comparison conservative. In the expression frame every arm is scored against the same half-sample truths.

SEE chapter 3 lists every reference in the chapter.

THE TIE TERM IS NOT COSMETIC. Discard it as bookkeeping and the metric breaks in residual space (§ 4.7), where a predictor of the generic response is exactly the zero vector, so every comparison is a tie. A strict inequality scores it 0.000 instead of the 0.500 it deserves, below every random predictor.

THE COMPARISON SET MUST BE DECLARED TOO. Conditions are grouped by (cell line, plate) in the expression frame and by cell line in the residual frame, and the two are not interchangeable. Widening J across plates moves a drug-agnostic baseline from 0.161 to 0.399 (§ 4.2), more than double the score for the same predictor. And because the generic response sits in every truth in equal measure, it cancels, leaving only the drug-specific difference to decide the score — the component the reversion artefact exploits.

THE EXPRESSION FRAME COMPARES TWO RECONSTRUCTIONS. The model emits a cell sentence, which is gene ranks and nothing else, so no distance in expression units can be taken on what it produces. The prediction and the truth it is scored against are both decoded to an expression vector before any distance is taken, through the functional form of the map in § 2.2, fitted once for the whole run as a single slope and intercept on the base-10 logarithm of a gene's position in the sentence, clamped at zero, with every panel gene the sentence omits set to zero. The expression frame's σ is therefore a distance between two reconstructions and not between two measured profiles, and the clamp and the zero fill make it one the underlying rank profiles do not determine.

SEE § 2.2 states the map and the reconstruction argument C2S makes with it

THREE SIMILARITY FUNCTIONS, FIVE DECLARED PAIRS. Three similarity functions are computed and stored: a negated Euclidean distance, a correlation between rank profiles, and a cosine. Naming the function does not name the metric. What fixes the metric is σ composed with the representation it reads, and the negated Euclidean is computed on three different vectors here. Table 4.2 declares the pair (σ, J) behind every quoted figure.

WHY EACH FRAME TAKES THE SIMILARITY IT DOES. In the expression frame the number attached to a gene is a statement about where that gene sits in the ordering. The decode reads a gene's position in the sentence and nothing else, so a large value means near the front and a small one means far down, and magnitude is not a scale sitting on top of the signal but part of what the vector says. Two profiles agreeing in shape while differing in scale are therefore making different claims — one puts its genes near the top of the list, the other puts the same genes in the middle — and a cosine, which divides the scale out, would call

them the same. The truths in a comparison set really do differ this way, listing different numbers of genes and so differing in magnitude as well as in pattern, and the Euclidean distance keeps both. What it keeps is a fact about the representation rather than about abundance, which is the price of scoring reconstructions.

IN THE RESIDUAL FRAME THE CHOICE DOES NOT ARISE. The expression frame demanded a judgement. Magnitude carries something there, so keeping it or discarding it is a decision with a cost either way, and the paragraph above argues for keeping it. In the residual frame there is no such judgement to make, and not because the cost happens to be small. The two candidate similarities return the same number.

What makes that possible is that the residual frame never compares raw residuals. Both arguments pass through ϕ first, and ϕ assigns its weights by rank within a block rather than by anything about the condition: the r -th most up-regulated gene receives $1/\log_2(r+1)$ whichever gene it happens to be, the r -th most down-regulated receives its negative, and every other gene receives zero. Two conditions whose blocks are full therefore hold the same multiset of values and differ only in which gene holds which. The encoding has standardised the magnitude before any comparison takes place, so every truth in the comparison set has the same norm — and a quantity that is constant across J cannot separate one member of J from another. There is nothing in the length for a cosine to throw away.

That alone makes the cosine harmless. The equality is stronger than harmlessness, and it follows in two lines. Write $\hat{u} = \phi(\hat{y})$ and $u_j = \phi(y_j)$, and let $\|u_j\| = C$ for the scored truth and every truth in J , with $\hat{u} \neq 0$. Expanding the squared distance,

$$-\|\hat{u} - u_j\|^2 = \underbrace{-\|\hat{u}\|^2 - C^2}_{\text{independent of } j} + 2\langle \hat{u}, u_j \rangle, \quad (4.6)$$

and because $t \mapsto t^2$ is strictly increasing on $t \geq 0$, ordering J by σ_{euc} is ordering it by $-\|\hat{u} - u_j\|^2$, hence by $\langle \hat{u}, u_j \rangle$ alone. The cosine gives

$$\sigma_{\cos}(\hat{u}, u_j) = \frac{\langle \hat{u}, u_j \rangle}{\|\hat{u}\| \|u_j\|} = \frac{1}{C\|\hat{u}\|} \langle \hat{u}, u_j \rangle, \quad (4.7)$$

a positive constant multiple of the same inner product, so it orders J the same way and breaks the same ties. The two similarities do not take equal values — one is a distance and the other is bounded in $[-1, 1]$ — but NIR reads σ only through the ordering it induces, and on that they are indistinguishable. Reporting the residual frame under either yields the identical score.

The equal-norm hypothesis is what carries the result, and it is a property of the encoding rather than of the biology: it holds wherever both blocks are full, and a residual too sparse to fill one is the only case in which the two could come apart. That is also why the expression frame’s argument does not transfer. There the truths genuinely differ in norm, because they list different numbers of genes, so magnitude is available to separate them and discarding it costs something real.

GRADING THE METRICS, NOT THE MODELS. The framework is Miller et al.’s [22]. Ask not which model scores highest but which *metric* can see quality at all. No model enters the audit; every predictor in it is built from measured expression.

DEFINITION Dynamic Range Fraction (DRF)

A metric is graded by where it places a predictor of known quality on its own scale. Fix three reference predictors and write $m(\cdot)$ for the metric’s value on each:

$$\text{DRF}(m) = \frac{m(\text{pos}) - m(\text{neg})}{m(\text{perfect}) - m(\text{neg})}. \quad (4.8)$$

The denominator is the entire range the metric has to work with, from a predictor carrying no drug information to one that is exactly right. The numerator is how much of that range a real but imperfect signal occupies. A DRF near 1 is a metric that scores genuine signal almost as highly as perfection, 0 one that cannot separate it from nothing at all, and a negative value one that placed the uninformative baseline *above* the real signal.

TABLE 4.2 Every (σ, J) pair behind a quoted **NIR** figure, declared once. A negated Euclidean distance appears in three of the five rows and reads a different vector in each, so what fixes a metric here is the row and not the similarity function.

NAME	σ	VECTORS COMPARED	COMPARISON SET J	WHERE QUOTED
nir_expr	negated Euclidean	reconstructed expression pseudobulk, prediction and truth both decoded	by (cell line, plate), and by cell line in the cross-plate arm of Table 5.1	the blindness verdict (§ 4.2), Table 5.1, Figure 4.5
NIR as the DRF audit computes it	negated Euclidean	measured pseudobulk expression, no decode	by cell line, and by (cell line, plate) in the same-plate arm	Table 4.3
token retrieval	negated Euclidean	undecoded log-expression, as full, shift and residual profiles	the group's other drugs, query and truth disjoint halves of one condition	the retrieval rows of Table 3.2 and Table 4.6
nir_rank	correlation of rank profiles	rank profiles over the truth's expressed genes	by (cell line, plate)	the admissibility screen below, and nothing else
residual frame	cosine	signed rank-discounted residual vectors	by cell line	the residual-frame positive and negative controls of this section, and every residual-frame figure (§ 4.7)

The token-retrieval row is the one departure from the definition above: its code counts strict wins over the comparison set and carries no half-tie term. Exact ties between Euclidean distances on continuous vectors are vanishingly rare, so no figure quoted from that row moves under the tie term, but the row is not an instance of the equation as written.

THE THREE REFERENCE PREDICTORS. All three are pseudobulk profiles of measured expression, stratified by cell line. For each (drug, cell line) the cells are split in half: S_{GT} supplies the target every predictor is scored against, and S_{TD} supplies a technical duplicate of the same condition, disjoint in cells.

Perfect is S_{GT} itself, which fixes the top of the scale at whatever value the metric returns on an identical pair rather than assuming it is one.

Negative is the drug-agnostic mean, the leave-one-out average over the other drugs in that cell line. It is a real expression profile and not

noise, which is the point: a metric that rewards it is rewarding what every drug does, not what this drug did.

Positive is the *interpolated duplicate*, the technical duplicate pulled gene by gene toward the negative control in proportion to how weakly that gene distinguishes the drug. Per gene,

$$\mu_{ID} = \alpha \mu_{TD} + (1 - \alpha) \mu_{\text{mean}}, \quad \alpha = 1 - p_{\text{DEG}}, \quad (4.9)$$

where p_{DEG} is the gene's differential-expression p -value for this drug against the other drugs in the same cell line, computed on the duplicate half. A gene that separates the drug from its neighbours keeps its measured value; a gene that does not is pulled back to what any drug would have produced. What results is a predictor carrying real drug signal with the noise-only genes damped — imperfect, but unambiguously better than knowing nothing, which is exactly the predictor a working metric must rank above the negative control. Every metric in Table 4.3 sees the same three profiles.

READING A NEGATIVE VALUE AS INVERSION. Taking a negative DRF to mean the metric is inverted takes one further step, that the sign belongs to the metric and not to the run. Three of the four negatives survive it; the fourth is carved out where it appears.

inversion A metric placing the drug-agnostic baseline above the positive control, as a property of the metric and not of one run.

HOW THE NUMBER IS FORMED. DRF is a ratio whose numerator and denominator are both estimated from the same data, and that makes three things decisions rather than defaults: how the point estimate is formed, how it is given an interval, and against what it is tested.

The estimate is a ratio of means and not a mean of ratios. Each of the three reference predictors is scored on every (drug, cell line) row, the three scores are averaged across rows, and the ratio is formed once

CORRECTION The DRF bootstrap resampled the wrong unit and imposed no null; both repaired and both arms rerun before any figure here was read.

from those averages. Forming a ratio per row and averaging afterwards would let any row whose denominator fell near zero dominate the result. Within the rank-based metrics ties take average ranks, since otherwise a tied gene's rank would be settled by its position in the panel, which is a fact about how the panel was ordered and not about the cell.

The interval is a percentile cluster bootstrap resampling whole cell lines, the unit that repeats across rows, and it recomputes the entire ratio inside every draw. That second point is the one that matters. The denominator $m(\text{perfect}) - m(\text{neg})$ is not a known constant but an estimate, so an interval that resamples the numerator while holding the denominator at its observed value describes the uncertainty in the top half of the ratio and calls it the uncertainty in the ratio. It comes out too narrow. Letting the denominator move with the numerator keeps the interval about the quantity actually reported. The cluster is the cell line while the grouping unit is the (cell line, plate) pair, and the two are reported separately because they are different things; only the cell-line channel is clustered, so these intervals ignore the crossed well dimension and are narrower than fully crossed ones, a cost the declared exception below states in full.

The test imposes its null rather than assuming it. A p read straight off the observed resampling distribution asks how often the data disagree with themselves, which for a clearly positive statistic is never, so the p pins to the Monte Carlo floor whatever the truth is and nothing is tested at all. The distribution is therefore recentred on $\text{DRF} = 0$ before anything is counted, and the p is the fraction of null draws reaching the observed value; with 2,000 draws the smallest one attainable is $1/2001$. Holm control runs across the five metrics at once, since they are tested together and read as one family.

WHAT THE NUMBER SURVIVES. Table 4.3 carries the values. Every choice above guards against the estimator flattering itself; none of them says whether the configuration that produced the estimate is doing the work, and that is a separate question with separate answers. Under the imposed null no draw of 2,000 reaches NIR’s observed value, so it sits at the floor in both arms and its estimate stands some forty cluster-robust standard errors from zero — but a distance from zero is only as good as the setup that generated it.

Three checks bound the setup. Restricting the comparison set, and the drug-agnostic negative built from it, to the same plate changes no sign; the two arms cover 25 and 27 cell lines (the all-plates run was capped at 25 lines, the same-plate run admitted every line with six drugs on one plate, 27 in all), so the separation is not an artefact of comparing across batches. A stricter configuration, capped at 25 *groups* and admitting only 18 distinct cell lines, still gives NIR +0.519 [+0.479, +0.546] over 335 rows, so it is not an artefact of breadth either. And the inversion of the four rivals belongs to the data rather than to a leak: a leave-one-out correction barely moves the numbers, real cell lines carrying 20–204 drugs, and it survives holding the plate constant.

What the argument then uses is the sign pattern and nothing finer, drawn for the same-plate arm in Figure 4.3, panel b. No ranking among the negative metrics is claimed.

WHAT THIS REPLICATES, AND WHERE IT DOES NOT. The methodological core of Miller et al. replicates, and this audit sharpens it. Metric choice dominates, and the control-referenced correlation they flag is the worst performer in both arms ($DE-\Delta r$, -0.686 across plates and -0.515 within). Their calibrated set does not transfer intact. `weighted_r2`, this project’s counterpart of the weighted $R^2\Delta$ [23] they

TABLE 4.3 Under the hardened protocol, **NIR** is the only one of the five whose **DRF** is positive, and the sign pattern is identical in both arms. All four rivals score the drug-agnostic baseline above the positive control and exclude zero in both arms. The second column is what separates NIR.

METRIC	SCORED AGAINST	DRF ALL PLATES	DRF SAME PLATE	READING
NIR (measured expression)	other conditions' truths	+0.635	+0.545	the only positive
weighted_r2	its own truth	−0.163	−0.104	no directional claim
panel- τ	its own truth	−0.319	−0.250	inverted
spearman_expr	its own truth	−0.650	−0.500	inverted
DE- Δr	its own truth	−0.686	−0.515	inverted; the default

The NIR row is the audit's own construction, a negated Euclidean distance on measured pseudobulk expression with no decode, which is the second row of Table 4.2 and not the `nir_expr` of the blindness verdict.

The all-plates run has 1,820 rows (one drug within one cell line) clustered on the 25 cell lines they nest in; the same-plate run has 586 rows over 44 cell-line $\times$ plate groups spanning 27 cell lines, clustered on cell line. NIR reads +0.635 [+0.604, +0.663] and +0.545 [+0.516, +0.569], Holm $p = 0.002$ at the resampling floor in both arms. The four negatives are point estimates, each excluding zero on the same clustering in both arms; `weighted_r2` sits closest to zero and its sign tracks cell count, hence no directional claim.

endorse, is negative in both audited arms, though closest to zero and sign-tracking cell count, so it carries no directional claim; the inversions of `panel- $\tau$` and `spearman_expr` belong to neither calibrated set. Their endorsed NIR does survive. Being rank-based is not what protects it; refusing to score anything the generic response can supply is.

ADOPTING THE INSTRUMENT IS ALSO A DEFENCE. Miller et al. conclude that under calibrated metrics deep models outperform uninformative baselines; § 4.2 finds ours at chance on that metric, in the decoded representation the verdict deploys and not in the one the audit graded. Genetic against drug perturbation and a discriminative against a generative model are candidate explanations; § 4.11 supplies a third. Adopting their metric also puts the verdict beyond the objections their own paper raises.

SEE § 2.4 sets out the dispute this audit enters.

ONE DECLARED EXCEPTION ON THE INTERVALS. DRF is a ratio of means; the two-way estimator used everywhere else here is written for a mean. Both DRF intervals are therefore *one-way* percentile cluster bootstraps that resample whole cell lines and recompute the

RECALL two-way cluster-robust intervals on every mean chapter 3

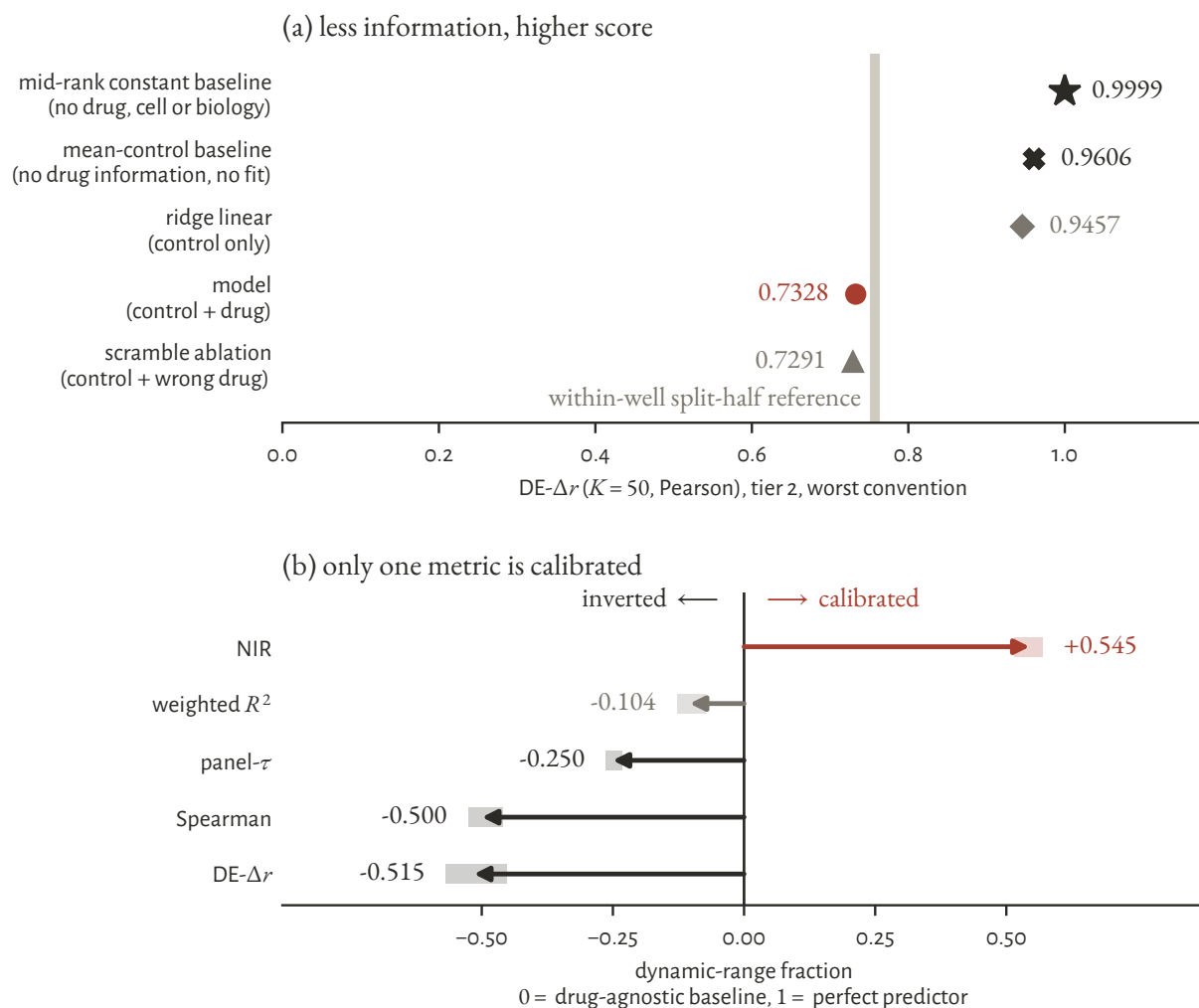


FIGURE 4.3 A predictor carrying nothing about drug, cell or biology scores the field's standard metric above both the within-well reference and the model, and of five metrics only **NIR** grades in the direction its name implies. (a) DE- Δr , $K = 50$, Pearson, tier 2 unseen drugs, worst convention, $n = 400$ conditions; the axis runs from 0 and is not truncated. The grey band marks the within-well split-half precision reference at 0.76, the two-real-cell positive control of Table 4.1: two halves of the same well, a precision figure and not a biological replicate. Its width is a drawing device, not an interval, and the arm marks are bare — the source carries a normal-approximation interval over conditions, and this document draws only the clustered bootstrap, so nothing is drawn rather than an interval of another kind. (b) DRF for the five metrics of Table 4.3, the same-plate arm only: 586 (drug, cell line) rows over 44 cell-line $\times$ plate groups spanning 27 cell lines, the pale block at each arrowhead the 95% percentile bootstrap interval clustered on cell line quoted there.

ratio inside each draw, respecting the cell-line dependence and ignoring the crossed well dimension, so they are narrower than fully crossed intervals. The separation they grade is far too large for that to change the sign, but the caveat belongs beside the number.

TWO QUESTIONS ABOUT THE INSTRUMENT REMAIN, AND THEY DIFFER. The first is whether the conditions are empirically

discriminable at the tested budget. A spike-in forced choice answers it. Within a plate, a reference pseudobulk from drug A's cells must be scored closer to a disjoint sample of A's own cells than to drug B's, chance 0.50, with contamination of B by A titrating the difficulty. At zero contamination three of the six metrics run on it separate the two populations essentially perfectly (panel- $\tau = 1.000$, DE- $\Delta r = 0.995$, top- $n \tau = 0.987$, plate held constant); at full contamination those same three return 0.501–0.502. The whole-profile similarities are much weaker even at zero contamination, cosine 0.740, Spearman 0.680, cosine on the shift 0.620, which matters because a whole-profile similarity is what NIR uses here. Split-half retrieval establishes discriminability, not predictability from untreated input, so discriminability is not the sole bottleneck at this budget.

The spike-in cannot answer the second question, whether NIR registers a true positive, since NIR was not among the metrics run on it. The evidence is the DRF audit's own positive control. Under NIR on measured expression the interpolated duplicate scores 0.802 against the drug-agnostic mean's 0.457 across plates, and 0.668 against 0.272 within plate; in the residual frame a split-half replicate reaches 0.854 against negatives at 0.501–0.506. None of those is the reference the verdict is measured against. That reference is the within-well split-half precision estimate recomputed on the tier-2 within-plate benchmark, where it sits above chance but not far above it; that, and not 1.0, is the bar § 4.2 states the model against.

The audit licenses NIR and nothing else. panel- τ is inverted in both audited arms, so no absolute claim rests on it and no ordering read off it counts as evidence of skill; an inverted metric preserves neither. Where DE- Δr appears later it is only as a paired comparison of two arms on one scale, never as an absolute score. And a metric is not calibrated by having a relative calibrated. What is graded here is a negated Euclidean distance on measured pseudobulk expression, the second row of Table 4.2, and the residual frame's cosine similarity in its own frame, each against its own controls. The nir_expr of the blindness verdict is that distance read on decoded pseudobulk, the first row, with controls

SCOPE LIMIT carried into § 4.2 and the rest of the chapter

of its own but no DRF of its own, so it stands on the assumption above. A third σ would need its own audit.

THE CLAIM. The field’s standard score is saturated by a predictor that carries no information, and of the five metrics audited only NIR grades in the direction its name implies. This audit grades five candidates, and NIR is the only one with a positive DRF (+0.635 [+0.604, +0.663] on a one-way cluster bootstrap over the 25 cell lines, the declared exception to this chapter’s two-way rule, Holm $p = 0.002$; +0.545 [+0.516, +0.569] over 27 cell lines with same-plate partners, Holm $p = 0.002$), and the other four are negative with intervals excluding zero in both audited arms. Three of them (panel- τ , spearman_expr and DE- Δr) are inverted rather than merely uninformative, each ranking the drug-agnostic baseline above the positive control; weighted_r2 sits closest to zero and its sign tracks cell count, so no directional claim is made for it. The saturating predictor is not a curiosity. It carries no information about drug, cell or biology and scores DE- $\Delta r = 0.9999$ in rank space, above the two-real-cell positive control of 0.76 and above the model’s 0.73. Under NIR on measured expression the positive control is retrieved well clear of the negative (0.802 against 0.457 across plates, 0.668 against 0.272 within), so the instrument registers a true positive. Every drug-use claim in the rest of this chapter that rests on a continuous prediction score is therefore made on NIR, in the audited construction where the audit reaches and in the decoded one that inherits its licence where it does not; DE- Δr appears below only where two arms are compared with each other on the same scale, never as an absolute score of the model’s drug use. The remaining instruments (forced-choice grading, output invariance, and the logit-space activation tests of § 4.5) are deliberately on scales of their own, each with a fixed chance value or a paired control.

4.2 Drug-blindness in the expression frame

■ EXPR

With a properly calibrated metric in hand, the investigation can ask the question it exists to answer. Does the model use the drug at all to make predictions? Four instruments answer it, and each exists because one specific objection would otherwise be fatal.

The answer is no, and reaching it takes more than a scoreboard. With the metric calibrated, the model can finally be scored on the tiers, and two readings follow that look nothing alike and settle nothing on their own.

Within plate on tier 2 ($n = 606$) the model scores 0.498 against a within-well split-half precision reference of 0.576. The gap is narrow, but so is the scale: the reference itself stands only a little above chance, and the model sits at chance inside it, matched by predictors that know nothing about the drug — a drug-agnostic linear map at 0.500 and a control-copy at 0.504, against the global mean's 0.180 (NIR, $n = 606$). That average also hides a split. Binning the same 606 conditions by their own reference gives 296 whose reference falls below 0.6 (stratum mean 0.287, itself below chance: on average not even a second sample of the same well ranks above chance); 106 marginal, between 0.6 and 0.8; and 204 identifiable, whose reference clears the pre-set 0.80 bar. On that last third the model looks far better and discriminates well above chance, reaching 0.768. A control-copy carrying no drug information whatsoever reaches 0.766.

Neither reading is causal, and that is the whole difficulty. A model scoring at chance might be ignoring the drug, or reading it and failing to turn it into anything; a model scoring 0.768 where a drug-blind predictor scores 0.766 might be adding nothing, or adding something too

RECALL one estimator, seven settings
Table 3.2

small for the metric to resolve. The two readings even point opposite ways, one suggesting the model has learned nothing and the other that it has learned a good deal, and both are consistent with the drug token never being consulted. No comparison of scores can separate these, because no comparison of scores intervenes on anything. What settles it is an experiment in which the drug changes and nothing else does.

FOUR INSTRUMENTS, AND EACH CLOSES A DIFFERENT OBJECTION. The four are not four attempts at one measurement. The second is the experiment the previous paragraph asked for, and the others exist because that experiment is still scored by something, and whatever scores it could be blamed for the result. Each successive instrument removes another part of the apparatus from the argument.

The first is the benchmark itself, run *within plate*. Drug and plate are partially confounded by the atlas’s design, so a comparison set free to span plates lets a batch effect stand in for drug identity and inflates every arm at once. Restricting the set to a single plate removes that channel before anything else is asked.

The second is the scramble ablation, the only manipulation among the four that is scored against a truth. The control cell is held fixed and only the drug name and mechanism in the prompt are replaced, by a different drug on the same plate; everything the model sees apart from the drug is unchanged. A gap between the two generations is caused by the drug or by nothing, which is what no comparison of scores could establish. Here the partner is drawn at random, and that carries a known risk: a swap to a drug with a similar true response makes an unchanged output correct rather than blind, which biases the test toward finding nothing. A swap-distance sweep bins (drug, swap) pairs by the response distance between the real and the swapped-in drug, turning the single test into a curve, and § 4.10 returns to the question

with partners stratified by residual cosine, where a neutral null can be identified rather than assumed.

The third is forced-choice grading. A prediction is scored against its own condition's truth and against another drug's truth from the same comparison set, and what is recorded is how often its own wins.

The fourth is output invariance, and it touches no truth at all. It asks only whether changing the drug token changes the generated text, comparing generations against other generations from the same fixed control cell. No similarity function against a ground truth, no comparison set and no decode enters it, so no scoring choice can confound it. It is the least informative of the four and the least confoundable, and those are the same property.

Taken together they share a model and little else. The first can be defeated by batch leakage, the second by a badly chosen swap partner, the third by a degenerate comparison set, the fourth by nothing except the model genuinely ignoring the drug — and it is that lack of a common failure mode, rather than any one of them, that carries the verdict.

ONE MEASUREMENT RULE COMES FIRST. Drug and plate are partially confounded by the atlas's design (chapter 3), so grouping by cell line alone lets the batch effect stand in for drug identity. A drug-agnostic mean baseline, containing no drug information whatsoever, scores 0.399 when the comparison set may span plates against 0.161 when it may not, and the precision reference falls from 0.625 to 0.575 alongside it, the batch effect having inflated that too. That is plate leakage, measured and not assumed. Being wider, its within-plate figures sit slightly off the ones above; what it licenses is the rule, not a substitute set of scores.

FRAME ■ EXPR all four instruments are scored in the expression frame

SEE Table 5.1 is the leak audit, a wider run than the benchmark

Every discrimination instrument gained a `-same_plate_only` mode and every discrimination result in this chapter was re-run within plate, except where a departure from that rule is named at the point of use, as with the swap-distance sweep that answers the fourth objection below, whose swap partners were not plate-restricted (§ A.3), and the residual-frame scores of § 4.7, whose comparison set is grouped by cell line and whose justification is given there. A third departure is the identifiability curve in Table 3.2: it is cross-plate because no within-plate sweep reaches the cell budgets it reports, the plate-matched version stopping at twenty cells per condition, and it is computed on raw pseudobulk rather than on the decoded expression the rest of that table uses. It is therefore read as a statement about aggregation and never set beside the within-plate references above it. Conclusions are unchanged; some magnitudes shrink.

SCOPE LIMIT carried into every discrimination result in this chapter

THE CENTRAL MANIPULATION IS THE SCRAMBLE ABLATION.

The prompt is left byte-identical except that the drug name and mechanism are replaced by another drug's, the control cell and the scoring truth unchanged. Control- and batch-derived leakage therefore cancels in the scramble gap $\Delta_{\text{scr}} = \text{NIR}_{\text{model}} - \text{NIR}_{\text{scramble}}$, the only difference in this chapter whose leakage cancels by construction. It does not displace the drug-agnostic comparisons above and below, which rest on the within-plate rule instead.

Where that ablation is applied is a choice, and this chapter makes the hard one first. Every NIR scramble in this section is scored on tier 2, held-out drugs the checkpoint never saw in training, and the substituted partner is drawn from that same held-out tier — so the model is asked to distinguish two drugs it knows nothing about. Nothing measured on NIR here touches trained conditions; the tier 1 arm of the NIR runs returned no scoreable conditions at all, since NIR needs a per-condition pseudobulk and that tier averages roughly two cells per condition. The one tier 1 figure in this section, the $\text{DE-}\Delta r$ row of Table 4.4, is a paired comparison that needs no pseudobulk, and it agrees. § 4.9 later runs the easier version of the same manipulation, scoring a model on conditions it did train on, where a drug-blind model has memorisation to fall back on and any real drug sensitivity has its best

chance of showing. This section establishes the expression-frame result on held-out drugs; § 4.6 tests the denoised expression-frame targets on that same tier. It does not establish their failure on trained conditions. § 4.9 establishes prompt sensitivity for the residual arm on trained conditions; the expression-frame controls reported there do not document their condition split and cannot close that separate question.

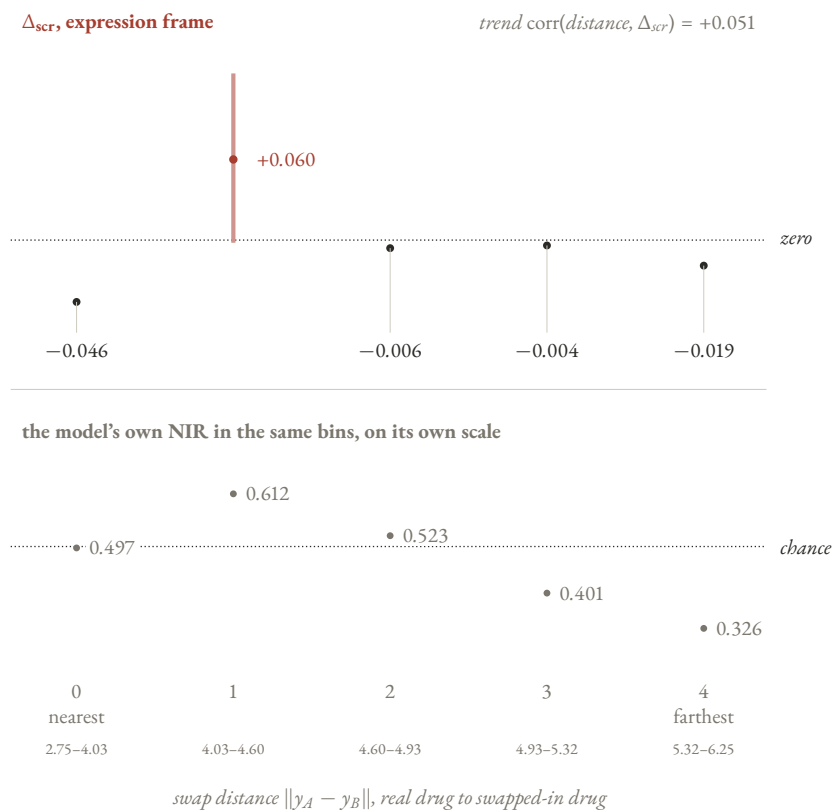


FIGURE 4.4 Naming a drug whose response is far from the truth damages the prediction no more than naming a near neighbour. Tier 2 results. Upper strip, the scramble gap across five bins of swap distance. A real dissimilarity effect would climb from left to right and peak at the farthest bin; instead the correlation between distance and gap is +0.051, and the only bin whose interval is quoted, +0.060 [–0.002, +0.124], is the second of five. Lower strip, the model's own NIR in those same bins, which does fall overall, though not monotonically, because drugs far from everything populate the far bins. The two strips are different quantities on different scales, and Δ_{scr} uses the same drug on both arms, so that drift cancels out of it. Non-canonical: $\max_new_tokens=1200, n = 40$ per bin, which excludes a large dissimilarity effect without establishing a null.

TWO CONSTRAINTS ON THE SWAP PARTNER. The partner must be a different drug. Allowing the same drug at another dose or plate substitutes the drug for itself, an unchanged prediction is then correct and not blind, and the contamination concentrates in the most similar stratum. It must also be on the same plate, since residuals retain plate structure and a cross-plate partner would differ in batch as well as in drug.

A scramble that swaps in a drug with a similar true response is a weak test, and that risk is structural here. In full-profile space the mechanism annotation is close to uninformative about transcriptional response. Same-mechanism drugs in Tahoe are just over two per cent closer to each other than arbitrary pairs are. Hence conditioning on mechanism fails to move the mean-shift grouping ladder quoted under the first of the four objections at the end of this section, and a partner of different mechanism cannot be relied on for a different response. Mechanism does carry signal in the residual frame (§ 4.12); in full-profile space the generic response swamps it, a frame difference and not a contradiction.

SEE § 4.9 quotes the distances behind that statement

Two remedies are used, one here and one later. The first turns the single test into a curve. Within a cell line — this earlier run drew partners without plate restriction (§ A.3) — each drug’s cells are split in half and the distance between two drugs is taken between their half-sample truth pseudobulks, $\|\gamma_A - \gamma_B\|$, the same negated-Euclidean geometry NIR uses to rank one condition’s truth against another’s. The sweep therefore inherits the benchmark’s similarity rather than importing one from elsewhere, and the axis is fixed before any generation: it is a property of the atlas, not of the model.

Partners are drawn to span the quantiles of that distance from A , so each real drug contributes swaps from across its whole available range instead of one partner of unknown similarity. Every swap rebuilds A ’s prompt with B ’s name and mechanism, leaves the control cell and the rest of the prompt intact, and scores the generation against A ’s truth. The target never moves; only the label in the prompt does. Pooling the pairs and binning them by distance gives five bins running from nearest to farthest (Figure 4.4, § A.3), and the gap is aggregated inside each bin with intervals clustered on cell line.

What makes the curve worth reading is that each outcome was given its meaning before the run. A gap rising with distance would say the model does discriminate dissimilar drugs and that the flat aggregate had been diluted by near-twin swaps — a correction to the claim, not a confirmation of it. A gap that stays flat into the far tail says the blindness holds at the hardest setting the plate affords, where the swapped-in drug’s real response is as unlike the truth as the data allows.

THE THIRD INSTRUMENT IS FORCED-CHOICE GRADING. A prediction is scored against its own condition’s truth and against another drug’s truth from the same comparison set, recording how often its own wins. Chance is 0.50 by construction, and the precision reference lands between 0.67 and 0.83 depending on cells per condition.

THE FOURTH IS OUTPUT INVARIANCE. It asks whether the drug token changes the generated text at all, without reference to any truth. It is run at 3800 tokens, the longest budget in the chapter, because truncation would cut the comparison itself. For a fixed control cell it compares

SEE chapter 3 records the four generation budgets

$$\text{invariance gap} = \underbrace{\overline{\text{sim}}(\text{same prompt, resampled})}_{\text{sampling noise alone}} - \underbrace{\overline{\text{sim}}(\text{prompts differing in the drug})}_{\text{noise plus any drug effect}} \quad (4.10)$$

with similarity as top-100 Kendall τ and as Jaccard overlap, over 8 contexts $\times$ 12 drugs at temperature 0.8.

TABLE 4.4 Four independent instruments agree, and the two that return intervals bound the drug’s effect instead of merely covering zero. Near-ceiling $DE-\Delta r$ is achieved identically with a different drug’s name and mechanism substituted into the prompt, so that score is drug-independent.

INSTRUMENT	QUANTITY	VALUE	READING
Forced-choice grading	own-truth win rate	0.48 ($\approx$ scramble; reference 0.67–0.83)	chance
Output invariance	invariance gap, top-100 τ	0.000 [−0.019, +0.016]	bounded, ≤ 0.016
	invariance gap, Jaccard	+0.000 [−0.008, +0.008]	bounded, ≤ 0.008
Scramble, tier 1	$DE-\Delta r$	0.740 real vs 0.739 scrambled	indistinguishable
Scramble, tier 2	$DE-\Delta r$	0.733 real vs 0.729 scrambled	indistinguishable
Scramble (identifiable subset)	Δ_{scr} (NIR)	+0.014 [−0.016, +0.042]	bounded, $\leq +0.042$

The two $DE-\Delta r$ rows are paired comparisons on one scale, the only use § 4.1 licenses for that metric.

AN INTERVAL THAT COVERS ZERO EARNS ITS KEEP BY WHAT IT EXCLUDES. Not by the word *null*. Two predictions from a byte-identical prompt agree at $\tau = 0.198$, so the invariance interval caps any drug-driven divergence at roughly a twelfth of that agreement, and the Jaccard version is tighter still, +0.000 [−0.008, +0.008]. On the identifiable stratum the scramble gap of +0.014 [−0.016, +0.042] runs against the 0.170 separating the model’s 0.768 from that stratum’s 0.938 reference; at the very top of the interval, naming the right drug would buy a quarter of what is there to be bought, and the point estimate puts it at 8% of that. Neither figure shows the effect is zero, and no experiment here could. Both put it far below the headroom these conditions leave.

FOUR WAYS THE VERDICT COULD BE AN ARTEFACT. Each is a way the result could be about the apparatus rather than about the model, and each is settled by a measurement rather than by argument.

That there is nothing to learn. A ladder of mean-shift baselines, each predicting the average shift of a progressively narrower group, says otherwise. On the $n = 400$ tier-2 generation-evaluation population

SEE a $DE-\Delta r$ diagnostic on the $n = 400$ generation-evaluation population, distinct from the $n = 606$ benchmark above and from the residual-frame ladder of § 4.11

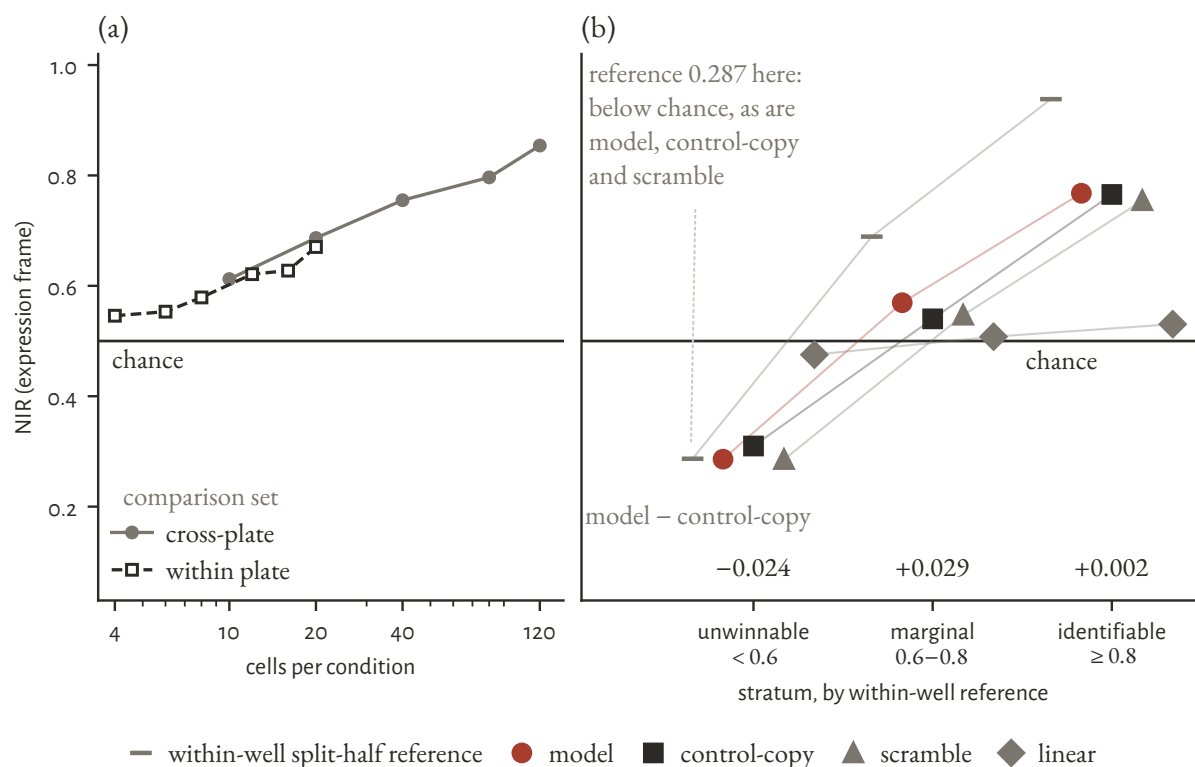


FIGURE 4.5 Conditions are more empirically discriminable at the tested pseudobulk budget where the precision reference is high, and precisely there, a predictor containing no drug information matches the model. Both panels are NIR in the expression frame, on one shared axis. (a) The within-well split-half precision reference against cells per condition, for two comparison sets: cross-plate, and within plate. (b) The same reference used to bin conditions, five arms per stratum, $n = 296$ unwinnable, 106 marginal, 204 identifiable. On the unwinnable stratum the reference itself is 0.287, below chance, as are the model, control-copy and scramble marks. On the identifiable stratum the model reaches 0.768 and a control-copy baseline 0.766, paired contrast $+0.002$ [-0.026 , $+0.028$], clustered over cell lines. Plotting the model alone would suggest it does better on higher-reference drugs. The arm marks are bare: the source carries no interval for a stratum mean, and this document draws only the clustered bootstrap, so nothing is drawn rather than an interval of another kind.

of Table 3.1, block C, read on $DE-\Delta r$ as a ladder of arms on one scale (the use § 4.1 licenses) and not on the within-plate NIR benchmark whose global mean is the 0.180 above, the global mean scores 0.056 and conditioning on cell line raises it about two-and-a-half-fold to 0.142, so the atlas carries real structure. What it does not carry is structure the drug label reaches: mechanism leaves the global figure untouched at 0.052, and mechanism together with cell line does no better than cell line alone (0.120 against 0.142). There is something to know, and the variable that knows it is the cellular context.

That the benchmark is too weak to say anything. Barely half the atlas clears the pre-set identifiability bar (Table 3.1), so the verdict is

re-scored where the wells are demonstrably adequate. It does not survive the move. On that stratum the model appears to beat the linear baseline, 0.768 against 0.531, and then a control-copy carrying no drug information reaches 0.766 on the same conditions; the paired contrast is +0.002 [−0.026, +0.028] and the leak-immune Δ_{scr} stays bounded as above.

That control-copy carries a second finding, and it is easy to walk past. Across the three strata its score tracks the reference almost step for step (Figure 4.5, panel b): 0.310 against a reference of 0.287 where the reference is lowest, 0.540 against 0.689 on the marginal stratum, 0.766 against 0.938 on the identifiable one. A predictor that knows nothing about any drug does best exactly where the ranking problem is easiest. So what makes a condition identifiable is largely what makes its untreated cells distinctive among their plate-mates, and the distance from the model to a high reference is not all room that better drug modelling would fill. The reference does rise with aggregation (Figure 4.5, panel a, within plate: 0.55 at 4 cells, 0.58 at 8, 0.67 at 20); what the strata add is that most of the ranking can be done without the drug at all.

That the encoding discards the drug. A cell sentence keeps the rank order of genes and throws away expression magnitude, so if the drug lived in the magnitudes the following sections would be attacking the wrong object. This needs no model to test: ask whether true expression discriminates drugs better than rank does. It does not. At single-cell resolution, cosine, Pearson and Spearman similarities on true normalised expression all sit within 0.04 of chance (0.529–0.531 [0.515, 0.543]), the same near-chance regime as rank. What recovers the signal is aggregation: at fifteen cells the control-referenced shift cosine reaches 0.793 [0.775, 0.809]. The re-encoding of § 4.9 will therefore not be a restoration of magnitudes; it keeps rank throughout and changes only

what is ranked. The same measurement also bounds what a single cell can deliver, since the single-cell same-drug against different-drug gap is only $+0.006$ [$+0.003$, $+0.008$]: these instruments do not resolve drug identity from one observed cell at this sequencing depth and 946-gene panel.

That the scramble swaps in near-twins. The swap-distance sweep bounds this rather than settling it. It sits outside the within-plate rule, an earlier run at `max_new_tokens=1200` on the unseen-drug tier of a different evaluation set, with swap partners not plate-restricted and $n = 40$ per bin — wide enough to exclude a large effect, not to establish a null (§ A.3). No distance bin has an interval excluding zero. The far tail reads -0.019 across a $2.3\times$ spread in swap distance with a trend correlation of $+0.051$, so a large dissimilarity effect is excluded for the expression-frame model and a small one is left open.

WHAT NONE OF THE FOUR CAN SEE. All four watch the output. They establish that the generated text does not move when the drug does, and that is the strongest behavioural statement available. It is also silent on why. A model that never encodes the drug and a model that encodes it and uses it incorrectly produce the same flat instruments, and no experiment that only reads generations can separate them. The ladder above already hints at where to look, since the asymmetry it finds between cellular context and drug class is one the activations turn out to repeat. § 4.3 therefore stops asking what the model emits and starts asking what it holds.

A FIFTH READING OF THE SAME SWEEP, WITH NO COMPARATOR IN IT. The four instruments above each contrast the model against something — a baseline, a control-copy, a scrambled prompt — and each is only as good as the thing it is set against. The swap-distance

sweep supports one further reading that contrasts nothing. Its partners were drawn to span the distance quantiles, five per condition, so within a single condition the same model answers the same question five times with a differently wrong drug named each time. Write y_{ik} for the NIR of condition i 's k th scrambled prompt and x_{ik} for the *negated* Euclidean distance between that condition's truth and the truth of the drug substituted into it — negated so that x rises with similarity, as the partner cosine does everywhere else in this thesis, and one predicted sign serves every version of the test. Then

$$y_{ik} = \alpha_i + \beta x_{ik} + \varepsilon_{ik}, \quad k = 1, \dots, 5, \quad (4.11)$$

with one intercept α_i per condition. The target-drug prompt is never a point on this line, so nothing is measured against a chosen baseline, and $\beta = 0$ is the null by construction: a model that ignores the name emits the same distribution whichever drug is substituted, putting its five points at one height whatever their similarities happen to be. Prompt sensitivity predicts $\beta > 0$, the score rising as the substituted drug's real response moves closer to the target's.

Which intercepts are fitted decides the answer. A single pooled intercept draws one line through all 200 points at once, so conditions are compared with each other as much as with themselves, and fitted that way the slope is +0.0986 [+0.0180, +0.1792] clustered on drug — a clear effect, and not a real one. A condition whose drug is unlike anything else on its plate has distant partners *and* an unusual truth that any prompt will struggle to rank, so the line bends without a name having been read. Giving each condition its own intercept removes that comparison and leaves only what varies inside a condition: one model, one target, five differently wrong drugs. Almost the whole effect is in the part that discards — the between-condition slope is +0.1563 against

a within-condition slope of $+0.0055$, $[-0.0472, +0.0582]$ clustered on drug and $[-0.0484, +0.0594]$ on cell line, over 200 prompts across 40 conditions. The tighter interval excludes the pooled estimate it replaces. Compared with itself, no condition's score moves with how unlike the substituted drug is.

What this adds is not more of the same evidence. The scramble is a paired intervention — one condition, one thing changed — and is confound-free by construction, but it needs a comparator, and § 4.10 shows how much of a gap the choice of comparator can carry. This regression needs none. What it gives up in exchange is what it can license: every point in it is a *wrong* prompt, so it can show only whether the model tells wrong answers apart, never whether it produces a right one. A model could pass it and predict nothing. Here that trade costs nothing, because the model does not pass it.

THE CLAIM. The model is at chance overall (0.498 against a 0.576 within-well split-half precision reference), matched on the identifiable stratum by a zero-drug-information control-copy (0.768 against 0.766, paired contrast $+0.002$ $[-0.026, +0.028]$), and no more moved by a maximally dissimilar swap than by a similar one in the single-cell column of a non-canonical `max_new_tokens=1200` sweep outside the within-plate rule (§ A.3).

4.3 Decoding drug identity from activations

■ EXPR

A model that ignores a drug it was told about can fail in two ways. Either the drug never enters its computation, an activation-side encoding failure that architectural fixes might reach, or it enters and nothing reads it, a decoder failure only an objective-side fix could reach. Behaviour cannot tell them apart, since both predict the same scores. The weights can. A linear decode probe, read as a floor test and not as evidence of use, settles the first half.

§ 1.4 left three explanations open for why generation ignored the drug: it is absent from the data, absent from the model's representation, or present there and simply not consulted when the model generates. The first closes here. § 4.2 measured a substantial stratum, selected by a bar set on the data alone, where the true separation between drugs is real and large — the identifiable stratum. On exactly that stratum the model does no better than a baseline copying the untreated cell, 0.768 against 0.766. What is missing there is not signal to find; it is the model's use of it. That is the failure the question above already split in two: does the drug never reach the model's computation, or does it reach it and go unread? No behavioural score, however this benchmark is sliced, can tell those two apart.

The three sections that follow can, and they answer in one direction. Drug identity is written into the activations and stays written, more legibly with depth once batch structure is projected out: the failure is not on the activation side. Generation then consults that identity too weakly to account for the drug-blindness already measured. Both halves of that answer carry a discount, and not the same one. Presence is close to guaranteed by construction — a category named verbatim in the prompt is not hard for a linear probe to find, which is why this section's own instrument is built to be read as a floor test and nothing more. The low-gain reading is the opposite case. Twenty-six tests were run across training arms and depths, and after correcting for testing that many at once, exactly one stays significant. Even that one is small, a shift of one to two per cent in the model's existing preference, against a 10% margin pre-registered as the threshold for calling an effect material. It is the weakest fundamental measurement in the chapter, and it is carried below with that weakness attached, in three steps. This section asks only whether the drug is in the computation at all. § 4.4 asks

identifiable stratum conditions whose own within-well split-half reference clears the pre-set bar

FRAME ■ EXPR scores here are full-profile, § 4.1

whether generation is causally sensitive to that region, and at what weight beside cell line. § 4.5 asks whether it is the drug's identity that gets read there, or merely the region's presence.

THE PROBE, AND WHY ITS POSITIVE IS NEARLY FREE. The prompt names the drug in plain text (chapter 3), so recovering it from the activations establishes only that the model did not discard what it was handed. The informative outcome would have been a negative, and what the probe buys is the depth at which one would appear.

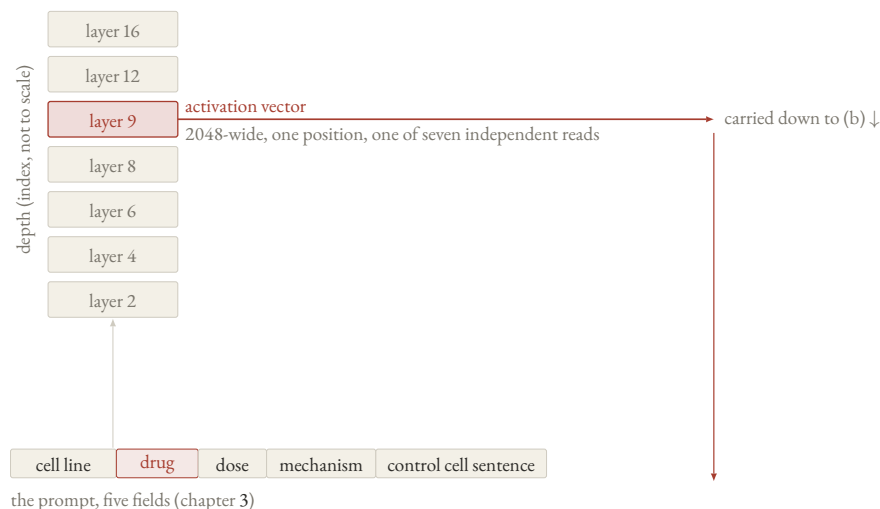
decode probe a classifier trained to name the drug from the residual stream at one layer

HOW THE PROBE IS BUILT, AND WHAT "ROBUST" IS CHECKED AGAINST. No text is generated for this instrument; every reading is a single forward pass over a real prompt. Tier-2 (unseen-drug) evaluation prompts are pooled by drug, and twelve drugs are drawn at random from those with at least forty scored cells; the eligible cells are pooled without regard to which well or plate they came from, and forty are drawn per drug, giving 480 prompts in total. For each prompt the residual-stream activation is read at the last prompt position, the point at which generation would begin, and this is done independently at each of seven depths (2, 4, 6, 8, 9, 12, 16).

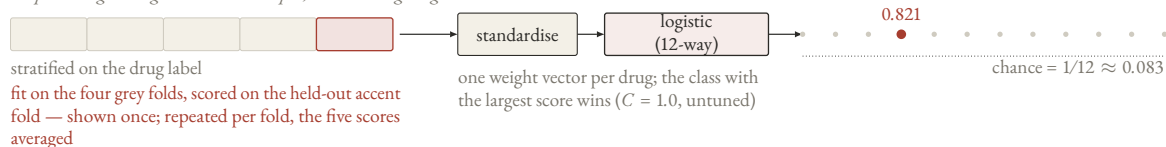
At each depth a separate multinomial logistic-regression classifier is fit on that depth's activations alone and asked to name which of the twelve drugs the prompt carried, its features standardised first. Accuracy is cross-validated: the 480 prompts are split into five folds, each preserving the twelve-way class balance, the classifier is refit on four folds and scored on the held-out fifth, and the five scores are averaged. Twelve balanced classes put chance at $1/12 \approx 8\%$ exactly, an analytic floor and not an empirical one; the same twelve-drug label set is what caps the drug subspace at eleven directions in § 4.4, since twelve centred class means span at most eleven.

(a) the sample and the tap

12 drugs $\times$ 40 cells, pooled without regard to well, 480 prompts

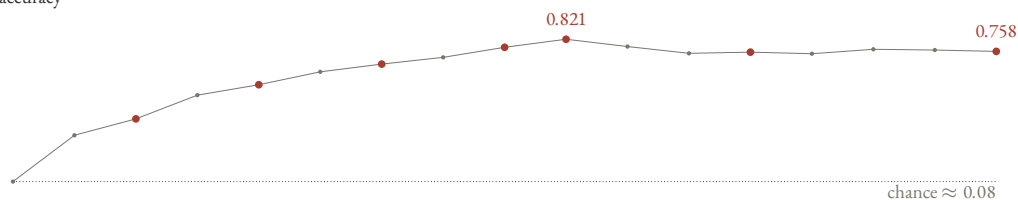
**(b) fitting one layer's probe**

a separate logistic regression at each depth, none sharing weights with another

**(c) accuracy holds up; the geometry does not**

every layer, 0 to 16 — a denser sweep than the seven tapped above

probe accuracy



between/within variance ratio, same activations

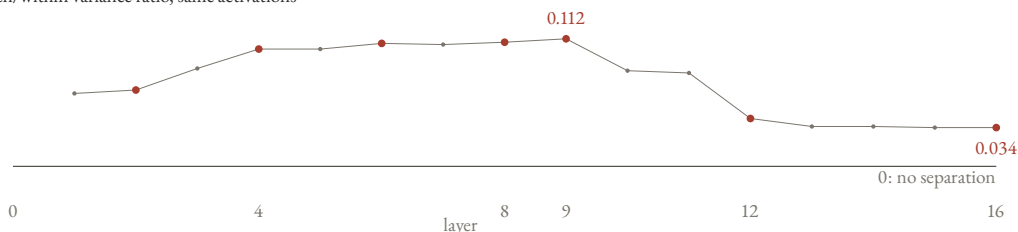


FIGURE 4.6 The prompt names the drug in plain text, so a positive here establishes only that the layer did not discard what it was handed. (a) The residual stream is tapped at the last prompt position, independently at seven depths. (b) At each depth a fresh logistic regression — no weights shared across layers — is fit to name the drug from that layer's activation alone, evaluated by five-fold cross-validation stratified on the drug label. Layer 9 reaches 0.821 against a chance floor of $1/12 \approx 0.083$. What this number can and cannot support is discussed in the text. (c) The same measurement repeated at every layer, 0 to 16 — accent marks the seven depths tapped in (a) and (b). Accuracy holds up almost to the output, 0.821 to 0.758. A second quantity, the ratio of between-drug to within-drug variance in the same activations, collapses over the same span, 0.112 to 0.034; what that divergence means is in the text.

The folds are stratified on the drug label only, not grouped by well, which is the design’s one unrepaired limitation: a classifier trained and tested with no regard for well identity could in principle be reading which well a prompt came from rather than which drug it named. The instrument attempts the obvious check — the identical probe, on the same activations, relabelled by cell line — and it returns no value, because the pooled sample leaves at least one cell line with too few prompts for the fold split to run. The check is therefore unavailable rather than passed, and the confound stays open here. What closes it is not this probe but the purification of § 4.4, which subtracts the cell-line, plate and dose directions explicitly and makes every causal claim on what survives.

THE NEGATIVE NEVER ARRIVES. The probe reads 82% at layer 9 and 76% at layer 16, the deepest measured, against the $\approx 8\%$ floor. Read alone, that is a probe that barely fades. A second quantity, computed over the same 480 prompts but at every layer rather than only the seven tapped above, reads differently. For each layer, group the activations by drug and take, over all 2048 dimensions at once, two sums of squares: how far each drug’s own cluster mean sits from the grand mean, weighted by cluster size, against how far individual activations sit from their own drug’s mean. The first is large exactly when the twelve drug clusters sit far apart; the second is large exactly when each cluster is loosely spread rather than tight. Their ratio is the between/within variance ratio, and it is answering a different question from accuracy: not whether a classifier can still tell the clusters apart, but how much of the layer’s total spread that separation actually accounts for. It peaks at 0.112 around layer 9, where accuracy also peaks, and collapses to 0.034 by layer 16 — about threefold — over the same span in which accuracy falls only from 82% to 76%. The falling ratio shows that between-drug

separation decreases relative to within-drug spread, while the probe still distinguishes the drugs. It does not establish that the absolute between-drug separation shrinks: the within-drug contribution could instead grow faster. The de-emphasis is therefore relative, and measured in the raw geometry. Decodability, by contrast, rises with depth once cell line, plate and dose are projected out, reaching 0.883 at layer 16 (Table 4.5); the variance ratio itself was not recomputed after projection. The survival carries the weight, not the peak, and that is the end of what a probe can say.

4.4 Ablating the drug subspace

■ EXPR

Knowing the drug is written somewhere says nothing about whether the network reads it. Intervening on the region requires localising it, purifying it of the batch structure it is confounded with, and proving it really holds the drug before any null about it can be believed.

The instrument is an ablation. A subspace of the residual stream carrying drug identity is located, cleaned of the batch structure it is confounded with, and then deleted mid-forward-pass, and the question is how much the model's own output moves when it goes. Deleting something the model does not consult changes nothing, so the measurement only means anything if two things are established first: that the subspace really holds the drug, and that a comparable deletion does move the output. Both are built in, the first as a gate that must pass before any null is reported, the second as a cell-line subspace ablated the same way.

The answer is that the region is read, and read weakly. Removing it costs between 3.6 and 18.6 times what removing a matched-random subspace of the same dimension costs, so the model is not indifferent

to it. Against the cell-line control the same removal costs 9 to 37% at six of the seven depths (98% at layer 12, where both costs collapse to about 0.13 nats), so whatever generation takes from the drug it takes at a fraction of the weight it gives to which cell line it was handed. Neither number is the interesting part on its own; the pair is, because a result that clears its null and still falls this far short of its positive control is the shape of a signal that is present, consulted, and not doing much work.

STEP ONE, WHERE THE DRUG LIVES. Write $x_i \in \mathbb{R}^d$ for the residual-stream activation of prompt i , read at the last prompt position at one layer, with d the model width; these are the rows of $X \in \mathbb{R}^{n \times d}$ over $n = 480$ tier-2 prompts (twelve drugs, forty real cells each). Let ℓ_i be the drug of prompt i , C the set of twelve drugs, μ_c the mean activation of drug c , and $\bar{x}$ the global mean,

$$\mu_c = \frac{1}{|\{i : \ell_i = c\}|} \sum_{i : \ell_i = c} x_i, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (4.12)$$

Stack the centred class means as *rows*,

$$\mathcal{M} = \begin{bmatrix} (\mu_{c_1} - \bar{x})^\top \\ \vdots \\ (\mu_{c_{12}} - \bar{x})^\top \end{bmatrix} \in \mathbb{R}^{12 \times d}, \quad (4.13)$$

so that the row space of $\mathcal{M}$ is exactly the set of directions along which the twelve drugs' mean activations differ from one another. Take the singular value decomposition $\mathcal{M} = U\Sigma V^\top$ and keep the leading right-singular vectors, which are the columns of V ,

$$V_{\text{drug}} = [v_1 \cdots v_k] \in \mathbb{R}^{d \times k}, \quad (4.14)$$

$$k = \min(10, |C| - 1, \#\{j : \sigma_j > 0.05 \sigma_1\}).$$

The three terms of that minimum are three separate refusals to overclaim. Ten is the dimension budget. Twelve centred class means span at most eleven directions, so the second term is what the data can support at all. The third is a *relative* spectral cutoff, and it is the one that matters: were the rank fixed at ten regardless, any direction the labels do not actually vary along would be filled with estimation noise — and a noise direction is not orthogonal to the context subspace of step two, so it would carry batch leakage into the very slab that step is meant to purify. In practice the budget binds and the slab keeps essentially the whole span, noisiest surviving direction included, which for a removal claim errs the conservative way. The layer grid is even (2, 4, 6, 8, 12, 16) plus layer 9, added after the raw probe peaked there.

THE POSITIVE CONTROL IS THE SAME CONSTRUCTION ON A DIFFERENT LABEL. V_{cell} is built by the identical procedure — same activations X , same layer, same rank rule — with ℓ_i read as prompt i 's cell line rather than its drug, so the two subspaces differ in nothing except which partition of the same 480 rows defines the class means. That is what makes the comparison in step four legible: when removing V_{drug} moves generation less than removing V_{cell} does, the difference cannot be an artefact of how either subspace was fitted. Cell line is the right control to hold that role because § 4.2 already established, from the data alone and with no model involved, that cellular context is the conditioning variable this atlas actually carries.

SEE the grouping ladder in § 4.2 is what makes cell line usable as a positive control

STEP TWO, PURIFICATION. The subspace step one returns is partly a batch subspace. Drug, plate and dose are confounded by the atlas's design, so directions fitted to separate the twelve drugs will also separate whatever those drugs were plated and dosed alongside. That this has happened is not assumed but measured: projecting the activations

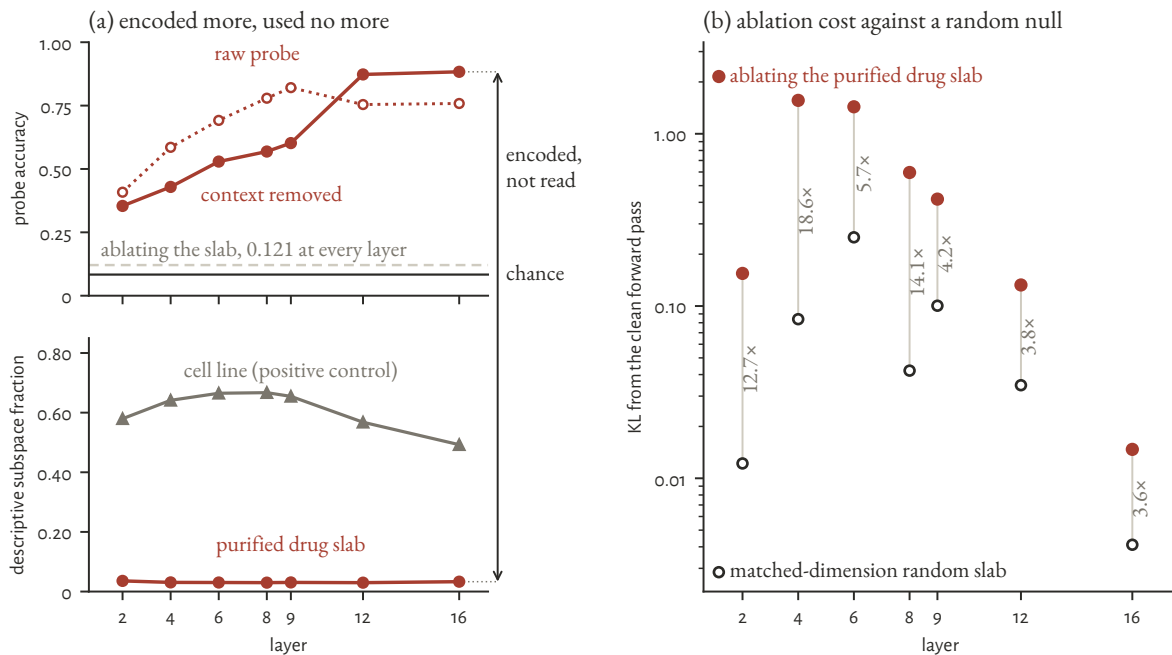


FIGURE 4.7 Drug-label information becomes more decodable with depth while interventions on the purified slab have limited effect. Layer is a true numeric axis in both panels. (a) *Top*: probe accuracy over twelve drug classes, chance $1/12$. The raw series and the context-removed series *cross* between layers 9 and 12, which no other figure shows; the flat grey line is the same probe after the drug slab is projected out. *Bottom*: subspace fraction of the purified slab against the cell-line control, on its own scale — an activation-space summary, not a decomposition of biological variance. (b) Mean KL in nats when the slab is ablated, against a matched-dimension random slab, $n = 60$ per entry. Log axis, so the vertical gap between a pair is their ratio. No intervals are drawn in either panel. These accuracies are twelve-class and must not be read against the twenty-four-class sweep of § 4.5.

onto the slab and fitting the same cross-validated logistic probe to the projected coordinates $XV_{\text{drug}} \in \mathbb{R}^{n \times k}$ — ten numbers per prompt, nothing else — recovers each prompt’s *dose* at 0.90 to 0.95 at every layer. A subspace built only to carry drug identity should not be able to do that, and a null obtained by removing it would not distinguish deleting the drug from deleting the batch it arrived in.

The remedy is to subtract that shared structure. A context subspace V_{conf} is built exactly as V_{drug} was, but from three labels at once rather than one. Cell line, plate and dose are each categorical labels carried by every prompt, and each cuts the same 480 rows a different way: by which cell line the cell came from, by which physical plate it was assayed on, by which concentration it received. Write $\mathcal{L} = \{\text{cell line, plate, dose}\}$, and for each $L \in \mathcal{L}$ let C_L be the dis-

tinct values that label takes in this sample — the individual cell lines that appear, the individual plates, the individual dose levels — so that a single $c \in \mathcal{C}_L$ picks out one cell line, or one plate, or one dose. For each such c write $n_c^L = |\{i : L(x_i) = c\}|$ for how many of the 480 prompts carry it, and

$$\mu_c^L = \frac{1}{n_c^L} \sum_{i : L(x_i)=c} x_i \quad (4.15)$$

for their mean activation, computed over the same rows and centred on the same global mean $\bar{x}$ as in step one. Collect one row per qualifying value, across all three labels:

$$\begin{aligned} M_{\text{conf}} &\in \mathbb{R}^{m \times d}, \\ \text{rows} &= \{(\mu_c^L - \bar{x})^\top : L \in \mathcal{L}, c \in \mathcal{C}_L, n_c^L \geq 5\}, \end{aligned} \quad (4.16)$$

$$m = \sum_{L \in \mathcal{L}} |\{c \in \mathcal{C}_L : n_c^L \geq 5\}|, \quad (4.17)$$

so m counts every cell line, plate and dose level with at least five prompts behind it, summed across the three labels — of order tens of rows against the drug slab’s twelve, and each of length d . Its singular value decomposition is truncated by the same relative spectral rule as step one,

$$\begin{aligned} V_{\text{conf}} &= [w_1 \cdots w_{k'}] \in \mathbb{R}^{d \times k'}, \\ k' &= \min(3k_{\text{max}}, \#\{j : \sigma_j > 0.05 \sigma_1\}), \end{aligned} \quad (4.18)$$

with $k_{\text{max}} = 10$ the drug slab’s dimension budget. Stacking the three label sets together rather than fitting them separately matters: cell line, plate and dose are themselves correlated in this atlas, and three independently fitted subspaces would subtract their shared directions three times over. One matrix spans their union once. The budget is three times the drug slab’s for the opposite reason — context genuinely needs more directions than drug does, since it is carrying three partitions rather than one, and a budget that bound here would leave real

batch structure unsubtracted and inside V_{pure} . The drug slab is then projected off it and re-orthonormalised. First subtract the component lying inside the context subspace,

$$\tilde{V} = V_{\text{drug}} - V_{\text{conf}} V_{\text{conf}}^{\top} V_{\text{drug}} \in \mathbb{R}^{d \times k}, \quad (4.19)$$

whose columns are no longer orthonormal, and in general no longer of unit length, since each has lost whatever part of itself the context could account for. Recover an orthonormal basis by QR decomposition,

$$\begin{aligned} \tilde{V} &= QR, \\ Q &\in \mathbb{R}^{d \times k} \text{ orthonormal}, \\ R &\in \mathbb{R}^{k \times k} \text{ upper triangular}, \end{aligned} \quad (4.20)$$

and keep only the columns whose surviving component was substantial,

$$V_{\text{pure}} = [q_j : |R_{jj}| > 0.2]. \quad (4.21)$$

The diagonal of R measures the size of the new orthogonal component at step j , after both context projection and removal of components along the earlier QR directions. Because V_{drug} 's columns begin at unit norm, $|R_{jj}|$ near one indicates a direction that largely survives both steps. A value near zero can indicate either removal by context or redundancy with the earlier projected directions; it does not identify context alone. Keeping only values above 0.2 prevents a tiny remaining component from being renormalised to unit length and amplifying estimation noise. Two screens guard the same step from the other side. A candidate context label is dropped before it can enter V_{conf} if it is more than 90% pure as a proxy for the drug, since orthogonalising against a relabelled copy of the drug would delete the drug axis by construction and manufacture any null that followed; and classes with fewer than five members are skipped, so that a near-continuous field like dose

cannot degenerate the context subspace into a generic top-variance basis.

What remains, V_{pure} , is the part of the drug slab that no context label can account for, and it is what every causal claim below is made on — never the raw slab. Figure 4.8 draws the construction.

TABLE 4.5 Drug-label information becomes more decodable with depth while the same slab carries little functional weight. All seven layers pass the removal gate, scored on the raw slab V_{drug} ; chance is 0.083. Columns under *Decode* and *Ablation* are labelled by which subspace was projected out of the residual stream before measuring: the two decode columns are the same twelve-class probe on the same activations, once with nothing removed and once with the context subspace V_{conf} — cell line, plate and dose — projected out, and neither is restricted to the drug slab. The random null matches V_{pure} ’s rank, not the variance it removes — the slab’s subspace fraction is roughly six times the null’s — so that ratio mixes drug content with the difference.

LAYER	DECODE ACCURACY		ABLATION COST (KL, NATS)			V_{pure} AGAINST	
	ORIGINAL	V_{conf} PROJECTED OUT	V_{pure}	RANDOM	V_{cell}	RANDOM	V_{cell}
2	0.408	0.354	0.155	0.012	1.115	12.7×	14%
4	0.585	0.429	1.563	0.084	4.180	18.6×	37%
6	0.692	0.529	1.435	0.251	4.307	5.7×	33%
8	0.779	0.569	0.596	0.042	5.419	14.1×	11%
9	0.821	0.602	0.418	0.100	4.689	4.2×	9%
12	0.754	0.873	0.133	0.035	0.135	3.8×	98%
16	0.758	0.883	0.015	0.004	0.059	3.6×	25%

The two decode columns answer different questions and are not two estimates of one quantity: the first includes whatever cell line, plate and dose contribute, the second is the same probe once those directions are gone.

Every ablation cost collapses at layers 12 and 16, the positive control most of all — V_{cell} falls from 4.689 nats at layer 9 to 0.135 at layer 12. Layer 12’s 98% is therefore a small numerator over a smaller denominator, not a depth at which the drug slab matters more.

STEP THREE, THE REMOVAL GATE. A null from an ablation is only evidence if the thing removed was really there. Delete a subspace the model never encoded the drug into, and generation will of course not move — and that outcome is indistinguishable, from the output alone, from a subspace that carried the drug and simply went unread. The gate settles which of the two is happening, before any null is reported.

The test reuses the probe of § 4.3 and runs it twice at each depth. First on the activations as they are, giving accuracy a ; then on the acti-

vations with the drug subspace projected out, $X \leftarrow X - (XV_{\text{drug}})V_{\text{drug}}^\top$, giving a_{abl} . What the gate scores is not the drop itself but the *fraction of above-chance signal* it accounts for, measured against the twelve-class floor $\pi = 1/12$ rather than against zero,

$$f = \frac{a - a_{\text{abl}}}{a - \pi}, \quad (4.22)$$

and f must exceed 0.8 for the depth to pass. The gate is scored on the raw slab V_{drug} ; the purified slab V_{pure} of step two, which is what step four removes, is not separately gated, so the nulls below are conditional on purification not having removed the drug axis along with the context, and the context-removed decode column of Table 4.5 is the indirect evidence that drug information survives outside V_{conf} . Measuring against π is what makes f interpretable: a probe cannot fall below chance in expectation, so $a - \pi$ is the whole of what the subspace could possibly have been carrying, and f asks how much of that the projection actually took.

The gate passes at all seven depths. It is the raw accuracy a that varies across depth, from 0.41 at layer 2 to 0.82 at layer 9; the ablated accuracy a_{abl} does not vary at all, sitting at 0.121 wherever the projection is applied, a floor just above the 1/12 the label set puts chance at. Projecting the subspace out therefore does not merely degrade the probe, it returns it to the same near-chance state at every depth, whatever it was reading before.

Because a_{abl} is fixed, the whole spread in the removed fraction comes from a . Where the raw probe is strongest the subspace accounts for almost everything it was reading — 0.949 at layer 9 — and where it is weakest the fraction is smallest, 0.885 at layer 2, giving the 88 to 95% band. The ordering matters more than the range: the gate is hardest to clear exactly where the probe has least signal to begin with, since a

fixed floor leaves proportionally more of a small gap standing, and it clears there too. Figure 4.8, panel (c), draws all seven; the shaded region below the rule is the set of outcomes that would have left every null in this section untestable, and it is empty. The nulls reported below are therefore properties of the model rather than of a hook that missed the drug slab as located; whether purification preserved it is taken up in step four.

(a) localise the slab

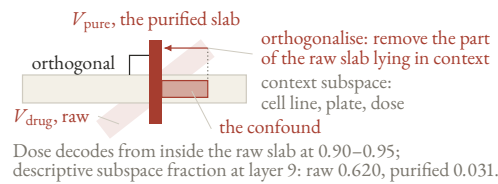
the twelve per-drug means



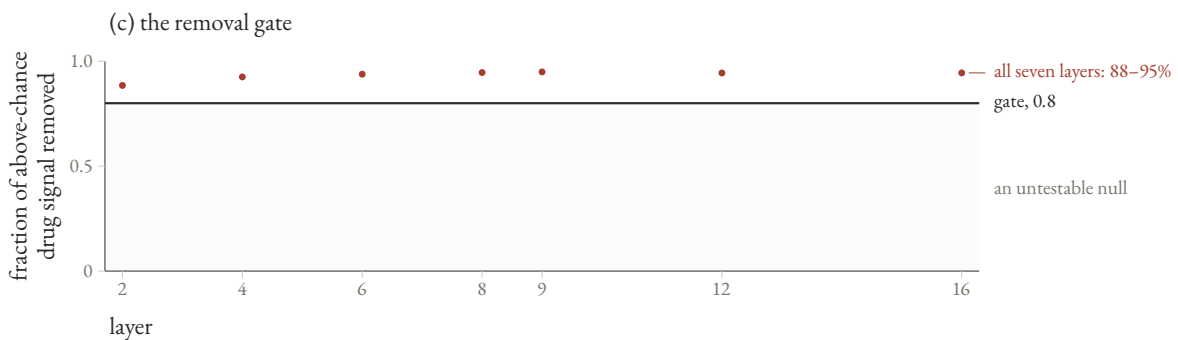
480 prompts: twelve drugs, forty real cells, seven layers

Twelve centred class means span at most eleven directions; ten are kept.

(b) purify it against context



V_{pure} is the part of the raw slab that is orthogonal to the context subspace.



Probe accuracy before 0.41–0.82, after 0.121 at every layer, chance 0.083, exactly 1/12.

The fraction is (before minus after)/(before minus chance). Intervals: none returned in (c); (a) and (b) carry no measured mark.

FIGURE 4.8 The slab is built, not found — ten directions fitted to twelve centred class means, then orthogonalised against the batch structure they are confounded with — and the raw slab earns the right to be purified and ablated only because projecting it out destroys 88 to 95 per cent of the decodable drug signal at every depth. (a) Where the twelve drug classes differ at one depth, giving V_{drug} . (b) Why the raw slab may not be used: it is partly a batch subspace, and the darker patch on the band is the component projected away to leave V_{pure} , on which every causal claim in this chapter is made. Neither panel has a scale — nothing in them is a measurement, and the four printed numbers are the only sourced quantities there. (c) The removal gate, and the only panel with a scale: each mark is one depth's share of above-chance drug-label accuracy destroyed when the subspace is projected out, over twelve classes at chance 1/12. The shaded region below the rule is the set of outcomes that would have left every null in this section untestable, and it is empty at all seven depths. No interval is drawn: the quantity is a single ratio of two accuracies over 480 prompts, with no stored dispersion. These twelve-class accuracies must not be read against the twenty-four-class sweep of § 4.5, whose chance is 1/24.

STEP FOUR, THE CAUSAL MEASUREMENT, AND THE DISSOCIA-

TION IT SHOWS. The first three steps built a subspace and earned the right to delete it. This one deletes it and measures what the model’s own output does in response. Let $y = (y_1, \dots, y_T)$ be a condition’s true target sentence and p its prompt. The sentence is teacher-forced — scored token by token, never generated — so that the same T positions are read in both passes and nothing is left to sampling. Write P_t for the model’s next-token distribution at position t on the clean forward pass. A forward hook then projects a subspace V out of the residual stream at *every* position, prompt and target alike,

teacher forcing the true sentence is scored token by token, never generated

$$h \leftarrow h - (hV)V^\top, \quad (4.23)$$

and the same positions are read again, giving P_t^V . The measure is the mean divergence between the two readings over the target tokens,

$$\text{KL}(V) = \frac{1}{T} \sum_{t=1}^T D_{\text{KL}}(P_t \parallel P_t^V), \quad (4.24)$$

averaged over 60 prompts. It is a distance between two distributions over the same tokens, not between two generated sentences, which is what makes a null here interpretable.

Three subspaces are ablated at each depth, and the comparison between them is the result. V_{pure} is the headline. A matched-dimension random subspace is the null, fixing what deleting k arbitrary directions costs. The cell-line slab V_{cell} of step one is the positive control, fixing what deleting something the model demonstrably uses costs. Figure 4.9 draws the construction and the three readings.

The model is sensitive to the drug subspace, and the sensitivity is small. Against the null, removing V_{pure} costs 3.6 to 18.6 times what removing a matched-dimension random subspace costs, so the slab is not an arbitrary set of directions and generation is not indifferent to it. Against the positive control, the same removal costs 9 to 37% of what

removing cell line costs at layers 2 to 9 — layer 12 the single exception at 98%, before layer 16 falls back to 25%. Read the other way, the cell-line control is 1.0 to 11.2 times the drug's.

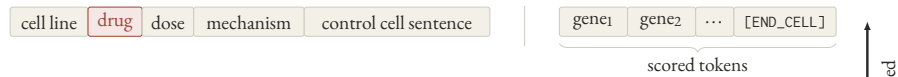
The same three subspaces can also be compared descriptively, by how much of the activations they occupy at all. Let $X_c = X - \mathbf{1}\bar{x}^\top$ be the activations of step one with the global mean removed, and for any orthonormal V define

$$s(V) = \frac{\|X_c V\|_F^2}{\|X_c\|_F^2} = \frac{\sum_{i=1}^n \|V^\top(x_i - \bar{x})\|^2}{\sum_{i=1}^n \|x_i - \bar{x}\|^2}, \quad (4.25)$$

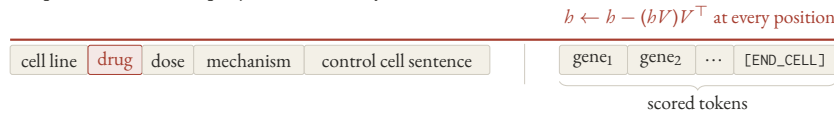
the fraction of the total squared deviation from the mean that lies inside V 's span. Because the projection is orthogonal, $s(V) \in [0, 1]$, and a k -dimensional subspace drawn at random from a d -dimensional stream has $\mathbb{E}[s] = k/d$ — here $10/2048 \approx 0.005$, which is what the matched-dimension null in fact measures. s is a summary of activation geometry and nothing more: it says how much of the stream's variation a direction set accounts for, not how much of the drug response it explains, and it must not be read as a decomposition of biological variance.

Measured this way the purified drug slab is negligible and stays negligible. It holds $s \approx 0.03$ at every depth — 0.030 to 0.036 across all seven — barely six times the random null's 0.005, while cell line holds ≈ 0.6 , or 15 to 22 times the drug slab's footprint. Yet over those same depths the drug's decodability from the same activations, once context is projected out, climbs from 0.354 at layer 2 to 0.883 at layer 16. That is the dissociation: the drug becomes steadily *more* legible with depth while the region carrying it remains a fixed and tiny fraction of what the stream is doing, and deleting that region moves generation a fraction of what deleting cell line moves. Figure 4.7 draws it whole.

(a) pass one, clean

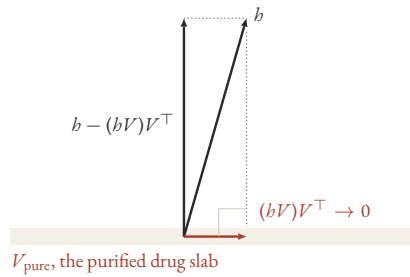


(b) pass two, the slab projected out at layer 4



one sentence, one set of weights, scored twice

(c) what the hook does to one vector



Intervals: none in (d) — the stored dispersion is a normal-approximation standard error, not this document's convention. (a) to (c) carry no measured mark.

(d) what comes out

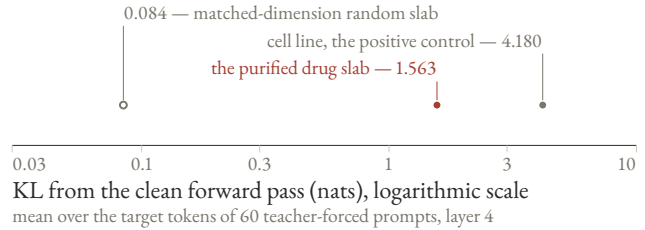


FIGURE 4.9 One true sentence, scored twice by the same weights with the purified drug slab projected out in between — removing it costs $18.6\times$ what removing a matched-dimension random slab costs, and still only 37% of what removing cell line costs. (a) and (b) are the same teacher-forced sentence, box for box; the only difference is the hook, whose rule spans prompt and target alike because $b \leftarrow b - (bV)V^T$ is applied at every position. The braces mark the target tokens the mean KL is read over. (c) draws the decomposition and is not to scale. (d) is logarithmic, so the gap between two marks is their ratio: purified slab 1.563, random null 0.084, cell-line control 4.180 nats, $n = 60$ prompts each, at layer 4 of the seven in Table 4.5. No intervals are drawn — the stored dispersion is not this document's convention, so these are sign-and-magnitude readings.

WHERE ABLATION STOPS. It cannot show that the slab's drug-specific content is what is read. The purified directions could carry generic strong-programme structure that merely co-varies with drug across the twelve. That needs a test which replaces what the slab says.

4.5 Activation substitution: the identity test

■ EXPR

Removal cannot distinguish a region whose content is read from one whose mere presence matters. The last question needs an instrument that replaces what the region says. Push the activations from the drug the prompt names towards another drug, and ask whether the model moves its preference towards that other drug's own held-out sentence. It moves, by a sliver, in one arm at one depth.

VERDICT The generation decoder is sensitive to the drug-label activation signal at very low gain, in one training arm and at one rung of the displacement ladder.

The verdict first. On the single-cell arm — the model trained to predict the treated cell's own sentence, and the arm every measurement in this chapter has been made on — this instrument finds nothing. Across the six depths where it could be run it returns a median displacement effect of -0.0012 : the wrong sign, and two orders of magnitude beneath the 10% of the model's own preference gap pre-registered for calling an effect material. Nothing about which drug the region names reaches what the decoder emits.

That null carries more than the two before it, because this instrument asks a harder question than either. Removal, in § 4.4, establishes that a region matters without establishing *what about it* matters: a model whose output shifts when a subspace is deleted might be reading the drug's identity out of it, or might simply require the region to be occupied by something at all. The two are indistinguishable by deletion, and only one of them is knowledge of the drug. Separating them means not removing what the region says but *replacing* it — pushing the activations from the drug the prompt names towards a different

drug, and asking whether the model’s preference follows towards that other drug’s own held-out sentence, against an identically constructed push naming a third drug.

This is the weakest fundamental measurement in the chapter and it is carried as such. No argument downstream rests on it alone, and a reader who discards it entirely loses nothing that the drug-blindness result depends on. What the following section tests is a prediction of the *interpretation* of this result, not the result itself, and the inference there is written to run one way only for that reason.

SCOPE LIMIT carried into § 4.6 and chapter 5

THE CONTRAST SCORE. Let g_A and g_B be *measured* cell sentences — real transcriptomes of held-out cells treated with drugs A and B , taken from the evaluation data and not produced by the model — and let prompt_A be the prompt naming drug A .

Scoring a sentence means asking how probable the model finds it, and nothing is generated to find out. Write $g = (t_1, \dots, t_T)$ for its tokens. The prompt and the sentence are concatenated into a single sequence and one forward pass is run over it. At each position the model emits a distribution over the whole vocabulary for the next token, and what is recorded is the mass it placed on the token that actually appears there, given the true tokens before it:

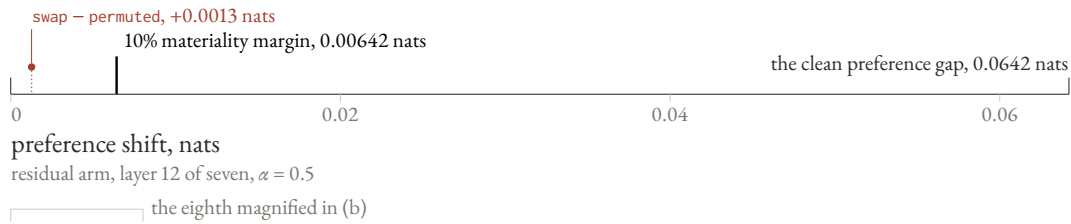
$$\log P(g \mid \text{prompt}) = \frac{1}{T} \sum_{i=1}^T \log P(t_i \mid \text{prompt}, t_1, \dots, t_{i-1}). \quad (4.26)$$

The conditioning is on the true prefix at every step, never on what the model would itself have written, so a wrong prediction early does not corrupt the positions after it; and the model is never required to produce g , only to score it. The result is a per-token mean, comparable across sentences of different lengths.

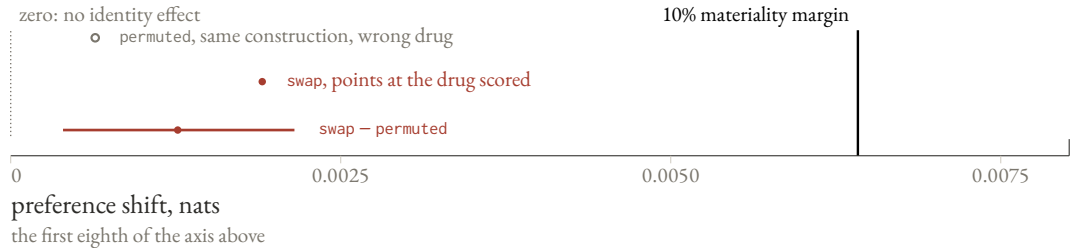
The quantity the test scores is the difference between two such readings under the same prompt,

$$\Delta = \log P(g_B \mid \text{prompt}_A) - \log P(g_A \mid \text{prompt}_A), \quad (4.27)$$

(a) the whole clean preference gap



(b) the first eighth, magnified 8×



Intervals: (b) 95% bootstrap over 58 ordered pair clusters, on the difference, drawn once. None is stored for permuted or swap alone, and (a) carries none.

FIGURE 4.10 The model does move its preference towards the other drug's own held-out sentence, but by one fiftieth of the preference it already has and one fifth of the margin set in advance. Residual arm, layer 12, $\alpha = 0.5$; normalised, the same estimate is the $+0.0197 [+0.0063, +0.0335]$ behind the one survivor Figure 4.11 rings; repeated at seeds 43 and 45 it returns $+0.0118 [+0.0060, +0.0173]$ and $+0.0085 [+0.0029, +0.0143]$ (§ 5.2). (a) The model's clean preference gap between the two sentences compared, measured with no intervention, drawn end to end, so that the axis's full width is the denominator every effect in this section is divided by; the effect is the accent disc near the origin, dotted down onto the rule and named above it, and the bracket beneath the axis label marks the eighth of it that (b) opens out. (b) That first eighth at exactly $8\times$ — exactly, because the lower scale is the upper one divided by eight and not by a rounded number, which is why the two tick lanes fall on the same four positions — in the same units, where the swap arm, the permuted control and their difference separate. The black rule in both panels is the 10% materiality margin pre-registered for calling an effect material, drawn in the same weight and colour as in Figure 4.11. **Which bootstrap draw this is.** The interval is the top-level draw, $p = 0.001$ over a 4000-resample cluster bootstrap on 60 rows in 58 pair clusters over 24 anchor drugs; a second draw of the same point estimate, stored inside the $\alpha = 0.5$ rung of the ladder and drawn in Figure 4.11's lower strip, gives $p = 0.004$, and the two fall either side of the $0.05/26$ admitted at rank one, so which draw is quoted decides whether anything in the family survives correction (§ 4.5). This figure quotes the top-level draw because Figure 4.11 and the correction itself both rest on it. **What the interval shows, and what it does not.** It excludes zero and lies wholly inside the margin: under textbook TOST that configuration is returned as equivalence and read as a null, whereas this chapter's rule returns a directional effect smaller than the margin — one fiftieth of the gap, one fifth of the margin. **No interval is drawn on the two arms or in (a).** None is stored for effect_swap or effect_perm alone, so their marks are bare; the one stored interval is on the difference and is drawn at the scale where it is legible, which is (b), and nowhere else. What may not be read across: this is one arm at one depth at one rung of the displacement ladder, and because the five arms emit different target formats their clean gaps run from this arm's 0.064 nats to 0.314 in the consensus arm, nearly fivefold, so a raw-nats axis is legitimate here and nowhere across arms — Figure 4.11's lower strip is a single-arm object for the same reason.

that is, how much the model prefers the wrong drug's real response to the right drug's, under a prompt naming the right one. Any offset uniform across positions cancels exactly in the subtraction, an inflated normaliser included. What does not cancel is its state-dependent part,

since once g_A and g_B diverge the two sequences visit different states, and that residue is what the second injection controls.

TWO INJECTIONS SHARING AN ANCHOR, SCORED OUT OF

SAMPLE. Write Π for the orthogonal projector onto a purified slab built exactly as in § 4.4 but on this test’s own fitting slice over twenty-four drugs, and μ_c for drug c ’s class mean computed on a second, disjoint slice. The treatment adds

$$\alpha \Pi(\mu_B - \mu_A) \tag{4.28}$$

to the residual stream, displacing the activations from the drug the prompt names towards the drug whose sentence is being scored. Its control adds $\alpha \Pi(\mu_C - \mu_A)$ for a third drug C : same construction, same anchor at μ_A , matched norm, and a displacement pointing at a drug that has nothing to do with either sentence in the contrast. The two differ in nothing but which drug they point at, so the difference between them isolates the model’s preference for B ’s response attributable to being pushed *towards* B rather than to being pushed at all. The subtraction $\mu_B - \mu_A$ cancels the global mean exactly, and nothing is ablated, so no removed term enters either.

Three disjoint slices of cells keep the test out of sample: one fits the subspaces, one supplies the class means, and a third supplies g_B , which therefore never comes from the rows defining μ_B . Were it otherwise, the injected vector and the scored sentence would share cells and the test would be partly circular. Because the training arms emit different target formats, every effect is reported as a fraction of that model’s own clean preference gap, measured with no displacement applied.

FOUR GUARDS CARRY THE INFERENCE. Each guard exists because one specific way of being wrong is live here, and each is fixed before the test is run rather than chosen after seeing it.

The uncertainty model. The natural unit to resample is the row, but rows are not independent here: every row is an ordered pair (A, B) , and two rows sharing a pair share both the injected vector and the scored sentence, so treating them as separate observations would count the same evidence twice and narrow the interval accordingly. The bootstrap therefore resamples whole pairs, not rows. At the anchor seed's layer 12 this is 60 rows in 58 pair clusters over 24 anchor drugs — close to a row bootstrap, because few pairs repeat — and it guards only against sharing within a pair, not across pairs that happen to share an anchor drug or a target drug.

cluster bootstrap resamples whole (A, B) pairs, so two rows sharing a pair count once

The equivalence criterion. A small interval that contains zero can mean two different things, and they must not be conflated: the effect is genuinely absent, or the instrument was too blunt to see it. Two one-sided tests, the standard remedy, declares equivalence whenever the whole interval falls inside a margin of practical irrelevance. That is too generous here, because an interval can lie inside the margin while sitting entirely above zero — a real directional effect, merely a small one, which TOST would return as a null. This section therefore requires both conditions: the interval must lie inside a margin pre-registered at 10% of the model's own clean preference gap, *and* it must contain zero. The conjunction runs one way only. It can withhold a verdict of equivalence that TOST would grant; it can never manufacture one.

TOST two one-sided tests: declares equivalence when the whole interval lies inside a pre-set margin

The headroom gate. An intervention that produces no effect proves nothing if the measurement could not have registered one, which is the same falsifiability problem the removal gate of § 4.4 answers for ablation. Here it is answered per layer: any depth where the cell-line positive

control shows less than $5 \times$ dynamic range is excluded outright, because at that depth the instrument is saturated and can support neither a positive verdict nor a null.

The magnitude ladder. A single displacement gives a single number, and a single number cannot distinguish a real effect from a coincidence of scale. If what the injection moves is genuinely the drug’s identity, then displacing further should move the output further — the effect should grow with α . The test is therefore repeated at $\alpha \in \{0.5, 1, 2, 5, 10\}$ times the natural displacement, and a monotone dose-response across those rungs is pre-registered as what a real effect would look like. This is the guard that bites.

THE RESULT OF THE EXPERIMENT. Assembled, the instrument runs as follows. At a given depth, take an evaluation prompt naming drug A ; retrieve the measured cell sentences g_A and g_B from the held-out slice; score, using Δ as defined above, both under prompt A with the residual stream displaced by $\alpha \Pi(\mu_B - \mu_A)$, and again with it displaced by the control $\alpha \Pi(\mu_C - \mu_A)$; take the difference of the two contrast scores; repeat over ordered (A, B) pairs; express the result as a fraction of that model’s clean preference gap; put an interval on it by resampling whole pairs; and admit the depth only if the cell-line control had the headroom to register an effect there. Then repeat the whole of it at each rung of the magnitude ladder (that is, for each value of α).

Run that way on the single-cell arm — the model trained to predict the treated cell’s own sentence, and the arm this chapter has measured throughout — it returns nothing at any depth where it could be run. The median effect across the six admitted depths is -0.0012 : the wrong sign, and two orders of magnitude beneath the margin. Displacing the activations towards another drug does not move what the model emits towards that drug.

The sweep that produced this section’s verdict was run on the single-cell arm alone. After the fact, with all five training arms of Table A.3 in hand, the identical intervention was re-run across every one of them at up to seven depths each — twenty-six measured tests in all, drawn in Figure 4.11. That wider sweep was not part of the pre-registered design, and it is reported here because it fails in a more instructive way than a second null would have: it fails at each guard in turn. One test survives Benjamini–Hochberg correction at the $\alpha = 0.5$ rung, the residual arm at layer 12. *It does not survive its own uncertainty model*: the q admitting it rests on a bootstrap p that the probe file records two draws of for the same effect, $p = 0.001$ and $p = 0.004$, falling either side of the $0.05/26$ admitted at rank one, so on the second draw nothing survives anywhere. *It does not survive the ladder*: applying the identical correction one rung up, at $\alpha = 1$, leaves nothing surviving; applying it at $\alpha = 2$ promotes a different arm at a different depth and demotes this one. The two arms that trade places are the same residual target construction at two training-set sizes, the second withholding 738 conditions on a (drug, cell line) split, so the ladder relocates the survivor within one recipe rather than between two, and the family of twenty-six is correspondingly less varied than five arms suggests. A pre-registered monotone dose–response was what a real identity effect was supposed to look like, and what the ladder produces instead is a different winner at each rung.

Taken together, the rung decides not merely whether a survivor exists but which one, and beneath that the bootstrap draw decides whether one exists at all. That is the signature of a family dominated by noise rather than one containing a single robust effect, and it is the reading this section takes: the wider sweep does not overturn the single-cell verdict, and the one candidate it raises does not hold up.

At the level of the mechanism — with the drug’s own representation localised, purified, and displaced directly towards another drug — the model’s output does not follow. Behaviour said the drug was not being used; the weights now say the same thing under intervention. What that leaves open, and what it does not, is the subject of § 4.6.

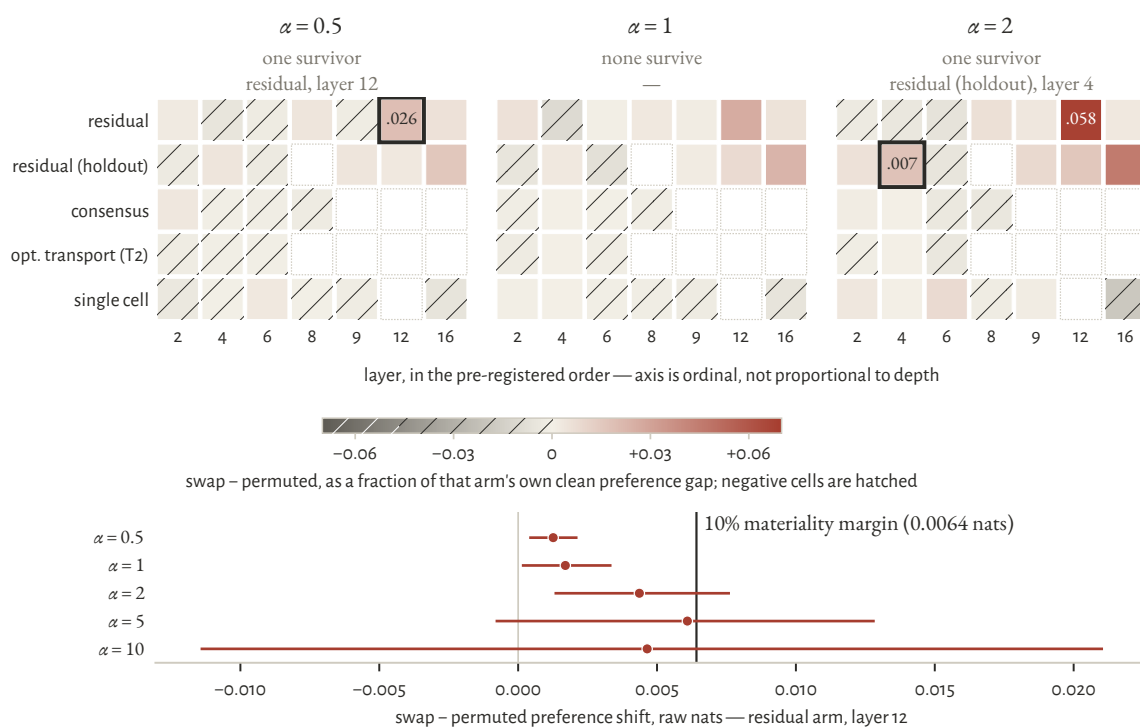


FIGURE 4.11 The one test that survives multiplicity correction changes arm and depth with the rung of the displacement ladder, and at one rung there is no survivor at all. Each panel is one slice of the ladder: the five training arms of Table A.3 (rows) by the seven probed depths (columns), $n = 26$ measured tests per slice, on one diverging scale shared across the three panels. Colour is the swap minus permuted effect as a fraction of that arm’s own clean preference gap; negative cells are hatched. The layer axis is ordinal, drawn at equal spacing, and is not proportional to depth. Nine of the 35 grid positions were never measured and are drawn as empty dotted outlines, not as zeros. Benjamini–Hochberg is pooled across arms within a slice, as the text applies it and not as the pipeline stores it; q is printed in the three cells that fall below 0.10 and ringed where $q < 0.05$. That criterion admits the residual arm at layer 12 at $\alpha = 0.5$ ($q = 0.026$, on the stored bootstrap draw the text discusses), the only survivor among this chapter’s twenty-six headline tests; nothing at $\alpha = 1$; and residual_holdout at layer 4 at $\alpha = 2$, a rung at which the residual arm’s own layer 12 falls to $q = 0.058$. Beneath, the same residual arm at layer 12 in raw nats across all five rungs, with the 10% pre-registered materiality margin as the vertical rule. Intervals throughout are 95% cluster bootstraps over ordered (A,B) pairs, 4000 resamples, which at this layer is 60 rows in 58 pair clusters over 24 anchor drugs.

4.6 Denoised targets: consensus and optimal transport

■ EXPR

If the defect is a decoder that barely consults the drug region whatever is written there, then improving what sits in that region should not help, and improving the training target is exactly that kind of intervention. Three target constructions, one scoring, one stratum. None separates from a prompt naming the wrong drug.

The repair a reader reaches for first is to write something better in that region: give the model a training target that carries the drug more legibly, and the region may start earning the attention it currently does not get. That is a target-side intervention, and it is the cheapest one available. The inference runs one way only. A positive here would tell against the low-gain reading; a negative is evidence about target-side fixes, not confirmation of the mechanism. Because one failed construction proves nothing, two more were built and compared with it.

THE ORIGINAL TARGET, AND WHAT IT STARVES. The single-cell arm trains on a measurement and nothing else. A prompt names one control cell together with the drug, its dose and its mechanism annotation; the response is the cell sentence of a real treated cell drawn from that same condition. Nothing is averaged, imputed or reconstructed, so the target is exactly something that was observed. That is also the hypothesised defect. A single treated cell's rank order is dominated by the same abundant genes whichever drug was applied, and the drug displaces that cell by less than sampling noise does, so under next-token cross-entropy the genes that distinguish one drug from another contribute almost nothing to the gradient. On this reading the drug is not absent from the target; it is present and outweighed.

CONSENSUS, BECAUSE AGGREGATION IS WHERE THE DATA SEPARATES DRUGS AT ALL.

The motivation is measured rather than assumed. § 4.1’s spike-in forced choice separates two real drug populations essentially perfectly once their cells are aggregated into a pseudobulk (panel- $\tau = 1.000$, DE- $\Delta r = 0.995$, top- $n \tau = 0.987$, plate held constant), and the within-well precision reference climbs monotonically with the number of cells standing behind a condition, 0.61 at ten cells, 0.76 at forty, 0.85 at 120 (Figure 4.5, panel a). Whatever in this data distinguishes one drug from another is a property of the averaged profile, not of any individual cell. Consensus aggregates the existing cell sentences. Cells are grouped by (drug, cell line, dose), dose kept separate because most drugs are run at several. For a gene at rank r in a sentence of length L , the construction assigns the proxy $1 - (r - 1)/L$, with absent genes contributing zero, and averages that proxy across the group’s cells. Genes passing a detection-frequency threshold are ordered by their mean proxy, truncated to the group’s median sentence length and closed with [END_CELL]; if no gene passes the threshold, the union of detected genes is used. Groups with fewer than two cells retain their original response. This is an aggregate of rank proxies, not a mean of measured expression. Every original prompt is retained and only the response is swapped, so the example count is preserved exactly and the target is the sole variable. The cost is structural: the map from prompt to target becomes many-to-one, every control cell in a condition receiving the identical sentence, so the heterogeneity is discarded along with the noise and there is nothing per-cell left to learn. That objection is what the third construction answers.

OPTIMAL TRANSPORT, THE MIDDLE THE FIRST TWO LEAVE

EMPTY. The measurement destroys the pairing it would be most useful to have. A condition yields a cloud of control cells and a cloud of

treated cells, and no cell appears in both, so which control cell became which treated cell is not observed and cannot be looked up. Optimal transport supplies the cheapest pairing consistent with both clouds under an explicit cost, which gives a target that is denoised like consensus and control-specific like the single-cell arm.

Conditions are keyed on the full tuple carried by the source, (cell line, plate, drug, dose/treatment well), and each is solved separately, its treated cells coupled only to the shared DMSO controls from that condition’s own (cell line, plate). Treatment conditions are never pooled, so batch identity cannot drive a coupling. The coupling π minimises the entropy-regularised cost $\sum_{ij} \pi_{ij} \|x_i - y_j\|^2 - \varepsilon H(\pi)$, with $H(\pi) = -\sum_{ij} \pi_{ij} \log \pi_{ij}$, under balanced uniform marginals, which impose mass conservation between the empirical control and treated measures — a modelling assumption, not an observed biological constraint. Sinkhorn iteration [33] solves it by alternately rescaling rows and columns of the kernel $\Gamma = \exp(-C/\varepsilon)$, 250 iterations, with C the pairwise squared-distance matrix. The coupling becomes a prediction through the barycentric projection $T(x_i) = \sum_j \pi_{ij} y_j / \sum_j \pi_{ij}$, a soft transport map answering where *this* control cell lands after treatment.

WHERE THE COUPLING LIVES, AND WHY THE DECODER IS NEVER CALLED. Optimal transport’s sample complexity degrades badly in high dimensions, so the coupling is not computed on raw expression. It is computed in the released Tahoe scVI latent [35] ($n_{\text{latent}} = 10$), following the optimal-transport pipeline Kaan [36] built on this atlas, which established that the shipped ten-dimensional latent, and not raw expression, is the space in which per-condition couplings are both tractable and faithful to cell-line structure. scVI is a variational autoencoder for single-cell counts: an encoder maps a cell’s counts to a low-dimensional coordinate meant to capture its state, and a decoder

maps such a coordinate back to a distribution over gene counts. Tahoe’s authors trained one on the full corpus and distribute its output with the data — the minified AnnData object carries, in `obs['X_latent_qzm']`, the posterior mean of the latent for each of its 95.6 million cells. The encoder has therefore already been run once, by them, over every cell this work uses, and its result is a stored per-cell 10-vector rather than something to be recomputed. No scVI model is instantiated anywhere in this pipeline. Each cell in the cache carries a `BARCODE_SUB_LIB_ID`, which is also the key indexing the shipped object’s rows, so attaching the latent is a lookup on that key: the 95.6-million-row index is streamed in chunks, the rows whose barcode appears in the cache are retained, and their stored vectors are copied into cache order. All 620,564 cells were found.

The alternative — running scVI locally over our own cells — was tried first, and it is worth recording because it failed without raising anything. Re-encoding produced a latent in which a linear probe recovered the cell line at 0.11 against 0.975 for the shipped one: a latent that had lost the single strongest structure in the data while remaining a perfectly well-formed array of the expected shape. The cause is a schema trap rather than a modelling one. Tahoe’s per-cell genes field holds identifiers into the model’s gene vocabulary, not positional indices into the gene metadata table, and its values run past that table’s 62,710 rows, so reading them positionally permutes gene identities across the entire matrix. Nothing about the resulting expression looks wrong — the sparsity, the library sizes and the value distribution all survive — and every downstream quantity would have been computed on scrambled genes.

The mapping is therefore made explicit rather than assumed, and then tested. An array indexed by vocabulary identifier is assembled

from the gene metadata's own `token_id` and `gene_symbol` columns and composed with the symbol list of the 946-gene L1000 panel the targets are built in; all 946 panel genes resolve. The test is the same probe that exposed the failure, run this time on the panel rather than on a latent: principal components are fitted to $\log(1 + x)$ of the panel expression, 50 components estimated on a random subsample of 60,000 cells and used to project the remainder, and a multinomial logistic regression on those coordinates is asked to name the cell line, three-fold cross-validated across the 50 lines holding at least ten cells. It reaches 0.864 against a chance rate of 0.02. This establishes that cell-line structure is retained in the panel, not that every gene identifier is correct: a consistent permutation of feature columns would preserve the information available to this refitted classifier. Identifier alignment rests on the explicit metadata mapping. The same components double as the coupling space held in reserve, computed in parallel with the scVI latent so that the choice of embedding is a setting rather than an assumption.

The latent's role ends with the cost matrix. It supplies the geometry in which one control cell is judged near or far from one treated cell, and once π has been solved it is not used again: the coupling weights are applied to the condition's observed treated expression vectors in the 946-gene panel, library-normalised to CP10K. The alternative would have been to average latent coordinates and map the average back to counts, which is the one operation that would have required the decoder, and it is declined on the grounds that a generative model fitted with no knowledge of the drug is a poor thing to pass a subtle drug effect through — it would return whatever it holds a cell of that state to look like, which is a prediction about the target rather than a measurement of it. Applying the weights to observed cells instead makes every target a convex combination of real measured cells: it lies inside the

convex hull of that condition’s treated cloud, though not necessarily on the data manifold. Each is ranked descending and truncated to the top K , where K is the median number of expressed panel genes among that condition’s treated cells, and closed with [END_CELL]; the prompt is the same control cell in the same trainer format, so the examples are drop-in compatible with the other arms. Applied at $\varepsilon = 0.5$ to the 3,856 conditions holding at least sixty treated and sixty control cells, this yields 381,485 training examples. Two properties were checked before any training ran: the targets vary across control cells within a condition, which is the property consensus lacks, and their mean is close to the measured treated mean. Before ranking or truncation, the CP10K target pseudobulk sits 0.006 from the real treated pseudobulk. This check concerns the expression vectors used to build the targets, not the decoded sentences used at evaluation.

THREE CONSTRUCTIONS, NOT A REGULARISATION SWEEP.

The OT parameter connects two limits that are analogous to the other training targets. As $\varepsilon \rightarrow \infty$, the coupling becomes uniform, every control cell is matched equally to every treated cell, and the barycentric target collapses to the measured treated mean. As $\varepsilon \rightarrow 0$, the coupling sharpens towards an unregularised sparse transport solution, though the limit need not map each control cell to one observed treated cell.

The analogy has limits. The trained consensus arm averages rank proxies rather than measured expression, and the single-cell arm draws an observed treated cell without an OT coupling. The three arms therefore do not form an exact one-parameter family. They compare a measured cell, an aggregate of rank proxies and a state-matched barycentric target. The single-cell target came first, consensus second for the aggregation reason above, and optimal transport last to combine aggregation with control-specific targets.

Three constructions were trained and scored, with OT at $\varepsilon = 0.5$; no regularisation sweep was run. The results do not cover intermediate ε , a different cost or embedding, or other target constructions.

ONE SCORING, NAMED BEFORE THE RESULT. The three arms differ in their training target and in almost nothing else about how they were trained: each is cold-started from the same base checkpoint under the identical hyperparameter set (Table A.3), and each is scored on the same held-out-drug conditions through one instrument fixed before any arm was read. The exception is length. Single cell and optimal transport each complete their epoch; consensus was scheduled for one and stopped by the wall-clock at 61.5% of it (Figure A.1), so its null carries an under-training confound the other two do not. Holding the evaluation fixed is what leaves the target as the variable of interest, so it is worth being explicit that none of the three is scored on its own target. All are scored against a pseudobulk of real treated cells from the drug’s own well — a tier-2 condition throughout, as in § 4.2 — split in two: one half the scoring truth, the other the within-well split-half precision reference.

What the two new targets were supposed to buy follows from where the drug signal lives. § 4.1 found it emerges only under aggregation: at single-cell resolution two drugs are indistinguishable, and it is the pseudobulk that separates them. The single-cell arm is trained on individual treated cells and then graded on aggregates, so it is asked to learn in the resolution where the drug is invisible and perform in the one where it is not. Consensus and optimal transport attempt to close that gap from the training side — an aggregate of rank proxies and a state-matched barycentric combination, respectively. Both aggregate observations before writing the target, although neither operation is identical to averaging decoded evaluation sentences. If drug-specific

structure only becomes legible after averaging, a target that is already averaged should be the one that teaches it.

The common scoring truth makes the arms comparable, but does not guarantee that their native targets coincide in the scoring representation. One exact identity holds before encoding. Write Y for a condition's n_t treated CP10K expression vectors and π for the entropic coupling described above. With exactly balanced uniform marginals, $\sum_j \pi_{ij} = 1/n_c$ and $\sum_i \pi_{ij} = 1/n_t$. The optimal-transport target for control cell i is $\hat{y}_i = (\pi Y)_i / \sum_j \pi_{ij}$, and averaging it over the condition's control cells gives

$$\frac{1}{n_c} \sum_i \hat{y}_i = \sum_i \sum_j \pi_{ij} y_j = \sum_j y_j \sum_i \pi_{ij} = \frac{1}{n_t} \sum_j y_j, \quad (4.29)$$

the condition's treated mean exactly in CP10K expression space. The finite Sinkhorn run approximates those marginals, and the 0.006 discrepancy above checks the resulting mean before encoding. Ranking, truncation and decoding are nonlinear, so the identity does not establish equality between the encoded training targets and the decoded evaluation pseudobulk. Consensus also groups by (drug, cell line, dose), without plate in its key, whereas the scoring truth is plate-specific. The common evaluation therefore compares the arms against one biological reference while leaving representation and grouping differences as limitations. Its paired model–scramble comparisons remain on the same scoring truth and representation within each arm.

The instrument is read twice, and the two readings ask different questions. The first asks whether naming a different drug changes the prediction at all: one number per arm, with the substituted drug drawn at random, taken on the identifiable stratum of § 4.2 — the 204 of 606 held-out-drug conditions whose own within-well reference clears the pre-set 0.80 bar, where the ranking is known to be winnable. The gap

is $+0.014 [-0.016, +0.042]$ for single cell, $-0.001 [-0.018, +0.016]$ for optimal transport and $-0.010 [-0.022, +0.003]$ for consensus; none separates.

A randomly drawn partner, though, sits at an arbitrary distance from the real drug, sometimes a near-twin whose response is almost the right answer anyway, sometimes an opposite. Each of those three numbers averages over that whole range, and the near-twins pull it toward zero, so an arm that responded to the drug only when the substitute was very unlike it would still read null.

The second reading separates those cases instead of averaging them. Every partner was recorded when the scramble was built, so each condition carries a measured distance between its own truth and its partner's, and its gap can be read against that distance rather than pooled across it; an arm that uses the drug should show a gap that grows with it. Binning by distance needs more conditions than a single number does, so this reading runs over all 454 tier-2 conditions whose substituted partner also has a truth profile, not the 204 identifiable ones. Figure 4.12 draws the two new constructions only: the single-cell arm's sweep was run directly, with partners chosen to span the distance range rather than reconstructed from random draws, and is Figure 4.4.

One limit travels with every number here. The identifiable stratum is exactly where a zero-drug-information control-copy already matches the model, so the scope limit of § 4.3 applies throughout: a null is a null about drug use, not a statement that the arms predict nothing.

OPTIMAL TRANSPORT REACHES THE FAR END AND NOTHING BEFORE IT. Read against how dissimilar the substituted drug was, the two aggregated targets do not behave identically, and the difference is worth stating precisely rather than collapsing into a shared null. Consensus is flat and negative at every distance, drifting between -0.016

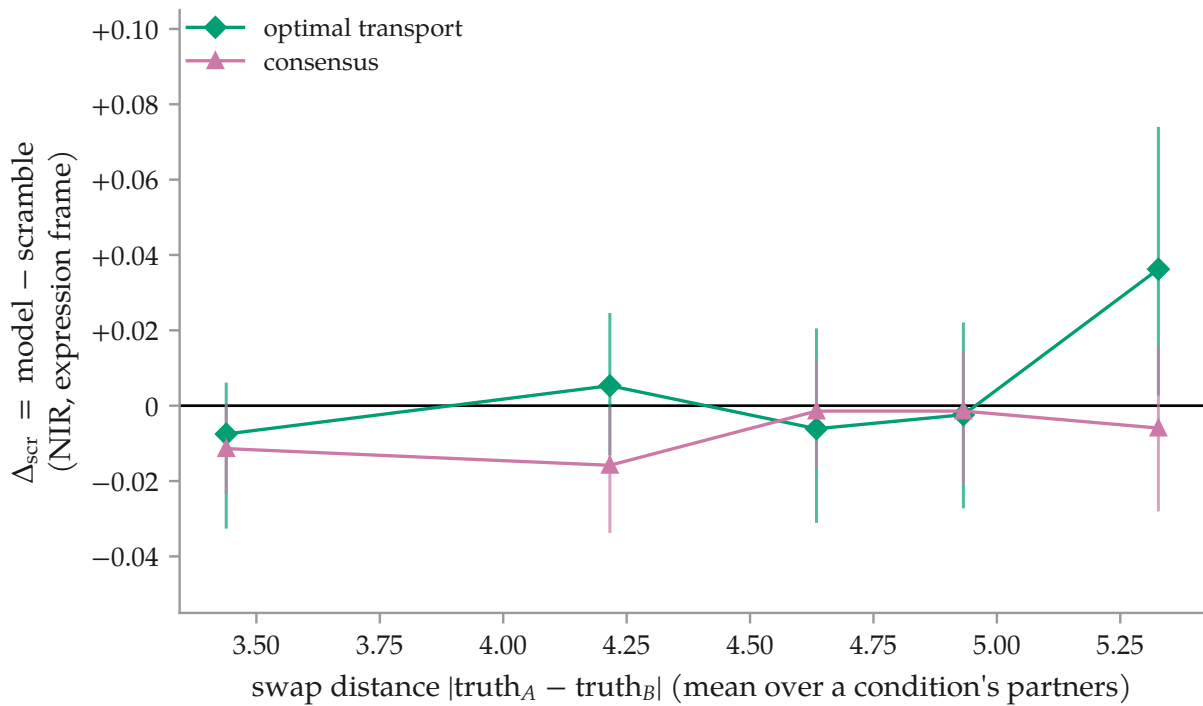


FIGURE 4.12 Optimal transport separates from its scramble only at the most dissimilar partners, and consensus not at all; neither produces the ordering across the range that drug sensitivity would give. Each point is a quintile of conditions binned on swap distance, the Euclidean distance $\|\text{truth}_A - \text{truth}_B\|$ between the real drug's half-A pseudobulk and the substituted drug's, computed within one (cell line, plate) group so the control cancels — the same distance Figure 4.4 uses, over a comparable range. The single-cell arm is not drawn: its sweep was run directly, with partners chosen to span the distance quantiles rather than reconstructed from random draws, and is reported in Figure 4.4. An ordered effect would climb across all five bins. Consensus is flat and negative throughout, never crossing zero. Optimal transport changes sign twice across the first four bins and then reaches $+0.036 [+0.003, +0.074]$ in the farthest — one of two intervals among the ten drawn here that exclude zero, and the only one to do so from above; the other is consensus's second bin, $-0.016 [-0.034, -0.0005]$. The per-condition correlation between distance and gap, which no choice of binning affects, is $+0.074$ for optimal transport and $+0.045$ for consensus. The scramble partner was drawn independently *per cell*, a median of six distinct partners per condition, so each condition's score already averages over a mixture of distances and is placed at its mean partner distance; the design cannot isolate a single sharpest partner the way § 4.9's ladder does. $n = 454$ of the 606 tier-2 conditions, those whose substituted partner also has a truth profile in the same group; intervals are 95% cell-line-clustered bootstraps, 2,000 resamples.

and -0.001 without once crossing zero: on this evidence it never distinguishes the drug it was named from any other, however unlike them it is — though it is also the arm that stopped short of its epoch, so its flat line is the one reading here with two explanations rather than one. Optimal transport does something, but only at one end. It changes sign twice across the first four bins and then reaches $+0.036 [+0.003, +0.074]$ against the most dissimilar partners — so when the substituted drug's true response is as far from the real one as this atlas allows, the model's output does register the difference.

That is a real if narrow finding, and it is also not what drug sensitivity looks like. A model reading the drug should separate a little from a near-twin and more from an opposite, with the gap growing across the range; what optimal transport gives is nothing across four bins and a single excursion in the fifth. The per-condition correlation, which no choice of binning affects, is $+0.074$ — indistinguishable from no relationship. One endpoint clearing zero out of ten unadjusted intervals is the weakest form the evidence could take while still being non-zero.

The same regression settles the ordering question for this arm directly. A purpose-built sweep exists for it, with partners chosen to span the distance quantiles rather than drawn at random, five per condition — which is what the reading of § 4.2 needs. For condition i and its k th scrambled prompt, regress that prompt’s NIR on the negated Euclidean distance between the condition’s own truth and the truth of the drug substituted into it,

$$y_{ik} = \alpha_i + \beta x_{ik} + \varepsilon_{ik}, \quad k = 1, \dots, 5, \quad (4.30)$$

with one intercept per condition, the target-drug prompt excluded so that no comparator enters, and x negated so that it rises with similarity. A model that never reads the name puts its five points at one height, so $\beta = 0$ is the null by construction and $\beta > 0$ is what sensitivity would look like.

The arm does not establish a positive within-condition slope. Pooled across conditions the slope is $+0.0485$ $[-0.0287, +0.1256]$ clustered on drug, already covering zero; fitted within condition it is -0.0008 , $[-0.0305, +0.0290]$ clustered on drug and $[-0.0247, +0.0232]$ on cell line, over 200 prompts across 40 conditions. The between-condition comparison the intercepts remove carries a slope of $+0.0790$ on its own, larger than the pooled figure itself. The within-condition

interval excludes the pooled estimate it replaces, but no positive within-condition slope is established at this precision. A slope per unit of similarity is not directly comparable with a margin on a raw NIR gap, so that gap margin does not establish equivalence here. One scope limit travels with it. The sweep was scored at 1200 tokens, and the separate reading that this arm is inert has been withdrawn on exactly that ground. The regression supports the narrower reading that no graded response to how unlike the substituted drug is was established at that budget.

That leaves a genuine question about what the training did accomplish, since it was plainly not inert. The consensus model stops echoing the control: on the identifiable stratum it scores 0.653 against a zero-drug-information control-copy at 0.766, where the single-cell model sits on the control at 0.768. Its predictions are not mode-collapsed either; there is real between-condition variation. But that variation is uncorrelated with real drug identity — the Mantel correlation between the model’s between-drug distance structure and the true one averages +0.03 over cell-line and plate groups. The failure this literature names first for a perturbation model that underperforms on well-calibrated metrics is mode collapse, where the prediction ignores the input and reverts to a mean, and the repair is to restore diversity in what the model emits [23]. This arm is not that. It moved off the control, kept the variation, and spent it on the wrong axis. Producing no structure and producing structure that does not track the drug are different failures with different repairs.

4.7 The residual target

▲ RESID

Every target compared in the last section preserves an expression-order representation rather than ranking a drug-specific residual. Those targets largely share their abundant genes across drugs. The apparatus chapter defines a target that ranks something else instead. Does it in fact expose the drug, and can it be estimated without the evaluation leaking into the fit? The instrument counts how many top target positions carry a different gene for two drugs in one context.

§ 4.6 ended on a distinction it did not resolve: producing no structure and producing structure that does not track the drug are different failures. This section takes up the second, and it opens by conceding the first. Three target constructions were tried there and none separated, so a fourth arrives owing an explanation of why it is not simply a fourth point on the same ladder.

It is not, because that ladder held one thing fixed. Every target compared in § 4.6 preserves an expression-order representation rather than ranking a drug-specific residual; the constructions differ in how they aggregate observations before writing that target. Their aggregation rules need not produce identical rankings, but none explicitly ranks the change specific to the drug. The prevalence of the same abundant genes across drugs motivates the construction outside that family which § 4.6 left open. § 3.5 defines one. What follows is the measurement that motivated it and the analysis of whether it can be estimated at all.

Encoding here means how the target string is written, not what the model's internal states carry (§ 4.4). The loss weights every position equally, so the measurement is a count over the 200 target positions (Table 4.6).

TABLE 4.6 83% of the standard target's top-200 gene positions are identical between any two drugs in the same context; under the residual, 40% are — and the residual's exposure does not track how similar the two drugs' responses are. Rows are quoted at the plate scope this build uses (§ 3.5); at cell-line scope the residual row reads 118.2 overall and 0.7432, and the other two rows are unchanged. The three stratum columns bin all 212,528 exhaustive within-group drug pairs on the cosine between the two residuals — near above +0.1 ($n = 23,024$), orthogonal within ± 0.1 ($n = 152,421$), opposite below -0.1 ($n = 37,083$) — on the same similarity axis the later swap evaluation ranks partners on; the ± 0.1 cut points are this table's own, since § 4.9 fixes its three partners by extremum (most similar, closest to orthogonal, most anti-correlated) rather than by threshold. The full profile and the shift climb monotonically across those strata and correlate negatively with residual cosine; the residual stays high in all three and does neither. That weakens the reading that the later swap-distance pattern is inherited trivially from target divergence rather than being an independent test of it. The counts are where the three encodings differ, and they do not on their own identify tokenisation as the binding constraint.

REPRESENTATION	TOP-200 TOKENS DIFFERING BETWEEN DRUGS					ρ	RETRIEVAL REFERENCE
	ALL PAIRS	NEAR	ORTH.	OPPOSITE			
full profile (the standard target)	34.6 (17%)	32.5	34.6	35.8	−0.149		0.742
shift (treated — control)	111.4 (56%)	107.4	111.3	114.5	−0.282		0.742
residual (shift — generic)	120.3 (60%)	118.4	121.0	118.7	+0.019		0.7467

The count columns and the retrieval column hold different quantities, a tally out of 200 positions and a retrieval score, so rows compare within a column and never columns with each other. ρ is the Spearman correlation between a pair's residual cosine and its differing-token count, taken over all pairs and not within a stratum.

The retrieval column pits disjoint cell subsets of one drug, in one cell line, on one plate, against each other. It is a precision reference, not a biological replicate ceiling, and it is near-constant across the three rows: the rewrite exposes the drug without buying that exposure from a coarser measurement.

The stratification is descriptive. Pairs reuse drugs and experimental groups, so no pair-level inferential claim is made and no interval is drawn.

WHICH SUBTRACTION DOES THE WORK. Removing the control moves the between-drug token difference from 34.6 to 111.4; removing the generic adds a further 8.9 at the plate scope this build uses, and 6.8 at cell-line scope. So the control subtraction alone buys 90% of the recoverable difference, 92% cell-line-scoped. That is the first thing the table says and the least comfortable one for the construction it motivates: almost all of the exposure comes from the cheaper of the two subtractions. The measurement is an overlap of gene sets on one atlas at $K = 200$ (6,602 conditions of the earlier divergence cache, 125 groups against the 119 of Table 3.1), so no margin is claimed for other atlases or other K . It squares with § 4.1, where what saturates $DE-\Delta r$ is reversion toward a central point, `revert_center` scoring 0.9999 without fitting anything.

Read as a verdict on the generic, though, that 90% proves less than it looks like, and the reason is in what the count counts. It tallies how many of the 200 positions carry a different gene between two drugs, so a subtraction is credited only where it reorders the two drugs' rankings differently *from each other*. Removing a component the two drugs share does that weakly by construction, however large the component is. The later squared-norm measurement gives a different comparison: normalised by the shift's squared norm, the observed residual contributes 0.621, the estimated generic 0.390 and the cross term -0.011 . The generic therefore carries about two fifths of that squared-norm scale, with little aggregate alignment to the residual. These are estimated components, not identified fractions of biology; the residual can still contain sampling noise, baseline-estimation error and context effects.

The two subtractions are not competing for one credit, then. The control subtraction is what exposes positions to the drug at all, and the generic subtraction changes the reference against which those positions are ranked. A target that skipped it would retain the estimated generic component, whose squared norm is about two fifths of the observed shift's. Neither subtraction is dispensable to this target definition, and Table 4.6 is simply the wrong instrument for asking whether the second one is.

One threshold choice belongs beside these counts. The inclusion minima of § 3.5, at least 40 treated and 20 plate-matched control cells, sit where the identifiability curve of § 4.2 flattens: the within-well split-half precision reference reads 0.61 at 10 cells, 0.76 at 40 and 0.85 at 120.

THE COUNT STOPS SHORT OF THE TOKENISATION CLAIM.

The natural inference, that only a small fraction of the gradient carries drug identity, is a hypothesis about the optimisation; what is measured is the overlap of target gene sets, not a token-level loss or a gradient attribution. Gene positions are not loss-token positions under byte-pair encoding, and the residual target alters aggregation, control referencing, generic subtraction, sign coding and sequence length at once. Separating those five factors needs a matched-budget factorial this thesis does not run.

WHAT THE PLATE-LOCAL GENERIC COSTS, MEASURED. § 3.5

states the limit as part of the definition: what survives both subtractions is not what makes this drug different from all drugs, but what makes it different from the other training drugs sharing its plate and cell line. Stated there it is a definitional caveat. Here it is a quantity, and mechanism is where it bites. § 4.12 finds mechanism-related drugs co-plated 0.404 of the time against 0.362 for a random draw, so for a co-plated mechanism class part of the class's own shared programme sits inside the generic and is subtracted out of every member's residual by construction.

That bias was tested rather than assumed, and the mechanism gate of § 4.12 is where it can be, because that gate is scored in this frame.

Two properties of this frame travel with every number measured in it. First, the generic is a plate-local mean, so where a mechanism class is co-plated the mechanism channel is later graded in a frame that has already removed some of what it is asked to find, which makes an inconclusive verdict on mechanism the conservative reading and not a generous one. Second, the plate-scoped generic removes plate structure from the truths and, in expectation, from the predictions, but that is an expectation and not an identity. An individual residual may retain some plate structure, and the size of what remains was never measured. The absolute NIR values of this frame are read within it and never set beside the within-plate tables elsewhere in the chapter.

SCOPE LIMIT carried into § 4.12 and chapter 5

WHY THE SCOPE IS THE PLATE, AND THE ARGUMENT THAT NO LONGER CARRIES IT. § 3.5 takes the generic at plate scope. The original argument for the cell-line alternative was retention: 61.7% of conditions reproduced across a half-split at cell-line scope against 19.3% at plate scope (cosine threshold 0.2, $n = 6602$ on that same earlier cache), on the reasoning that a plate holds too few drugs to estimate a mean. Both figures were computed with a shared control pseudobulk subtracted from both halves, which makes the halves correlate through their common control error, and the inflation favours cell-line scope precisely because the cell-line generic is itself less noisy. That alone withdraws the retention argument. A disjoint-control recomputation exists but was never written to an artifact, and the two surviving records disagree on the plate figure, so no corrected percentages are quoted; both agree that plate scope retains at least as many conditions, so retention is no reason to prefer cell-line scope. What settles the scope is § 4.8: the same-plate, different-drug control that produced the earlier false alarm reads +0.485 under a cell-line-scoped generic and -0.017 under the plate-scoped one, and a control built to read zero is the diagnostic that decides between them.

CORRECTION the retention argument for cell-line scope is withdrawn here; the scope decision rests on § 4.8

ESTIMATING THE GENERIC IS WHERE LEAKAGE WOULD ENTER. The residual is a subtraction, and the quantity subtracted is not handed down by the design: it is estimated from the same data the model is trained and scored on. That makes the generic the one part of the construction capable of going wrong in a way the target string never shows. This is particularly concerning if we wish to perform preprocessing on conditions and admit only conditions that carry signal and not noise. A leaky generic still produces well-formed sentences of plausible genes; it simply produces them against a baseline that has already seen what the model is about to be tested on. § 3.5 states two properties of

D_c as fundamental, leave-one-drug-out and split-before-fit, without arguing for either. They are argued here, because between them they settle the two questions an estimated subtrahend raises: what the mean is taken over, and which conditions are allowed into it.

THE MEAN IS OVER DRUGS, AND A CONDITION'S OWN DRUG LEAVES IT ENTIRELY. The generic is built in two steps rather than one. Within a group, each drug's conditions are first averaged into a single per-drug shift; the generic is then the mean of those per-drug shifts. Both steps carry weight. Averaging within a drug first is what makes the result a mean over drugs at all: otherwise a drug measured at five doses in the group would contribute five terms and count five times as heavily as a single-dose drug, and the generic would be a mean over conditions wearing the name of a mean over drugs.

Leave-one-out then names a removal that could be drawn at several levels, and the level is what the rule is about. What is removed is the drug. When the generic for a condition is formed, that condition's drug is struck from the group's roster altogether — every dose of it and every condition of it, not merely the one condition whose residual is being computed — and the mean is taken over the drugs that remain. The narrower alternative, dropping only the condition itself, sounds equivalent and is not. A drug run at five doses would keep four of its own conditions inside the mean subtracted from its fifth, so part of what that drug does would be subtracted from itself, and the residual would answer a different question: what makes this dose unlike this drug's other doses, rather than what makes this drug unlike the other drugs. Removing at drug level is what guarantees that the quantity subtracted from a condition carries no information about that condition's drug.

The rule also keeps the two splits commensurable, which is the reason less often noticed. The generic is fitted on training conditions only, so a training condition's drug sits in the fit inventory and, without the rule, would appear in the mean subtracted from itself. A held-out condition's drug is struck from its group's roster too, because the leave-one-drug-out rule is applied at transform time whether or not that drug still has training wells elsewhere. The two splits would therefore be scored against differently defined targets, and a comparison between them would be confounded by the definition rather than by generalisation. Leave-one-drug-out removes the asymmetry by making both sides alike: neither has its own drug in its own generic.

WHICH CONDITIONS MAY ENTER THE GENERIC, AND WHAT MAKES THE QUESTION HARD. The second question has an answer that raises no difficulty whatever: every condition the split assigns to training. Fitting the generic on that set is one pass over fixed data, nothing in it refers to any quantity computed later, and there is no circularity to speak of. The difficulty is introduced deliberately, by us, and it is introduced for a reason worth keeping.

Not every condition carries a residual worth training on. At this cell budget the two subtractions can leave a vector that is almost entirely sampling noise, and such a condition still emits a full-length target string — ranked, closed with [END_CELL], indistinguishable in form from any other — which the loss weights exactly as heavily as a condition whose residual is real. Training on it teaches the model to reproduce noise in the voice of a drug effect. The scale of that is not marginal: the line finally adopted retains half the training conditions, which is another way of saying that the median training condition sits almost exactly on the line: its two halves agree only as well as the best

unrelated pair in twenty. A filter is worth having, and the natural test for one is self-consistency.

That test is the half-split, and its content is precise. A *condition* — one drug, at one dose, on one cell line, on one plate, the unit of chapter 3 — has its treated cells shuffled and cut in two, and the plate's control cells are cut in two alongside them; the same halving is applied to every other condition in the same (cell line, plate) group. Each half yields a shift, its treated mean minus its control mean, computed inside that half alone. Each half also yields a generic, the leave-one-drug-out mean just described taken over the other training drugs' shifts on the matching half. Subtracting the one from the other leaves two residuals that share no cell and no estimate, and the condition is kept only if the cosine between them clears a threshold. Two independent measurements of the same condition must agree before it is allowed to teach anything.

Adding that test is what closes the loop. It decides membership of the training set, and it decides it from the residual — which is a shift minus the generic, and the generic is a mean over the other *training* drugs. Membership is therefore settled by a quantity that is itself a function of membership. Note where the problem is not: nothing in the test is recursive, since a shift needs only cell means and a generic needs only other drugs' shifts, so for any fixed list of training drugs every quantity computes in a single pass and a condition's residual never requires its own residual. The circularity sits one level up and it is about *membership* alone. Written as edges: the reliability line selects the training set, the training set fits the generic, the generic is subtracted to give the residual, and the residual is what the reliability line is applied to (Figure 4.13, left). Every edge is defensible on its own, the four close, and nothing in the definitions says where to start.

What that would cost is not abstract. Suppose the generic were refitted on the conditions the filter retains, which is the natural reading of “the mean over the drugs we are actually training on”. Dropping one condition would then change the generic of every other condition in its group, hence their residuals, hence which of *them* pass — so whether a condition survived would depend on when it was evaluated, and the procedure would have to be iterated to a fixed point that nothing guarantees to exist or to be unique.

So the design problem is not whether to filter but where to let the filter act. It must be allowed to decide what the model trains on, which is the whole point of it, and it must not be allowed to decide what the generic is fitted on, which is what would close the circle. Those two can be separated, because they are separate sets: the fit inventory is fixed by the metadata split, and the filter operates downstream of it on conditions whose residuals have already been computed. The generic is fitted once, on every condition the split assigns to training, before a single reliability test has run; the filter is one terminal pass against that fixed estimate; and dropping a condition moves no other condition’s residual. The retained set is a function of the data and not of the order it was walked in.

The hazard is set out at this length because an earlier build of ours was in it, estimating generic and filter over every condition and assigning the holdout afterwards. That adds a fifth edge to the figure, held-out conditions entering the fit, and makes each held-out target partly a function of the other held-out conditions. The failure is quiet: it survives every check that inspects the targets themselves, since the targets it produces look exactly like the ones it should have produced. Any construction that estimates what it subtracts, and then filters on what is left, is exposed to some version of it.

THE LOOP IS CUT BY ORDERING, AND THE ORDERING IS ENFORCED. The build runs as *inventory*, *split*, *fit*, *transform*, and each stage may read only what the stage before it is allowed to expose (Figure 4.13, right). Inventory enumerates conditions from metadata and cell counts alone — the (drug, cell line, plate, dose) tuples clearing 40 treated and 20 plate-matched control cells — and aggregates no expression at all. Split then assigns train and holdout from that metadata, keyed on the treated well (and, for *unseen_drug*, on the drug), so that every cell line of a held-out well leaves with it and no well contributes conditions to two splits (§ 4.10), and it does so before a single pseudobulk exists. That removes the first edge: nothing downstream can have defined the split, because at the moment the split is written there is nothing downstream to define it with.

Fit estimates the generic, and is handed the training keys explicitly rather than inferring them. Its inventory is every condition the split assigned to training, taken whole and before any reliability test has run, which removes the second edge — admission cannot select the fit inventory, because the fit inventory is fixed before admission is decided. Within that inventory the estimator is the one argued above, drug-weighted and leave-one-drug-out, scoped to the (cell line, plate) group and failing closed where a group holds fewer than three other training drugs, in which case the condition is dropped and counted rather than quietly falling back to a coarser scope. Transform then projects every condition, held-out ones included, through that fitted generic. Held-out conditions are scored by it and never enter it.

The training set is made last, and in a single pass. Each training condition's two half-residuals are formed against the fixed generic, their cosine is taken, and the condition is kept if it clears the line, 2,523 of the 4,999 training conditions or 50.5%. Because the generic is never

refitted on the survivors, a drop moves no other condition's residual and the retained set does not depend on the order the pass walked in. Held-out conditions are transformed but deliberately not filtered. Each retained condition then emits one training example per plate-matched control cell, the prompt carrying that control cell together with the drug, its dose and its mechanism annotation, and the response being that condition's residual string.

The staging is checkable rather than merely asserted. A poison test corrupts every held-out expression vector with large noise, rebuilds, and requires both the training JSONL and a hash of every fitted quantity to be unchanged. If one held-out cell reaches the fit, one of the two moves, and restoring the earlier build's one-line defect does make it fail on the digest, so the check is not vacuous. Poisoning a training condition instead must change both, which is the negative control that stops it passing for the wrong reason. One quantity the staging does not settle is the threshold the filter applies, and that is the last piece.

poison test Corrupt the held-out side; the training targets and the fit digest must not move. Corrupt a training condition; they must.

HOW THE RELIABILITY LINE WAS CHOSEN. The threshold decides more of the training set than any other single choice, so it is measured rather than picked. The builder constructs a matched null by pairing half *A* of one condition with half *B* of a *different* condition — same encoding, same dimensionality, same noise level, no shared biology — reports the whole retention curve against it, and places the line at that null's 95th percentile, +0.109. What that implements is a criterion rather than a setting. A condition is kept when its two halves agree more than two unrelated conditions' halves do, the line being the level of agreement only one unrelated pair in twenty reaches. It retains 2,523 of the 4,999 training conditions, 50.5%.

The filter is applied to the training side alone. Dropping held-out conditions for failing their own reproducibility bar would select the

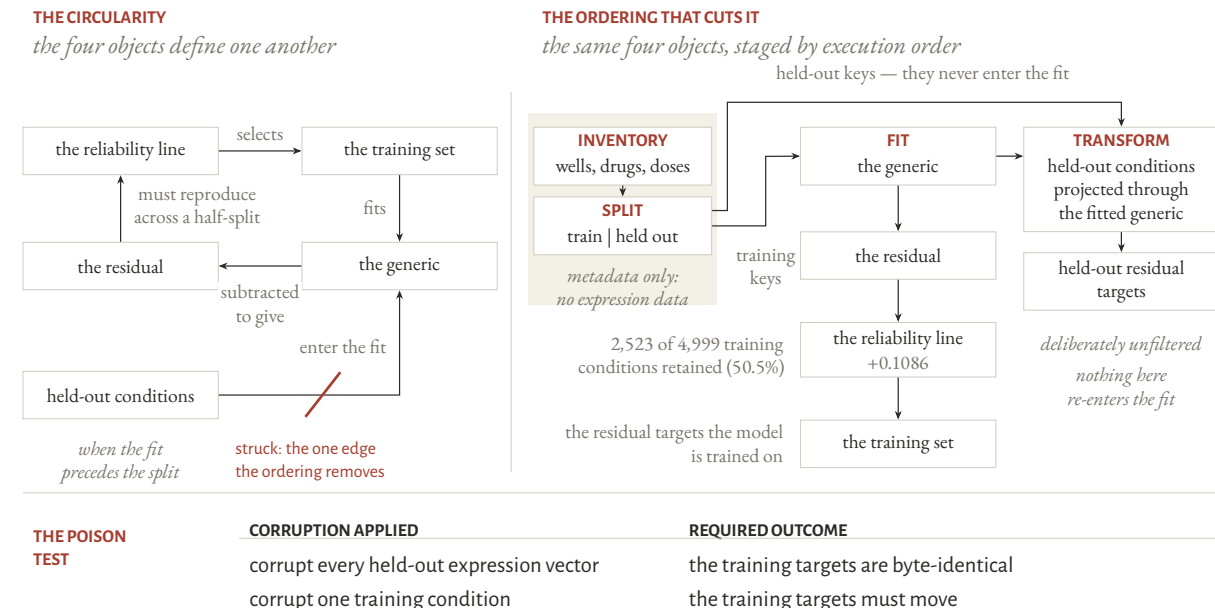
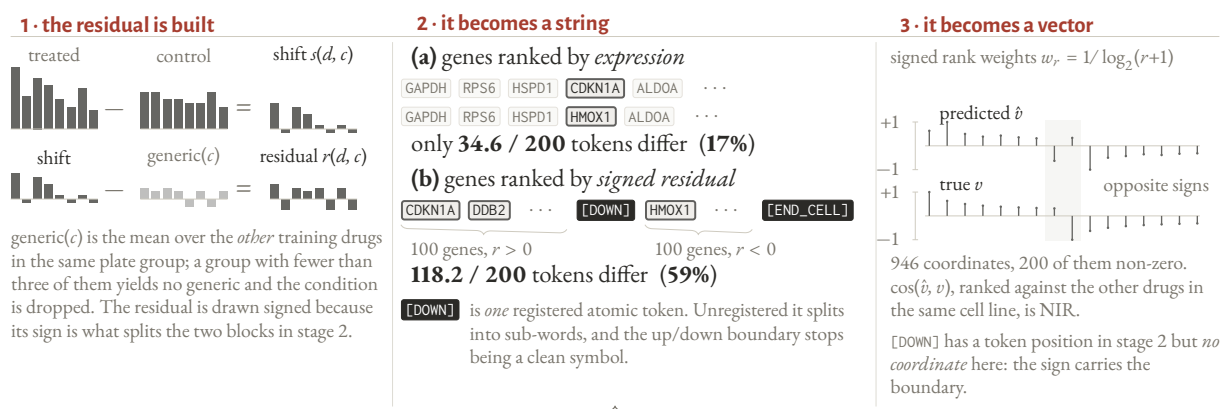


FIGURE 4.13 The definition of the residual could have closed on itself, and what stops it is execution order, not an argument. *Left, the circularity.* A condition enters training only if its residual reproduces across a half-split; the residual exists only once the generic is subtracted; the generic is fitted on the training set; and the training set is what the reliability line selects. Estimate the generic and the filter over every condition and assign the holdout afterwards, and held-out conditions enter the fit — the struck edge — so each held-out target is defined partly by other held-out conditions. *Right, the ordering that cuts it,* the build this thesis reports. The same four objects, staged. Inventory and split read metadata only, before any pseudobulk exists, so nothing downstream of the split can have defined it; the fit is handed the training keys explicitly; and transform projects held-out conditions through the fitted generic in one direction, with no path back into it. The generic is the mean over training drugs other than this one, within plate. The two counts are the *training-set* retention at the resolved reliability line +0.1086, the 95th percentile of a null pairing half *A* of one condition with half *B* of another; the held-out sets are deliberately unfiltered, since dropping held-out conditions for failing their own reproducibility bar would select the evaluation set on its own outcome. *Beneath,* the two assertions that make the ordering checkable in both directions rather than merely asserted. This is a provenance diagram: no NIR value, arm or split-level result appears in it.

evaluation set on its own outcome, and the two errors are not symmetric: noise in a retained training condition costs learning efficiency, whereas noise admitted to the evaluation would cost validity.

What clearing the line certifies is sampling precision and not biology. Both halves are drawn from the same well, so a condition that clears it has a residual estimated stably at this cell budget, which is not the claim that the effect would reappear in a fresh experiment. 286 of the 287 distinct (drug, dose) treatments in this cache occur in exactly one well, so a biological replicate is not available to ask for.



and the within-well split-half retrieval reference is **unchanged**: 0.742 → 0.743

FIGURE 4.14 The information was always present; the standard target's tokens barely expose it — and the string the model writes is not the vector that scores it. *Stage 1.* The drug-specific residual is built by two subtractions. The control removes the cell's own state, leaving the *shift*; the mean over the other drugs in the same plate group removes the *generic* programme that every drug triggers, leaving what makes this drug different. The six gene profiles are illustrative shapes and carry no measurement — no axis is drawn — but the arithmetic within the figure is exact bar by bar, and the residual is drawn signed because its sign is what splits the two blocks in stage 2. *Stage 2.* That same measurement, written two ways. Ranking genes by expression (a) leads every drug's profile with largely the same abundant genes: only 34.6 of the top 200 tokens differ between two drugs in the same context; ranking by signed residual (b) raises that to 118.2. (Divergence counts quoted at cell-line scope; Table 4.6 quotes the plate scope this build trains in, 120.3, and gives the cell-line figure in its note.) Row (b) is the target string itself: a 100-gene up block, [DOWN], a 100-gene down block, [END_CELL]. *Stage 3.* What NIR ranks is not that string but a signed vector over the 946 panel genes, top- k up positive and top- k down negative. The stem heights are the exact weights $1/\log_2(\text{rank} + 1)$, so the eight drawn per sign are 1.000, 0.631, 0.500, 0.431, 0.387, 0.356, 0.333 and 0.316; which gene sits at which coordinate is illustrative, and no cosine value is printed, because these two vectors are drawn rather than measured. Stage 2 and stage 3 are different objects and only stage 3 is scored. The bracket is the point of the figure — the within-well split-half retrieval reference is essentially identical under both encodings, so no information has been added. What changed is how much of it the token sequence exposes to the loss — a property of the encoding, which is not the same as showing the encoding is what binds (Table 4.6).

READING A GENERATION BACK, AND WHAT IT IS SCORED

AGAINST. § 3.5 gives the target's two-block form and the map ϕ that turns a signed signature into a vector; Figure 4.14 draws the whole path, from the two subtractions to the vector that is scored. Three things are specific to scoring a *generation* and are fixed here. A generation supplies its ranks by token order within each block, and a token outside the panel or already placed is skipped without consuming a rank, so a block fills to at most 100 genes. The sampled generations of one condition are averaged as vectors and the mean is scored once, rather than scored one by one and averaged. And the score is NIR over the other drugs in the same cell line — a set that spans plates, in apparent violation

of the same-plate rule of § 4.2, and one that stands only because the plate-scoped generic has already subtracted each plate’s mean shift, in the expectation sense the scope limit above fences off.

4.8 Shift decomposition and the transfer coefficient

▲ RESID

The residual says what the model should predict, not how much there is to predict, and the same training result reads differently against a large reachable signal and a small one. Two measurements settle the scale: what share of the observed perturbation shift remains in the residual, and how much of that residual a table of per-drug constants would already capture. The first has a clean answer. The second has an honest one.

This section measures scale rather than skill, and it takes its two measurements in order. The first asks how much of the observed perturbation shift remains after the estimated generic is subtracted. The second asks how much of *that* a table of per-drug constants would already reproduce with no model involved.

The first is a decomposition, and it is available because the residual was defined as a subtraction. Since $r = s - \text{generic}$, the shift is the sum of its two parts, and squaring gives

$$\|s\|^2 = \|r\|^2 + \|\text{generic}\|^2 + 2 \langle r, \text{generic} \rangle. \quad (4.31)$$

Dividing through by $\|s\|^2$ gives three shares that sum to one by construction. Measured across the cache with the drug-weighted, leave-one-drug-out generic of § 4.7, $\|r\|^2/\|s\|^2 = 0.621$ (observed residual), $\|\text{generic}\|^2/\|s\|^2 = 0.390$ (shared component), and the cross term $2\langle r, \text{generic} \rangle/\|s\|^2 = -0.011$. These are shares of an observed squared norm, not identified fractions of a biological response. § 3.5 forms the generic as the mean shift of the *other* training drugs in the con-

dition’s own (cell line, plate) group, with the condition’s own drug struck out. It estimates the component shared across those drugs without including this drug in its own baseline. The residual is what remains after that estimate is subtracted; it can still carry sampling noise, baseline-estimation error and context effects. Leave-one-drug-out prevents direct self-inclusion, but does not by itself identify every residual coordinate as drug-specific biology.

The cross term adds a geometric check. Orthogonality between residual and generic is exactly the statement $\langle s, \text{generic} \rangle = \|\text{generic}\|^2$, which is the condition for the coefficient in a regression of the shift on the generic to equal one. Measured here that coefficient is 0.986, consistent with subtracting the estimated generic at approximately unit weight on this aggregate scale. It is an empirical check on the estimator of § 4.7, not a guarantee that every residual is free of shared biological or technical structure.

Two descriptive readings follow. The first is that the estimated components have little aggregate alignment: the cross term is -0.011 , rather than a large contribution along the generic. This is compatible with differences in response direction, but the observed residual includes noise as well as signal. The second reading is size. Treating that small cross term as approximately zero, $\|r\|$ and $\|\text{generic}\|$ stand at 0.788 and 0.624 of $\|s\|$ on this aggregate scale, roughly five to four in norm and three to two in squared norm, with an implied angle of about 38° between shift and residual. These describe the estimated components, not a noise-corrected partition of biology.

This is also what reconciles the decomposition with the counts of Table 4.6, where subtracting the control added 76.8 differing positions and subtracting the generic only 8.9 more. The generic is *common mode*. Two drugs in one group are handed nearly the same generic, dif-

fering only by which single drug leave-one-out withholds from a mean over the group, so subtracting it translates both shifts by almost the same vector and leaves the difference between them almost unchanged. A component can accordingly carry two fifths of the squared norm and still barely move a count of between-drug differing positions; the 8.9 it does move is reranking, since a ranking is not invariant under translation. Large magnitude and small contrast contribution are not in tension, and the two measurements say different things about the same object.

One qualification travels with all of it. What has been decomposed is the *shift*, treated minus control, whereas a cell sentence ranks the treated profile itself, so no term of this decomposition appears anywhere in the standard target (Figure 4.15).

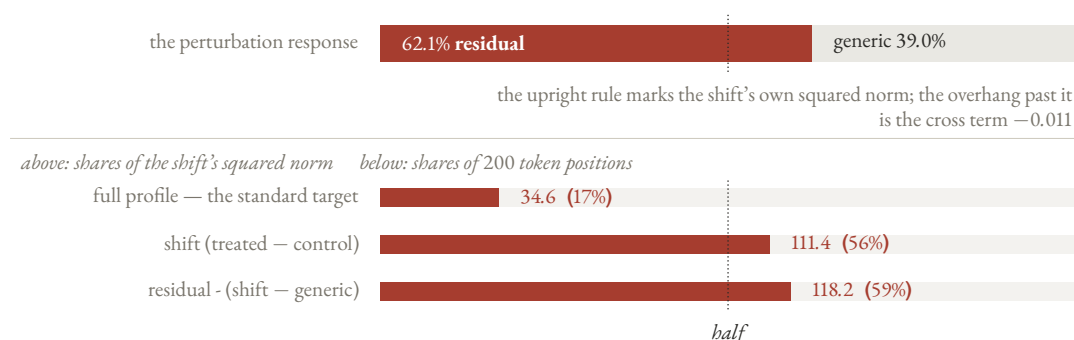


FIGURE 4.15 The observed residual is most of the shift's squared norm, while standard targets share most of their genes across drugs. Above the rule: the shift's squared norm decomposes exactly into an observed residual, an estimated generic and a cross term. The residual share is 62.1%; it includes measurement noise and unremoved context effects, so this is not a fraction identified as drug-specific biology. Below it: of the 200 positions a cell sentence spends on the standard target, only 34.6 carry a different gene for two drugs in the same context — the other 83% are the same string. Ranking the same measurement by signed residual raises that to 118.2, and at the plate scope this build trains in the residual row reads 120.3 (60%). The two halves of the figure are different quantities, a squared-norm share and a count of token positions; they share the axis and the half-way rule and nothing else, and no difference between them is a number. The residual carries a majority of the observed shift's squared norm; separately, residual targets differ across drugs at more positions than standard targets do (Table 4.6).

THE SECOND QUESTION IS WHETHER THAT MAJORITY HAS TO BE PREDICTED AT ALL. The first measurement says the observed residual is the larger component of the shift by squared norm. It says

nothing about whether that part is hard to produce. A residual could be large and still be a per-drug *constant* — the same signature for a given drug wherever it is applied — in which case it is reproducible from a table indexed by drug name, with no model, no prompt and no cell involved. So the second question is how much of the residual is that constant, and how much of it moves with the context the drug is applied in.

Writing

$$r(d, c) = \beta(d) + \kappa(d, c) + \varepsilon, \quad \mathcal{T} = \frac{\text{var } \beta}{\text{var } \beta + \text{var } \kappa}, \quad (4.32)$$

splits the residual into a per-drug constant, a context-specific remainder and noise. $\beta(d)$ is the drug *main effect* and is what a lookup table reproduces with no model at all; $\kappa(d, c)$ is the only component a conditional model could win on; and $\mathcal{T}$ is the share of the structured part that the constant carries. The baseline this sets up is not invented for the occasion. It is DrEval’s naive predictor [21] in transcriptomic space: their NaiveMeanEffectsPredictor predicts $\hat{y}_{ij} = \mu_i^c + \mu_j^d - \mu$, cell-line and drug main effects with no interaction term, and $\text{generic}(c) + \beta(d)$ is that quantity lifted from a scalar potency to a 946-dimensional profile. Asking what κ holds is therefore asking what a model could add over the field’s own no-skill predictor.

SEE § 4.11 builds that lookup and scores it.

THE DIRECT ESTIMATOR IS UNAVAILABLE, AND WHAT REPLACES IT. Fitting the variance components would be the obvious route: estimate $\text{var } \beta$ and $\text{var } \kappa$ and read $\mathcal{T}$ off them. It cannot be done on this cache. Separating $\text{var } \kappa$ from $\text{var } \varepsilon$ needs two independent measurements of the same drug in the same context, and 286 of the 287 distinct treatments here occur in exactly one well, so the model is unidentified on its own data. The route taken instead buys identification from a quantity already computed for the reliability line. Two

residuals of one drug in two cell lines are compared by cosine; that cosine is depressed by measurement noise in both, and dividing by the geometric mean of the two conditions' split-half reliabilities removes the depression.

DEFINITION The transfer coefficient

With 286 of 287 treatments in exactly one well there is no replicate from which to separate var κ from var ε , so the variance-components model is unidentified on its own. The cosine route buys identification through split-half reliability:

$$\hat{T} = \frac{\cos(r(d, c_1), r(d, c_2))}{\sqrt{\rho(d, c_1)\rho(d, c_2)}}, \quad \rho(d, c) = \frac{2 \cos(r_A, r_B)}{1 + \cos(r_A, r_B)}, \quad (4.33)$$

where r_A and r_B are disjoint halves of condition (d, c) . Here $\rho(d, c)$ is the Spearman–Brown reliability of that condition's residual at its full cell count.

We name explicitly two terms in the formula. The *cross-line cosine* is the numerator alone: $\cos(r(d, c_1), r(d, c_2))$, the plain cosine similarity between one drug's two full residuals in two different cell lines, computed with nothing else applied to it. *Disattenuation* is a correction, not a description, and it targets a specific bias. Each residual entering that cosine is itself a noisy estimate — it carries the same sampling error the reliability line was built to detect — and noise in two vectors being compared pulls their cosine toward zero independently of whatever the vectors would agree on if measured exactly. So the raw cross-line cosine understates the true agreement between contexts by an amount set by how noisy each residual is, and it understates it more where reliability is lower. Dividing by $\sqrt{\rho(d, c_1)\rho(d, c_2)}$, the geometric mean of the two conditions' own split-half reliabilities, removes exactly that shrinkage: this is the classical correction for attenuation, applied here to a cosine instead of the correlation coefficient it was derived for. $\hat{T}$ is what the formula returns after that division, and calling it *disattenuated* names the correction it has had applied, not a new quantity being introduced.

Whether the correction actually removes the shrinkage, rather than just relabelling it, is checkable, because a case exists where the true

SEE § A.1's planted-truth check licenses using it on a cosine

answer is known. Data were simulated with $\kappa = 0$ built in — every context gets the same residual for a given drug, so by the definition of $\mathcal{T}$ the truth is exactly 1. Measurement noise is simulated too, so the raw cross-line cosine on this synthetic data still comes out below 1: it reads 0.734, which a reader taking it at face value would misread as 27% context-specific structure, $1 - 0.734$, when by construction there is none. Dividing that same 0.734 by the geometric-mean reliability recovers $\hat{\mathcal{T}} = 1.001$ (Table 4.7), against a truth of 1. The correction is therefore shown to work on a case where the answer is known, not merely assumed to work because the derivation is standard, and every number below rests on that check.

WHAT THE COEFFICIENT IS, AND WHAT IT IS NOT. Four qualifications attach before any estimate is quoted, and together they make $\mathcal{T}$ a descriptive sensitivity analysis rather than a fundamental measurement. First, κ is written as a drug $\times$ cell-line interaction because the decomposition assumes one, but the estimator does not identify such a term; κ is therefore called the *context-specific remainder* and $\mathcal{T}$ a disattenuated cross-context residual-similarity coefficient, never a variance share. Second, it might be tempting to read $1 - \mathcal{T}$ as the headroom above the constant baseline, however we advise against that: the cross-line pairs entering it may differ in dose and in treatment well, so it mixes several axes into one number. Third, the scoped calculation retains conditions at a reproducibility threshold of 0.2 and takes each residual against a *self-including* generic its own drug helped define, which is not the training frame. Fourth, the reliability in the denominator is a within-well split-half precision, optimistic relative to biological reproducibility — and that optimism has a direction. An over-large ρ makes $\hat{\mathcal{T}}$ too small, so $1 - \hat{\mathcal{T}}$ errs *high* and bounds from above what a per-drug constant leaves on the table, without measuring it.

One further caution is methodological rather than interpretive. Spearman–Brown is derived for correlations between parallel halves, not for cosines, and what licenses its use here is the planted-truth validation of § A.1.

QUANTITY	VALUE
ground truth, by construction ($\kappa = 0$)	$\mathcal{T} = 1.000$
uncorrected cross-line cosine, $\cos(r(d, c_1), r(d, c_2))$	0.734
that cosine read naively as an interaction share, $1 - 0.734$	27% (truth: 0%)
disattenuated estimator, $\hat{\mathcal{T}}$	1.001

TABLE 4.7 The disattenuation correction, checked where the truth is known. Residuals were simulated with $\kappa = 0$ built in, so by the definition of $\mathcal{T}$ the ground truth is exactly 1; every row is computed on that synthetic data, not on the atlas.

Measurement noise alone, with no cross-context difference anywhere in the simulation, is enough to pull two identical residuals' cosine down to 0.734. The correction recovers the true value to within 0.001 and licenses the reference column of Table 4.8.

TABLE 4.8 The transfer coefficient, estimated on real molar doses with dyadic cluster-robust intervals. The matched negative control sits on zero. The same-plate, different-drug control that produced the earlier false alarm reads +0.485 under a cell-line-scoped generic and -0.017 under the plate-scoped one this build uses; it differs structurally from the estimand and is voided as a matched control (§ 2.6). $\mathcal{T}$ is a cross-context similarity coefficient, not a variance share, and $1 - \mathcal{T}$ must not be read as a drug $\times$ cell-line interaction share.

QUANTITY	ESTIMATE	95% INTERVAL	REFERENCE
$\mathcal{T}$ (cross-context, plate-scoped generic)	0.553	[0.514, 0.593]	simulated $\kappa=0$ null = 1.001
similarity scale NOT shared across contexts, $1 - \mathcal{T}$	44.7%	[40.7%, 48.6%]	mixes cell line, dose, well and estimator
$\mathcal{T}$ across dose (same drug, same line)	0.707	[0.632, 0.782]	reference scale
structure-matched negative control	-0.008	spanning zero	must be ≈ 0

WHAT THE ESTIMATE SAYS. Two things about how this number was obtained are worth restating before it is read. *Real molar doses*: the dose entering this calculation is the resolved concentration of chapter 3, recovered from `drugname_drugconc`. *Dyadic cluster-robust*: $\hat{\mathcal{T}}$ is computed over *pairs* of conditions, and two pairs sharing either a drug or a cell line are not independent observations, so the interval is not a plain bootstrap over pairs but the Fafchamps–Gubert dyadic estimator [38] of § 3.7, built for exactly that dependence.

Under that estimator, on real doses, $\hat{T} = 0.553 [0.514, 0.593]$ (Table 4.8). Read with every qualification above in force: a little over half of the cross-context similarity scale is carried by the per-drug constant, and 44.7% [40.7%, 48.6%] of it is not.

That reading rests on the negative control holding, so the control is worth describing rather than only quoting. It is built by keeping everything the estimand depends on fixed — the same pair of cell lines, the same plate-scoped generic, the same disattenuation — and breaking only the one thing $\hat{T}$ is supposed to measure: which drug is compared. Two *different* drugs' residuals are compared across the same cell-line pair instead of one drug's two residuals, so any structure the control still detects would have to come from the pipeline itself, the generic construction or the dyadic clustering, and not from drug identity. It returns -0.008 , an interval spanning zero. Nothing in the construction manufactures apparent cross-context agreement on its own; what $\hat{T} = 0.553$ measures is therefore attributable to the shared drug, not to an artifact of how the estimator is built.

There is consequently a conditional target that a per-drug constant does not express, and its size is bounded from above at roughly 45% of a similarity scale — not a variance share, and not a quantity any model has been shown to reach.

DOSE IS THE Milder PERTURBATION. The estimand $\hat{T}$ is meant to say how much of the context-specific remainder survives a change of *cell line* specifically, but the cross-context pairs computing it also differ in well and, since doses are not matched across cell lines, in dose — so the 44.7% that does not transfer could in principle be about that mixture rather than about cell-line context as such. Dose supplies a cleaner comparison, because it is the one axis that can be varied while everything else the estimand depends on — same drug, same cell line,

same generic, same disattenuation — stays fixed. If a drug’s residual moved as much between two of its own doses as it does between two cell lines, cell line would be doing no special work; if it moved much less, dose would be shown to be the milder perturbation, and the part of the cross-context estimate that dose alone could explain away would be small.

Holding drug and cell line fixed and varying only the molar concentration, $\hat{T} = 0.707$ [0.632, 0.782] on 363 pairs, against $\hat{T} = 0.553$ [0.514, 0.593] pooled across cell lines; the intervals do not overlap. Dose is therefore the milder perturbation: a drug’s residual survives a change of concentration better than it survives a change of cell line, so the 44.7% that does not transfer across context is not merely an artefact of comparing different doses to each other. The comparison is not perfectly clean — the cross-context arm still differs in well and dose as well as cell line, so this is the closest the section comes to a single-axis test rather than being one — but the direction runs the way the estimand’s own logic predicts.

WHETHER ANYTHING COULD LEARN THE REMAINDER. The 44.7% a per-drug constant does not reach is not automatically a target only a full model could hit. A simpler explanation is available first. If a cell line modulated every drug’s response in a broadly consistent direction — a “cell-line personality” that is not any drug’s own signal but is nonetheless learnable from that line’s other conditions — then two *different* drugs measured in the *same* cell line should share more of that direction with each other than either shares with a drug measured in a *different* cell line, once each drug’s own main effect is removed. That prediction is specific enough to test directly.

Testing it means removing the main effect first. Write $\kappa(d, c) = r(d, c) - \hat{\beta}(d)$, where $\hat{\beta}(d)$ is estimated *leave-one-cell-line-out*: averaged

over every cell line except c , so condition (d, c) never enters its own baseline. This avoids fitting that baseline directly to the condition being scored; it does not remove every dependence introduced by centring. With m equally weighted cell lines, a leave-one-line-out residual is $m/(m - 1)$ times the residual centred on the full mean. The pairwise cosines, and any $-1/(m - 1)$ correlation induced by that centring under independent equal-variance observations, are therefore unchanged. With the estimated main effect removed, the test compares two cosines: $\cos(\kappa(d, c), \kappa(d', c))$, two different drugs' remainders in the same cell line, against $\cos(\kappa(d, c), \kappa(d', c'))$, two different drugs in two different cell lines. A cell-line personality predicts the first running higher than the second. Both cosines are disattenuated by κ 's own split-half reliability before the comparison is made, for the same reason $\hat{T}$ itself is: measurement noise can depress an uncorrected cosine, and κ is a noisier quantity than r .

Averaged over lines, neither arm shows the signature the hypothesis predicts. Under the plate-scoped generic the same-cell-line arm reads $-0.0149 [-0.0251, -0.0050]$ and the different-cell-line arm $+0.0003 [-0.0071, +0.0081]$, an excess within line of -0.0153 for which no interval was computed; both come from a cell-line-clustered bootstrap, 38 and 42 clusters respectively, over 4,000 same-line and 3,884 cross-line pairs drawn from 1,282 conditions. A real per-line direction predicts the same-line arm sitting *above* the cross-line one; instead it sits below it, and it is the only one of the two distinguishable from zero at all. That negative same-line reading is not itself read as evidence of some opposite structure. Group centring can induce negative dependence when the generic is estimated from the group members, and same-line pairs that also share a plate are exposed to that construction. The different-drug same-plate diagnostic also reads -0.017 under the

plate-scoped generic (Table 4.8), consistent with this concern, without measuring how much of the present difference it explains. Subtracting an arbitrary shared quantity does not by itself guarantee a negative cosine. The cross-line arm, whose pairs do not share a plate generic, is indistinguishable from zero. Neither arm supplies positive evidence for a common cell-line remainder under this construction; that is not proof that no such biological effect exists.

4.9 Prompt sensitivity on trained conditions

▲ RESID

Every target change tried so far has not shown robust counterfactual predictions. Does training on the residual change that? The instrument is the stratified scramble, the same prompt with another drug's name and mechanism substituted, read as a ladder of strata because the swapped-in drug is itself a drug the model knows and so is not a neutral comparison. The model moves. It is not the only target change that makes it move.

Every scramble so far has been run on tier 2: held-out drugs, ones the checkpoint never saw in training, with the substituted partner drawn from that same held-out set. That is a hard test and deliberately so, but it asks for two things at once. A model that passes it must respond to the drug it is named *and* have learned something transferable about a drug it has never encountered. Nothing has passed it. The single-cell arm is null under the random-partner comparison and shows no ordering under its own designed sweep; consensus is flat at every swap distance; optimal transport clears zero only against the most dissimilar partners and nowhere before them.

This section separates the two demands and asks only the first. The ablation is unchanged. Same prompt, same substitution, same comparison set, but it is run on conditions the residual checkpoint actually

trained on and, for the two controls, on a sample whose split its evaluation did not record — so that for at least the arm this section is about, nothing has to be extrapolated and memorisation is available. That makes it the easier question by construction, and the right one to ask next: if a model is sensitive to the drug token at all, this is where it shows, and a null here would be much harder to attribute to the difficulty of the split.

It also brings in a fourth target. The residual re-encoding of § 4.7 was not among the three constructions of § 4.6; it is what this section is about, and two of those three are carried forward with it — but not as a yardstick. Relaxing the split is a concession, and the obvious objection to anything found after making it is that the concession *is* the finding: that scoring a model on conditions it has already fitted is by itself enough to open a gap against a scrambled prompt. The controls are here to answer that objection, not to compete for a ranking. They are read out of the same evaluation cache, through a ladder built by the same rule, against the same kind of comparison. One thing about them has to be said straight away, because it bounds what they can settle: their evaluations carry no split manifest, so which of their conditions each checkpoint had already seen is not recorded, and they cannot certify that the relaxation was granted to them on the same terms. What they do bound is the instrument — whether this ladder, read out of this cache, returns a large positive gap as a matter of course — and that is the reason they are drawn. No ordering between arms is claimed anywhere below.

Carrying them forward also fixes where the earlier nulls do *not* come from, and the two controls answer that differently. § 4.2 and § 4.6 found neither construction able to order its gap by swap distance on held-out drugs, scored in the expression frame they were trained

to emit — flat throughout for single cell, and for optimal transport a single narrow excursion at the far end and nothing before it. Read here on seen conditions and in the residual frame, the single-cell arm is still not recovered: its series spans zero at every stratum. Neither control's split is recorded, so that is not the clean seen-condition null it would be worth having; what it does establish is narrower and still useful: pointed at this checkpoint, on this cache and this ladder, the instrument returns nothing at all, so it does not open a gap merely by being run. Optimal transport is partly recovered, and that is the second reason to draw it. Its orth and opposite gaps both clear zero here, small but real, and this section is where the reading that that arm does nothing is withdrawn. The withdrawal is budget-driven and internal to this instrument: on the same checkpoint against the same orth comparator, 600 tokens give $+0.0144$ $[-0.0032, +0.0323]$, spanning zero, and 1400 give $+0.0292$ $[+0.0144, +0.0441]$, which does not — though on a pool sharing only 10 conditions with the first. It is not a re-run of § 4.6's own sweep, which is a different frame, population and comparator; what it withdraws is that sweep's conclusion, not its number. What it still does not show is ordering across the ladder, and the limits below are explicit that this is a statement about the ladder its pool supports rather than about the arm. Neither reading turns on a comparison with the residual arm, and § 4.10 is where the harder question goes back to whichever arm survives this one: whether the response holds on conditions never trained on.

One consequence of that relaxation belongs up front, before any design detail or any number: the arms are no longer scored on a common set. Each is read on a scoring sample of a few hundred conditions rather than a full population, since three scrambled generations per condition is expensive, and those samples were drawn separately — three pools

in all: one 250-condition pool, shared by all three arms at 600 tokens and reused for the pre-repair checkpoint at 1400; the repaired residual arm's 300 at 1400; and the controls' 248 at 1400. At 1400 the two are mostly non-overlapping and they differ in size. What is held fixed is not the data but the shape of the instrument: the near/orth/opposite ladder below is built the same way wherever it is built, so the only reading any of it supports is an arm against its own scramble on its own conditions, never one arm's number against another's.

The instrument itself needs one design choice made explicit before any number is worth reading. For each condition, a partner drug is drawn from every other condition sharing its cell line *and* its plate but naming a different drug — same plate so a swap never differs from the target in batch as well as in identity, different drug so a drug's own other dose is never mistaken for a scramble — and three fixed partners are taken from that pool by measured residual cosine: the *near* partner is the most similar available, *orth* the one closest to orthogonal, *opposite* the most anti-correlated. The pool is ranked by residual cosine and not by the moa-fine mechanism annotation because the annotation does not track it: two drugs sharing a mechanism label sit a mean distance 2.322 apart against 2.377 for two that do not (1,244 same-mechanism against 17,563 different-mechanism pairs, from the drug atlas of § 3.5), just over two per cent closer, so “different mechanism” does not imply “different response”. A condition with fewer than three eligible partners is dropped rather than assigned a contaminated one. The three partners are fixed by this construction and reused, unchanged, across every arm scored within one evaluation run — though not across runs: the 1400-token control run draws on a different pool, whose opposite rung is much the weaker of the two, as Figure 4.16 sets out.

On a scoring sample of 300 conditions the holdout assigns to train, 293 of which the evaluation also flags as seen, while all 300 clear the resolved reliability line (minimum reproducibility cosine +0.1089 against +0.1086), so the seven unflagged conditions are a gap in the evaluation's training manifest rather than conditions the filter excluded — the train tier of the three-way holdout § 4.7 builds, scored on that same checkpoint's evaluation (one generation budget of 1400 tokens throughout, the only budget this checkpoint has been scored at) — the model separates from its own scramble on two of the three partners, and moves in the predicted direction on the third, in the order the construction predicts throughout. Against near, the hardest test, the gain is smallest and does not clear zero, +0.0229 [−0.0232, +0.0689]; against orth it is +0.0898 [+0.0382, +0.1415], clustered on cell line and well (+0.0898 [+0.0291, +0.1506] clustered on drug and well); against opposite it more than doubles again, +0.1942 [+0.1358, +0.2527]. One checkpoint, one evaluation, one generation budget, one population: on the conditions this checkpoint was trained on, the model separates from its own scramble in the order the construction predicts, and it is not the only target construction that separates at all.

scramble The same prompt with the drug's name and mechanism replaced by another drug's, in three strata by residual similarity.

WHY THREE STRATA, AND WHY ONE OF THEM IS QUOTED.

The three strata are not three attempts at the same measurement. They are three detection sensitivities, set deliberately. Every prediction is scored against the *real* drug's truth, so what changes across the strata is not how hard the prediction is to produce but how visible it is that the model followed the drug it was named. Substitute a near-twin and an unchanged prediction is nearly the correct answer anyway, so the scramble scores well and there is almost nothing to see. Substitute an opposite effect drug and a model that actually follows its prompt is pushed away from the truth in the one direction the metric measures,

so the scramble collapses and the separation is at its widest. The model's own score is fixed at 0.598 throughout; it is the scramble NIR across the stratum that falls, 0.575 to 0.508 to 0.404, and the gap that opens with it. A model doing genuine counterfactual work should therefore separate most on opposite and least on near, which is the ordering Table 4.9 reports. That ordering is the evidence. A metric artefact, or a model reproducing a generic response regardless of the drug, has no reason to arrange itself by how dissimilar the substituted drug happens to be, so the shape of the series across strata says something no single stratum can.

A single number is nonetheless needed, and not for anything internal to this section: everything downstream requires one. Reporting an arm compactly means one figure per arm, not three; § 4.10 compares the trained tier against two held-out ones, and its evaluation was run against one fixed comparator (`scramble_orth`) rather than re-run three times; and every later section that cites this effect — the lookup-table comparison, the channel gate, the chapter summary — inherits whatever one quantity is named here. The choice is therefore about which stratum's gap can honestly carry the label “the effect”.

`Orth` is the only candidate, and the reason is that it alone has a defensible null. Scored on its own, the near partner reaches 0.575, above the 0.500 chance line: naming a similar drug is partly *correct*, so a gap measured against it subtracts real signal and understates the effect. The opposite partner scores 0.404, below chance: that substitution is actively wrong, so a gap measured against it inflates. `Orth` sits at 0.508, on the chance line almost exactly, which is what makes its gap read as distance above chance rather than distance from a contaminated baseline. Reading the three partners as points on one line rather than as three separate comparisons makes the same point: their own scores fall

monotonically with the cosine between the substituted drug’s truth and the real one, and or th is where that line crosses chance. It is the one place on the ladder where subtracting the comparator neither gives the model credit for a partly right answer nor charges it for a wrong one.

HOW THE SCORING SAMPLE IS BUILT FOR EACH ARM, AND WHAT MAKES THE THREE COMPARABLE.

Nothing here is scored on a full population. For each arm, a *scoring sample* is drawn from the evaluation cache, for the residual arm from conditions it had already seen during its own training — a few hundred conditions, not the whole set — and every number in this section is read off that sample. The residual arm’s is the 300-condition train tier of the three-way holdout § 4.7 builds, and its trained status is recorded per condition. The two controls are read on one shared sample at each budget — 250 conditions at 600 tokens and a near-disjoint 248 at 1400, with only 10 conditions common to the two — drawn from the same evaluation cache the residual arm is itself scored out of. Neither control evaluation carries a split manifest; both record the split as unknown for every condition, so which of them each checkpoint had already trained on is not established. Their rows are nulls of unverified split rather than demonstrated seen-condition nulls, and the seen-condition framing of this section is secure only for the residual arm, whose 300 come with the three-way holdout attached.

The three samples are not three views of one population, and this is the point worth being plain about: the three arms were trained on different data. Each target construction came with its own corpus, built at its own time, under its own inclusion rules, and no arm was retrained to match another’s. A scoring sample drawn this way therefore means a different set of conditions for each arm, and the samples differ in size as well: the residual arm’s 300 and the controls’ 248 have just 15

conditions in common. Nothing survives that in the form a cross-arm comparison would need. What each sample does support is one arm read against its own scramble on its own conditions, and that is the only reading taken below.

TABLE 4.9 On the conditions the checkpoint trained on, the gap runs from a partly-correct partner to an actively wrong one, at one checkpoint and one generation budget. Model NIR is 0.598 in every row; only the comparator changes. $n = 300$ throughout.

STRATUM	PARTNER IS	PARTNER COS.	SCRAMBLE NIR	GAP	95% INTERVAL (CELL LINE, WELL)
near	most similar available	+0.248	0.575	+0.023	[−0.023, +0.069]
orth	closest to orthogonal	−0.0001	0.508	+0.090	[+0.038, +0.142]
opposite	most anti-correlated	−0.261	0.404	+0.194	[+0.136, +0.253]

Partner cosine is the mean cosine, in the residual frame, between a condition's own residual and the partner substituted for that row's stratum. Scramble NIR is the NIR of the prediction when swapping the name of the drug with that of what is in the stratum (near, orth, opposite). The orth row additionally excludes zero clustered on drug and well, +0.0898 [+0.0291, +0.1506]; no drug-clustered interval was computed for the other two rows.

TWO CONTROLS, READ AT BOTH BUDGETS, AND ONE RESIDUAL CELL THAT IS A DIFFERENT MODEL. Every run this section has is in Table 4.10, and the residual arm contributes three of them rather than two. Two are the *pre-repair* checkpoint — trained on targets built before the split-before-fit correction of § 4.7, when the generic and the reliability filter were still fitted over every condition and the holdout assigned afterwards — scored at both budgets on one 250-condition pool. Those two rows are the chapter's one clean budget comparison: one checkpoint, one pool, one comparator, with the budget the only thing that moves: the gap roughly doubles at orth and opposite, and at near moves from a zero-spanning −0.005 to +0.035. The third is the repaired checkpoint at 1400 tokens, which is where this chapter's canonical result comes from and which has never been scored at 600; it cannot be read against either pre-repair row without confounding a target rebuild with a budget change. The two control arms have no such problem at the level of the checkpoint: each was scored at

both budgets on its own single checkpoint. Their two budgets are not read on one condition set, though — the 600- and 1400-token pools share only 10 conditions — so those rows move budget and population together, and are not budget comparisons alone.

Read at 1400 tokens, where the residual arm's canonical checkpoint actually lives, neither control shows a three-stratum pattern — though on their pool the three strata are not three distinct tests, so only the near-versus-rest half of that is a real null. The single-cell control is flat and null at every stratum. The optimal-transport control is not flat, and clears zero on two of the three strata — but at orth and opposite almost equally, which on this pool is close to being the same measurement twice: in 138 of its 248 conditions the two strata name the same partner drug. Both arms' full three-stratum results at both budgets are in Table 4.10.

One asymmetry in scrutiny attaches to the optimal-transport orth value. Unlike the residual arm's own headline number, it has not been re-checked under drug-and-well clustering; the drugs behind those conditions repeat across them the same way the residual arm's do, so a wider drug-clustered interval is not ruled out, only untried.

Three limits stop any of this being read as an ordering between arms. The first is the frame. The residual checkpoint's generations are scored where they are emitted, while the two controls emit cell sentences, whose decoded profile is a function of gene rank order alone; a drug effect can reach their gaps only by reordering top-ranked genes, which is the channel Table 4.6 measures as nearly closed — 34.6 of 200 top positions differing between two drugs under the standard encoding, against 120.3 under the residual one. Their gaps are attenuated by construction, and an attenuated gap cannot be read against a native one.

The second is that the ladder is not equally built in the two pools, and this is measurable rather than a matter of interpretation. Across the residual arm's 300 conditions, every opposite partner is a different drug from that condition's orth partner, and all 300 opposite cosines are negative, mean -0.261 . Across the controls' 248, opposite and orth name the *same* drug in 138 of them — 56% — and only 133 of the opposite cosines are negative at all, mean -0.009 . The controls' top two rungs are largely one rung, which is the plainest reading of why their orth and opposite gaps come out almost equal, and it means an absence of near-to-opposite ordering in those arms is not evidence that they lack it: the instrument that would have shown it was not there to be failed.

The third is that these are separate evaluations over different, mostly non-overlapping condition sets drawn from separately built corpora, not a paired comparison, and the arms were not held to the same training discipline.

One thing that could have been a further limit is not one, and it is worth saying so rather than leaving it to be assumed. The residual arm's 300 conditions are the reliability-surviving population of § 4.7, which invites the objection that its result is a property of a pre-screened sample. The controls' 248 are the more reliable pool of the two, not the less: their lowest reproducibility cosine is $+0.202$ against the residual tier's $+0.109$, their median $+0.278$ against $+0.183$. Whatever this section's numbers are, they are not the product of the residual arm having been handed the easier conditions.

A pattern that survives the sharpest stratum on one arm is evidence about that arm. Its absence on an arm whose sharpest stratum was never built is not evidence about that one, and neither fact is a ranking. The comparative case in this chapter is not made in this section at all:

it is made by Table 4.6, which counts how far two drugs' target strings diverge under each encoding and needs no model to do it.

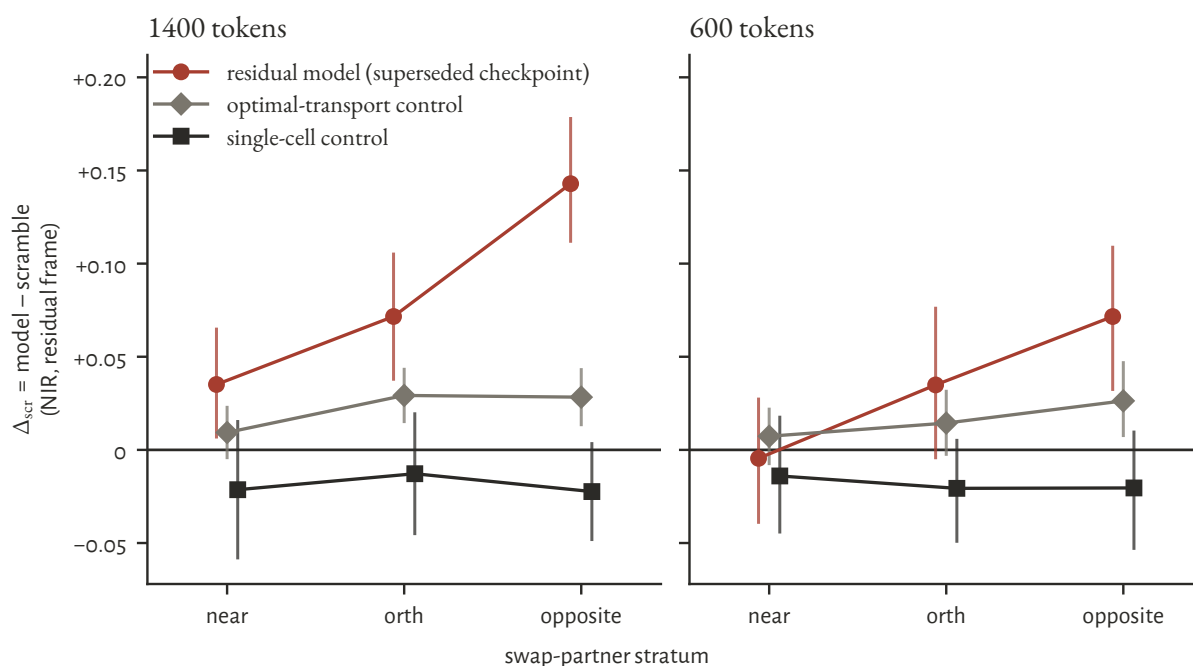


FIGURE 4.16 Raising the generation budget from 600 to 1400 tokens sharpened a real pattern rather than manufacturing one: the residual checkpoint's advantage over its own scramble grew at every stratum, the optimal-transport control's orth gap grew from spanning zero to excluding it (its opposite gap already excluded zero at 600 tokens and its near gap spans zero at both), and the single-cell control stayed flat at both. Residual frame throughout; the optimal-transport and single-cell series are the implied residual described above, since neither checkpoint was trained to emit one. Both panels share the residual checkpoint's superseded, pre-repair run, the only one with a 600-token measurement; the current checkpoint's canonical result (Table 4.9) is not drawn here. The x-axis is the stratum by name, not by measured cosine: at 600 tokens all three series share one 250-condition partner pool (mean residual cosine +0.406, -0.0016, -0.329 for near, orth, opposite); at 1400 tokens the residual series keeps that same pool while the two controls were scored on a different, smaller 248-condition pool whose near and orth partners shift by less, +0.403 and +0.036, but whose opposite partner is only weakly anti-correlated, -0.009, so the two panels' opposite points are not a matched pair. Intervals are 95% bootstrap, $n = 250$ per point at 600 tokens and $n = 248$ at 1400 for the two controls; clustering differs by series and is not uniform across this figure — the two controls' 1400-token intervals are clustered on cell line and well, matching the number quoted for them in the text, while every other series here (the residual arm at both budgets, both controls at 600) has only a coarser cell-line-only interval computed and is plotted with that instead.

THE SLOPE CARRIES ITS OWN NULL. One further check supports reading orth as the honest point on that ladder, and it does so without selecting any comparator at all. Each condition contributes three scrambled prompts, and each names a partner drug whose true residual sits at a measured cosine from the real drug's — on this tier +0.25, 0 and -0.26 for near, orth and opposite. Regressing each prompt's NIR

TABLE 4.10 Every arm at both generation budgets, model minus its own stratum-matched scramble.

The four control rows hold one checkpoint each across the two budgets, but not one condition set: the 600- and 1400-token pools share only 10 conditions. The two pre-repair residual rows are the cleanest pair here — one checkpoint, one 250-condition pool, budget the only thing that moves. The repaired row is a different checkpoint from those two, so reading it against them would mix a target rebuild with a budget change.

ARM	BUDGET	<i>n</i>	NEAR	ORTH	OPPOSITE
residual, repaired ^a	1400	300	+0.023 [−0.023, +0.069]	+0.090 [+0.038 , +0.142]	+0.194 [+0.136, +0.253]
residual, pre-repair ^b	1400	250	+0.035 [+0.006, +0.066]	+0.072 [+0.037, +0.106]	+0.143 [+0.111, +0.179]
residual, pre-repair ^b	600	250	−0.005 [−0.040, +0.028]	+0.035 [−0.005, +0.077]	+0.072 [+0.032, +0.110]
optimal transport	1400	248	+0.009 [−0.005, +0.024]	+0.029 [+0.014, +0.044]	+0.028 [+0.013, +0.044]
optimal transport	600	250	+0.007 [−0.008, +0.023]	+0.014 [−0.003, +0.032]	+0.026 [+0.007, +0.048]
single cell	1400	248	−0.021 [−0.059, +0.016]	−0.013 [−0.046, +0.020]	−0.022 [−0.049, +0.004]
single cell	600	250	−0.014 [−0.045, +0.018]	−0.021 [−0.050, +0.006]	−0.020 [−0.054, +0.010]

^aThe canonical result, repeated from Table 4.9 for comparison. It is the only row scored on a population with a split manifest attached, and the only one the chapter's claim rests on.

^bPre-repair checkpoint, trained on targets built before the split-before-fit correction of § 4.7, scored at both budgets on one 250-condition pool. These two rows are the series Figure 4.16 draws for the residual arm. The repaired checkpoint has never been scored at 600 tokens, and is not drawn in that figure.

on that cosine, over all three prompts of every condition at once, asks a question none of the individual gaps can: does the score track how similar the named drug actually is? The regression is pooled least squares with one intercept, three points per condition and no condition-level intercepts absorbed, and the target-drug prompt is never a point on the line — only the three scrambles are, so nothing in the fit contrasts the model against a chosen baseline.

Zero is the null by construction. A model that ignores the drug it is told emits the same prediction whichever partner is substituted, so its three points sit at the same height and the line is flat, whatever the partners' cosines happen to be. On the trained tier the slope is +0.3728 [+0.2084, +0.5139]: positive, excluding zero, and steep enough that

moving a partner from anti-correlated to similar — a span of roughly 0.5 in cosine — buys about 0.19 in NIR, which is the same order as the spread between the three comparator scores the ladder is built from. What that establishes is not orth directly but the ladder underneath it: the comparators really do fall in the order their labels claim, and by an amount that scales with the cosine, so orth’s position at the chance crossing is a property of the measured geometry and not an artifact of which three partners happened to be picked. § 4.10 refits this regression on all three tiers with one intercept per condition, the specification whose null is exact within a condition, and reports the pooled-intercept variant beside it; on the trained tier the two agree, +0.361 against the +0.3728 quoted here. One caveat on reading its magnitude rather than its sign: with a single pooled intercept the zero null is exact within a condition, but the fitted slope also carries a between-condition component.

BOTH BUDGETS, SIDE BY SIDE. The budget is not a neutral runtime setting here. Raising it from 600 to 1400 tokens moved the optimal-transport control’s orth gap from spanning zero to excluding it, very nearly doubling the point estimate on an identical checkpoint and an identically constructed comparator, though on a nearly disjoint sample of conditions, while leaving the single-cell control null at both budgets — so the budget sharpens a gap that is already there and does not manufacture one where there is none. Figure 4.16 puts all three arms at both budgets side by side, one panel each, across the three strata; Table 4.10 carries the same numbers.

The figure’s x -axis is the stratum by name rather than measured cosine, because the two panels do not rest on one partner pool. At 600 tokens all three arms share a single 250-condition pool. At 1400 the two controls were re-run on a smaller 248-condition pool whose opposite

partner is barely anti-correlated, -0.009 against -0.329 in the pool the residual series keeps; near and orth move far less, $+0.406 \rightarrow +0.403$ and $-0.0016 \rightarrow +0.036$. The controls' 1400-token opposite point is therefore a weaker test than any other point on either panel, and plotting all three series on one continuous cosine axis would draw a shared difficulty ladder the data does not support.

4.10 Held-out wells and held-out drugs ▲ RESID

A model that moves when the drug token moves has cleared a low bar, because reproducing a training average requires that much too. Everything scored so far came from conditions the model trained on, so what has been shown is that the token is consulted, not that anything transferable was learned. Separating the two needs a holdout, and the atlas's own randomisation decides which one means anything: whole treatment wells. Prompt sensitivity survives that test. Whether the residual itself transfers comes back unsettled.

§ 4.9 established one thing, and deliberately only one. On conditions the checkpoint had trained on, re-encoding the target made the model respond to the drug it was named: the separation opens in the order the construction predicts, it holds clustered on the cell line and on the drug, and the comparator-free slope excludes zero. That was the easy question by design. Nothing had to be extrapolated, memorisation was available, and a model that did no more than reproduce a training average would have cleared much of the same bar. What it establishes is that the prompt is consulted, not that anything was learned that travels.

Where the response stops is the question worth asking, and this atlas supports two ways of asking it. One recombines what the model

has already met: a trained drug seen again in a treatment well held out whole. The other withholds the drug from fine-tuning altogether. The first is the weaker test and the second is the one the claim has to survive. Both are reported below, and they fail in different places.

The physical layout of chapter 3 forces the choice. A (drug, cell line) holdout, the field’s standard Leave-Pairs-Out split and the one an earlier version of this work used, leaves the held-out condition’s treated well sitting in the training set through its other ≈ 48 cell lines. Measured against that holdout, 124 of the 255 treated wells for which split information exists placed a held-out condition and a training condition in the same well. That estimand measures within-atlas matrix completion, not independent generalisation, and no processing of the targets changes it.

SEE § 4.11: what a held-out gap has to beat.

Holding out whole wells is the only assignment-respecting choice the design offers, and it answers a narrower question, generalisation to an unseen treatment well of a seen drug. Fixed-dose cross-context transfer, present in one line and absent in another, is structurally unidentifiable in this atlas.

WHAT EACH SPLIT HOLDS. `train` takes every condition of the remaining wells, 4,999 before the reliability line and 2,523 after it. `unseen_combo` holds 36 treated wells out whole (904 conditions, 900 after inventory filters), and it never takes a drug’s last training well; without that guard a two-well drug could silently become an unseen drug while sitting in the unseen-combination arm, and the two arms would measure the same thing. The guard acts on wells before the reliability line, which only `train` is filtered at. Of the split’s 34 drugs, 30 also appear in the sampled train evaluation tier; the other four are absent from that sample, not from training. Their training-only lookups in § 4.11 confirm that retained training conditions exist for

them. `unseen_drug` holds every condition of 10 held-out drugs (714 conditions, 711 after inventory filters). Zero wells contribute conditions to both sides.

`unseen_drug` is a control and not a third result. Tahoe fine-tuning never sees the token, so a large effect there would indict the comparator, not report a biological finding. “Unseen” means held out from fine-tuning with mechanism supplied, not unknown to the pipeline; the Pythia-1b backbone was pretrained on natural language and the prompt still carries the Mechanism: field.

Two warnings ride with the design. DrEval reports, citing [27], that apparently good performance on pair-holdout splits can be reached simply by predicting each drug’s average response across the training cell lines, which is why the baseline ladder of § 4.11 is not optional. And the evaluation draws under a per-split quota, because a uniform draw reproduces the pool’s proportions and starves the split the experiment exists to test; in a first run, 250 constructed held-out combinations yielded only 33 scored conditions. Under the quota the evaluation scores 1,394 conditions in all, 300 trained and 199 `unseen_drug`, the rest unseen combinations, almost the whole pool of 900 that arm affords, while `unseen_drug` remains a quota of the 711 available.

THE FIRST RESULT OF THIS DESIGN IS ABOUT THE COMPARATOR. Which partner defines the gap decides the size of every number in this section, and only one of the three has a claim to the role. A gap splits at chance into how far the model rises above it when handed the right drug and how far the comparator falls below it when handed a wrong one,

$$\Delta_{\text{scr}} = \underbrace{(\text{NIR}_{\text{model}} - 0.5)}_{\text{how much naming the truth helps}} + \underbrace{(0.5 - \text{NIR}_{\text{scramble}})}_{\text{how much naming the lie hurts}}, \quad (4.34)$$

and only *orth* makes the second term vanish: pooled over all 1,394 scored conditions its own score is 0.501, against 0.550 for *near* and 0.423 for *opposite* (on the trained tier alone, Table 4.9, the three read 0.508, 0.575 and 0.404).

That second term is not noise — it is the model being pushed away from a truth it was told to miss — but how far it is pushed depends on which partner the prompt named. Each condition’s three partners are drawn from the other drugs on its own plate and ranked by the cosine between their true residual and the target’s (§ 4.9): *orth* is the partner closest to zero, and *opposite* the most anti-correlated available, meaning the drug whose measured response points most nearly against the target’s. That cosine is a fact about which drugs shared a plate, not about the model, and it does not hold still across the splits. Averaged over conditions, the *opposite* partner reaches -0.2615 on *train* but only -0.1842 on *unseen_combo*, because the most anti-correlated drug *available* is whatever happened to be plated alongside; *orth* sits within 0.0005 of zero on all three splits, since a partner near zero is one every pool can supply.

That is what decides the choice. This section’s claim is a comparison *between* splits — whether a separation found on trained conditions survives on held-out ones — so the comparator has to mean the same thing in every row, or part of any change from one row to the next is the plate rather than the model. On the numbers above, *opposite* fails that condition and *orth* meets it. Every gap in Table 4.11 is therefore taken against *orth*, and what is reported is the difference itself, $\text{NIR}_{\text{model}} - \text{NIR}_{\text{orth}}$, never one term of the split above.

WHAT THE TWO HELD-OUT ROWS REPORT. The gap is a paired contrast — the same checkpoint on the same conditions, differing only in the drug the prompt names — so it can be informative where an ab-

TABLE 4.11 Whole-well holdout on the corrected evaluation, with the truth verified against the build's own fit digest, `unseen_combo` scored almost whole (895 of the 900 it affords) and `unseen_drug` under a per-split quota (199 of 711). The two right-hand columns carry the SAME point estimate and differ only in what they treat as the unit of generalisation. On `unseen_combo` the gap excludes zero clustered on cell line and does not clustered on drug, which is why that row is reported as not established. The `unseen_drug` row corroborates but does not prove: its orth-partner gap is negative while its opposite gap is +0.0379. The strips share one axis, -0.12 to $+0.16$, with a hairline at zero; a filled point excludes zero, a hollow one does not.

SPLIT	n (LINES; DRUGS; WELLS)	MODEL NIR	Δ_{scr} , CLUSTERED ON CELL LINE	THE SAME, CLUSTERED ON DRUG
train	300 (39; 76; 136)	0.598 [0.549, 0.647]	 +0.0898 [+0.0382, +0.1415]	 +0.0898 [+0.0291, +0.1506]
<code>unseen_combo</code>	895 (42; 34; 36)	0.533 [0.492, 0.575]	 +0.0332 [+0.0069, +0.0594]	 +0.0332 [−0.0031, +0.0695]
<code>unseen_drug</code>	199 (40; 10; 28)	0.459 [0.404, 0.514]	 −0.0315 [−0.0836, +0.0206]	 −0.0315 [−0.1049, +0.0419]

Every gap printed here is measured against the geometry-defined orth partner, whose own score is empirically near chance on these conditions; opposite figures never appear as a column. Model NIR intervals test against the chance value 0.5 with no comparator. Intervals are two-way cluster-robust over cell lines and wells, with a Student t reference on $\min(G_{line}, G_{well}) - 1$ degrees of freedom, which is why the well counts are printed. The `unseen_drug` arm spans 28 wells, and $t_{27} = 2.05$ against 1.96 widens its interval by five per cent (chapter 3).

solute score is not. On `unseen_combo` naming the right drug does help, by +0.0332: clustered on cell line and well the interval excludes zero, +0.0332 [+0.0069, +0.0594], but clustered on drug and well it does not, +0.0332 [−0.0031, +0.0695], and drug is the axis a claim about transferring a *drug*-specific residual has to survive. That first interval is also the tighter of the two measurements Table 4.11 affords on this row — a half-width of ± 0.026 against the ± 0.042 on the model's own score against chance quoted below, both two-way cluster-robust on cell line and well — because differencing against the scramble removes what the two prompts share and buys a precision the absolute score cannot have. Positive for 20 of 34 drugs at a sign test of $p = 0.39$, the effect is a majority tilt and not a uniform one, so transfer to held-out combinations is not convincingly established. On `unseen_drug` no benefit to the name is established: the gap is -0.0315 [−0.0836, +0.0206], a point estimate favouring the *scrambled* prompt, and though the interval covers zero

the direction is the one a model with no knowledge of the held-out drug would give.

A second limitation sits beside that one and is not the same. The model's own NIR is 0.533 [0.492, 0.575] on unseen_combo and 0.459 [0.404, 0.514] on unseen_drug, both covering chance: on a well it did not train on, an above-chance ranking of that condition's truth against the other drugs' truths is not established. Responding to the name and predicting the response are separate properties, and this table does not establish the second on held-out data whatever it finds of the first. Whether the first survives is asked below, on an instrument that needs no comparator and reaches all three splits.

The three splits are not scored on comparable populations either, and the difference is one of selection. Their raw within-well split-half precision references, the grey dashes of Figure 4.17a, are 0.956, 0.824 and 0.836, a spread of 0.132; train is filtered at a reliability line the two held-out splits are not, and imposing that line on all three brings the references to 0.956, 0.961 and 0.960, a spread of 0.005. The held-out shortfall therefore cannot be softened by appeal to a lower noise floor.

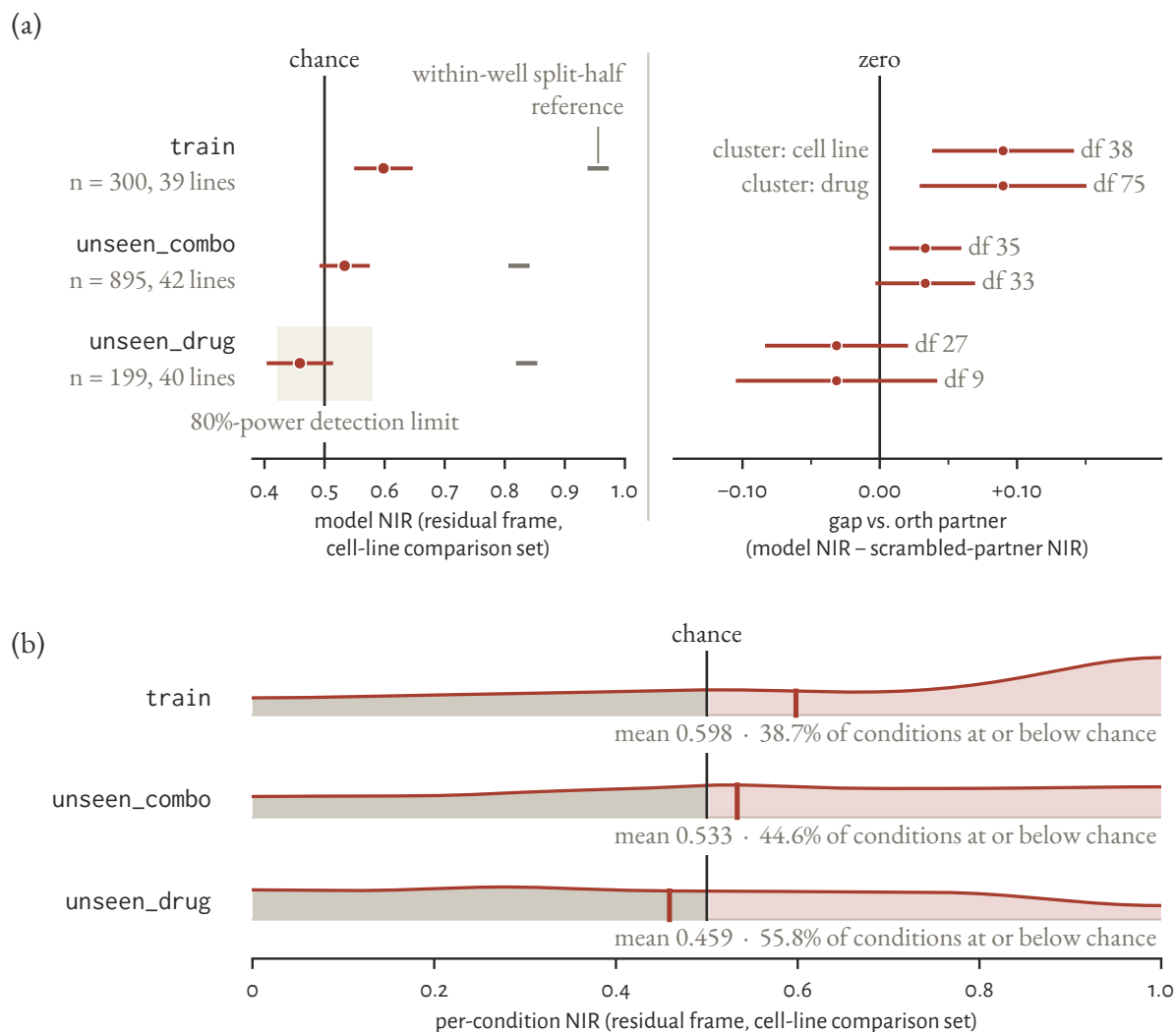


FIGURE 4.17 The residual-frame model clears chance on trained conditions only; the held-out-combination gap excludes zero clustered on cell line but not clustered on drug, and the unseen-drug gap flips sign. Residual frame throughout, and NIR ranks each prediction against the other drugs measured in the same cell line. (a) Left, model NIR against the chance value 0.5 (solid rule), with 95% two-way cluster-robust intervals; $n = 300, 895$ and 199 conditions on train, unseen_combo and unseen_drug. The grey dash on each row is that split's raw within-well split-half precision reference, a measurement-precision figure computed from two halves of the same well and not a replicate experiment; it carries no interval and is drawn as a bare marker, and only train is filtered at the reliability line. Right, on its own axis, the gap against the geometry-defined orth partner. Each row has one point estimate, drawn twice with two intervals because the verdict depends on which unit of generalisation is clustered: cell-line-clustered above, drug-clustered below, each labelled with the Student t degrees of freedom it is carried on. The grey band on the unseen_drug row is that arm's approximate departure from chance associated with 80% power under a normal approximation on the absolute-NIR scale, ± 0.079 ; it is neither a hard detection threshold nor a confidence region. (b) The per-condition NIR distributions behind those three means, as reflected-boundary kernel densities on a shared axis, each with its mean marked and the region at or below chance shaded. The pooled 0.537 is a mean over nearly flat distributions rather than a location: even on train, 38.7% of conditions sit at or below chance, against 44.6% and 55.8% on the two held-out splits.

The difference between the trained and held-out estimates is $+0.0567 [-0.0014, +0.1148]$ with cluster-robust $p = 0.0556$. An unclustered permutation returns $p = 0.0142$, quoted only for contrast, because ignoring the clustering this chapter mandates everywhere else flatters the result. It does not clear the conventional threshold, and the two arms differ in three ways at once in any case, in whether the condition was trained on, in which drugs they contain, and in how their scored populations were selected. Restricting the comparison to the 30 drugs present in both arms reverses the sign, with the trained arm sitting *behind* the held-out one, $-0.0606 [-0.1506, +0.0295]$, $p = 0.1797$, on 79 trained and 750 held-out conditions, and the pooled difference is a composition effect: 83.8% of held-out-combination conditions sit on those shared drugs against 26.3% of trained ones. The claim of a training-exposure or memorisation advantage is therefore withdrawn. None is established, the drug-matched estimate points the other way, and the two arms are scored on populations selected by different rules.

✗ **CORRECTION 1** a training-exposure or memorisation advantage

REGRESSION RECOVERS PROMPT SENSITIVITY. Every condition i contributes three scrambled prompts, one per stratum, each naming a partner drug $p(i, s)$ whose true residual sits at a measured cosine from the target's. Pooling all three prompts of every condition,

$$\begin{aligned} \text{NIR}_{i,s} &= \alpha_i + \beta \cos(r_{p(i,s)}, r_i) + \varepsilon_{i,s} \\ s &\in \{\text{near, orth, opposite}\}, \end{aligned} \tag{4.35}$$

fitted by ordinary least squares with one intercept per condition, so that each condition is compared only with itself and no part of the slope can come from conditions differing from one another. The target-drug prompt is never a point on this line — only the three scrambles are — so nothing in the fit contrasts the model against a chosen baseline, and β is not a gap. It answers a different question: does the score move with how similar the drug the model was *told* is to the drug the condition actually received? And $\beta = 0$ is the null by construction, since a checkpoint that ignores the name emits the same distribution whichever partner is substituted, putting its three points at one height whatever their cosines happen to be. No comparator can be blamed for the answer, because none enters it.

Measured, β is $+0.361 [+0.249, +0.472]$ on train, $+0.321 [+0.207, +0.434]$ on unseen_combo and $+0.324 [+0.146, +0.502]$ on unseen_drug, clustered on the drug, and all three still exclude zero clustered on the cell line ($[+0.240, +0.482]$, $[+0.235, +0.406]$ and $[+0.234, +0.414]$), on 900, 2,685 and 597 points across 300, 895 and 199 conditions. Pooling the conditions instead of giving each its own intercept moves the estimates barely, to $+0.373$, $+0.336$ and $+0.342$, so unlike the arms of § 4.2 and § 4.6 this one does not depend on the choice. The prediction tracks the drug it is named on every split, held-out ones included. One restriction attaches to the last: on unseen_drug the partners named are overwhelmingly drugs the model *was* trained on, so the slope there shows it responding to a named trained drug while scoring a condition whose own drug was withheld, and not that it knows the withheld drug.

A POSITIVE SLOPE BESIDE A CHANCE-LEVEL SCORE IS NOT A CONTRADICTION. The two instruments conflict only if they ask the same question, and they do not. The slope asks whether the prediction moves with the drug it is named. NIR asks whether it lands nearer that condition’s own truth than the truths of the ≈ 40 other drugs in the same cell line. A model passes the first and fails the second by knowing what a drug does on average and not what it does *here*, and the size of that shortfall is itself measured: § 4.8 puts the share of a drug’s residual that transfers across contexts at $\mathcal{T} = 0.553 [0.514, 0.593]$, leaving 44.7% [40.7%, 48.6%] that does not, a remainder mixing cell line, dose, well and estimator noise. The slope reads the transferable part; winning a ranking needs the rest. It is also why the lookup wins in § 4.11, since drug_lookup is that same main effect, estimated straight from data far more precisely than the model learned it. The condition-level picture agrees: on trained conditions the cosine

between prediction and own truth averages $+0.0389$ (sd 0.1119 , median $+0.0220$), spanning $[-0.3046, +0.3976]$ with 33.7% clearing $|0.1|$ — predictions pointing the right way on average, and seldom far enough to win.

ONE ASYMMETRY ATTACHES TO EVERY HELD-OUT NUMBER IN THIS CHAPTER.

The evaluation's truths are not refitted. The generic is reconstructed from the build's own fit inventory and verified against the build's `fit_digest`, and a mismatch aborts the run. The reliability line selects the *training* split, so train is by construction the reliability-surviving population, and the two held-out splits are not filtered at all. In the unseen-combination arm 499 conditions sit below the $+0.1086$ line and 74 have a negative split-half cosine; in the unseen-drug arm, of 199 conditions, 88 sit below the line and 18 are negative. This is deliberate, since filtering the evaluation set on its own outcome is the selection this chapter removed, but it leaves the comparison conservative in the direction that matters. The held-out arms are scored on their full, noisier populations while the trained arm is not.

THE CLAIM. Re-encoding the target made the model respond to the combined drug-and-mechanism prompt. That much is established and survives every stress: on trained conditions it holds clustered on the cell line and on the drug, it holds against chance with no comparator at all, and the comparator-free slope excludes zero on all three splits under both clusterings — $+0.361$ [$+0.249$, $+0.472$], $+0.321$ [$+0.207$, $+0.434$] and $+0.324$ [$+0.146$, $+0.502$], fitted with one intercept per condition so that each condition is compared only with itself. Transfer is a different question, and the two instruments answer it with very different confidence. The scramble intervention answers it barely. On `unseen_combo` the gap is $+0.0332$, excluding zero clustered on cell line and well, $+0.0332$ [$+0.0069$, $+0.0594$], but not clustered on drug and well, $+0.0332$ [-0.0031 , $+0.0695$], and it is the drug axis a drug-specific claim has to survive. The split is weaker than its name — 30 of its 34 drugs and 39 of its 42 cell lines also appear in the scored train split, and 82% of its conditions recombine a scored-train drug with a scored-train cell line — and the effect is a majority tilt rather than a uniform one, positive for 20 of 34 drugs (sign test $p = 0.39$). The model's own NIR covers chance on both held-out splits. Transfer is not established.

The regression answers the same question far more confidently, and the distance between the two verdicts is itself the finding. Every point in it is a scrambled prompt, so it never asks whether the model is right, only whether it tells wrong answers apart — and by that measure it tells them apart cleanly on every split, held-out ones included. Intervening directly on the prompt does not recover an effect of the same weight, perhaps because ranking a condition against its neighbours needs the part of the residual that does not transfer across contexts, which § 4.8 measures at 44.7% [40.7%, 48.6%].

What the slope on the *unseen-drug* split does and does not add has to be stated exactly. At $+0.324$ [$+0.146$, $+0.502$] it shows that the prediction still tracks the drug *named in the prompt* when the condition's own drug was held out. But the drugs named in that regression are the three scrambled partners, which on this split are overwhelmingly training drugs, and the real target-drug prompt contributes no point to it. The slope is evidence that the model responds to named trained drugs in an unseen-drug scoring context, and not evidence that it knows the held-out drugs.

4.11 The baseline ladder and the per-drug lookup

▲ RESID

What does a drug-response prediction have to beat before it is worth having? The instrument is a ladder of predictors scored through one pipeline, the lowest rung of which needs no model at all, only a table of what each drug did elsewhere. That table wins.

Every section so far has asked whether the model uses the drug at all — whether the token is read, whether the prediction moves when it moves, whether any of that survives a holdout. Those are questions

about mechanism, and they set the lowest bar that a model that claims to make predictions based on context must clear. This section changes the question to what the prediction is worth, which is not a question about the model alone but a comparison against what can be had without one. A model can respond to the drug in its prompt and still predict badly, so the pre-registered bar was never “beats chance”. Residual targets are plausibly memorisable, and a predictor that looks up what this drug did elsewhere needs no checkpoint at all.

WHAT THE LADDER IS, AND WHAT KEEPS IT FAIR. Every arm in Table 4.12 predicts the same object — a condition’s drug-specific residual — and is scored through one pipeline, on the same truths, the same encoding and the same comparison set, so that the only thing separating the rungs is what each is allowed to know. Scores are NIR in the ▲ RESID frame, where chance is 0.50; Figure 4.18 draws the same numbers against chance and the reference.

The rungs run from most informed to least. At the top sits the within-well split-half reference, which is not a predictor at all but a precision ceiling: a disjoint half of the same well’s cells substituted for the prediction, scoring 0.854. Below it come two lookups that use no model, no prompt and no checkpoint. `drug_lookup` predicts a condition’s residual as that drug’s mean residual across *all* the other cell lines it was run in. `drug_lookup_1` narrows that to one other cell line, chosen at random: it still averages the drug’s conditions *within* that line, which are usually several doses, but it never averages across lines. The pair exists to separate memorising a drug from being denoised by repetition, since `drug_lookup` is an average over many lines while the reference it is read against is one noisy half of a well against another. `moa_lookup` drops from the drug to its class, predicting from other drugs that share this drug’s mechanism annotation. The model sits

lookup Predicts this drug’s residual as its mean residual in the *other* cell lines.

below these, given a control cell and the drug. Beneath it are three arms that know nothing about the drug at all and fix the floor: `control_copy` returns the untreated cell unchanged, `generic` predicts the average response, which in this frame is the zero vector, and `random` is what its name says.

Three supports are in play, and rows may not be read across them. The reference, the model and the drug-agnostic arms are scored on all 1,394 conditions. A drug lookup is undefined wherever a drug has no *training* condition in a cell line other than the one being scored, which drops the 199 unseen-drug conditions, whose drug was withheld from fine-tuning altogether, together with three whose only training conditions sit in the cell line under test. That leaves the lookups a common support of 1,192, on which the model and the reference score 0.549 and 0.857 rather than the 0.537 and 0.854 they score on the full set. `moa_lookup` is narrower again at 218. Even the reference moves between the first two, which is why a mark in one column of Figure 4.18 may not be set beside a mark in another.

Two disciplines keep the lookups from becoming oracles. They are fitted from training conditions only: fitted from the full cache, a lookup scoring a held-out condition would read residuals the model was never shown, an oracle and not a baseline, so that construction is excluded. And `drug_lookup_1`, having picked its one cell line, averages that line's conditions rather than drawing a single condition from it, because 62% of drugs in the streamed characterisation sample carry a real dose series (Table 3.1, block A) and the cache behaves the same way, so one condition would confound “different cell line” with “different dose”. `moa_lookup` is not an oracle in any split either: its partner pool is restricted to *training* conditions in the same cell line, so a held-out condition never enters the prediction made for it.

DEFINITION Coverage of the reference

A predictor's *coverage* is the fraction of the distance from chance to the within-well split-half precision reference that it closes,

$$(\text{NIR}_{\text{arm}} - 0.50) / (\text{NIR}_{\text{ref}} - 0.50).$$

Coverage above 100% therefore means the arm scores above the reference, a statement about the reference and not a headroom estimate. Reference and coverage are computed on the same support as the arm they describe.

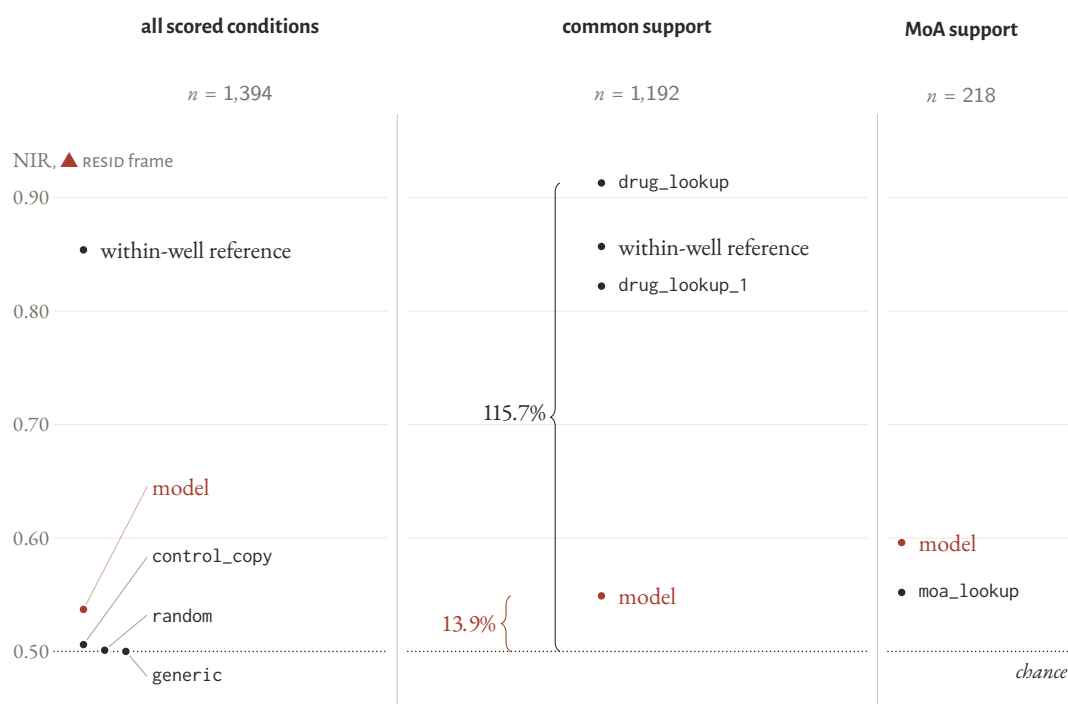


FIGURE 4.18 A predictor that needs no model, no prompt and no checkpoint scores above the reference the model is measured against, and the three columns may not be read across. Marks are NIR in the $\blacktriangle$ RESID frame, where chance is 0.50. Each column is scored on its own support ($n = 1,394, 1,192$ and 218), so a mark in one may not be set beside a mark in another; even the reference differs between the first two. On the common support the model covers 13.9% of the distance from chance to the within-well split-half precision reference, and a training-only per-drug lookup covers 115.7% of it, which is a property of the reference and not headroom (Table 4.12). On the MoA support the model scores 0.596 against moa_lookup's 0.552, a +0.0436 the interval does not separate from zero. No interval is drawn: the source gives them on differences between arms, not on the level of one arm, and those are in Table 4.12. *Reading the labels:* monospace names an arm exactly as Table 4.12 names it, roman names the reference in prose, the accent marks the model, and there is no fourth style. A label a rule would strike or a neighbour would crowd is lifted clear on a hairline; the dot carries the value. Horizontal position inside a column means nothing, and arms whose scores nearly coincide are dodged sideways so each keeps its own dot.

ABOVE CHANCE, AND BELOW THE TABLE. The outcome predicted before the comparison was run was that the model would land above chance and below the lookup, and that is where it lands. On the 1,192 conditions where both are defined it scores 0.549, above the

TABLE 4.12 A training-only per-drug lookup is the bar to beat. Reference, model and drug-agnostic arms use all 1,394 conditions; the drug lookups exist on a common support of 1,192, where model and reference score 0.549 and 0.857. There, model minus drug_lookup is -0.3631 [-0.3982 , -0.3280]; the lookup scores 0.913 pooled, 0.980 on trained conditions and 0.890 on held-out wells, where its same-well advantage is absent. moa_lookup has narrower support ($n = 218$) and is not separated from the model ($+0.0436$ [-0.0480 , $+0.1352$]); its pooled row must not be compared with rows on other supports. The bar is each arm’s distance above chance (0.50) on a common axis to 1.0.

ARM	WHAT IT KNOWS	NIR	ABOVE CHANCE	
within-well split-half reference	disjoint cell halves, one well	0.854		reference precision, not a replicate; 0.857 on the lookups' support
drug_lookup (training-only)	drug’s mean residual, other lines	0.913		drug_lookup the bar to beat; $n = 1,192$ drug_lookup_1 still beats the model; same support
drug_lookup_1	the same, from one other cell line	0.822		moa_lookup narrower support; not separated
moa_lookup ($n = 218$)	drugs sharing this drug’s MoA	0.552		model 0.549 on the lookups' $n = 1,192$ control_copy drug-agnostic
model	control cell + drug	0.537		generic the zero vector random floor
control_copy	nothing (copies the control)	0.506		
generic	nothing (predicts the average)	0.500		
random	nothing	0.501		

Rows on different supports are not comparable. Chance is 0.50 in the ▲ RESID frame under the exchangeability condition audited in § 4.1.

0.50 chance line and well below drug_lookup’s 0.913; the difference is -0.3631 [-0.3982 , -0.3280]. What that licenses was fixed in advance and is narrow: the model recovers something of a drug’s average residual and adds no tailoring to the cell line in front of it, since a table holding that average already beats it. Transfer to held-out combinations is not established (§ 4.10).

THIS IS DR EVAL’S FINDING, IN TRANSCRIPTOMIC SPACE.

The finding travels furthest of anything here, needing no checkpoint, no prompt and no cell-sentence encoding. DrEval reports that deep learning models barely outperform a naive predictor of mean drug and cell-line effects [21], and our drug_lookup is the $\beta(d)$ term of that additive model, measured where the generic(c) term has already been

subtracted. On a 946-gene transcriptomic response, not a scalar potency, that drug main effect reaches 0.913, above the 0.857 reference itself. DrEval’s inherited warning [27], that good pair-holdout performance is reachable by predicting each drug’s training average, is here the hypothesis. It holds.

SEE § 4.8 measures what that constant leaves out

THE MISSING HEAD-TO-HEAD, AND WHY THE BAR SURVIVES

IT. A third objection is that this chapter’s model is the wrong one. A conditional model built for perturbation response [9, 8, 11] was never run on this split, so the head-to-head a reader wants is missing, and that is a real gap. But the ladder is architecture-free, so a competitor can be dropped in and scored, and what it has to clear for a seen drug in a new context is `drug_lookup`, 0.913 pooled, or 0.890 on the held-out split. Not 0.500. For an unseen drug no lookup exists, so that regime needs a different bar and gets one in § 4.12.

The two regimes must not share a number. `drug_lookup` is the bar for a *seen* drug in a new context. No lookup exists for a drug that never appeared in training, so nothing here sets a bar for that regime. Neither may rows on different supports be combined: 1,394, 1,192 and 218 select conditions by different rules, and a distance to the reference is comparable within a support and not between supports.

SCOPE LIMIT carried into § 4.12 and chapter 5

WHY THE LOOKUP BEATING THE REFERENCE IS NOT HEAD-

ROOM. On the common support the lookup scores *above* the reference, `drug_lookup` — reference = +0.056, tempting to read as “the modelling target has vanished”. Three reasons, all favouring the lookup, block that inference. The reference is scored against a half-sample truth and every baseline against the full-sample truth; `drug_lookup` averages ≈ 38 conditions against that single noisy half; and NIR near 0.91 compresses most of the cosine range into a few thousandths. The coverage figures are themselves distances to a split-half reference chapter 3 calls optimistic; that optimism sits in the denominator and if anything makes both coverage figures too small, while the half-sample truth of

the first reason pushes the other way. The net magnitude is unknown and the ordering carries the claim.

AVERAGING IS NOT WHY THE LOOKUP BEATS THE MODEL.

Of those three reasons only the averaging could also deflate the comparison with the model, and `drug_lookup_1` tests it by drawing on one other cell line. It scores 0.822, and paired on the same support the model trails it by $-0.2723 [-0.3074, -0.2372]$, clustered two-way on cell line and well. This drug's mean residual in one other cell line beats a 1B-parameter model handed this condition's own control cells. Neither gap is an average over a mixed population (Figure 4.19): condition by condition, the model scores above `drug_lookup` on 85 of the 1,192 and above `drug_lookup_1` on 214, 7.1% and 18.0%, counts over conditions carrying no interval.

THE BAR IS NOT EQUALLY CLEAN ON BOTH SIDES OF THE

SPLIT. The pooled 0.913 covers two populations. On trained conditions the lookup scores 0.980. On held-out wells, where it cannot borrow the target's own well, it scores 0.890 against the model's 0.533 on the 893 of its 895 held-out-well conditions on which a training-only lookup is defined; the paired gap there is $-0.357 [-0.403, -0.311]$. Nothing in the argument turns on the exact size of that population. The ordering holds on both sides, and the held-out side, where the well advantage is absent, is the side a claim about transfer has to survive.

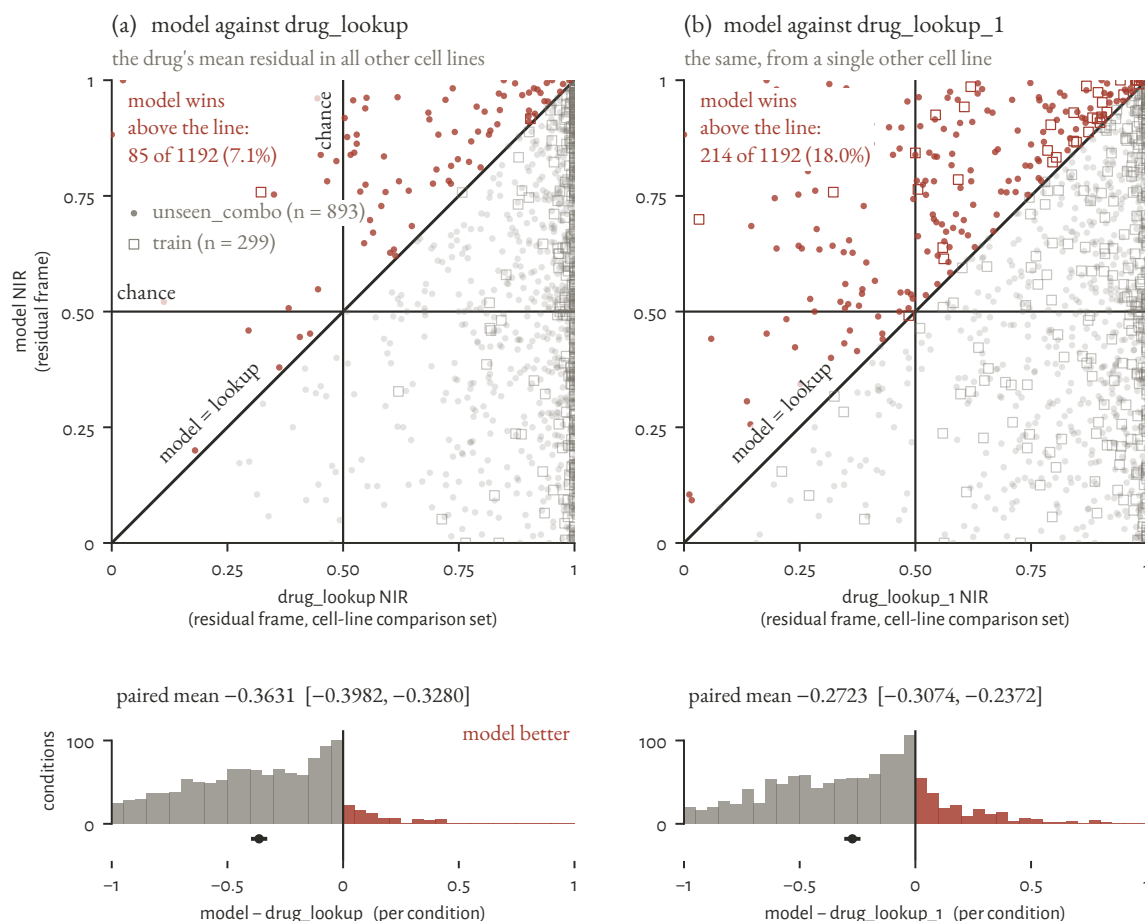


FIGURE 4.19 The headline gap is a tilt of nearly the whole population and not a mean over a mixed one: the model scores above drug_lookup on 7.1% of conditions, and above the single-cell-line lookup on 18.0%. Every mark is one of the 1,192 conditions of the common support—those on which a training-only lookup exists—and both axes are NIR in the $\blacktriangle$ RESID frame, comparison set other drugs in the same cell line. On this support the model scores 0.549 and the within-well split-half precision reference 0.857; the 1,394-condition figures quoted elsewhere (0.537 and 0.854) are a different population and are not combined with these. **(a)** Against drug_lookup, the drug's mean residual in all other cell lines, 0.913 pooled: the model is above the identity line on 85 of 1,192 conditions, 7.1%. **(b)** Against drug_lookup_1, the same quantity from a single other cell line, 0.822 pooled: 214 of 1,192, 18.0%. Marker shape gives the split, 299 train and 893 unseen_combo; no unseen_drug condition appears, because no training lookup exists for a drug held out of training, and that regime is therefore not addressed by this figure at all. The lower panels are the per-condition paired difference on a shared count axis, with the paired mean drawn below the counts as a point and bar, the bar a 95% bootstrap interval clustered two-way on cell line and well: -0.3631 $[-0.3982, -0.3280]$ in (a) and -0.2723 $[-0.3074, -0.2372]$ in (b). The win counts are counts over conditions and carry no interval; ties (23 and 28 conditions) count as not a win. The 0.50 rules are chance on both axes, labelled once in (a).

4.12 Channels for unseen drugs

▲ RESID

The bar just set exists only for a drug that appeared in training. A drug that never did can be reached at all only through some property it shares with drugs that were seen, a protein target, a mechanism, a chemical shape. The instrument predicts each drug from its annotated neighbours in the same cell line, against a null matched on everything but the channel itself. One channel carries a lead, on a very short roster of drugs.

drug_lookup predicts a condition from that same drug's residual measured in other cell lines, so it is undefined the moment the drug is new — which is the case a model would be worth having for in the first place. This section asks what can replace it there: with a drug's own history unavailable, can the drug be reached through the drugs it resembles?

Resemblance has to be defined along something, and that something is called a *channel* here. Tahoe's metadata annotates each drug several ways — the protein it targets, its mechanism class, its chemical structure — and each annotation induces a different set of neighbours. For a channel X , a drug's residual is predicted as the mean residual of the other drugs in the same cell line that share property X with it, the construction of moa_lookup (§ 4.11) generalised from mechanism to any annotated property. A channel along which the response really varies should predict better than one built from arbitrary neighbours; a channel carrying nothing should not.

THE NULL, AND WHAT IT HAS TO HOLD FIXED. That comparison is the entire measurement, so the control decides what it is worth. Each channel arm is paired with a *plate-matched random null*: partners drawn from the same cell line, in the same number, with the same split between the target's own plate and other plates, but chosen at random instead of by the annotation. The matching is not decoration. Drugs

sharing a mechanism or a protein target are co-plated more often than chance, so a null matched on partner count alone would differ from its arm in plate membership as well as in the annotation — two differences where the estimand allows one, and the gap would then be reporting batch structure as channel information. A second, count-matched null is reported alongside it as a sensitivity check, and it is the weaker control precisely because it leaves that second difference standing. Where a plate-matched null cannot be built across enough cell lines the channel is reported UNTESTABLE rather than closed, because not being able to construct a control is not a negative result. Coverage is read before any gap for the same reason: a channel annotated on a handful of drugs was never really tested, and a null on it closes nothing.

TABLE 4.13 Coverage first, then the gap. The drugs behind each channel are stated before any verdict, because a null result on a channel never really tested is not a closure. The plate-matched null is the estimand-correct control; the count-matched column shows sensitivity to plate membership, not a causal share. The gate asks the plate-matched lower bound to clear a pre-set margin of 0.03 clustered on cell line and well. On that axis, and only there, protein target clears it (+0.0745) while mechanism (+0.0279) and chemistry (+0.0035) do not. All three exclude zero, so the two that fail are inconclusive rather than equivalent to their nulls, and neither may be written as closed.

CHANNEL	DRUGS (AN- NOT./SCORED)	<i>n</i>	VS PLATE-MATCHED NULL	(VS COUNT- MATCHED)	VERDICT
protein target	264 / 18	712	+0.1351 [+0.0745, +0.1958]	+0.1498	live, cell-line axis [†]
mechanism (MoA)	180 / 26	921	+0.0805 [+0.0279, +0.1332]	+0.0623	inconclusive
chemical struc- ture	377 / 94	4,131	+0.0261 [+0.0035, +0.0487]	+0.0122	inconclusive

[†]The axis is written into the target verdict because “live” is not usable without it. Clustered on the drug, the target interval falls short of this margin against both estimand-correct nulls and clears it only against the weaker count-matched one.

The counts are annotations in Tahoe’s drug metadata and not testable coverage here; within this cache only 18, 26 and 94 of the 94 scored drugs received a channel arm at all.

WHAT THE THREE CHANNELS RETURN. All three arms exclude zero against their plate-matched nulls on the cell-line axis, so each carries something; what separates them is whether the effect clears the relevance margin of 0.03 set before the gate was run. Clustered on cell

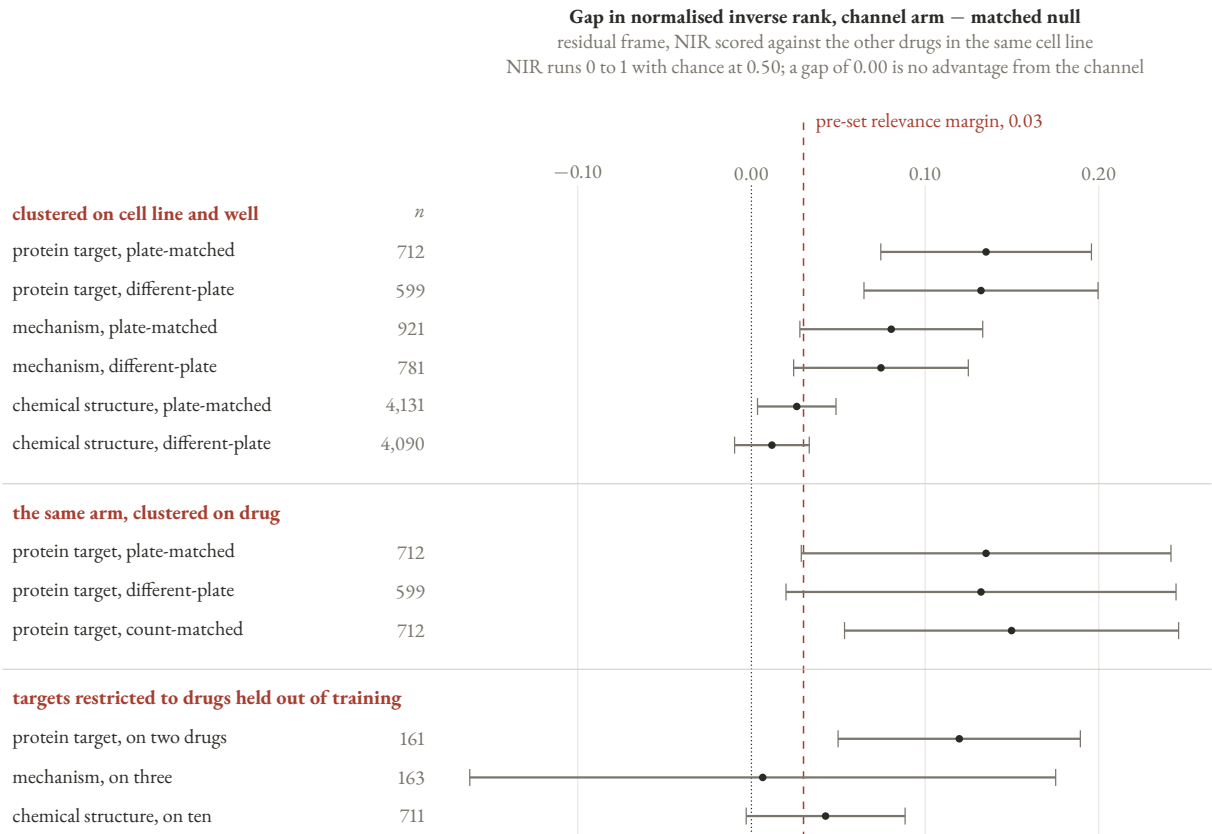


FIGURE 4.20 The gate's verdict is decided by what the interval is conditioned on, not by where the point estimate falls. Every row is residual-frame NIR scored against the other drugs in the same cell line: the arm predicts a drug's residual from its annotated neighbours there, and its matched null draws the same number of partner drugs from that cell line. The dot is the estimate and the bar its 95% cluster-robust interval. Blocks one and three cluster two-way on cell line and well; block two clusters one-way on drug, since wells nest within drugs. The dashed line the pre-set relevance margin of 0.03 that the gate asks the *lower* bound to clear. The count beside each row label is the number of conditions that row scores, not of drugs; the first block's plate-matched counts are the *n* of Table 4.13. The second block re-clusters the same target arm on the drug, the unit a claim about unseen drugs has to generalise across, leaving the point estimates untouched; the lower bound now falls short against both estimand-correct nulls and only the weaker count-matched one clears it. The third restricts the targets to the ten drugs explicitly assigned to unseen_drug, using all eligible conditions of those drugs rather than the model evaluation's quota. Its target, mechanism and chemistry rows cover two, three and ten drugs, respectively. Rows may be read within a block and not across blocks.

line and well, protein target does, at +0.1351 [+0.0745, +0.1958] over 712 predictions. Mechanism reaches +0.0805 [+0.0279, +0.1332] and falls 0.0021 short of the margin; chemical structure reaches +0.0261 [+0.0035, +0.0487] and straddles it. Neither of the two is equivalent to its null, and neither is closed — inconclusive is a different verdict from negative, and the distinction is the reason coverage was printed first.

WHICH UNITS THE INTERVAL IS CLUSTERED ON DECIDES THE VERDICT. The intervals above are two-way cluster-robust on the cell line and the treated well, as elsewhere in this chapter, which is right for a statement about this experiment. But the claim a channel is asked to support — that an *unseen drug* is reachable through its annotation — generalises over drugs, and only 18 annotated drugs carry the target arm. Wells nest inside drugs rather than crossing them, so on the drug axis the estimator clusters one-way on the drug and takes its reference from the drug count (§ 3.7). Read that way the target gap is $+0.1351$ [$+0.0287$, $+0.2416$], on 18 drugs and so $df = 17$, whose lower bound falls 0.0013 short of the margin; against a null drawing every partner from other plates it is $+0.1322$ [$+0.0199$, $+0.2445$]. Both exclude zero and neither clears 0.03. Only the count-matched null clears it on that axis, $+0.1498$ [$+0.0536$, $+0.2460$], and that is the control already set aside as the weaker one. Mechanism, which excludes zero on the cell-line axis, spans it on the drug axis under all three constructions (plate-matched $+0.0805$ [-0.0038 , $+0.1649$], count-matched $+0.0623$ [-0.0152 , $+0.1398$], different-plate $+0.0746$ [-0.0036 , $+0.1528$]; 26 drugs, $df = 25$). The axis is chosen by the claim and not by which interval comes out narrower, so the honest summary is narrower than *live*.

THE POPULATION THE QUESTION IS ACTUALLY ABOUT. The channels are scored on all retained conditions, spanning 94 drugs. The held-out target subset is defined by the 10 drugs assigned to `unseen_drug`, whose conditions never enter training. Absence from the sampled train evaluation tier does not define this subset: four additional drugs absent from that sample have retained training conditions (§ 4.10). Restricting the targets to the 10 assigned held-out drugs puts the question where it belongs, and only one channel

survives it. Clustered on cell line and well, the target gap is +0.1197 [+0.0499, +0.1894]; mechanism falls to +0.0065 [−0.1622, +0.1752] on 163 conditions from three drugs, and chemistry reaches +0.0427 [−0.0030, +0.0885] on 711 conditions from all 10 drugs. Both intervals span zero (Figure 4.20, third block). The target arm survives on 161 conditions from two drugs.

WHAT THOSE EIGHTEEN DRUGS ARE MADE OF. The target arm’s 18 annotated drugs are almonertinib, amsacrine, cepharanthine, clonidine, elagolix, gemfibrozil, goserelin, idarubicin, lumateperone, mitoxantrone, neratinib, norepinephrine, olanzapine, rapamycin, relugolix, temsirolimus, tofacitinib and tofacitinib citrate. Their mean partner count is 1.19: of the arm’s 712 predictions, 579 come from a *single* partner drug and 133 from two, against 2.46 partners for mechanism and 5.00 for chemical similarity. And sometimes that partner is an unusually close relative. Tofacitinib and Tofacitinib (citrate), the free base and the citrate salt of one drug, are scored in the same 20 cell lines and have one partner each; so are rapamycin and its ester prodrug temsirolimus, in 22 lines. Neratinib and almonertinib pair the same way in 20 lines and are chemically distinct, so the pattern is not only duplicate chemistry. What the channel establishes is that a same-target neighbour predicts the drug. That a target *annotation* generalises across a class is the stronger statement, and testing it would mean dropping the nearest analogue from each pool.

THE TWO DRUGS THE HELD-OUT VERDICT RESTS ON. They are **amsacrine** and **relugolix**, and since the duplicate-chemistry pattern above is the obvious objection to them, their pools are worth naming. Amsacrine is annotated to TOP2A and its partners, across its 39 cell lines, are drawn from doxorubicin, epirubicin, idarubicin and mitoxantrone,

never more than two in any one line, because only training conditions retained in that cell line qualify; relugolix is annotated to GNRHR and its partners are elagolix and goserelin. Neither pool contains a salt, prodrug or stereoisomer of the drug it predicts, so this is not the tofacitinib case in another guise — though amsacrine’s pool is three anthracyclines and a close structural analogue of them, which is a drug class and not a duplicate. The support is 161 conditions over 39 cell lines and 5 treated wells, 70 of them amsacrine in two wells and 91 relugolix in three; the five wells are the smaller cluster count and are where this interval’s t_4 reference comes from.

SEE chapter 3: why the smaller cluster count sets the reference distribution

SENSITIVITY TO PLATE MATCHING, MEASURED RATHER THAN ARGUED AWAY. Mechanism partners are co-plated with the target 0.404 of the time against 0.362 for a random draw; elsewhere the diagnostic is negligible, protein target 0.368 against 0.366 and chemical similarity 0.370 against 0.367. Under a third construction drawing every partner from other plates, target and mechanism keep intervals clear of zero and chemistry’s spans it (Table 4.14). Absolute NIR there is 0.638, 0.575 and 0.499 against nulls of 0.505, 0.500 and 0.487, so target and mechanism stay well clear of the 0.50 chance line while chemistry sits just below it, where its plate-matched arm already sat. Removing the shared plate moves target and mechanism by two and seven per cent of their plate-matched gaps, and attenuates chemistry again. That construction also changes support and available partners, so the attenuation does not identify co-plating as its cause.

THE ORDERING IS DESCRIPTIVE, AND ONE OF ITS TERMS IS A FLOOR. Target, then mechanism, then chemistry, under every construction on the full channel supports. On the held-out targets, the point estimates instead order target, chemistry, then mechanism.

On the full supports the target and chemistry intervals do not overlap under any of the three nulls; each adjacent pair's do, and on the drug axis even the two ends overlap. The arms differ in support, drug set, partner count and annotation coverage, so the order does not identify a biological ranking. One correction runs in a known direction: the residual frame subtracts a plate-local generic, and § 4.7 shows that for a co-plated mechanism class part of that class's own shared programme goes with it, so mechanism's middle position is a floor rather than an estimate. Below target the two failures differ in kind. Mechanism's lower bound falls 0.0021 short, a knife-edge that is not a negative result in either direction; chemistry's interval is the *narrowest* of the three, 0.045 wide against 0.121 for target, but straddles the margin.

WHAT THE LEAD IS, AND WHAT BOUNDS IT. Two design documents concluded that unseen-drug generalisation was out of reach, and both inferred it from the chemical-structure channel alone. The same-target result shows that the inference from chemistry to every channel was not licensed: the protein a drug hits is what transcriptional response follows, and structure encodes it only indirectly. Three things bound the lead. Coverage first — the arm reaches 18 of the 94 scored drugs. Then scale: a shared property buys a fraction of what the drug itself buys, the target arm sitting +0.14 above chance where a same-drug lookup for a seen drug sits +0.41 (§ 4.11, on a different condition set, so a comparison of scale and not a paired one). Then overlap: whether target adds anything over mechanism is unsettled here, and settling it would need a leave-one-mechanism-out split, since a held-out drug's mechanism class is generally still represented in training. What this section measures throughout is whether the information a channel carries is present in the atlas at all, not unseen-drug predictive performance, which two drugs cannot establish.

4.13 Drug name versus mechanism field

■ EXPR / ▲ RESID

The previous section asked whether the information an unseen drug would need is present in the atlas at all. The complementary question is whether the model reads what it is already handed, and which part of it, the name or the mechanism line beside it. The instrument swaps one span of the prompt at a time and leaves every other byte identical. It does not settle the question, and what follows is why, and what would settle it.

The chapter has established that the prediction responds to that information as a block; it has not established *which* part of it is read. Saying so precisely matters, because the next step writes itself. Append the protein targets to the prompt, retrain, and see whether unseen-drug performance rises. That arm assumes the bottleneck is information availability. Everything in this chapter points the other way, but the direct evidence is weaker than earlier drafts claimed.

THE INSTRUMENT, AND THE ONE GUARD IT NEEDS. The scramble has until now replaced the drug name and the mechanism together, so a non-null gap establishes sensitivity to the combined prompt without attributing it to either span. A field decomposition swaps exactly one span to the opposite partner, the name with the mechanism kept, the mechanism with the name kept, and both together. One guard is essential. If the swap partner happens to share the original's mechanism, swapping only the mechanism produces a prompt identical to the model's own and the arm scores exactly zero by construction, which read naively is indistinguishable from "the model ignores the mechanism", the very claim under test. Such conditions are dropped, which is why the mechanism arm carries fewer conditions.

WHAT THE ARMS SAID, AND WHY IT DOES NOT STAND. On the earlier target build the decomposed arms suggested the prompt sensitivity is carried almost entirely by the name span. Swapping only the drug name gave Δ_{scr} of +0.0809 [+0.0592, +0.1010] ($n = 570$); swapping only the mechanism gave +0.0091 [−0.0174, +0.0383] ($n = 325$); both together gave +0.1024 [+0.0827, +0.1236]. All were scored against the opposite comparator that § 4.10 shows inflates every gap read against it. They *were* re-run under the corrected evaluation (Table 4.14, Panel B), but the defect survives it, being a property of the swap partner and not of the target build. The claim that the model “reads the name and not the mechanism” is therefore withdrawn pending comparators whose own scores are empirically near chance. What the current run establishes is sensitivity to the combined prompt, with the mechanism field’s contribution unresolved.

CORRECTION the name-over-mechanism reading, withdrawn pending neutral comparators

WHY THE TWO PANELS CANNOT BE ADDED TOGETHER. Panel A says a channel carries information in the atlas; Panel B asks whether the model uses what it is handed. Neither answers the other’s question, and an ordering that reverses when the comparator is removed is not an attribution.

THE PRACTICAL CONSEQUENCE, NARROWER THAN EARLIER DRAFTS DREW. The target-prompt arm is not retired by measurement. It is untested. What argues for building it carefully is the shape of the failure above. The model’s deficit is not confined to missing information about unseen drugs, because on seen drugs it covers 13.9% of the distance from chance to the within-well reference, against a training-only lookup that covers all of it and more (§ 4.11). What argues for building it at all is the one thing a lookup cannot do, handle a drug it has never seen, the regime where § 4.12 finds same-target infor-

TABLE 4.14 Availability does not establish use. Panel A shows target and mechanism retrieval surviving the different-plate null while chemistry does not; only target clears the relevance margin, and only on the cell-line axis, on an arm covering 18 drugs (§ 4.12). Panel A intervals are two-way cluster-robust over cell line and treated well; Panel B intervals are clustered on drug and well (one-way on the drug, since wells nest in drugs, § 3.7). Panel B cannot attribute the combined-prompt effect between name and mechanism, because the two single-span arms’ partners lie on opposite sides of chance (0.434 and 0.520), inflating one gap and suppressing the other. Read without a comparator, as each arm’s own NIR minus chance on its own rows, the ordering reverses (+0.0367 for name, +0.0506 for mechanism), so the split is unresolved pending comparators whose own scores are empirically near chance. Identical mechanism swaps are omitted, which is why that row has fewer conditions.

PANEL A: CHANNEL AVAILABILITY — METADATA-NEIGHBOUR RETRIEVAL

CHANNEL	DRUGS (ANNOT./SCORED)	<i>n</i>	GAP VS PLATE-MATCHED RANDOM NULL
protein target	264 / 18	712	+0.1351 [+0.0745, +0.1958]
mechanism (MoA)	180 / 26	921	+0.0805 [+0.0279, +0.1332]
chemical structure	377 / 94	4131	+0.0261 [+0.0035, +0.0487]

the same three, against the stricter DIFFERENT-PLATE null

protein target	599	+0.1322 [+0.0648, +0.1996]
mechanism (MoA)	781	+0.0746 [+0.0243, +0.1249]
chemical structure	4090	+0.0118 [−0.0097, +0.0333]

PANEL B: FIELD USE — ONE SPAN SWAPPED, EVERY OTHER BYTE IDENTICAL

SPAN SWAPPED	<i>n</i>	Δ_{scr}	COMPARATOR DIRECTION (PARTNER NIR)
drug name, mechanism kept	1394	+0.1026 [+0.0625, +0.1427]	inflated (0.434)
mechanism, drug name kept	738	+0.0303 [−0.0078, +0.0685]	suppressed (0.520)
both spans (the standard arm)	1394	+0.1139 [+0.0781, +0.1496]	inflated (0.423)

The plate-matched rows of Panel A are repeated from Table 4.13 and the different-plate rows from Figure 4.20, so the two panels can be read against each other; both are the same measurement, not a second one.

mation present in the atlas on the cell-line axis, on 18 annotated drugs of which only two are held out of training; a Leave-Drugs-Out result is not established by it.

THE OPPORTUNITY IS NOT THE PART THAT FAILS TO TRANSFER. § 4.8 puts $1 - T$ at 44.7% [40.7%, 48.6%], and it is tempting to call that the model’s headroom. It is not. That figure is a shortfall in cross-context similarity, mixing cell line, dose, well and estimator; it is neither a variance share nor an interaction. With 286 of 287 treatments in exactly one well there is no replicate from which to separate a

context-specific remainder from measurement noise, which is why the coefficient is a disattenuated cross-line cosine and not a fitted variance decomposition. Whether any of that shortfall is learnable structure and not noise is separated nowhere here. The opportunity is the unseen drug, and whether the model can be made to take it is not answered in this chapter.

SEE § 4.8 for the estimator and its five checks

4.14 Coda: a Jacobian-lens view of the drug swap

■ EXPR / ▲ RESID

A coda to the nine beats, not a tenth. The background chapter took the swap from the workspace account and left its central claim alone: that a model reads from a small privileged set of directions, and that content outside it can be richly represented and still inert. The expression-frame results of this chapter have that shape, identity written into the activations and barely consulted. The account's other instrument, the lens that finds such directions, was held back for here. Point it at the drug. Swap the drug in an otherwise identical prompt, take the change the swap makes where the answer begins, and ask how much of that change lies along the directions the lens says the model's last layer reads.

VERDICT In the expression frame, at the deepest layer read, the change a drug swap makes where the answer begins is small and lies along the lens's token directions barely more than along random ones, and less than a larger control edit does; the residual recipe is associated with a drug change that reaches the output, and with more co-location only in the ten-epoch run, where it cannot be separated from the change's size.

The verdict first. In the expression frame the lens does not single the drug out. Where the answer begins, swapping the drug moves the layer-

12 activation by a median 1.7% of its norm, against 5.4% for swapping the control cell’s sentence, and the directions that code the two prompts capture that small change barely better than random directions of the same number: φ_{adj} is $+0.0013$ $[+0.0004, +0.0023]$, a 95% drug-node bootstrap interval, as every interval in this section is unless noted. The drug change lies along the lens’s token directions less than the control change does, though it is also a third of its size, and in this frame the two share too narrow a range of size to say whether the ordering would survive at matched size. It moves the model’s own first-token logits about half as far, 39.5 against 73.4. Nothing in this geometry singles the drug out, which fits what § 4.5 found by intervention on the single-cell arm, where the substitution instrument finds nothing.

The residual checkpoints respond to the drug at the output, and the ten-epoch run’s excess of drug over control response is by far the largest. The geometry separates them from the expression frame in one cell of six, the ten-epoch run at layer 12, and there the difference cannot be told apart from the size of the change it measures. The residual recipe is associated with a larger drug change that reaches the output. That it moves the drug into the directions the lens names is not established.

THE LENS, AND THE DIRECTIONS IT NAMES. Write $h_\ell \in \mathbb{R}^d$ for the residual-stream activation at layer ℓ and one position, with $d = 2,048$. The Jacobian lens of Gurnee et al. [31] summarises how the last layer responds to an earlier one. Writing $dh_{16,t}/dh_{\ell,s}$ for the Jacobian of layer 16’s activation at position t with respect to layer ℓ ’s at position s , every other position held fixed, the lens sums it over the positions a source position can reach and averages over source positions,

$$J_\ell = \mathbb{E}_s \left[\sum_{t \geq s} \frac{dh_{16,t}}{dh_{\ell,s}} \right] \in \mathbb{R}^{d \times d}, \quad (4.36)$$

Jacobian lens The average derivative of the last layer’s activations with respect to an earlier layer’s.

estimated for each checkpoint on 64 windows of 128 tokens cut from 32 calibration prompts on 16 drugs, none of them an evaluation drug, the first 16 positions of each window excluded as sources. Carried back through J_ℓ , the output head's row w_v for token v , scaled by the final LayerNorm gain and centred, becomes a direction at layer ℓ ,

$$d_v = \frac{J_\ell^\top w_v}{\|J_\ell^\top w_v\|}, \quad (4.37)$$

along which a change raises token v 's logit at the output, to first order. Over the 50,252 token ids all four checkpoints share, these are the lens's token directions: 50,252 of them in a space of 2,048 dimensions, of which centring removes exactly one. The same construction with J_ℓ replaced by the identity is the logit lens, kept as a baseline dictionary D_0 with no Jacobian in it. The source layers are 4, 8 and 12. The lens J_ℓ is unrelated to the comparison set J of the reference apparatus.

THE CODE HOLDS LITTLE OF ANY ACTIVATION. Each activation b is read at the last prompt token, the colon of `Response cell:`, where the answer begins, and coded by non-negative orthogonal matching pursuit: at most $k = 16$ directions chosen greedily, their coefficients refitted jointly under a non-negativity constraint after each choice, with $k = 8$ and 32 as sensitivity. These codes stand in for the workspace account's handful of concepts active in the J-space at a position. Sixteen directions reconstruct on average between 2.9% and 5.8% of the activation's squared norm at this position, depending on checkpoint and layer, and sixteen from the logit lens between 1.9% and 3.3%. That is a sparse reconstruction score, $1 - \|b - \hat{b}\|^2 / \|b\|^2$, not variance explained, and it fixes how everything below must be read. Most of every activation lies outside its code, so the numbers are comparisons, a drug edit against a control edit and one checkpoint against another, never a measure of how much of the drug is in the workspace.

THE PAIRS. Twenty-four evaluation drugs, four base prompts each, and every base prompt paired four times: twice with a different drug, once with a different control sentence, and once with itself. In a *drug pair* the drug's name and mechanism line are replaced by another evaluation drug's and nothing else changes; 84 of the 192 drug pairs keep the mechanism text, because the partner shares it. In a *control pair* the control cell's sentence is replaced by another real control sentence from the same cell line, the drug untouched. And each base prompt is paired with itself, run twice. The repeats return identical activations, so their floor is exactly zero, and no drug or control pair falls below the exclusion threshold set from the numerical noise of control pairs. That gives 192 drug pairs and 96 control pairs per checkpoint and layer.

DEFINITION Co-location score φ

For two prompts a and b that differ in one span, let $\delta = h_b - h_a$ at one layer, and let U be the union of the directions chosen to code h_a and h_b . Then

$$\varphi = \frac{\|P_U \delta\|^2}{\|\delta\|^2},$$

the share of the change's squared norm that lies in the span of U , and φ_{adj} is φ minus its mean over 200 random spans of $|U|$ directions from the same dictionary. $\varphi_{\text{adj}} = 0$ means the change lies along the directions that code the two prompts no more than along random ones. It measures geometric co-location with the lens's token directions, not use and not causal involvement.

FOUR CHECKPOINTS, AND WHAT THEY LICENSE. The expression-frame checkpoint is the one-epoch single-cell arm of Table A.3 (42,198 updates). Two residual checkpoints bracket it in training amount on one training file: the canonical residual, repaired checkpoint of Table 4.10, after its one epoch (9,231 updates), and the ten-epoch run of § A.4 (92,310 updates), whose held-out loss is best after its first epoch and worse at every later one. The pretrained base is a reference only. It has no atomic [END_CELL], so it reads the same texts under its own token ids, and no claim rests on it. The checkpoints differ in data, filtering and target format as well as in updates, so a difference between them is associated with the residual training recipe and never attributed to the target encoding. The co-location score is computed the same way in every checkpoint, against no comparison set and no reference, so unlike the two frames' scores it may be compared across them; only the first-token logit change inherits the frames' different targets (Table 4.15, note f).

SEE § A.4 for the ten-epoch run's losses

THE STATISTICS WERE FIXED BEFORE THE ACTIVATIONS WERE CODED. Pair scores are averaged within each base drug, then with equal weight over the 24 drugs. Every drug is both a base and other bases' partner, so intervals come from a drug-node bootstrap that reweights drugs and weights each drug pair by both of its members, over 2,000 resamples. The analysis plan names one primary family, the

six contrasts of each residual checkpoint against the expression-frame checkpoint on drug pairs at each depth, Holm-adjusted at a familywise 0.05. Each is refitted with two half-lenses estimated on disjoint calibration drugs and called inconclusive if the halves disagree in sign or differ by more than the contrast itself. A difference may be called drug-specific only if, once the same contrast on control pairs is subtracted, the difference-in-differences excludes zero and passes the same gate. The plan also fixes its sensitivity analyses in advance: other values of k , the logit-lens dictionary, a support capped at k directions, a second read position at the end of the drug name, and restrictions to pairs whose mechanism text changes and to drugs absent from residual training.

WHERE THE EXPRESSION FRAME PUTS THE DRUG. At layers 4 and 8 the drug edit lies along the lens's directions less than the control edit in all three trained checkpoints (Table 4.15, Panel A); the pretrained reference agrees at layer 4 and is inconclusive at layer 8. Those depths decide nothing, for two reasons. There the expression frame's drug edit is an eighth and a sixth of its control edit in relative size, and the two share no range of size at all, so the ordering cannot be separated from size. And the lens says little there: its linear prediction of how a change reaches the output agrees with the true change at a Spearman correlation of at most 0.073 at those depths, against 0.30 to 0.63 at layer 12.

Layer 12 carries the argument. There the expression-frame checkpoint's drug edit sits below its control edit by -0.0043 $[-0.0061, -0.0026]$, as the pretrained reference's does, by -0.0036 $[-0.0062, -0.0010]$. In pair means the scale is plain: 1.16% of the drug change's squared norm lies in the span of its directions against 1.03% for random spans of the same size, where the control change reaches 1.78% against

TABLE 4.15 The drug edit lies along the lens's token directions less than a control edit, except in the ten-epoch run at layer 12, whose drug change is seventeen times the expression frame's in relative size. Panel A is $\mathcal{W}$, the drug pairs' ϕ_{adj} minus the control pairs', per checkpoint and depth: negative means the drug edit is the less co-located. Panel B is the pre-registered family, each residual checkpoint minus the expression-frame checkpoint on drug pairs, Holm-adjusted over its six members. Panel C gives the size of the two edits at layer 12 and at the output, and the drug-minus-control difference within the range of size the two kinds share. ϕ values are $\times 10^{-3}$; unless noted, intervals are 95% drug-node bootstrap intervals over 24 drugs. The pretrained base is a reference arm, outside the family.

PANEL A: DRUG EDIT MINUS CONTROL EDIT, $\mathcal{W}$ ($\times 10^{-3}$)

CHECKPOINT	LAYER 4	LAYER 8	LAYER 12
expression frame	−2.3 [−3.4, −1.3]	−4.2 [−5.7, −2.5]	−4.3 [−6.1, −2.6]
residual, one epoch	−2.9 [−3.8, −2.0]	−2.8 [−4.1, −1.4]	−1.1 [−2.9, +0.7] ^a
residual, ten epochs	−2.1 [−3.3, −0.9]	−3.6 [−4.6, −2.6]	+6.3 [+3.9, +8.7]
pretrained (reference)	−3.0 [−3.8, −2.1]	−0.5 [−1.5, +0.5] ^a	−3.6 [−6.2, −1.0]

PANEL B: RESIDUAL CHECKPOINT MINUS EXPRESSION FRAME, DRUG PAIRS ($\times 10^{-3}$)

residual, one epoch	−0.1 [−1.4, +1.2] ^a	−0.5 [−1.9, +0.9] ^b	+1.1 [−0.8, +3.3] ^b
residual, ten epochs	−0.4 [−1.9, +1.1] ^a	−0.1 [−1.4, +1.2] ^a	+7.0 [+4.4, +9.5] ^c

PANEL C: SIZE OF THE EDIT, AND THE DIFFERENCE IN THE SHARED SIZE RANGE

CHECKPOINT	RELATIVE CHANGE ^d		DRUG — CONTROL ^e SHARED RANGE ($\times 10^{-3}$)	FIRST-TOKEN LOGIT CHANGE ^f	
	DRUG	CONTROL		DRUG	CONTROL
expression frame	1.7%	5.4%	−2.5 [−8.0, +2.4]	39.5	73.4
residual, one epoch	8.7%	6.9%	−1.8 [−2.8, −0.6]	105.4	81.4
residual, ten epochs	30.1%	6.3%	no shared range	501.5	118.6
pretrained (reference)	5.4%	7.9%	−2.7 [−4.8, −0.6]	98.8	129.6

^aInconclusive: the half-lens gate fails, the two half-lenses disagreeing in sign or differing by more than the estimate itself.

^bHolm-adjusted $p = 1$: no difference from the expression frame is established.

^cRaw $p = 0.001$, Holm-adjusted $p = 0.006$. Drug-specific under the plan's rule: the control-pair contrast is −3.6 [−5.2, −2.1], the difference-in-differences +10.6 [+7.6, +13.6], and both pass the gate. Not established once the support is capped at eight or sixteen directions, a sensitivity analysis the plan fixes in advance (see text).

^dMedian over pairs of $\|\partial\|/\|b_a\|$ at layer 12. Descriptive, outside the analysis plan.

^eLayer 12. Within each base drug, the mean ϕ_{adj} of its drug pairs minus that of its control pairs, among pairs in the range of relative size both kinds occupy (bounds fixed from the full sample), averaged with equal weight over the drugs that have both kinds there: 7 drugs in the expression frame, whose shared range, 3.4% to 4.7%, holds 15 of its 192 drug pairs; all 24 in the one-epoch checkpoint and the pretrained reference. The ten-epoch run's drug and control changes share no range of size. The interval resamples those drugs, 2,000 times, and ignores the dependence through partners. Exploratory, outside the analysis plan.

^fMean L2 norm of the change in the model's own first-token logits over the shared vocabulary, no lens involved. The first response token means different things under the two targets, so read drug against control within a row, not rows against each other across frames.

1.21%. The lens’s own prediction points the same way. It predicts how a drug swap moves the output less well than how a control edit does, by $-0.118 [-0.154, -0.086]$ in Spearman correlation.

Whether that ordering is more than size cannot be settled in this frame. The control edit is three times the drug edit, and the two kinds share only a narrow range of relative size, 3.4% to 4.7% of the norm, which holds 15 of the 192 drug pairs, from 10 drugs. Averaged within drug first, as the plan aggregates, over the 7 drugs with both kinds in that range, the drug edit sits lower by $-0.0025 [-0.0080, +0.0024]$, a difference that is not established. Where the shared range covers every drug, in the one-epoch checkpoint and the pretrained reference, the drug edit sits lower within it, by $-0.0018 [-0.0028, -0.0006]$ and $-0.0027 [-0.0048, -0.0006]$ (Table 4.15, Panel C). In the one-epoch checkpoint the drug changes in that range are slightly the larger, so the size mechanism below would push the other way; in the pretrained reference the control changes are still the larger, so part of its gap could be size. None of these comparisons is in the analysis plan.

THE RESIDUAL CHECKPOINTS RESPOND, AND ONE IS PLACED DIFFERENTLY. Both residual checkpoints move their first-token logits more for a drug swap than for a control edit, the reverse of the expression frame’s $-33.9 [-43.1, -20.0]$: by $+23.9 [+8.9, +39.9]$ after one epoch and by $+382.9 [+363.5, +403.6]$ after ten. That is a drug sensitivity at the first response token, on a panel of which 17 of the 24 drugs are recorded absent from residual training, and it is larger in the ten-epoch run. Whether it is the prompt sensitivity § 4.9 scores, with a different metric and on trained conditions, is not tested here. At layer 12 the lens also predicts the drug swap’s effect on the output better than the control edit’s in both, by $+0.094 [+0.066, +0.120]$ and $+0.100 [+0.074, +0.124]$.

Where the change lies is another matter. For the canonical one-epoch checkpoint a difference from the expression frame is not established at any depth: at layer 12 it is $+0.0011 [-0.0008, +0.0033]$ and at layer 8 $-0.0005 [-0.0019, +0.0009]$, both with Holm-adjusted $p = 1$, and at layer 4 the half-lens gate fails. Its own drug-minus-control W at layer 12 fails the gate too, so for it the comparison in the shared size range, outside the plan and without a stability gate, carries no weight on its own. For the ten-epoch run the gate fails at layers 4 and 8, and at layer 12 the contrast is $+0.0070 [+0.0044, +0.0095]$, raw $p = 0.001$ and Holm-adjusted $p = 0.006$, the only member of the family to survive. Its control-pair companion runs the other way, $-0.0036 [-0.0052, -0.0021]$, so the difference-in-differences is $+0.0106 [+0.0076, +0.0136]$; it passes its gate, and by the plan's rule the difference may be called drug-specific. It holds under most of the variations the plan made. It is $+0.0034 [+0.0016, +0.0049]$ with $k = 8$ and $+0.0146 [+0.0107, +0.0181]$ with $k = 32$; the logit-lens dictionary gives $+0.0102 [+0.0074, +0.0135]$; pairs whose mechanism text changes give $+0.0078 [+0.0043, +0.0108]$, and the 17 evaluation drugs recorded absent from residual training give $+0.0076 [+0.0040, +0.0113]$; read at the end of the drug name, a position the lens was mostly not fitted at, it is $+0.0167 [+0.0133, +0.0197]$. Leaving out any one drug keeps it between 0.0064 and 0.0073, and the two half-lenses give 0.0077 and 0.0070. The exception is the plan's own capped support at eight and sixteen directions, taken up next.

WHY THAT DIFFERENCE CANNOT BE READ AS THE DRUG MOVING INTO THE LENS'S DIRECTIONS. Three observations are consistent with a size effect rather than with the drug being placed differently.

Capping the support shrinks it. Keeping only the sixteen directions of U with the largest summed coefficients, and drawing the null at that size, gives $+0.0003 [-0.0012, +0.0018]$; coding with $k = 8$ and capping at eight gives $+0.0001 [-0.0012, +0.0012]$; with $k = 32$ and a cap at thirty-two a smaller difference survives, $+0.0025 [+0.0003, +0.0046]$, though not with the logit-lens dictionary, $+0.0016 [-0.0009, +0.0040]$. At the primary k a difference on the drug pairs is then not established, though the interval still admits a quarter of the primary estimate. The difference-in-differences stays positive under the cap, $+0.0046 [+0.0024, +0.0068]$, but through the control pairs, which sit lower in the ten-epoch run, $-0.0043 [-0.0058, -0.0028]$. The excess lies in the directions beyond the leading few, which enter U because the two prompts' codes part company: the ten-epoch run's drug pairs have a median $|U|$ of 25 at layer 12, against 18 in the expression frame.

It does not need the Jacobian. The logit-lens dictionary, which never applies J_ℓ , shows it as strongly, so it is not something the lens's transport finds.

It tracks size, and size cannot be matched. The ten-epoch run's drug change is a median 30.1% of the activation's norm, seventeen times the expression frame's 1.7% and more than three times the canonical checkpoint's 8.7% (Table 4.15, Panel C). Within its own drug pairs ϕ_{adj} rises with that relative size at a Spearman correlation of $+0.66$, and with $|U|$ at $+0.64$; in the expression frame the same two correlations are -0.006 and $+0.010$. No range of size is shared. Every one of the ten-epoch run's 192 drug pairs exceeds a tenth of the activation's norm and 182 exceed a fifth, while no expression-frame drug pair reaches a twentieth. Apart from the capped support, none of these comparisons is in the analysis plan.

One mechanism would produce all three. When a change is a large part of the activation, the directions chosen to code h_b are partly chosen to code the change itself, which random directions are not, so ϕ_{adj} rises with the change's relative size whatever the model then does with it. Run the other way, the same mechanism means that the expression frame's W , where the control edit is the larger, cannot separate placement from size either, which is why the comparisons in the shared size range are reported, and why they speak only where that range covers every drug. The plan's rule licenses calling the ten-epoch difference drug-specific in the narrow sense it defines, that a control edit does not show it. It does not license reading it as the drug moved into the set of directions the model reads from.

Three limits bound every number above. The lens is applied where it was not fitted: the last prompt token closes every tail calibration window and is never used as a source, so its directions there are applied outside the positions they were estimated at, though at layer 12 they still predict the true output change at a Spearman correlation between 0.30 and 0.63. The directions are token directions, not genes: only 8 of the 946 panel genes are single whole tokens, so no reading here is about a gene, and a positive coefficient is not up-regulation. And co-location is geometry, not use. No workspace vector is swapped, so the causal claim of the workspace account remains untested, as § 2.7 records.

SCOPE LIMIT takes up the limit carried from § 2.7

THE CLAIM. In the expression frame, at the deepest layer read, the change a drug swap makes where the answer begins is small and lies along the lens's token directions barely more than along random ones, and less than a larger control edit does; the residual recipe is associated with a drug change that reaches the output, and with more co-location only in the ten-epoch run, where it cannot be separated from the change's size. The expression-frame result is negative, and it rests on the pre-registered measurements at layer 12. Where the answer begins, the drug swap's change there is a third of a control edit's in relative size, lies along the lens's token directions barely more than along random ones and less than that control edit does, and moves the output about half as far. That fits the single-cell null of § 4.5; it measures co-location, not use. Whether the drug edit would sit below a control edit of its own size is not settled in this frame, where the two kinds share only a narrow range of size. Layers 4 and 8 decide nothing. The residual result is a statement about the output and a caution about the geometry. The residual training recipe is associated with a drug change that reaches the first response token, larger in the ten-epoch run than after one epoch. Whether the drug is thereby placed in the directions the model reads from is not established for the canonical checkpoint; for the ten-epoch run the one difference the family finds is confounded with a change seventeen times the expression frame's in relative size.

Limitations and future research directions



IN THIS CHAPTER

- 5.1 Populations and selection effects
- 5.2 What the intervals do not cover
- 5.3 The unseen-drug arm
- 5.4 Future work

■ EXPR / ▲ RESID

each section declares the frame in force beside its heading

Every measurement in this thesis was made on one atlas, one check-point and one slice of both. Which of its conclusions are properties of the model, and which are properties of that slice?

A limitation is worth writing down when it changes what a number licenses.

5.1 Populations and selection effects

ONLY REPRODUCIBLE CONDITIONS. A residual that does not reproduce between two halves of the same well is not a target anything could learn, so the residual arm trains on the 50.5% of conditions whose residual does reproduce, 2,523 of 4,999 training conditions kept at $\cos > 0.109$, the 95th percentile of a null pairing half A of one condition with half B of another. The model was taught the empirically reproducible fraction. Split-half retrieval establishes discriminability of treated populations under this sampling design and cell budget; it does not establish predictability from untreated input or learnability.

A reliability filter might be expected to select *toward* context-independent conditions and bias the transfer coefficient upward. It does the opposite. Unfiltered 0.597 [0.563, 0.632], filtered 0.553 [0.514, 0.593]; the filtered value is quoted, because disattenuation is unstable at the very low reliabilities the filter removes.

reliability line The half-split cosine threshold above which a condition's residual counts as reproducible.

That calculation's own filter is the inherited $\cos > 0.2$ line, where the training set uses 0.109, so the two estimates bracket the trained-on population without reproducing it. The generation evaluation mismatches differently again, applying 0.109 to train and leaving the held-out splits unfiltered. That one is fundamental and is taken up below.

THE SPLITS ARE SCORED ON DIFFERENT POPULATIONS. The within-well split-half precision reference differs sharply across the three splits as scored, 0.956 on train, 0.824 on unseen_combo and 0.836 on unseen_drug. The obvious reading is that held-out conditions carry less recoverable signal. That reading is wrong.

It is the selection asymmetry noted above. train is drawn from a pool already filtered at the 0.109 line, and none of its 300 scored conditions falls at or below it, while the held-out splits are unfiltered and still contain conditions whose two halves disagree outright. Impose the training line on all three and the references agree at 0.956, 0.961 and 0.960, a spread of 0.005 where there had been 0.132 (Figure 5.1). Matched on reproducibility, a held-out condition is as recoverable as a trained one.

Those conditions carry less signal is therefore not a reason to discount a held-out result, and this thesis nowhere uses it as one. What survives is narrower and binding: the matched-population rescore covered the precision reference only. The model's own arms were never rescored, so where its held-out scores would sit once the populations agree is not reported here, and the train-against-held-out comparison of § 4.10 inherits the mismatch — which matters most for the score that section reports as covering chance on both held-out splits.

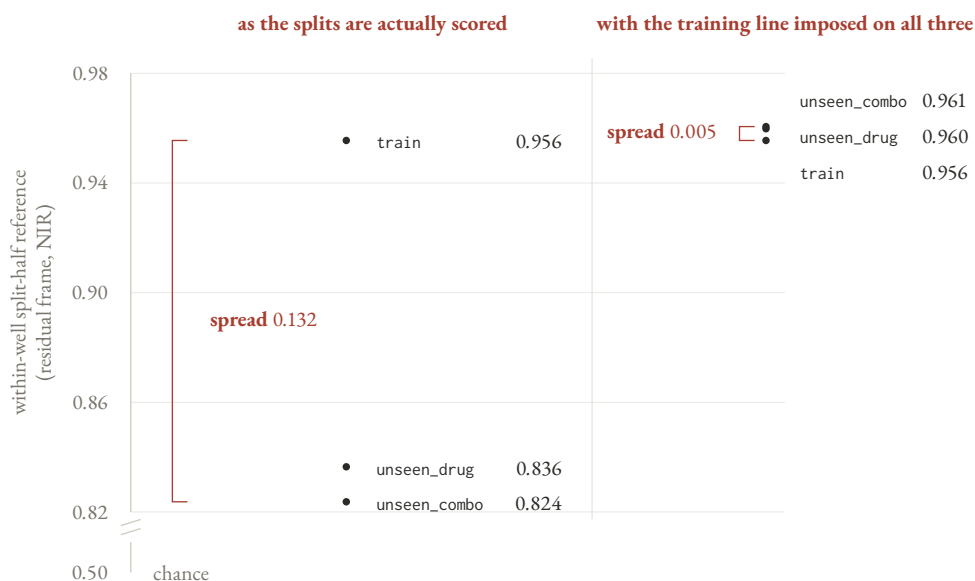


FIGURE 5.1 The three splits do not differ in recoverable signal; they differ in which conditions were allowed into them. The vertical axis is the within-well split-half precision reference — one half of a condition’s cells scored against the other half’s truth, in the residual frame, on the NIR scale — and it is *truncated* to 0.82–0.98, a sixth of the scale; chance is 0.50, marked below the axis break and far under everything drawn. Both panels share that truncated scale, so the two spans may be read against each other. Left, the reference as each split is actually scored, 0.956 on train against 0.824 and 0.836 on the held-out splits, a spread of 0.132. Right, the same reference with the resolved +0.1086 reliability line imposed on all three, 0.956, 0.961 and 0.960, a spread of 0.005; at this scale the three sit inside a bracket two millimetres tall, so they are listed beside the cluster in their own order rather than labelled in place. The left-hand spread is a selection artefact. The markers are bare: no interval is carried for this reference in the source record, and none is drawn rather than invented.

Three selection rules are in force at once, and no result computed under one of them may be set beside a result computed under another. A raw score read across the three splits is not answering the same question on each, because the splits were filtered by different rules. An absolute NIR computed with the comparison set grouped by cell line is not comparable with one computed within plate. And a per-drug lookup is the bar for a drug that was *seen* in training and is being asked about a new context: no lookup exists for a drug that never appeared in training, so that regime keeps its own bar, set in § 4.12.

SCOPE LIMIT discharges the limit carried from § 4.11

5.2 What the intervals do not cover

NO TRAINING RUN WAS REPEATED. Seeds in this work move prompts, partners and resampling. None of them moves the weights. Every arm here was trained once, from one base checkpoint under identical hyperparameters (Table A.3), so each dissociation this thesis rests on — the residual re-encoding against the standard target, and the optimal-transport target against both — compares one training run with one training run. A second run of the same arm under a different

data order would land on a different checkpoint and score differently, and the size of that difference was never measured here. It is therefore not covered by any interval in this document: the intervals quantify sampling over conditions, cell lines and wells, and a between-arm gap of the size reported would survive or fail a training replication that nobody has run. The ten-epoch diagnostic varies the horizon and not the seed, so it bounds undertraining and not run-to-run spread. What this costs is specific rather than general: the *direction* of the re-encoding effect is corroborated by the token-divergence measurement, which needs no model at all, but the *magnitude* of every arm-to-arm difference is an n-of-one comparison.

A SWING THAT LOOKED LIKE GENERATION VARIANCE AND WAS NOT. Unseeded generation is the kind of defect that surfaces as an unstable number, and once it seemed to.

Two superseded evaluations of the held-out-drug control returned values of Δ_{scr} far enough apart to be recorded as evidence of seed instability. Only one of the pair survives in the results archive, so the size of the swing is no longer quotable. The diagnosis was wrong. It named the trigger and missed the cause.

✗ **CORRECTION 2** seed instability in the held-out-drug control

Both of the superseded figures are gaps against the *opposite* swap, which § 4.10 shows is not a null; it carries a term for how far the *partner* drug was pushed the wrong way, which changes with the draw. On the canonical evaluation the same control reads +0.0379 against *opposite* and −0.0315 [−0.0836, +0.0206] against the neutral stratum, clustered two-way on cell line and well.

SEE § 4.10, where a neutral stratum replaces *opposite* as the comparator

Two mechanisms were at work, and both are now addressed. `do_sample` advanced from global torch state, so the two arms of a contrast were drawn from different points of one stream; each call now seeds its own generator from (condition, arm, batch), and the comparator has been corrected. The general caution stands regardless; this swing is no longer evidence for it.

THE INTERVALS OMIT GENERATION VARIANCE. The intervals here are cluster-robust with each condition’s score treated as fixed: two-way over *cell line* and *treated well* on a $\min(G_{\text{line}}, G_{\text{well}}) - 1$ Student t reference for the headline gaps, reported again clustered on the *drug*; multiway dyadic for the transfer coefficient; a one-way clustered bootstrap, unit named, in a few tables.

They capture between-cluster variance and leave out the variance from sampling the model’s predictions. With four sampled generations per condition at temperature 0.8 in the residual-frame arms and eight in the expression-frame ones (chapter 3), that source is not negligible, and no across-seed replication of the generation arms was run, deferred or planned. Every interval quoted from a generation arm covers the sampling of conditions, cell lines and wells, and stops there. That includes the within-condition slope regressions of § 4.2, § 4.6 and § 4.10, whose points are themselves sampled generations, so the comparator-free instrument this thesis leans on hardest is bounded the same way as the rest.

One measurement *was* repeated across seeds, recorded so it is not mistaken for the replication that was not run. The activation probe of § 4.5 returned +0.0197 [+0.0063, +0.0335] on the anchor seed 42 and +0.0118 [+0.0060, +0.0173] and +0.0085 [+0.0029, +0.0143] on seeds 43 and 45, each excluding zero. Its spread does not transfer to NIR and does not establish a scale for generation variance. Those runs share a checkpoint and score teacher-forced log-probabilities, so their seed moves the prompts, the partners and the bootstrap, and leaves both the weights and the sampled prediction untouched. They are scored on 60, 240 and 240 prompts in that order, so the largest of the three effects is also the smallest- n one. Each of those three is a single-test interval that carries no correction for the twenty-six tests

of Figure 4.11, and the first is the draw whose survival that correction leaves in doubt (§ 4.5); three uncorrected seeds excluding zero therefore do not reinstate the effect.

RESOLUTION AND CROSS-PLATE ABSOLUTES. Residuals are condition-level pseudobulks over at least 40 treated cells, so single-cell heterogeneity of the response is not modelled in this frame at all; § 4.2 measures what a single cell can and cannot deliver. Those pseudobulks are then scored against comparison sets grouped by cell line rather than by plate. The *difference* Δ_{scr} is leak-immune, since both arms share the same control cell, but the absolute NIR values are a different matter. The plate-scoped generic removes plate structure in expectation only, and it is estimated from a small plate group, so some may survive in an individual residual. The rescore that would size what survives, regrouping the generation-free arms within plate, needs no GPU time and was not run.

What exists instead is one-sided, and Table 5.1 bounds it from above. In the expression frame with no generic subtracted, holding the plate constant moves a drug-agnostic mean baseline from 0.399 to 0.161 and the precision reference from 0.625 to 0.575: the yardstick falls with the baseline, so the leak is not an offset one side of a comparison could be corrected by. Residual-frame absolutes are read within their own frame and never set beside a within-plate number. The audit's within-plate figures sit slightly off the benchmark figures of § 4.2: control-copy 0.510 against 0.504, mean baseline 0.161 against 0.180, precision reference 0.575 against 0.576.

FRAME ■ EXPR the plate-leak audit

THE SIMILARITY IS A CHOICE TOO. NIR is parameterised by a similarity σ (§ 4.1), and the two frames instantiate it differently: the expression frame scores $\sigma(\hat{y}, y) = -\|\hat{y} - y\|$, the residual frame the

PREDICTOR (DRUG-AGNOSTIC)	CROSS-PLATE	WITHIN PLATE
mean baseline	0.399	0.161
control-copy	0.551	0.510
precision reference (within-well split-half)	0.625	0.575
fraction of drugs at chance	43%	49%

cosine. What that choice carries was measured on the run the blindness verdict is stated on. The same 606 stored predictions and truths, the same 80 comparison groups, tier 2 within plate, were rescored under both similarities using the definition above, half-tie rule included: the aggregate reads 0.4977 under the negated Euclidean distance against 0.5007 under the cosine, both at chance. The Euclidean value reproduces the published figure on that support, which is what validates the rescore.

What does not hold across the substitution is the ordering beneath the aggregate. The two scorings agree per condition at Spearman 0.6217, and 352 of the 606 conditions move by more than 0.10. That neither strengthens nor weakens the aggregate verdict: neither similarity establishes aggregate discrimination on this run. The per-condition changes need not be noise; an aggregate near chance can conceal subgroup gains and failures, and the rescore does not establish which conditions carry either.

Two properties of the published convention belong beside it. A Euclidean distance is dominated by its largest coordinates, so the high-abundance genes carry the comparison. And the vectors compared are reconstructions: the prediction is a cell sentence, decoded to expression through a fitted linear map before any distance is taken, and that map was fitted for absolute fidelity and never validated for discrimination (§ 2.2). The residual frame carries no such measurement, because its

TABLE 5.1 Expression-frame NIR, the leak audit. A drug-agnostic mean baseline more than doubles its score when the plate is free to vary. The precision reference falls once the plate is held constant, because batch structure was inflating it too.

A wider diagnostic run than the $n = 606$ tier-2 benchmark the blindness verdict is stated on, at $n = 1,163$ cross-plate and $n = 998$ within plate. Holding the plate constant costs every condition with no same-plate comparison set, so the within-plate column is a subset of the cross-plate one rather than the same conditions rescored.

predictions and truths were not stored in a form that can be rescored, so nothing is claimed about its sensitivity to σ in either direction.

An at-chance aggregate NIR in the expression frame holds under either similarity, negated Euclidean or cosine, on the one run rescored here, tier 2 within plate. Expression-frame aggregates that carry signal were not rescored, so their standing under σ is untested. No per-condition or per-stratum reading has that standing either: on the run that carries the verdict the two orderings agree only at Spearman 0.6217, so no figure below the aggregate may be presented as independent of σ . The residual frame carries no such measurement, so its scores are covered neither way.

SCOPE LIMIT carried into § 4.2 and every per-condition reading of an expression-frame score

5.3 The unseen-drug arm

Two limits on the unseen-drug arm must be stated: what the prompt reveals and what the intervals resolve. A separate defect, the comparator, was found and corrected, and § 4.10 reports the arm against the neutral stratum throughout.

THE DESIGN LEAKS THE DRUG’S CLASS. Holding out a drug does not remove everything the model knows about it. The prompt contains Mechanism: {moa}, so a held-out drug still arrives with its class label. The correct design is leave-one-*mechanism*-out.

The leak is measured, and the measurement is not clean. Scrambling the Mechanism field alone moves the model’s NIR by +0.0303 [−0.0078, +0.0685] ($n = 738$, clustered on drug and well), an interval containing zero. That understates the channel; it does not bound it, because this row’s swap partner scores 0.520, above chance, and a comparator above chance suppresses the gap (Table 4.14, Panel B). A mechanism-only lookup is the same measurement from the other side and equally inconclusive. Pooled over the $n = 218$ conditions on which both are scored, model − moa_lookup = +0.044 [−0.048, +0.135], clustered two-way on cell line and well, spans zero, so the comparison settles nothing in either direction.

On `unseen_drug`, the split where it would matter, the mechanism lookup is ahead of the model by 0.099 on 44 conditions; in the contrast's reported direction, model minus lookup is $-0.099 [-0.222, +0.008]$, clustered on cell line. The interval spans zero and does not establish which predictor is better on that support. It also does not establish how much the mechanism field contributes to either score. The arm tests held-out drug identities with mechanism information still supplied, so it cannot establish transfer to a new mechanism family.

THE GAP DOES NOT ESTABLISH EQUIVALENCE TO ZERO.

On the $n = 199$ `unseen_drug` conditions, the model-minus-neutral-scramble gap is $-0.0315 [-0.0836, +0.0206]$ clustered two-way on cell line and well, and $-0.0315 [-0.1049, +0.0419]$ clustered on drug (Figure 5.2). Both intervals include zero and a range of small positive and negative effects. Neither establishes a positive unseen-drug gap, but neither is an equivalence test or evidence that transfer is absent.

The pooled mechanism-only lookup scores 0.552, or 0.052 above chance. That is a baseline result on different support, not an upper bound on the effect a mechanism-conditioned model could achieve. It is also a different contrast from the model-minus-scramble gap; it cannot supply the gap's maximum plausible effect. Likewise, an approximate power calculation based on the model's absolute NIR interval does not determine which effects lie inside the observed gap intervals.

As an *instrument control*, this result is compatible with the neutral comparison in § 4.10; compatibility with zero alone does not validate a comparator or confirm a null effect. “Unseen” is narrow in a second way too. Pythia-1b's pretraining is natural-language text in which drug names occur, and the prompt still supplies Mechanism.

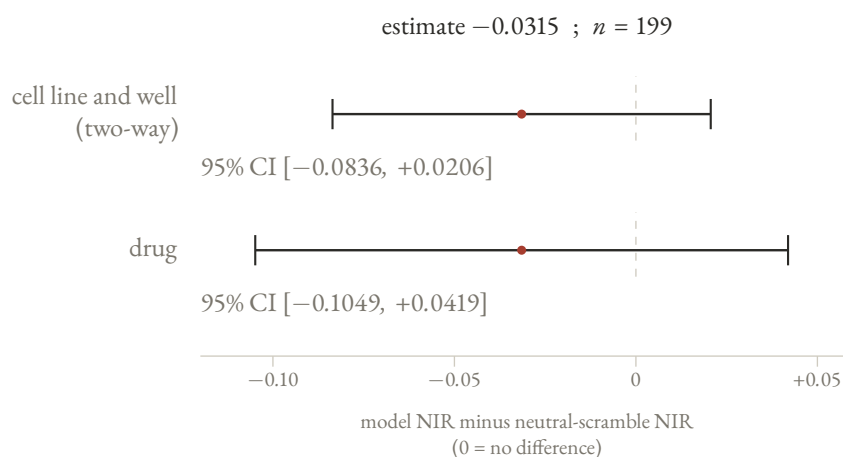


FIGURE 5.2 The unseen-drug gap includes zero under both clusterings. The points show the same model-minus-neutral-scramble estimate on $n = 199$ conditions in the unseen_drug split, -0.0315 . The comparator is the geometry-defined orth scramble on those conditions. The upper row shows the 95% interval clustered two-way on cell line and treated well, $[-0.0836, +0.0206]$; the lower row clusters on drug, $[-0.1049, +0.0419]$. Both intervals include zero and a range of small effects. Neither establishes a positive gap or equivalence to zero. They quantify the reported cluster uncertainty, conditional on the trained checkpoint and sampled generations (§ 5.2).

SCOPE LIMIT carried into chapter 6

The unseen_drug arm is read as an instrument control. Its estimated gap is compatible with zero and a range of small effects under both reported clusterings. This does not establish equivalence to zero or the absence of unseen-drug transfer. “Unseen” means held out from Tahoe fine-tuning; it does not mean a new mechanism family or absence from natural-language pretraining.

5.4 Future work

A limitation earns its place when something could be run to remove it.

THE ARM TO BUILD CAREFULLY. The obvious continuation of § 4.12 is a channel-conditioned prompt that appends protein targets to the instruction, retrains, and evaluates on a leave-one-mechanism-out split. We do not recommend building it eagerly. § 4.13 leaves unresolved which span of the prompt is even read, so a new field would be added to an instruction whose existing fields are of unmeasured value; and the case against it does not rest on that anyway, but on the size of the readout deficit. On the common support the model covers 13.9% of the distance from chance to the within-well bar that a context-blind

lookup covers entirely, and adding a field to a prompt does not touch that deficit.

Worth building instead is an intervention that makes the drug *causal* rather than merely present: a learned per-drug modulation of the transformation from control to treated, so the drug parameterises the computation instead of sitting in the residual stream as an input the readout can round away. That is a research architecture, outside the scope of what remained here.

TWO CHANNEL QUESTIONS THAT REMAIN OPEN. Target and mechanism overlap, so whether target adds anything *over* mechanism is unresolved, and it matters: the prompt already exposes mechanism, and whether the model reads that field is itself unsettled. Second, the gate of § 4.12 measures *similarity transfer*, an unseen drug resembling a seen drug with the same target. It does not test a *compositional rule*, that a named target's own pathway moves, which would apply to any drug whose target is named, including one with no similar drug in training. That claim is the stronger one, it does not reduce to a lookup, and it is untested.

Protein target clears the relevance margin on the cell-line axis, misses it on the drug axis against both estimand-correct nulls, clearing it there only against the weaker count-matched one, and rests on 18 annotated drugs of which two are held out; mechanism misses its lower bound by 0.0021 on the cell-line axis and spans zero on the drug axis; chemistry's interval straddles the margin on both. Chemistry attenuates under the different-plate construction, but that construction also changes support and partner availability, so the attenuation cannot be assigned causally to co-plating. On its restricted support, same-target retrieval is the strongest observed lead, and stands only as a direction worth testing. The binding limitation is coverage: 18 drugs cannot settle it, and the

experiment that would be a target-annotated panel broad enough to cluster on.

UNSEEN CELL LINES. A cell line never seen with any drug is a different generalisation axis from the held-out wells tested here, and one the plating design does not compromise in the same way. It asks whether the control cell's state is used *compositionally* with the drug.

The transfer coefficient scopes it only loosely. 44.7% [40.7%, 48.6%] of the similarity scale $\mathcal{T}$ is measured on is not shared across the compared contexts, and no structure was recovered in that unshared part by tests that do not settle it. That figure is not a drug $\times$ cell-line interaction share, since it mixes cell line, dose, well and estimator (§ 4.8), so it bounds what a per-drug constant leaves on the table without establishing that a new cell line is the axis the remainder lies along.

No consistent per-line direction was found, so the achievable target on a new cell line plausibly reduces to recalling the drug's per-drug constant, which a training-only lookup already does at 0.890 on held-out wells, the nearest measured analogue of a new context (0.913 pooled, inflated on trained conditions by the well advantage). The axis is less promising than it appeared, though the argument is weaker than a coefficient-based one.

OBJECTIVES THAT WERE CONSIDERED AND REJECTED. Three training objectives were designed and then set aside on their own mechanics, recorded so they are not re-litigated.

- **Reinforcement learning with a contrastive reward.** Its per-sequence reward keeps the token-dilution ratio out of the gradient, attacking a bottleneck the residual encoding has already lifted (34.6 $\rightarrow$ 118.2 differing tokens at cell-line scope, 120.3 at the plate scope the trained build uses). Its reward optimises similarity to the

SEE § 4.8: $\mathcal{T}$, and what its complement does not measure

true residual, which drug_lookup already achieves at 0.913 pooled, 0.890 on held-out wells. Its best case is convergence to the lookup.

- **A differentiable profile-level contrastive loss.** The natural construction, $\text{score}(g) = \sum_i P(\text{token}_i = g) w(i)$, presumes $\text{gene} \approx \text{token}$, but 86.9% of the 946 panel genes share a first sub-token under this tokeniser (mean 3.254 BPE tokens per gene, only 8 single-token). Under teacher forcing the target *is* the drug-specific residual, so the true prefix identifies the drug within a few symbols and the gradient vanishes at the readout solution the loss exists to improve on.
- **Loss re-weighting toward differentially expressed genes.** A re-weighted cross-entropy against a single target contains no negatives, so nothing forces one drug’s prediction to differ from another’s. It leaves the discrimination failure where it was, and that failure is what the measurements establish, whether or not the readout is also blind to direction specifically, which § 4.5 declines to establish.

ONE MODEL SIZE, AND THE SMALLEST ONE. Every result here comes from a 1B-parameter backbone. C2S-Scale releases checkpoints up to Gemma-2 at 27B, but these larger models were not evaluated. This thesis therefore does not establish whether increasing model size improves counterfactual fidelity.

The constraint this work points to is not capacity. The target tokens are overwhelmingly shared across drugs — only 17% of them differ between two drugs in the same cell line under the standard target, against 59% under the residual encoding at cell-line scope and 60% at the plate scope the trained build uses (Table 4.6) — so very little of the training signal can carry drug identity. That step is a hypothesis about the optimisation; no token-level loss or gradient attribution was computed,

so no fraction of the gradient is claimed. A larger model receives the same diluted signal from the same objective, and re-encoding lifted the effect while leaving capacity untouched, which points away from capacity without establishing what it points towards. But the re-encoding changed five things at once, aggregation, control referencing, sign coding and length among them, and a budget-matched optimal-transport target lifts the effect too, so § 4.9 declines to identify the tokenisation as the constraint that binds.

The genuinely different argument for scale is *prior knowledge*. A 27B model plausibly knows, as a fact about the world, that a named drug inhibits a named kinase, which a 1B model may lack. § 4.13 leaves unresolved whether the model uses the *mechanism* field it already has, at $+0.0303 [-0.0078, +0.0685]$ ($n = 738$) with a comparator above chance suppressing the gap, and real pharmacological knowledge is exactly the case where that might change. Testing it needs only a low-rank adaptation of a mid-sized checkpoint. We did not run it, and it is the first thing we would run with more time.

ONE EPOCH, AND A THIN SLICE OF THE ATLAS. Data scale is the other half of that question and it is not the same one. Fine-tuning saw 620,564 cells, 600,000 treated and 20,564 control, drawn from the roughly one hundred million Tahoe-100M holds, covering 106 of its 379 drugs, for one epoch at one learning rate (Table A.3). Under a hundredth of the atlas was used, and no arm saw any cell twice. The diluted-signal argument above survives that: a model trained on ten times the cells receives the same target strings, so more data of the same kind does not make the drug easier to see in the loss. The ten-epoch diagnostic separately excludes undertraining as a trivial explanation — validation loss is best at the first epoch and worse at every epoch after it — though it tests one schedule at one learning rate, and no long-run

checkpoint was scored with NIR (§ A.4). Neither argument rules out the possibility that a drug must be seen in more cellular contexts than this sample provides before its response becomes learnable, and that possibility is not established either way. It is a cheaper experiment than a larger backbone, and it is not one we ran.

THE STARTING CHECKPOINT WAS NEVER PREPARED FOR THIS TASK. We fine-tune from C2S-Scale-Pythia-1b-pt, whose cell-sentence pretraining corpus is atlas data alone, CELLxGENE and the Human Cell Atlas, with no perturbation corpus of any kind, and whose released capabilities do not include perturbation prediction.

That is why the held-out splits are as clean as they are. A drug withheld here entered neither the fine-tuning data nor the generic that defines its target, the split being assigned from metadata before any expression was aggregated, so no Tahoe perturbation response for it can have been memorised. But the failure we report is then a failure of *this* starting point, and the fair follow-up is whether a checkpoint already tuned for perturbation prediction would behave differently.

Within the C2S-Scale line that question cannot be answered from what is public. The perturbation-specialised model there is a Gemma-3 4B checkpoint that was not released. Outside that line it is answerable, and we did not answer it. Tahoe-x1 [39] is a perturbation-trained foundation model on this same atlas, public before this thesis was submitted. It is not a cell-sentence model, so it does not isolate the representation variable, but it is the obvious external check on whether a readout failure of this kind survives perturbation-specific pretraining, and it was not run here. That is a gap in this work, not a gap in the public record.

Every arm starts from the same base and differs only in the variable under test, with one material exception, the consensus arm, stopped at

roughly 60% of an epoch with its tier-1 NIR unscorable (Table A.3). The *comparisons* between arms are therefore unaffected even where the absolute level is not, and that arm’s null is quoted with the qualification wherever it is used.

GENERALITY. With one backbone, atlas and representation, this study leaves open whether readout-specificity failure is specific to cell-sentence models or general to autoregressive conditional transcriptome generation. The protocol in §§ 4.4 and 4.5 (purified subspace, removal gate, positive control, matched-norm swap) transfers unchanged to models with an accessible residual stream.

Conclusions



IN THIS CHAPTER

Does a cell-sentence language model fine-tuned on this atlas learn what a particular drug does to a particular cell? The instrument was a chain built in order: a metric calibrated before it was trusted, controls a drug-blind predictor cannot pass, and a bar that needs no model at all. The bar was not cleared.

- 6.1 Summary of the argument
- 6.2 Prompt sensitivity on trained drugs
- 6.3 Transfer
- 6.4 The lookup baseline
- 6.5 Unseen drugs
- 6.6 The role of the controls

■ EXPR / ▲ RESID
each section declares the frame in force
beside its heading

Prompt sensitivity to trained drug names is established; biological transfer is not established; and nothing built here beats a training-only per-drug lookup on common support. The win condition was fixed in advance (chapter 1): a conditional model must beat that lookup, because only then has its access to the untreated cell supplied something a dictionary of per-drug averages lacks. A split-half reference cannot stand in for it, since it cannot separate a drug's characteristic response, which transfers, from this context's departure from it. That verdict ends a chain, and no isolated point estimate may leapfrog it.

6.1 Summary of the argument

THE METRIC HAD TO BE SETTLED FIRST. Four of five candidate metrics failed the calibration audit, and the field's standard one is saturated outright by a predictor carrying no drug information: on DE- Δr such a predictor scores 0.9999. Only NIR registered known differences in quality, at a dynamic range fraction of +0.635 [+0.604, +0.663], so it is the only metric this thesis grades anything on. That is a result

before it is an apparatus choice, and it decides what every later number is allowed to mean.

ON THAT METRIC THE EXPRESSION-FRAME CHECKPOINT IS DRUG-BLIND. It scores 0.498 against a within-well split-half reference of 0.576. On the stratum where the ranking is provably winnable, a control copying the untreated cell matches it outright, 0.768 against 0.766, a paired contrast of +0.002 [−0.026, +0.028]. Three instruments that fail in different ways agree: the scramble, the control copy, and a within-condition regression of score on how unlike the substituted drug is, which returns +0.0055 [−0.0472, +0.0582] once each condition is compared only with itself. That last one matters because its pooled version returns +0.0986 and looks like an effect; the difference between the two is condition-level structure, not the drug.

THE DRUG IS PRESENT, AND BARELY CONSULTED. The null is not a failure to measure. The drug label is decodable from the residual stream at 82% against an 8% chance rate, so it survives into the computation. Ablating the purified drug subspace costs the output 3.6 to 18.6 times a matched-dimension random null, and 9 to 37% of what ablating cell line costs at six of seven depths (98% at layer 12, where both costs collapse). The label is read, and read weakly, which is a different failure from absence and needs a different repair.

6.2 Prompt sensitivity on trained drugs

If the drug is in the data and in the activations, the next suspect is the training target. Two drugs in one context share 83% of their standard target's top-200 gene positions; the residual encoding drops that to 40%. Aggregating the target without re-referencing it does not work: of three constructions tried, none separates from a random-partner scramble, at +0.014, -0.001 and -0.010 , and the optimal-transport arm's within-condition slope is -0.0008 [-0.0305 , $+0.0290$]. No positive slope is established at this precision; the interval does not establish equivalence to zero.

Re-encoding does. On conditions the checkpoint trained on, the residual arm separates from its own scramble against the geometry-defined orth partner by +0.0898, excluding zero clustered on cell line and well, +0.0898 [$+0.0382$, $+0.1415$], and on drug and well, +0.0898 [$+0.0291$, $+0.1506$].

THE COMPARATOR IS PART OF THE MEASUREMENT. A prompt replacement is not automatically a null. Pooled over all 1,394 scored conditions, the scrambled arm itself scores 0.550, 0.501 and 0.423 as the substituted drug moves from aligned through orthogonal to anti-aligned, so a dissimilar lie inflates the apparent benefit of the correct name. Orth is the one comparator whose own score sits at chance, and it is the only one whose geometry holds still across the splits the next section compares.

THE GAIN IDENTIFIES A PACKAGE, NOT A MECHANISM. Residual encoding changes aggregation, control referencing, generic subtraction, sign coding and sequence length at once, and the factorial comparison that would separate them was not run. Nor is it uniquely re-

sponsible: at a matched 1400-token budget the optimal-transport arm also clears zero against its orthogonal comparator, $+0.0292$ [$+0.0144$, $+0.0441$], where its 600-token estimate spanned zero. A generation cap can manufacture a null, so any claim about the residual advantage carries its budget with it.

6.3 Transfer

One question is whether the prediction moves when the drug named in the prompt changes. The other is whether it moves *toward* that drug's response. The trained-condition effect settles the first for drugs represented in fine-tuning and leaves the second open.

Conditions in held-out treated wells score $+0.0332$, excluding zero clustered on cell line and well, $+0.0332$ [$+0.0069$, $+0.0594$], but not when clustered on drug, $+0.0332$ [-0.0031 , $+0.0695$] — and drug is the axis a transferred residual has to generalise over. That split is largely recombination in any case: 30 of its 34 drugs and 39 of its 42 cell lines also appear in the sampled train evaluation, and 82% of its conditions pair a drug and cell line present in that sample. Absence from the sample is not absence from fine-tuning: all four other drugs have training-only lookup support. The separate `unseen_drug` split contains the 10 drugs withheld from fine-tuning (§ 4.10). The model's own score covers chance on both held-out splits.

WHAT SURVIVES THE HOLDOUT IS SENSITIVITY, NOT ACCURACY. The within-condition slope is positive on all three splits and under both clusterings, $+0.361$ [$+0.249$, $+0.472$], $+0.321$ [$+0.207$, $+0.434$] and $+0.324$ [$+0.146$, $+0.502$], on an instrument that needs no comparator and whose null is fixed by construction. On the same instrument the single-cell and optimal-transport arms return

nothing. But every point on that line is a *wrong* prompt, so it shows the model tells wrong answers apart and never that it produces a right one; and on the unseen-drug split the drugs it names are overwhelmingly training drugs. The map is coarse and trained-drug-to-residual.

The unseen_drug arm is read here as an instrument control. Its estimated model-minus-neutral-scramble gap is compatible with zero and a range of small effects under both reported clusterings. This does not establish equivalence to zero or the absence of unseen-drug transfer, and the prompt's mechanism field limits the claim to held-out drug identities.

SCOPE LIMIT discharges the limit carried from chapter 5

6.4 The lookup baseline

A per-drug lookup needs no checkpoint, no prompt and no cell sentence. It predicts a drug's residual as its mean residual in other cell lines, fitted from training conditions only. On the 1,192 conditions where both are defined it scores 0.913 against the model's 0.549, with the within-well reference at 0.857. Restricted to a single other cell line it still scores 0.822, so the win is not an artefact of averaging away noise.

SEE § 4.11 scores the full ladder through one pipeline

The reading was fixed before the comparison ran: above chance and below the lookup means the model recovers something of a drug's average residual and adds no tailoring to the cell line in front of it. That is where it landed. This is DrEval's finding with a transcriptional vector in place of a scalar potency, and it sets the bar a conditional architecture has to clear to earn its complexity.

6.5 Unseen drugs

No lookup exists for a drug absent from training, so the last gate asks which shared property could stand in for one. Three channels were tested against a plate-matched random null, with partners restricted to training conditions.

Protein target clears the pre-set relevance margin clustered on cell line and well, +0.1351 [+0.0745, +0.1958]. Clustered on the drug — the axis an unseen-drug claim rides on — it clears the margin only against the weaker count-matched null and falls short against both estimand-correct ones. Mechanism excludes zero on the cell-line axis and misses the margin by 0.0021; chemistry straddles it. All three exclude zero, so the two that fail are inconclusive rather than equivalent to their nulls, and none of the three may be written as closed.

The lead rests on 18 target-annotated drugs, two of them held out. Protein target carries transcriptional signal on that support, and a Leave-Drugs-Out result is not established by it. Which span of the prompt the model actually reads is unresolved (§ 4.13): the two spans were swapped separately, but the results cannot be told apart, because each was scored against a comparator on a different side of chance (partner NIR 0.434 and 0.520; Table 4.14, Panel B), so a channel-conditioned prompt would add a field to an instruction whose existing fields are of unmeasured value. The next experiment needs a target-annotated panel broad enough to cluster on, preferably with held-out mechanism families, and a prompt whose spans have been told apart.

6.6 The role of the controls

In a literature where uninformative baselines score well, a negative result is worth what its controls are worth. Each surviving claim here rests on a control added after a specific failure: plate-matched comparison sets after a zero-information baseline improved when plate was free to vary; a swap-distance sweep after a similar-drug scramble proved weak; a removal gate before any ablation null; matched-norm injection before any swap was interpreted; condition intercepts after a pooled slope returned an effect that was not there. Intervals were clustered on the units that repeat — cell line, drug and treated well — and never on cells.

THE CLAIM. On the 1,192 conditions where both are defined, a training-only per-drug lookup scores 0.913 against the model's 0.549, with the within-well reference at 0.857; from a single other cell line it still scores 0.822. Residual re-encoding produces a real trained-condition effect against the orth partner, excluding zero on both clustering axes, and establishes no transfer: held-out treated wells give an interval that excludes zero on the cell-line axis and spans zero on the drug axis, and the model's own score covers chance on both held-out splits. What survives the holdout is the comparator-free slope, which is sensitivity to the drug named and not accuracy about the drug given. No drug effect is shown to be zero by any instrument here. Of the three channels open to a drug with no lookup, protein target is live on the cell-line axis and inconclusive on the drug axis, mechanism and chemistry are inconclusive, and none is closed.

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Appendix



IN THIS CHAPTER

- A.1 Planted-truth checks of the estimators
- A.2 The absent-gene convention
- A.3 The swap-distance sweep
- A.4 Training adequacy
- A.5 Three failed identity-test designs
- A.6 Metric self-test

This appendix holds the apparatus the body leans on but does not stop to justify. Each section states its verdict in its heading. None is a new result; several are defects that changed one.

A.1 Planted-truth checks of the estimators

Three measurements rest on estimators built for the purpose, and a reader has no prior reason to trust them. The instrument is a world with a known answer planted in it, run twice, with the effect and without. All three recover what was planted, and nothing when nothing was.

The weight falls on the null direction.

WHAT THE TRANSFER-COEFFICIENT ROWS DO NOT SHOW.

They are a data-free unit test whose planted world has cell line as its only unshared axis, so recovery shows the estimator inverting its own generative model and no more. The canonical run, computed with `same_dose_only=false`, `loo_generic=false` and `repro_thr=0.2`, lets cell line, dose, well and estimator noise all enter one number. On synthetic data calibrated to its gene dimension, cell counts and empirical noise, the estimator returns 1.001, 0.747 and 0.494 for planted $\mathcal{T}$ of 1.00, 0.75 and 0.50.

■ EXPR / ▲ RESID

each section declares the frame in force beside its heading

planted world A synthetic run with the answer built in, no data, no network, one fixed seed.

TABLE A.1 All three estimators recover planted answers in both directions. Every row is a synthetic planted-world check, and none of these numbers comes from the evaluation run or is stored in a result artifact. The $\mathcal{T}$ and κ -structure rows reproduce from `variance_decomposition.py --self test`, the channel-gate rows from `channel_gate.py --self test`. The κ -structure check must separate two worlds, within-line consistency exceeding across-line where every cell line bends what it is given (+1.17), the excess vanishing where κ is specific to each drug-and-line pairing with no line-level direction (−0.00). The lower row of each pair is the null world, the one that rules out an instrument reporting structure wherever it looks.

PLANTED TRUTH	RECOVERED
TRANSFER COEFFICIENT $\mathcal{T}$	
$\mathcal{T} = 1.00$ (fully shared)	0.999
$\mathcal{T} = 0.70$	0.703
$\mathcal{T} = 0.40$	0.404
CHANNEL GATE, OVER 24 WORLDS	
a channel that genuinely predicts	+0.397 (sd 0.029)
a channel carrying nothing	−0.016 (sd 0.091)
κ -STRUCTURE	
a consistent per-line modulation	+1.173
an idiosyncratic interaction	−0.002

SCOPE LIMIT carried into § 4.8

Neither the planted rows nor the calibrated repeat licenses reading $1 - \mathcal{T}$ as a drug $\times$ cell-line interaction share. The quantity measures the fraction not shared across the compared contexts, and in the canonical run those contexts differ in more than cell line.

TWO OF THESE CHECKS CHANGED THE INSTRUMENT. The count-matched null originally drew random partners from a pool *excluding* the channel’s own selections, which depletes the null of the partners the channel chose and anti-correlates the two arms. And the useless-channel gap has a standard deviation of roughly 0.09 on one small world, so the original validation, asserting that a single draw fell within 0.10, tested luck; it now averages 24 worlds. That same variance is why the channel results are gated on a clustered interval and not on a point estimate.

A THIRD DEFECT WAS FOUND AFTER THE CANONICAL RUN.

The *retrieval* pool, whose partners’ mean residual forms the prediction,

was every retained condition in the cell line except the target's own drug, so a condition held out of training could enter the prediction for a held-out drug. That is not what § 4.12 says the gate was intended to do. The code now restricts the pool to the split's training conditions, logs its size, and never falls back silently.

The training subset is 2,523 of a retained total logged as 4,131 conditions, so the old pool was around 64% larger than it was entitled to be. What makes it a defect is that a held-out condition could enter at all, which holds at any tally. Every channel-gate figure in the main text is from the corrected re-run; the planted worlds are unaffected. The repair moved the channels in no common direction. Target fell, mechanism rose, chemistry roughly halved, so the pre-fix figures are not quoted alongside them.

CORRECTION The channel gate's retrieval pool, repaired and re-run after the canonical run. § 4.12

A.2 The absent-gene convention

A predicted cell sentence lists only the genes it expresses, so some panel genes reach the scorer with no rank and the scorer invents one. Does the invented rank decide the result? Two conventions were carried side by side. The choice does not matter, and it never reaches the metric the canonical results rest on.

worst assigns absent genes the last rank G ; francesca assigns them a fixed middle rank $G/2$. These are the extremes of the plausible range, so a convention that mattered anywhere would show up between them.

It does not. In an earlier spike-in run, not the same-plate titration of § 4.1, spike-in DE- Δr reads 0.952 under the first and 0.954 under the second, and no baseline or positive control scored under both moves by more than 0.01. That comparison was a one-off diagnostic, never written to an artifact, so it is a check that was run and not a reproducible result.

HOW FAR THE CONVENTION REACHES. It is an input to the rank-based scores DE- Δr , panel- τ and nir_rank, which fills absent genes at the last rank like the rest. It does not reach the NIR figures the canonical results rest on, scored by instruments that never receive the fill. All body figures use worst.

SEE § 4.1 for why nir_rank never entered the graded family

A.3 The swap-distance sweep

If the scramble control swaps in a drug that behaves like the real one, the null it produces says nothing. The instrument is an earlier, continuous version of that control, measuring the scramble gap against swap distance. It bounds a large effect; it does not establish a null, and only one of its two columns carries that bound.

None of these numbers is canonical. This is a separate, earlier run. Its columns are the single-cell and optimal-transport checkpoints, not the residual arm; it was scored on the unseen-drug tier of a different evaluation set; and its swap partners were not restricted to plate, whereas the canonical run draws within plate. Both artifacts, `scramble_sweep_single.json` and `scramble_sweep_T2.json`, record `max_new_tokens=1200`, against the 1400-token budget of the later optimal-transport result.

swap distance How far the swapped-in drug's response lies from that of the drug it replaced.

TABLE A.2 No bin in either model has an interval excluding zero in this historical 1200-token run, and the trend correlations are ≈ 0 . 200 (drug, swap) pairs over 20 cell lines, $n = 40$ per bin. The single-cell column bounds a large distance effect here. The optimal-transport column does not carry into the canonical result, where the same checkpoint clears zero at 1400 tokens under the within-plate stratified test (§ 4.9).

	SINGLE-CELL	OPTIMAL-TRANSPORT
BIN (SWAP DISTANCE)	MODEL	MODEL
0, nearest (2.75–4.03)	−0.046	−0.001
1 (4.03–4.60)	+0.060	+0.031
2 (4.60–4.93)	−0.006	+0.017
3 (4.93–5.32)	−0.004	+0.043
4, farthest (5.32–6.25)	−0.019	−0.005
trend, $\text{corr}(\text{distance}, \Delta_{\text{scr}})$	+0.051	+0.012

WHAT THE SWEEP CAN AND CANNOT SUPPORT. The five bins are drawn in Figure 4.4. $n = 40$ per bin gives half-widths of 0.06–0.10 in the single-cell column and 0.03–0.06 in the optimal-transport

column, so the sweep excludes a large effect without establishing a null. The later budget-matched result withdraws the optimal-transport column as evidence of a null, that verdict being generation-budget dependent and non-null at 1400 tokens (§ 4.9).

Only the single-cell column remains a bound from this earlier design, and it is a bound and not a null. At this sample size it excludes a large dissimilarity effect and leaves a small one open. The optimal-transport column has been withdrawn as evidence of a null, so nothing in this section rests on it.

SEE § 4.2 for the stratified scramble that replaced this design

CORRECTION The optimal-transport column of this sweep, withdrawn as evidence of a null. § 4.9

SCOPE LIMIT carried into § 4.2

A.4 Training adequacy

A negative result invites the objection that the model was simply not trained enough, and that objection deserves an answer from the logs. Two instruments give one, what the loss was doing while the learning rate was still large, and what a much longer schedule does to held-out cross-entropy. Under this schedule, longer optimisation does not rescue it.

TABLE A.3 Five arms share a base checkpoint and hyperparameter set. Learning rate 10^{-5} , batch size 1, gradient accumulation 16, maximum length 8192, bf16, weight decay 0.01, warmup ratio 0.03 throughout. Among the three target constructions in § 4.6, rank-proxy consensus has the weakest training evidence: its schedule ended early while loss was flat. These are distinct constructions, not endpoints of a regularisation sweep. At the residual arm's generation budget against a neutral partner, the optimal-transport gap excludes zero (§ 4.9). The single-cell arm therefore provides the strongest negative evidence among the three, while small effects remain open. These five arms (single_cell, consensus, ot_T2, residual, residual_holdout) are also the complete family over which the identity test of § 4.5 takes its multiplicity correction, five arms by up to seven depths and 26 measured tests, the family behind the q -values quoted there. One qualification attaches to the residual row. Its step count is the rebuilt target build's, on which the canonical behavioural results are scored, while the identity test of that arm ran on the earlier target build.

ARM	TARGET	TRAINING	NOTE
single cell	the observed treated cell	1 epoch, 42,198 steps	steps from the training state; loss log not retained
consensus	rank-proxy consensus	≈ 60% of an epoch	tier-1 NIR unscorable
optimal transport (T2)	barycentric, $\varepsilon=0.5$	1 epoch, 23,842 steps	
residual	signed residual	1 epoch, 9,231 steps	rebuilt build's step count
residual (holdout)	as above, 738 withheld	1 epoch	

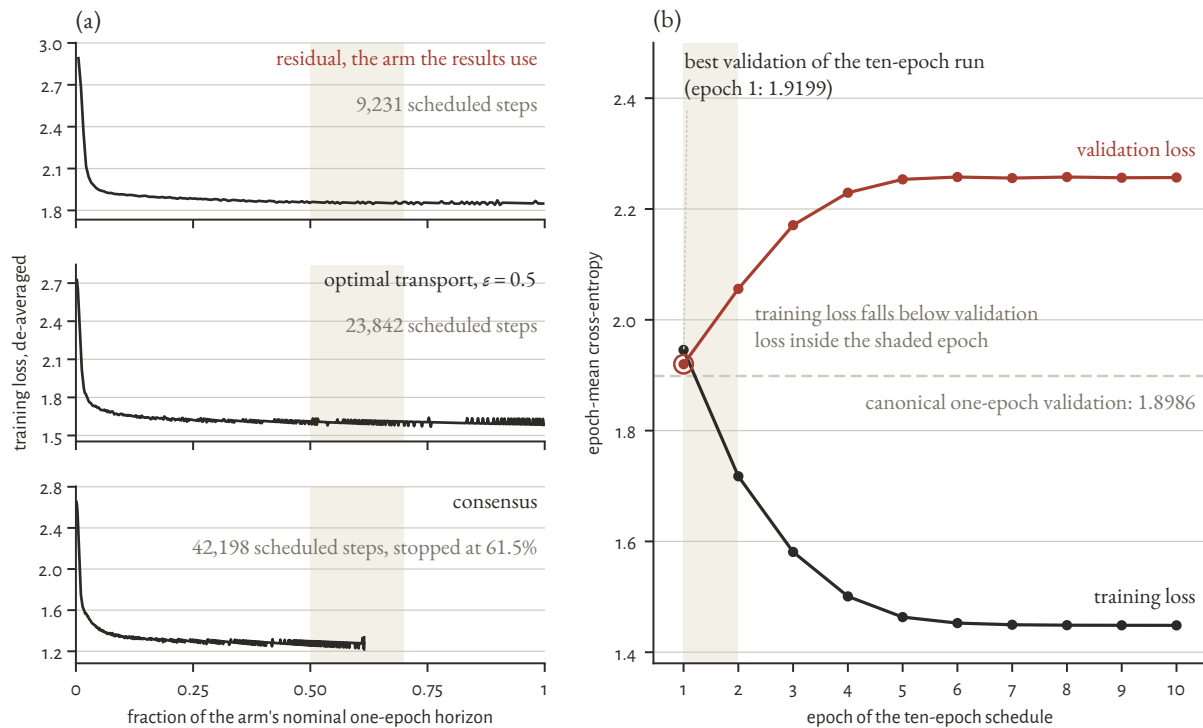


FIGURE A.1 The canonical short-run arms flatten, while a separate longer residual schedule overfits on held-out cross-entropy. (a) De-averaged training loss against fraction of the nominal one-epoch horizon, three target constructions, each on its own y -axis: absolute losses are not comparable between arms because their target strings differ, so no shared scale is offered. The residual and optimal-transport arms complete their epoch; consensus was scheduled for one but terminated by the wall-clock at 61.5%, where its line stops. Three superseded residual builds are omitted, as is the single-cell arm, whose training log was not retained. Every arm runs a cosine schedule annealing to ≈ 0 ; the shaded band marks 50–70% of the horizon, where the residual arm's learning rate is still 22–52% of its post-warmup peak of 10^{-5} and its de-averaged loss travels only from 1.859 to 1.854. (b) Epoch-mean losses for the ten-epoch residual schedule (nominal horizon 92,318; 92,310 optimizer updates). The ringed point is that run's best validation loss, at epoch 1. The dashed line is the canonical one-epoch run's validation loss on the same 3,600-example file, 1.8986, which the ten-epoch run does not reach at any epoch. Training loss is above validation at epoch 1 and below it at epoch 2, so the crossing falls inside the shaded epoch; where inside it is not logged, and no point is drawn for it. Two things this figure does not establish: that all possible learning-rate schedules are ruled out, and that NIR worsens with additional epochs — no long-run checkpoint was scored with NIR.

SCOPE LIMIT carried into § 4.5

The residual arm's two measurements are not on the same weights. The step count recorded above is the rebuilt target build's, which is the build the canonical behavioural results are scored on, whereas the identity test of that arm ran on the earlier target build. A behavioural result for that arm and an activation probe of it therefore come from two different builds, and this qualification travels with any statement that pairs them.

HOW PANEL (A) MUST BE READ. Two properties of the one-epoch logs behind Figure A.1 decide it. The trainer records a *cumulative* epoch mean, not an instantaneous loss, and that flattens by arithmetic whatever the model is doing, so every series is de-averaged to its per-

window value before plotting. And the cosine schedule anneals the learning rate to ≈ 0 by the end of the epoch, 2.96×10^{-10} at the last logged step against a post-warmup peak of 10^{-5} , so the final third of each curve is flat partly because the step size has gone to zero, and is not independent evidence of anything.

What is fundamental is the band between 50% and 70% of the epoch, where the learning rate is still 22–52% of peak, enough to move the weights, and the residual arm’s loss has already flattened, travelling only from 1.859 to 1.854. Its final held-out loss, on a matched validation file of 3,600 examples, is 1.8986 against a training loss of 1.8854.

PANEL (B) TESTS ORDINARY LONGER OPTIMISATION DIRECTLY. Both residual logs agree on base model and tokenizer, training and validation files and counts, seed, effective batch, and peak logged learning rate; weight decay and AdamW betas are not logged and are not claimed as verified matches (`training_adequacy_long.json`). The ten-epoch run has a nominal horizon of 92,318 but executes 92,310 optimizer updates, accumulation resetting at every epoch boundary so that each epoch contributes $\lfloor 147,710/16 \rfloor = 9,231$. It is a separate longer schedule, not a continuation; changing `num_epochs` stretches the cosine schedule across all ten epochs.

The result runs the wrong way for the objection. Validation loss is best at epoch 1 (1.9199), worsens to 2.0559 and 2.1708 at epochs 2 and 3, and ends at 2.2568, while training loss falls from 1.9456 to 1.4485. The canonical one-epoch run remains better on the same validation file (1.8986).

SCOPE LIMIT carried into chapter 5

That result does not show that every possible schedule would fail, and no long-run checkpoint was scored with NIR, so it says nothing about whether NIR itself worsens with additional epochs. Together with the one-epoch traces, it excludes undertraining as a *trivial* explanation, not as a general one.

A.5 Three failed identity-test designs

The substitution test the body reports is the fourth design, and the three before it each produced a result this thesis once quoted as a finding. The instrument here is the algebra of each failure, what the control arm was actually doing.

All three failures below are failures of the control arm, not of the read-out. Figure A.2 draws the geometry each of them misses, and its three insets are the three designs in the order they are argued here.

FAILED DESIGN ONE, MISMATCHED GEOMETRY. The first test overwrote drug A’s code with drug B’s inside a ≤ 10 -dimensional purified subspace, and compared the resulting KL against a random vector of identical norm drawn in the *full* 2048-dimensional stream. Roughly 99.5% of the control’s squared norm then lay where the intervention never reached, including the cell-line directions at ≈ 0.6 variance share. “The swap moves the prediction less than noise” is what that geometry predicts, whatever the readout does.

FAILED DESIGN TWO, ABLATE-THEN-ADD. The second drew the control inside the subspace, which looked like a repair. The hook ablates before adding, so what is delivered is (added — removed), and an uncentred class mean shares its global-mean component with the thing removed. The substitution arm was then close to a restoration, the control arm a real displacement. Simulated with no model present, the control delivers $5.3 \times$ the perturbation in 100% of draws, predicted

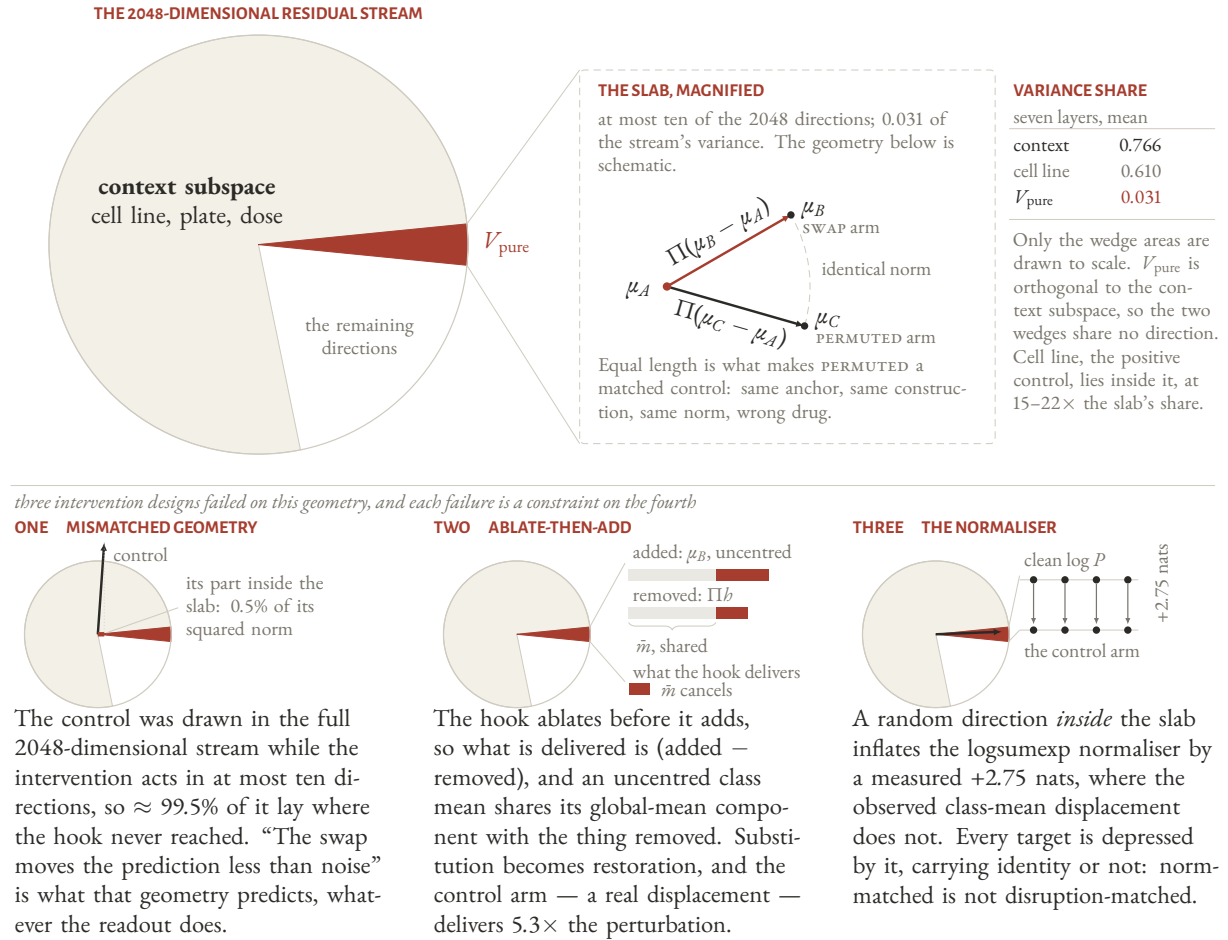


FIGURE A.2 The purified drug slab is a thin wedge of the residual stream beside the context subspace, and each of the three failed identity tests turns on that geometry. The disc is the 2048-dimensional residual stream at one layer. The two wedge areas are the only scaled marks in the figure, drawn to the measured subspace variance shares in `workspace_probe.json` (twelve drug classes, forty cells each, chance 0.083), averaged over the seven probed layers: 0.766 for the context subspace of cell line, plate and dose, and 0.031 for the purified drug slab V_{pure} . Cell line alone, the positive control, reads 0.610 and lies inside the context subspace, so it is quoted as a number rather than drawn as a third wedge. V_{pure} is what remains after the drug slab — spanned by at most eleven directions, since it is built from twelve centred class means, of which the top ten are kept — is orthogonalised against that context subspace, which is why the two wedges share no direction. The unshaded remainder closes the disc and carries no number. Everything else is schematic and has no scale: the slab is 11.3° wide and cannot hold three class means at true size, so its interior is magnified, where the SWAP and PERMUTED displacements are drawn at equal length because equal length is what makes PERMUTED a matched control. The one exception among the insets is inset one's angle, which is drawn true: a direction drawn isotropically in 2048 dimensions puts, on average, 10/2048 of its squared norm in a ten-dimensional slab, the appendix's $\approx 99.5\%$ lying outside it. No decode accuracy, layer axis or KL divergence appears here; those are Figure 4.7. The three insets are the failed designs of § A.5.

gap $2\langle \mu_B, u \rangle = 9.179$ against an observed 9.162, with μ_B the injected class mean and u the cell's component in the slab. The retracted result was geometry, not biology.

CORRECTION The ablate-then-add identity result, retracted as an artefact of the hook's own algebra.

FAILED DESIGN THREE, THE NORMALISER. The third injected the displacement and scored $\Delta \log P(g_B)$ against a norm-matched random direction, and was confounded the opposite way. A random direction inside the slab inflates the logsumexp normaliser (measured +2.75 nats) where the observed class-mean displacement does not, so every target is depressed under the control arm whether or not it carries identity. With the scored target stripped of all identity information, 36% of the effect still stood. Norm-matched is not disruption-matched.

That argument also retires the simplest *injection* test, “how far did the prediction move”. Distance moved is what norm-matching equalises, so the test has no power in precisely the world where the readout does read the drug. It does not reach the behavioural output-invariance instrument of § 4.2, which perturbs the prompt and matches no norms; that instrument’s own limitation, registering whether the generated text moves and not whether it moves correctly, is stated where it is used.

THREE FURTHER GUARDS ARE ENGINEERING, NOT INFERENCE. A *drug-proxy screen* refuses any context label that is a near-bijection with drug; it caught `dose_float` holding a sample identifier in the residual builders, a label that would have deleted the drug axis had the builders orthogonalised against it. An *injected-norm diagnostic* reports the displacement against the cell’s centred slab component (0.69–1.40), so the injection is the size of the thing it displaces and a null cannot be blamed on a whisper. A *selftest with teeth* runs three planted worlds (drug inert, drug read, and drug inert behind a readout sensitive to magnitude and not direction) through the headline code path, separates them (−0.042, +1.520, −0.010), and failed four times in development, each failure a real defect the guard list now covers.

SCOPE LIMIT carried into § 4.5

The third planted world is not a normaliser control. Each planted response is a single token, so nothing diverges, and the state-dependent part of the normaliser is precisely what the selftest cannot exercise. What separating the three worlds licenses is the code path the selftest runs them through.

A.6 Metric self-test

A metric audit whose own instrument is untested proves nothing, and the exploit it turns on was demonstrated with a single construction. The instrument here is a second, independent zero-information baseline put through the same framework. It saturates the standard metric too.

The self-test does the deterministic things first (a perfect prediction reaches the metric's maximum, a zero shift and a flat truth return no score), then the thing that matters. It asserts the exploit under a *second, independent* zero-information baseline, where a leave-one-out mean scores $DE-\Delta r = 1.000$ while NIR refuses to place it above chance. The coded assertion is $NIR \leq 0.6$; the recorded value is 0.000.

WHAT THAT 0.000 IS NOT. It is not the metric's chance value, 0.500 (§ 4.1), which is what the generic arm scores in Table 4.12. Nor is it the untied degeneracy that section describes, since the self-test uses the same half-tie convention as everything else. It is a property of this construction. The mean of the five other drugs resembles those five more closely than the held-out one, so it loses every comparison.

Two unrelated constructions saturate the standard metric, and the discrimination metric places neither above chance. One would leave the standard metric merely unlucky; two make the exploit a property of the metric.

Reference apparatus

Every term, symbol and convention that carries weight across more than one chapter is fixed here.

Where an interval exists, a number is quoted as a value followed by its bracketed lower and upper bounds, a two-way cluster-robust 95% interval unless stated otherwise.

Two measurement frames, one chance value

The two frames score different objects against different competitors, and they both put chance in the same place. Comparing a number from one with a number from the other is the most expensive misreading available in this thesis.

In the ■ **EXPR** *expression frame* the prediction is a cell profile, scored against other drugs on the same plate. In the ▲ **RESID** *residual frame* it is the drug-specific residual, what survives once the generic programme has been subtracted, scored against other drugs in the same cell line. In both frames the *comparison set* J , the conditions a prediction is ranked against, fixes chance at 0.50, and J is declared explicitly at every score.

WHERE THE TRAP IS. Because chance is 0.50 in both frames, two numbers that look alike invite subtraction. They may not be subtracted. The residual frame's within-well precision reference is 0.854 where the expression frame's is 0.576, so the 0.537 in one frame and the 0.498 in the other are not two attempts at one task (Table 4.12 and § 4.2).

SEE Figure 4.1 maps the nine beats in order

FRAME ■ **EXPR** ▲ **RESID** in every running head

SEE Figure 3.2 draws the residual construction

HOW TO TELL WHICH IS IN FORCE. Three redundant cues say so. Every section declares its frame in its opener as a chip, ■ **EXPR** or ▲ **RESID**, shape and word before colour so that the cue survives greyscale; the running head repeats it on every page; and a margin cue marks any change of frame inside a section. A head naming no frame asserts nothing in either, as here and in the apparatus of chapter 3.

A distance to a reference is comparable within a frame and not between frames. The same restriction holds *within* a frame across supports: populations selected by different rules are not one population. Where this thesis quotes two numbers side by side it names the population each was computed on; where no population is named, they are not being compared.

SCOPE LIMIT carried into every chapter

Three registers of qualification

There are exactly three registers and no fourth.

A scope limit fences in what a number *licenses*, and the reader has to carry it forward. The test: would a later section be entitled to say something different if this sentence did not exist? Its tag names where the limit carries to.

register The three kinds: a limit carried forward, a correction on record, and ordinary inline care.

A correction marks a withdrawal this thesis makes against itself. It may concern a claim or a defective piece of apparatus — a voided control, a mis-resampled interval, a replaced instrument, a leaking partner pool. Each mark of either kind stands where the withdrawal is argued.

Routine qualification stays inline and unmarked, in the sentence it qualifies, and most of the qualification in this thesis is of this kind. It is not a lesser statement for being unmarked, only one the reader need not carry anywhere.

TWO PHRASES THAT ARE NOT SYNONYMS. not established and shown to be zero are different statements. A measurement that fails to separate two arms has established nothing about their difference. It

has not shown the difference to be zero. Where an interval spans zero, this thesis says which of the two it means.

AND TWO MORE THAT ARE NOT. Excluding zero is not the same as clearing a pre-specified relevance margin, and spanning zero is not evidence of equivalence. Two margins are pre-registered, on different quantities and different instruments.

relevance margin The pre-set 0.03, in NIR units, that a channel's interval has to clear on the gate of § 4.12 before the channel is called live.

materiality margin The pre-registered 10% of the model's own clean preference gap, on the identity test of § 4.5. An effect inside it is too small to be called material.

Where a pre-registered margin exists the verdict vocabulary is fixed in advance, three states plus a fourth for a control that could not be built.

live The interval clears the pre-set 0.03 relevance margin on the clustering axis appropriate to the claim. The axis is written into the verdict, because the word is not usable without it.

closed The interval lies wholly below that margin.

inconclusive Neither. This includes an interval that excludes zero but straddles the margin, and it is not equivalence with the null.

untestable The plate-matched null spans too few cell lines to be built here. "We could not build the control" must never be written as "closed".

Reading the page

A number a sentence *asserts* goes in the body, always, with its interval, its comparison set and its reference beside it. A number it merely *recalls* may go in the margin column, bare, with a pointer to where it was established. Delete every margin note and the body must still be complete and correctly qualified.

margin column Definitions, landmark constants, frame cues and pointers; never an assertion.

FIVE SHAPES. A section *opener* states the question and the instrument built to answer it, in words, with no numbers; the beat strip that goes with it is the investigation's nine beats with the current one filled, and no cell filled means the section is not one of the nine, as here. A *claim block*, hairlined above and below, carries the verdict in language that could be shown wrong. § 4.13 closes without one, deliberately: it asks which span of the prompt the model reads, cannot answer, and ends in the open question. A *definition* sits in a tinted panel with no rules. A *scope limit* and a *correction* are the two marked hedge registers above.

Symbols

Fixed for the whole document, never redefined in a section.

NIR *Normalised inverse rank.* The fraction of a declared comparison set whose truth a prediction resembles less than it resembles its own, with ties counted at one half. Chance is 0.50 under the exchangeability condition audited in § 4.1. Its meaning is not complete without a frame: the similarity σ and the comparison set J are fixed per frame in Table 4.2. In the expression frame the vectors compared are expression recon-

structed from ranks through a fitted map, and not the ranks the model emits.

DRF	<i>Dynamic range fraction</i> . Not a performance score but a meta-metric. It fixes a scale with an uninformative predictor at zero and a perfect predictor at one, then asks how far up that scale a candidate metric places a positive control of known quality (the interpolated duplicate in the expression frame, a raw split-half in the residual frame, chapter 3).
DE- Δr	The field's standard differential-expression delta-correlation, the correlation between predicted and true change from control, restricted to the genes with the largest true change.
panel- τ	Kendall's τ between predicted and true rank over the full 946-gene panel. It selects no gene subset.
Δ_{scr}	The scramble gap, NIR of the model minus NIR of the scramble.
$\mathcal{T}$	The disattenuated transfer coefficient, measuring how much of a drug's response is shared across cellular contexts. It is a similarity coefficient, not a variance share.
r	The drug-specific residual, $r(d, c)$: what is left of the shift once the generic programme has been subtracted from it. The scored object in  RESID.
s	The treated-minus-control shift, $s(d, c)$: the treated pseudobulk minus its plate-matched control pseudobulk, and nothing else. An arithmetic object, not a verdict about the drug.

$\|\cdot\|$ *Euclidean norm.* $\|x\|^2 = \sum_i x_i^2$, the L^2 norm, throughout; no other norm appears, and nothing is centred first, so a squared norm is a sum of squares about the origin and not a variance. A *squared-norm share* is one term of $\|s\|^2 = \|r\|^2 + \|g\|^2 + 2\langle r, g \rangle$, with $g = s - r$ the generic programme, divided by $\|s\|^2$: the three shares sum to one by construction, cross term included, and each is a mean over conditions.

[END_CELL] The registered atomic token that terminates a cell sentence.

[DOWN] The registered atomic token marking the up/down boundary in a residual target. Left unregistered it splits into sub-words and the boundary ceases to be a clean symbol.

■ EXPR Page furniture, not a quantity. The expression frame is in force here.

▲ RESID Page furniture, not a quantity. The residual frame is in force here.

Terms

Defined once, used in that sense everywhere.

SEE chapter 3 builds the apparatus these terms name

ONE REFERENT, ONE WORD. A synonym reads as a second object.

These are the terms used throughout the thesis.

drug The perturbagen, whatever its chemistry. *Compound*, *molecule* and *agent* are not used as synonyms for it; *small molecule* appears only where the class of thing the atlas contains is what is meant.

cell line The biological context. Two words as a noun, hyphenated only as a compound adjective, as in *cell-line-specific*. The bare *line* stands in where the paragraph has already fixed the referent.

shift, generic, residual Three objects, never interchangeable: the *shift* is treated minus control, the *generic* is the drug-agnostic mean response subtracted from it, and the *residual* is what remains. *Response* keeps its ordinary biological sense and *effect* its ordinary one; neither stands in for the residual.

prediction What the model produces and what gets scored. *Generation* names the act of decoding and only that. *Output* is not used as a noun for it, and survives only inside the fixed instrument name *output invariance*, which names a test and not the thing predicted.

readout Two senses, and the surrounding sentence fixes which. Biologically, what the sequencing reports. Of the model, the decoding stage that turns the residual stream into a sentence, the second term of the *representation and readout* distinction chapter 5 turns on, for which *decoder* is the alternative word. Neither sense names the prediction, which is what that stage emits.

prompt What the model is given at inference: a cell sentence, the drug name, the dose and the mechanism field.

profile A ranked expression vector. Not a synonym for *sentence*, which is the encoding of one cell.

UNITS OF ANALYSIS.

condition A (drug, cell line, plate, dose) tuple. It is the unit that gets scored.

treated well The physical treatment assignment. Conditions nest inside it, so a well is not itself a condition and the two are not counted against each other.

batch unit The pair (cell line, plate), never the plate label alone, because plates are numbered within cell line and not globally.

pseudobulk The mean expression profile of a group of cells. A *control pseudobulk* is the same object over plate-matched DMSO cells.

dose A molar concentration resolved from `drugname_drugconc`, carried as a concentration and never as a unitless float.

cell sentence A cell written as its expressed panel genes in descending order of expression, terminated by `[END_CELL]`. The encoding discards expression magnitudes and retains ranks.

FRAMES AND COMPARISON SETS.

expression frame Scores a predicted cell profile against other drugs on the same plate; conditions grouped by (cell line, plate).

residual frame Scores a predicted residual against other drugs in the same cell line, after the generic programme is subtracted; conditions grouped by cell line.

comparison set *J* The conditions a prediction is ranked against. It fixes chance at 0.50 in both frames and is declared wherever a score is quoted.

within-well split-half precision reference The score obtained when a disjoint cell subset of the same treated well is substituted for the prediction, recomputed for every stratum scored. One estimator, not several; what varies between settings is the frame, the comparison set and the cell budget. It is a sampling-precision estimate, optimistic relative to true biological reproducibility: not a biological replicate, and not the maximum predictable fraction of biology.

ceiling A shorthand for the reference above, used wherever a result is stated as a distance from it. The one exception in this document is the DRF positive control of § 4.1, which is not a row of Table 3.2.

identifiability A property of a single condition, not of a drug, and distinct from the aggregate references above. The condition’s own within-plate split-half NIR against its plate-mates clears 0.8.

TARGET CONSTRUCTION.

generic The drug-agnostic mean response subtracted to form the residual, operationally the mean shift over the other drugs at the stated scope. It is conventionally read as the stress and cell-cycle programme common to all drugs, a reading used for intuition and nowhere tested.

residual target What remains once the generic is removed. Built *split-before-fit* and *leave-one-drug-out*.

split-before-fit The holdout is fixed from metadata before any statistic is estimated, so no held-out target is defined partly by other held-out conditions.

leave-one-drug-out No condition’s generic contains its own drug, so the two sides of the split are scored against identically defined targets.

scope The set of drugs the generic mean is taken over, either `cell_line` or `plate`. The choice is consequential.

quota The explicit per-split sampling rule, used because a uniform draw reproduces the pool’s proportions and starves the held-out splits.

training arm A model variant differing from the others only in its training target or objective; five are used, cold-started from the same base checkpoint with identical hyperparameters, so that it is the only

variable. The bare word *arm* also names any one side of a paired comparison (a split, a run, a channel, a baseline), and is qualified wherever the surrounding sentence does not fix the sense. An unqualified *arm* is not necessarily one of the five.

BASELINES, SPLITS AND REFERENCES.

control_copy The baseline that predicts the untreated cell unchanged.

It carries zero drug information by construction, and it is the actual zero-shift predictor.

revert_center A predictor that places every gene at the middle rank.

It carries no information about drug, cell or biology, and it is *not* zero-shift.

drug lookup This drug's mean residual, estimated from training conditions in other cell lines. It needs no model, no prompt and no checkpoint.

moa_lookup The same construction over drugs sharing this drug's mechanism annotation, on its own narrower support.

channel An annotated drug property — protein target, mechanism, chemistry — an unseen drug might be reached through. For a channel X , a drug's residual is predicted as the mean residual of the other drugs in the same cell line sharing property X with it: `moa_lookup` generalised from mechanism to any annotation. It is called live, closed, inconclusive or untestable against a plate-matched null differing in the channel alone (§ 4.12).

scramble The prompt left byte-identical except that the drug name and mechanism are replaced by another drug's, while the control cell and the scoring truth are unchanged. It is the only manipulation that isolates drug use. Its near, orth and opposite strata are fixed by the measured cosine between the partner's residual and the target's,

so a stratum's neutrality is an empirical property of these data and not a consequence of its construction.

tiers The three evaluation tiers of the broad characterisation, defined by what the training set contained. *Tier 1*: the (drug, cell line) pairing was seen, so it measures fit, not generalisation. *Tier 2*: the drug never appeared (Leave-Drugs-Out). *Tier 3*: drug and cell line were each seen but never together (Leave-Pairs-Out).

train / unseen_combo / unseen_drug The residual arm's own three-part split. `unseen_combo` is an unseen *well* of a seen drug, so it tests recombination and not full novelty.

held out Held out from Tahoe fine-tuning, with the mechanism annotation still supplied in the prompt. Not absent from the model's prior world knowledge. Every unseen-drug result must be read under that narrower scope.

Statement on AI assistance

AI tools, including OpenAI Codex [40] and Anthropic Claude [41], were used to assist with formatting, wording and rephrasing parts of the text, and creating figures for visualisation. Further uses included generating scripts for submitting jobs to high-performance computing (HPC) systems, preparing documentation, bibliographic checks, and selected consistency checks of code and reported results. The author remains responsible for the final text, figures, analyses and conclusions.